1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

D8 MLB ULTIMATE

LAST_MODIFIED=Tue Jul 3 15:14:05 2012

REV	ECN	DESCRIPTION OF REVISION	CK APPD	DATE
				2012-07-03

DRAWING
TITLE=K72
ABBREV=DRAWING
LAST_MODIFIED=Tue Jul 3 15:14:05 2012

DRAWING TITLE
SCH,D8,MLB ULTIMATE

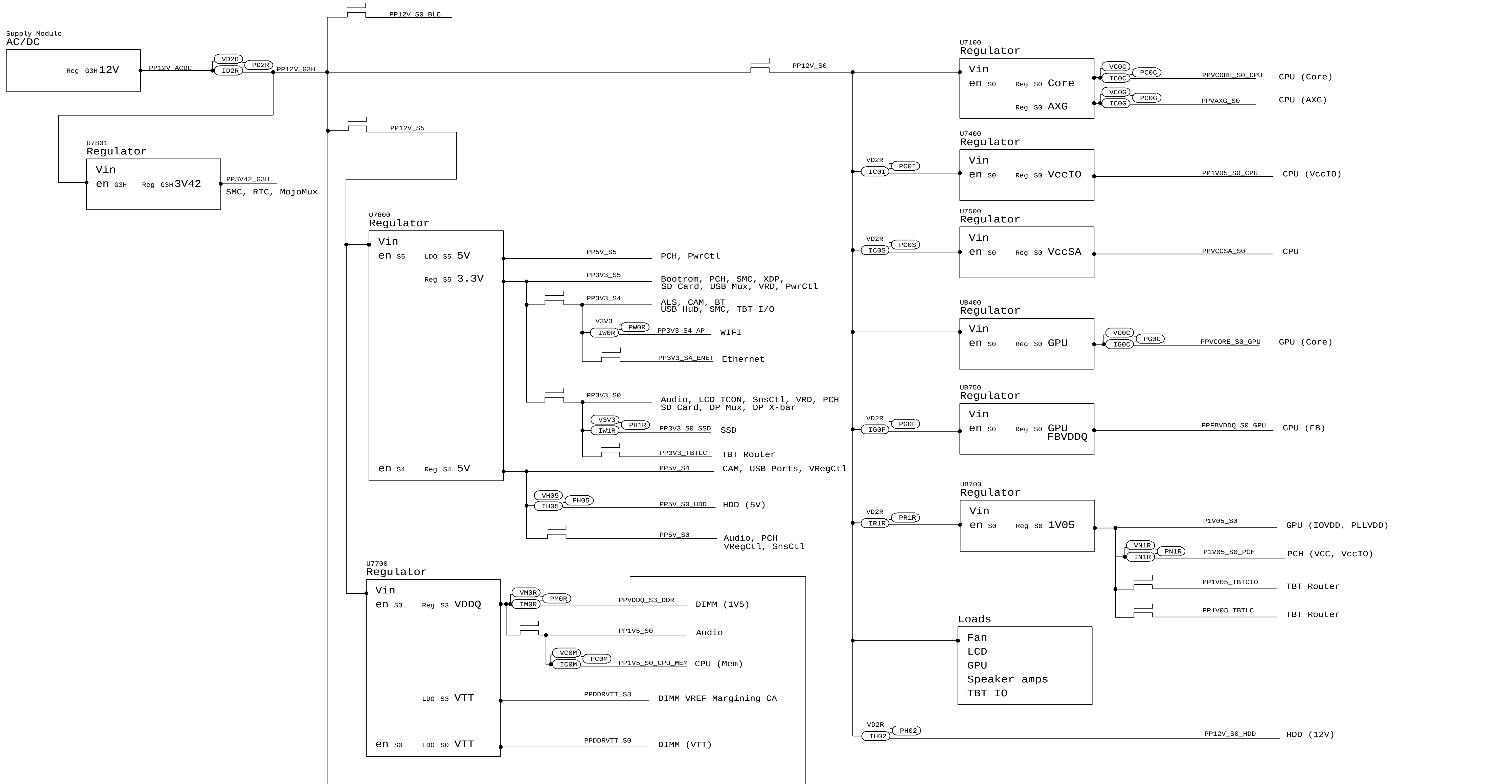
Apple Inc.

DRAWING NUMBER 051-9505 SIZE D
REVISION 6.1.0
BRANCH

System Block diagram can be found on Kismet

PATH: KISMET > K70/72 > BLOCK DIAGRAMS > K72 BLOCK DIAGRAM

PAGE TITLE	System Block Diagram
Apple Inc.	DRAWING NUMBER 051-9505 SIZE D
	REVISION 6.1.0
NOTICE OF PROPRIETARY PROPERTY:	BRANCH
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	PAGE 2 OF 144
	SHEET 2 OF 123



High-side Component Total Power Keys

PCTR = 1.176 * (PC0C + PC0G + PC0M) + PC0S + PC0I

PGTR = 1.176 * PG0C + PG0F + 5.7 (GK104/GK107_BLENDED_CONSTANT)

PH0R = PH02 + PH05

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

Power Block Diagram

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

PAGE 3 OF 144

SHEET 3 OF 123

8 7 6 5 4 3 2 1

BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-3662	PCBA,MLB,ULTIMATE,3.4G,GTX,SAM,2GB,D8	D8_COMMON,D8,CPU:4C_3P4GHZ,GPU:104GTX,FB:2G_SAMSUNG,EEEE:F0V5
639-3950	PCBA,MLB,ULTIMATE,3.2G,GTX,SAM,2GB,D8	D8_COMMON,D8,CPU:4C_3P2GHZ,GPU:104GTX,FB:2G_SAMSUNG,EEEE:F49R
639-3560	PCBA,MLB,ULTIMATE,3.4G,GT,SAM,1GB,D8	D8_COMMON,D8,CPU:4C_3P4GHZ,GPU:104GT,FB:1G_SAMSUNG,EEEE:DYW3
639-3949	PCBA,MLB,ULTIMATE,3.2G,GT,SAM,1GB,D8	D8_COMMON,D8,CPU:4C_3P2GHZ,GPU:104GT,FB:1G_SAMSUNG,EEEE:F49P
639-4087	PCBA,MLB,ULTIMATE,3.4G,GTX,HYN,2GB,D8	D8_COMMON,D8,CPU:4C_3P4GHZ,GPU:104GTX,FB:2G_HYNIX,EEEE:F64W
639-4091	PCBA,MLB,ULTIMATE,3.2G,GTX,HYN,2GB,D8	D8_COMMON,D8,CPU:4C_3P2GHZ,GPU:104GTX,FB:2G_HYNIX,EEEE:F652
639-4086	PCBA,MLB,ULTIMATE,3.4G,GT,HYN,1GB,D8	D8_COMMON,D8,CPU:4C_3P4GHZ,GPU:104GT,FB:1G_HYNIX,EEEE:F64V
639-4090	PCBA,MLB,ULTIMATE,3.2G,GT,HYN,1GB,D8	D8_COMMON,D8,CPU:4C_3P2GHZ,GPU:104GT,FB:1G_HYNIX,EEEE:F651
085-4435	PCBA,MLB,DEV,D8.ULTIMATE	DEVELOPMENT,D8_DEVEL

Bar Code Labels / EEEE #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7122	1	MLB LABEL,48.0X4.8	EEEE_DYW3	CRITICAL	EEEE:DYW3
825-7122	1	MLB LABEL,48.0X4.8	EEEE_F0V5	CRITICAL	EEEE:F0V5
825-7122	1	MLB LABEL,48.0X4.8	EEEE_F49P	CRITICAL	EEEE:F49P
825-7122	1	MLB LABEL,48.0X4.8	EEEE_F49R	CRITICAL	EEEE:F49R
825-7122	1	MLB LABEL,48.0X4.8	EEEE_F4MW	CRITICAL	EEEE:F4MW
825-7122	1	MLB LABEL,48.0X4.8	EEEE_F4TY	CRITICAL	EEEE:F4TY
825-7122	1	MLB LABEL,48.0X4.8	EEEE_F64W	CRITICAL	EEEE:F64W
825-7122	1	MLB LABEL,48.0X4.8	EEEE_F652	CRITICAL	EEEE:F652
825-7122	1	MLB LABEL,48.0X4.8	EEEE_F64V	CRITICAL	EEEE:F64V
825-7122	1	MLB LABEL,48.0X4.8	EEEE_F651	CRITICAL	EEEE:F651

BOM Groups

BOM GROUP	BOM OPTIONS
D8_COMMON	COMMON,ALTERNATE,D8_COMMON1,D8_PROGPARTS
D8_COMMON1	XDP,RSMRST:GATE,SPEAKERID,VREF:CPU,TBTHV:P12V,PRE_BOOST:Y
D8_PROGPARTS	SMC:PROG,BOOTROM:PROG,TBTROM:PROG,CIVROM:PROG,CAMROM:PROG,BLCMCU:PROG
D8_DEVEL	XDP_CONN,LPCPLUS,VREFMRGN:EXT,BKLT_PWM,DEVEL_AUDIO,TEMPSNSDEV

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4356	1	IVB,SR0T8,PRQ,N1,3.2,77W,4+1,1.1,6M,LGA	CPU	CRITICAL	CPU:4C_3P2GHZ
337S4247	1	IVB,SR0PK,PRQ,E1,3.4,77W,4+2,1.15,8M,LG	CPU	CRITICAL	CPU:4C_3P4GHZ

ASICs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4277	1	IC,PANTHER POINT,C1,SLJC7,PRQ,BD82Z77	U1800	CRITICAL	
338S1113	1	IC,TBT,CR-4C,B1,PRQ,288 FCBGA,12X12MM	U3600	CRITICAL	
343S0616	1	IC,BCM57766A1,ENET&SD,8X8	U3900	CRITICAL	
337S4333	1	IC, GPU, NV GK104 7-4-PS-A2	UA000	CRITICAL	GPU:104GT
337S4333	1	IC, GPU, NV GK104 7-4-PS-A2	UA000	CRITICAL	GPU:104GT2
337S4332	1	IC, GPU, NV GK104 8-4-PS-A2,	UA000	CRITICAL	GPU:104GTX

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION	
341S3640	1	IC,EEPROM,CR,V11.1 (B1),D8	U3690	CRITICAL	TBTROM:PROG	
335S0865	1	IC,EEPROM,SERIAL,8KB,MLP8	U3690	CRITICAL	TBTROM:BLANK	
341S3641	1	IC,PROGRMD,EFI ROM,PIB,D7/D8	U5110	CRITICAL	BOOTROM:PROG	ALTERNATE:335S0812
335S0807	1	IC,64 MBIT SPI SERIAL FLASH	U5110	CRITICAL	BOOTROM:BLANK	
341S3394	1	IC,PROGRMD,SMC,A3,V2.2A32,D8	U4900	CRITICAL	SMC:PROG	
338S1098	1	IC,SMC,LX4FS1AH5BBCIGA3	U4900	CRITICAL	SMC:BLANK	
341S3646	1	IC,CAMERA FLASH,V7226,D7/D8	U4202	CRITICAL	CAMROM:PROG	
335S0852	1	IC,FLASH,SPI,1MBIT,3V3	U4202	CRITICAL	CAMROM:BLANK	
341S3645	1	IC,ENET 1MBIT, SPI,ROM, V1.13 D8	U3990	CRITICAL	CIVROM:PROG	ALTERNATE:335S0854
335S0862	1	IC,SERIAL FLASH,2MBIT, 2.7V, REF F	U3990	CRITICAL	CIVROM:BLANK	
341S3627	1	IC,BLC,MCU, PRPOGRAMMED, V0203, D8	U9700	CRITICAL	BLCMCU:PROG	
337S3978	1	IC,BLC MCU LPC2132FBD64/01, LQFP64	U9700	CRITICAL	BLCMCU:BLANK	

CPU SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0073	1	SOCKET.MOLEX,LGA1155,CPU-LF	U1000	CRITICAL	

CPU SOCKET ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
511S0071	511S0073		ALL	TYCO SOCKET
511S0072	511S0073		ALL	FOXCONN SOCKET

D8 SCHEMATIC / PCB #'S

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-9505	1	SCH,MLB,D8,ULTIMATE	SCH1	CRITICAL	D8
820-3299	1	PCBF,MLB,D8,ULTIMATE	PCB1	CRITICAL	D8

D8 ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0147	377S0126		ALL	USB diodes
157S0084	157S0058		ALL	Enet Magnetics
341S3644	341S3645		U3990	CIVROM
376S0975	376S1081		ALL	P/NCH DUAL FET
152S1668	152S1591		ALL	33UH BLC INDUCTOR
128S0365	128S0368		ALL	150UF CAPS BLK

VRAM Module Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
333S0619	4	IC,SGRAM,GDDR5,32MX32,1.5GHZ,G-DIE,HF	UA300,UA350,UA400,UA450	CRITICAL	FB:1G_SAMSUNG
333S0619	4	IC,SGRAM,GDDR5,32MX32,1.5GHZ,G-DIE,HF	UA500,UA550,UA600,UA650	CRITICAL	FB:1G_SAMSUNG
333S0620	4	IC,GDDR5,32MX32,1.5GHZ,VEGA 44NM,B-DIE	UA300,UA350,UA400,UA450	CRITICAL	FB:1G_HYNIX
333S0620	4	IC,GDDR5,32MX32,1.5GHZ,VEGA 44NM,B-DIE	UA500,UA550,UA600,UA650	CRITICAL	FB:1G_HYNIX
333S0631	4	IC,SGRAM,GDDR5,64MX32,D-DIE	UA300,UA350,UA400,UA450	CRITICAL	FB:2G_SAMSUNG
333S0631	4	IC,SGRAM,GDDR5,64MX32,D-DIE	UA500,UA550,UA600,UA650	CRITICAL	FB:2G_SAMSUNG
333S0630	4	IC,GDDR5,64MX32,A-DIE	UA300,UA350,UA400,UA450	CRITICAL	FB:2G_HYNIX
333S0630	4	IC,GDDR5,64MX32,A-DIE	UA500,UA550,UA600,UA650	CRITICAL	FB:2G_HYNIX

SYNC_MASTER=D8_MLB_ULTIMATE SYNC_DATE=06/15/2012

PAGE TITLE

BOM Configuration

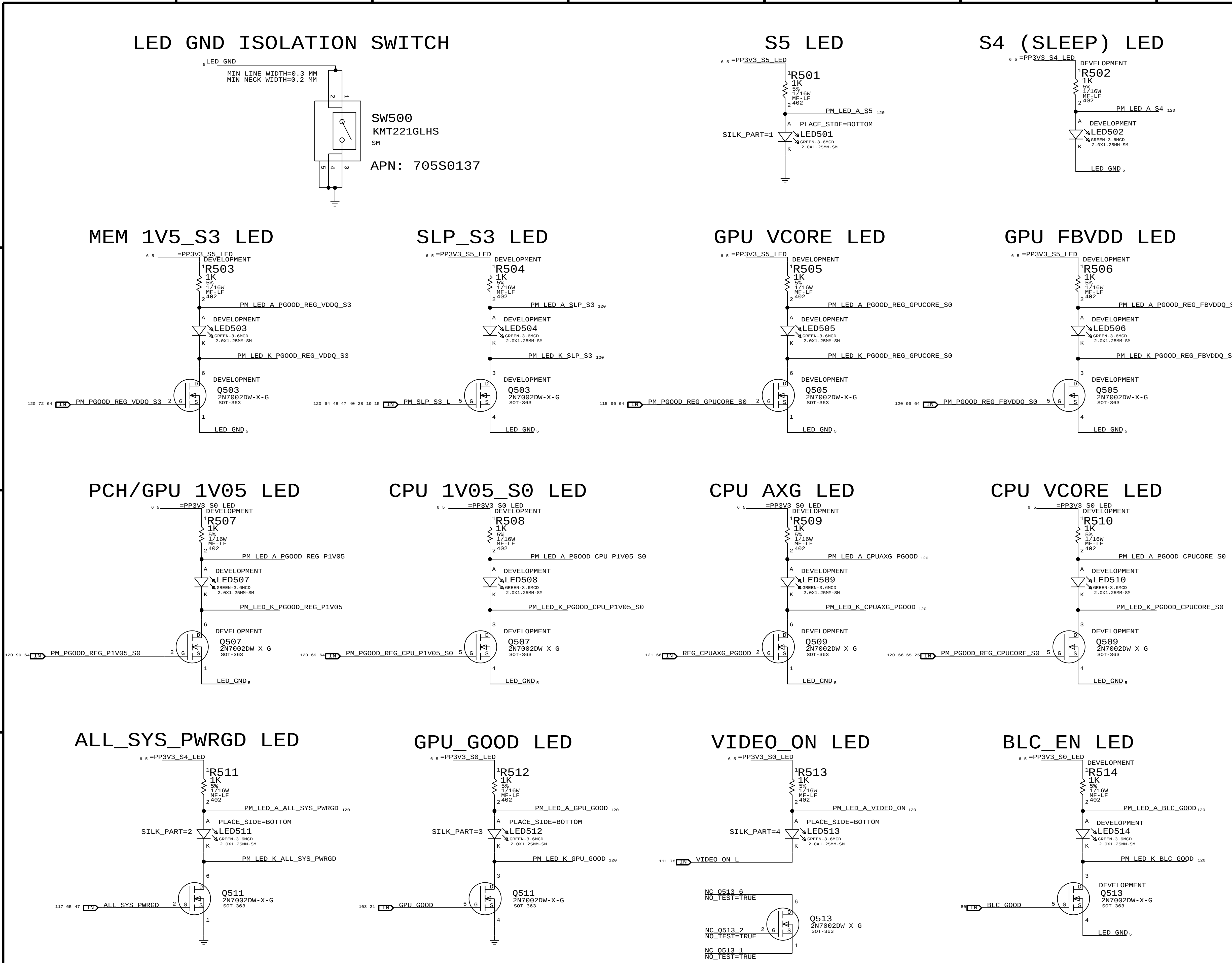
Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

8 7 6 5 4 3 2 1



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE: DEBUG LEDS

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

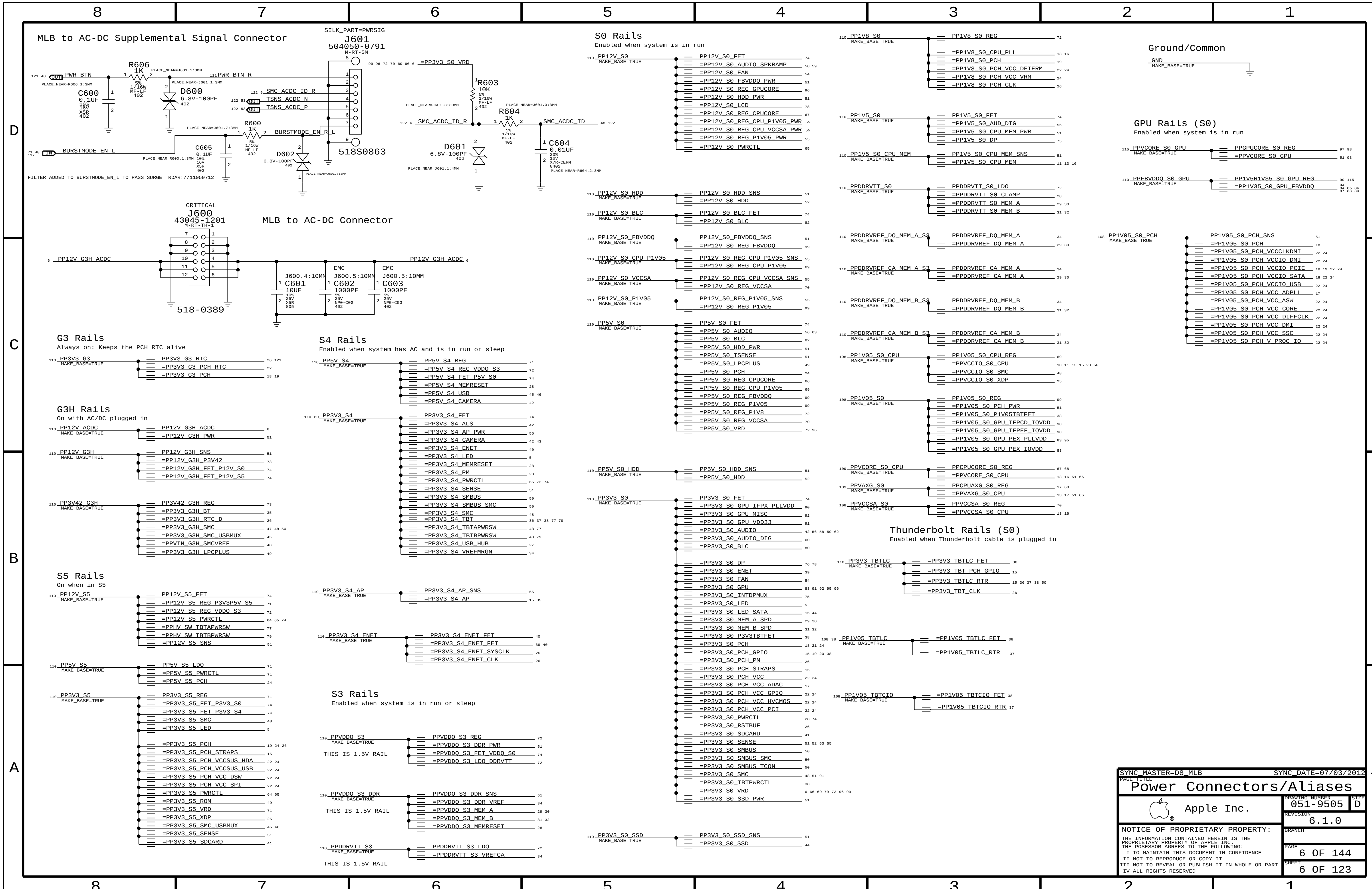
REVISION: 6.1.0

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

PAGE 5 OF 144

SHEET 5 OF 123



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

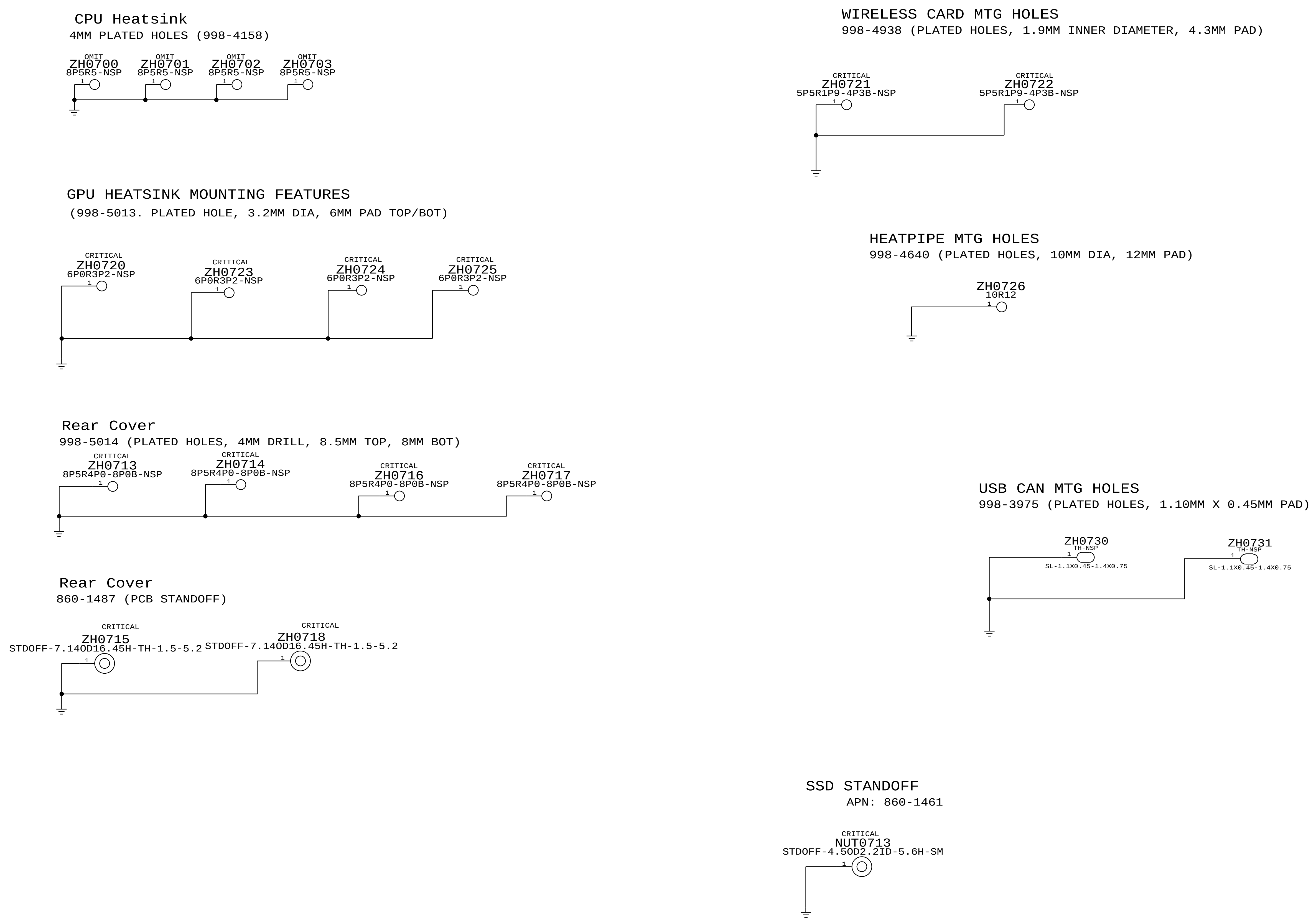
Power Connectors/Aliases

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

Holes/PD parts

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

CPU Reserved

Page	Net	Alias	Properties
10	TP_CPU_RSVD<16..1>	NC_CPU_RSVD<16..1>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_CPU_RSVD<46..19>	NC_CPU_RSVD<46..19>	MAKE_BASE=TRUE NO_TEST=TRUE
10	CPU_CFG<15..12>	TP_CPU_CFG<15..12>	MAKE_BASE=TRUE

CPU Memory

Page	Net	Alias	Properties
12	TP_MEM_A_DQ_CB<7..0>	NC_MEM_A_DQ_CB<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_MEM_A_DQS_N<8>	NC_MEM_A_DQSNX<8>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_MEM_A_DQS_P<8>	NC_MEM_A_DQSPX<8>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_MEM_B_DQ_CB<7..0>	NC_MEM_B_DQ_CB<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_MEM_B_DQS_N<8>	NC_MEM_B_DQSNX<8>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_MEM_B_DQS_P<8>	NC_MEM_B_DQSPX<8>	MAKE_BASE=TRUE NO_TEST=TRUE

PCH PCIe

Page	Net	Alias	Properties
18	TP_PCIE1_D2RN	NC_PCIE1_D2RNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE1_D2RP	NC_PCIE1_D2RPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE1_R2D_CN	NC_PCIE1_R2D_CNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE1_R2D_CP	NC_PCIE1_R2D_CPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE2_D2RN	NC_PCIE2_D2RNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE2_D2RP	NC_PCIE2_D2RPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE2_R2D_CN	NC_PCIE2_R2D_CNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE2_R2D_CP	NC_PCIE2_R2D_PNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	DMI_MIDBUS_CLK100M_N	NC_DMI_MIDBUS_CLK100NX	MAKE_BASE=TRUE NO_TEST=TRUE
18	DMI_MIDBUS_CLK100M_P	NC_DMI_MIDBUS_CLK100PX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE_CLK100M_PE0N	NC_PCIE_CLK100M_PE0NX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE_CLK100M_PE0P	NC_PCIE_CLK100M_PE0PX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE_CLK100M_PE4N	NC_PCIE_CLK100M_PE4NX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE_CLK100M_PE4P	NC_PCIE_CLK100M_PE4PX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE_CLK100M_PE5N	NC_PCIE_CLK100M_PE5NX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE_CLK100M_PE5P	NC_PCIE_CLK100M_PE5PX	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCIE_CLK100M_PE6N	NC_PCIE_CLK100M_PE6NX	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCIE_CLK100M_PE6P	NC_PCIE_CLK100M_PE6PX	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCIE_CLK100M_PE7N	NC_PCIE_CLK100M_PE7NX	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCIE_CLK100M_PE7P	NC_PCIE_CLK100M_PE7PX	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_PE_TX_N<3..0>	NC_PE_TNX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_PE_TX_P<3..0>	NC_PE_TPX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_PE_RX_N<3..0>	NC_PE_RNX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_PE_RX_P<3..0>	NC_PE_RPX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE

PCH USB

Page	Net	Alias	Properties
20	USB_PCH_4_N	NC_USB_PCH_4NX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_4_P	NC_USB_PCH_4PX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_5_N	NC_USB_PCH_5NX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_5_P	NC_USB_PCH_5PX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_6_N	NC_USB_PCH_6NX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_6_P	NC_USB_PCH_6PX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_11_N	NC_USB_PCH_11NX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_11_P	NC_USB_PCH_11PX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_12_N	NC_USB_PCH_12NX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_12_P	NC_USB_PCH_12PX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_13_N	NC_USB_PCH_13NX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_13_P	NC_USB_PCH_13PX	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Clocks

Page	Net	Alias	Properties
18	TP_PCH_CLKOUT_DPN	NC_PCH_CLKOUT_DPNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_CLKOUT_DPP	NC_PCH_CLKOUT_DPPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	PCH_CLK25M_XTALOUT	NC_PCH_CLK25M_XTALOUT	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_GPIO64_CLKOUTFLEX0	NC_PCH_GPIO64_CLKOUTFLEX0	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_GPIO65_CLKOUTFLEX1	NC_PCH_GPIO65_CLKOUTFLEX1	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_GPIO66_CLKOUTFLEX2	NC_PCH_GPIO66_CLKOUTFLEX2	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_GPIO67_CLKOUTFLEX3	NC_PCH_GPIO67_CLKOUTFLEX3	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Unused Display

Page	Net	Alias	Properties
19	TP_CRT_IG_RED	NC_CRT_IG_RED	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_GREEN	NC_CRT_IG_GREEN	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_BLUE	NC_CRT_IG_BLUE	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_HSYNC	NC_CRT_IG_HSYNC	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_VSYNC	NC_CRT_IG_VSYNC	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_DDC_CLK	NC_CRT_IG_DDC_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_DDC_DATA	NC_CRT_IG_DDC_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_MLN<3..0>	NC_DP_IG_B_MLNX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_MLP<3..0>	NC_DP_IG_B_MLPX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_AUX_N	NC_DP_IG_B_AUXNX	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_AUX_P	NC_DP_IG_B_AUXPX	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_HPD	NC_DP_IG_B_HPD	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_DDC_CLK	NC_DP_IG_B_CTRL_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_DDC_DATA	NC_DP_IG_B_CTRL_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_MLN<3..0>	NC_DP_IG_C_MLNX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_MLP<3..0>	NC_DP_IG_C_MLPX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_AUX_N	NC_DP_IG_C_AUXNX	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_AUX_P	NC_DP_IG_C_AUXPX	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_HPD	NC_DP_IG_C_HPD	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_CTRL_CLK	NC_DP_IG_C_CTRL_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_CTRL_DATA	NC_DP_IG_C_CTRL_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_MLN<3..0>	NC_DP_IG_D_MLNX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_MLP<3..0>	NC_DP_IG_D_MLPX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_AUXN	NC_DP_IG_D_AUXNX	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_AUXP	NC_DP_IG_D_AUXPX	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_HPD	NC_DP_IG_D_HPD	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_CTRL_CLK	NC_DP_IG_D_CTRL_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_CTRL_DATA	NC_DP_IG_D_CTRL_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_TVCLKINN	NC_SDVO_TVCLKINNX	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_TVCLKINP	NC_SDVO_TVCLKINPX	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_STALLN	NC_SDVO_STALLNX	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_STALLP	NC_SDVO_STALLPX	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_INTN	NC_SDVO_INTNX	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_INTP	NC_SDVO_INTPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_L_BKLTCTL	NC_PCH_L_BKLTCTL	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_L_BKLTEN	NC_PCH_L_BKLTEN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_L_VDD_EN	NC_PCH_L_VDD_EN	MAKE_BASE=TRUE NO_TEST=TRUE

PCH SATA

Page	Net	Alias	Properties
18	TP_SATA_C_R2D_CN	NC_SATA_C_R2D_CNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_C_R2D_CP	NC_SATA_C_R2D_CPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_C_D2RN	NC_SATA_C_D2RNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_C_D2RP	NC_SATA_C_D2RPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_D_R2D_CN	NC_SATA_D_R2D_CNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_D_R2D_CP	NC_SATA_D_R2D_CPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_D_D2RN	NC_SATA_D_D2RNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_D_D2RP	NC_SATA_D_D2RPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_E_R2D_CN	NC_SATA_E_R2D_CNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_E_R2D_CP	NC_SATA_E_R2D_CPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_E_D2RN	NC_SATA_E_D2RNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_E_D2RP	NC_SATA_E_D2RPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_F_R2D_CN	NC_SATA_F_R2D_CNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_F_R2D_CP	NC_SATA_F_R2D_CPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_F_D2RN	NC_SATA_F_D2RNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_F_D2RP	NC_SATA_F_D2RPX	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Reserved

Page	Net	Alias	Properties
19	TP_PCH_RESERVE_0	NC_PCH_RESERVE_0	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_1	NC_PCH_RESERVE_1	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_2	NC_PCH_RESERVE_2	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_3	NC_PCH_RESERVE_3	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_4	NC_PCH_RESERVE_4	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_5	NC_PCH_RESERVE_5	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_6	NC_PCH_RESERVE_6	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_7	NC_PCH_RESERVE_7	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_8	NC_PCH_RESERVE_8	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_9	NC_PCH_RESERVE_9	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_10	NC_PCH_RESERVE_10	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_11	NC_PCH_RESERVE_11	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_12	NC_PCH_RESERVE_12	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_13	NC_PCH_RESERVE_13	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_14	NC_PCH_RESERVE_14	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_15	NC_PCH_RESERVE_15	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_16	NC_PCH_RESERVE_16	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_17	NC_PCH_RESERVE_17	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_18	NC_PCH_RESERVE_18	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_19	NC_PCH_RESERVE_19	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_20	NC_PCH_RESERVE_20	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_21	NC_PCH_RESERVE_21	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_22	NC_PCH_RESERVE_22	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_23	NC_PCH_RESERVE_23	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_24	NC_PCH_RESERVE_24	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_25	NC_PCH_RESERVE_25	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_26	NC_PCH_RESERVE_26	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_27	NC_PCH_RESERVE_27	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_28	NC_PCH_RESERVE_28	MAKE_BASE=TRUE NO_TEST=TRUE

PCH and CPU FDI

Page	Net	Alias	Properties
10	TP_CPU_FDI_TX_N<7..0>	NC_CPU_FDI_TNX<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_CPU_FDI_TX_P<7..0>	NC_CPU_FDI_TPX<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	PCH_FDI_RX_N<7..0>	NC_PCH_FDI_RNX<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	PCH_FDI_RX_P<7..0>	NC_PCH_FDI_RPX<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_CPU_FDI_FSYNC<1..0>	NC_CPU_FDI_FSYNC<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_CPU_FDI_LSYNC<1..0>	NC_CPU_FDI_LSYNC<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_CPU_FDI_INT	NC_CPU_FDI_INT	MAKE_BASE=TRUE NO_TEST=TRUE
19	PCH_FDI_FSYNC<1..0>	NC_PCH_FDI_FSYNC<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	PCH_FDI_LSYNC<1..0>	NC_PCH_FDI_LSYNC<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	PCH_FDI_INT	NC_PCH_FDI_INT	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Test Points

Page	Net	Alias	Properties
21	TP_PCH_TP1	NC_PCH_TP1	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP2	NC_PCH_TP2	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP3	NC_PCH_TP3	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP4	NC_PCH_TP4	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP5	NC_PCH_TP5	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP6	NC_PCH_TP6	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP7	NC_PCH_TP7	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP8	NC_PCH_TP8	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP9	NC_PCH_TP9	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP10	NC_PCH_TP10	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP11	NC_PCH_TP11	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP12	NC_PCH_TP12	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP13	NC_PCH_TP13	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP14	NC_PCH_TP14	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP15	NC_PCH_TP15	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP16	NC_PCH_TP16	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP17	NC_PCH_TP17	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP18	NC_PCH_TP18	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP19	NC_PCH_TP19	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP20	NC_PCH_TP20	MAKE_BASE=TRUE NO_TEST=TRUE

PCH PCI

Page	Net	Alias	Properties
20	TP_PCI_AD<31..0>	NC_PCI_AD<31..0>	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCI_C_BE_L<3..0>	NC_PCI_C_BE_L<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCI_PAR	NC_PCI_PAR	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCI_RESET_L	NC_PCI_RESET_L	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCH_PCI_GNT0_L	NC_PCI_GNT0_L	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_INIT3V3_L	NC_PCH_INIT3V3_L	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_LPC_DREQ0_L	NC_LPC_DREQ0_L	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Miscellaneous

Page	Net	Alias	Properties
18	TP_HDA_SDIN1	NC_HDA_SDIN1	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_HDA_SDIN2	NC_HDA_SDIN2	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_HDA_SDIN3	NC_HDA_SDIN3	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_PWM0	NC_PCH_PWM0	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_PWM1	NC_PCH_PWM1	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_PWM2	NC_PCH_PWM2	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_PWM3	NC_PCH_PWM3	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_SST	NC_PCH_SST	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_CL_CLK1	NC_PCH_CL_CLK1	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_CL_DATA1	NC_PCH_CL_DATA1	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_CL_RST1	NC_PCH_CL_RST1	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCI_CLK33M_OUT2	NC_PCI_CLK33M_OUT2	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCI_CLK33M_OUT3	NC_PCI_CLK33M_OUT3	MAKE_BASE=TRUE NO_TEST=TRUE

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE
Unused Signal Aliases

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

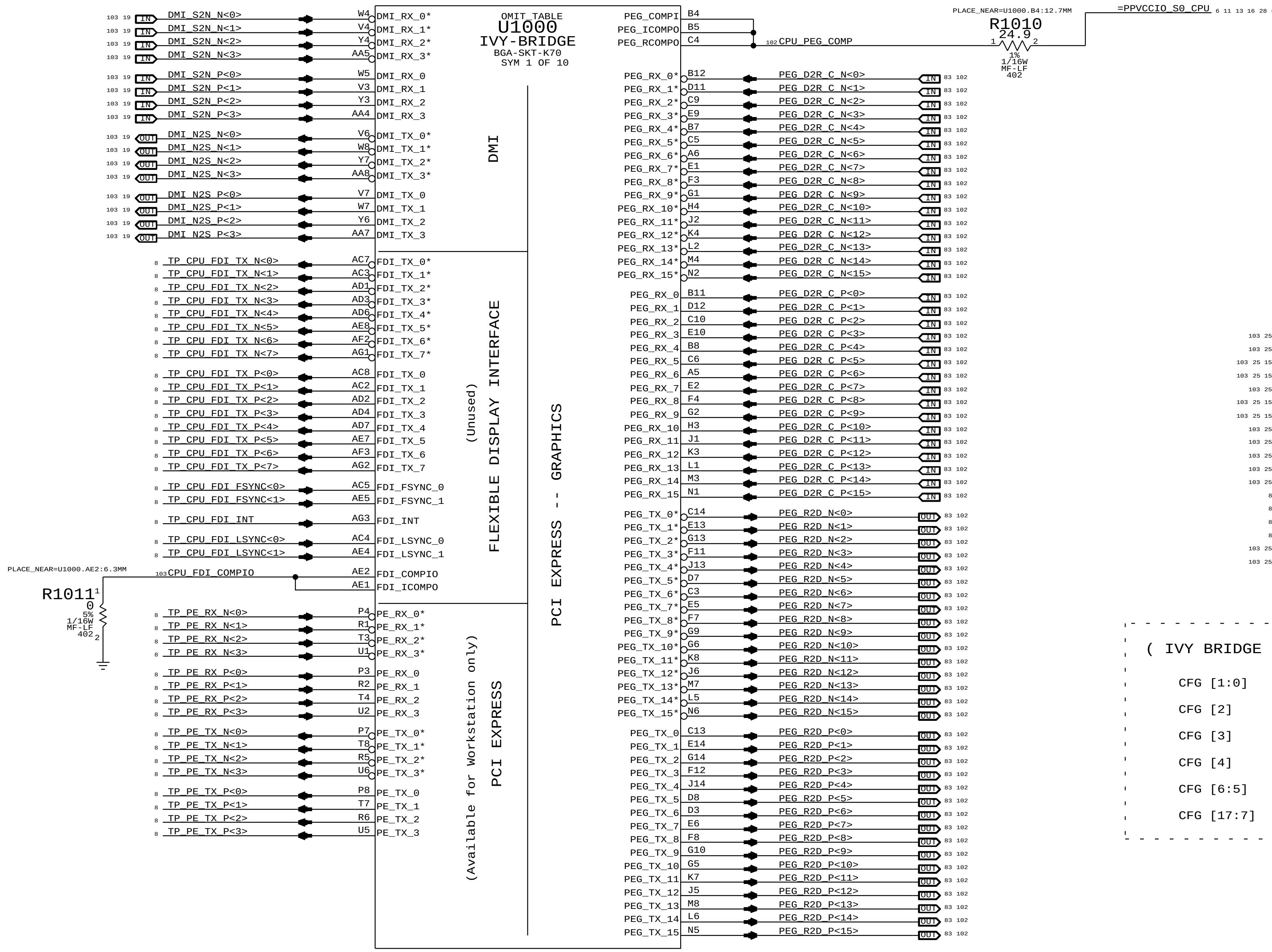
BRANCH

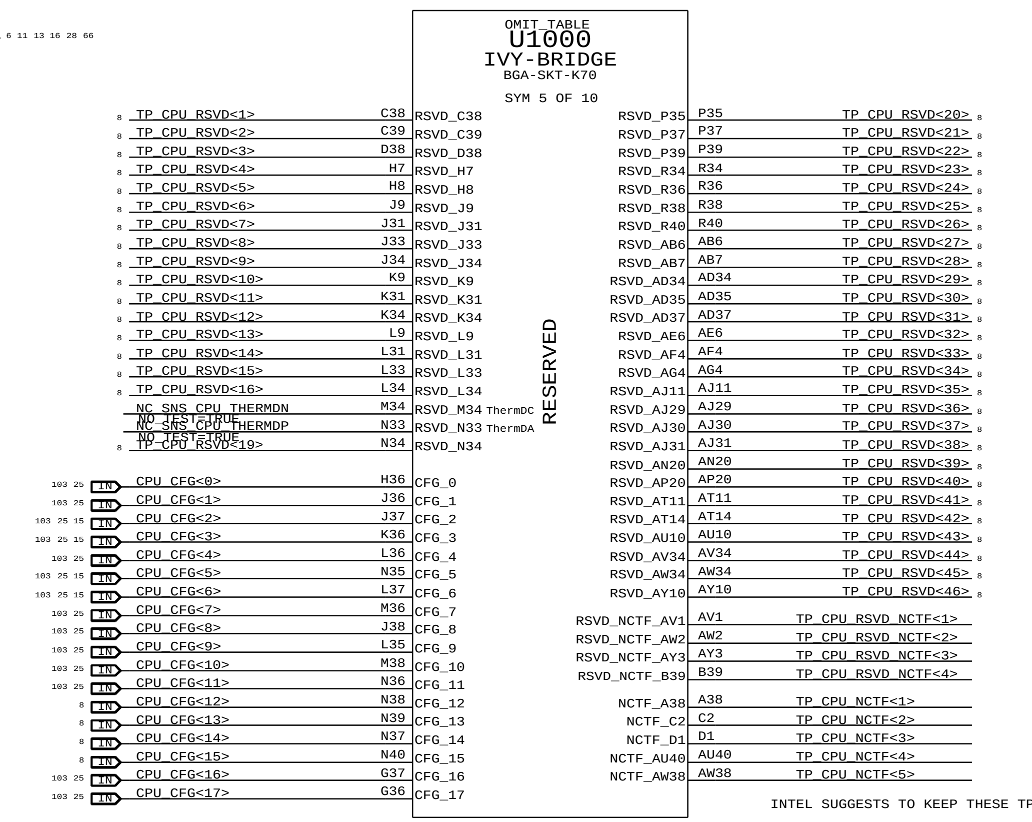
PAGE 8 OF 144

SHEET 8 OF 123

ALIASES (BLANK)

SHORT B4 & C4 TOGETHER, ROUTE AS A SINGLE 4 MIL TRACE TO R1010.1
ROUTE B5 TO R1010.1 AS A SEPERATE 12 MIL TRACE.





(IVY BRIDGE EDS #473717 TABLE 6-5)

CFG [1:0]	RESERVED CONFIGURATION LANE				
CFG [2]	PCIE STATIC X16 LANE REVERSAL	1 = NORMAL OPERATION	0 = LANES REVERSED		
CFG [3]	PCIE STATIC X4 LANE REVERSAL	1 = NORMAL OPERATION	0 = LANES REVERSED		
CFG [4]	RESERVED CONFIGURATION LANE				
CFG [6:5]	PCIE BIFURCATION	11 = 1 X16 (DEFAULT)	10 = 2 X8	01 = RSVD	00 = 1 X8, 2 X4
CFG [17:7]	RESERVED CONFIGURATION LANE				

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

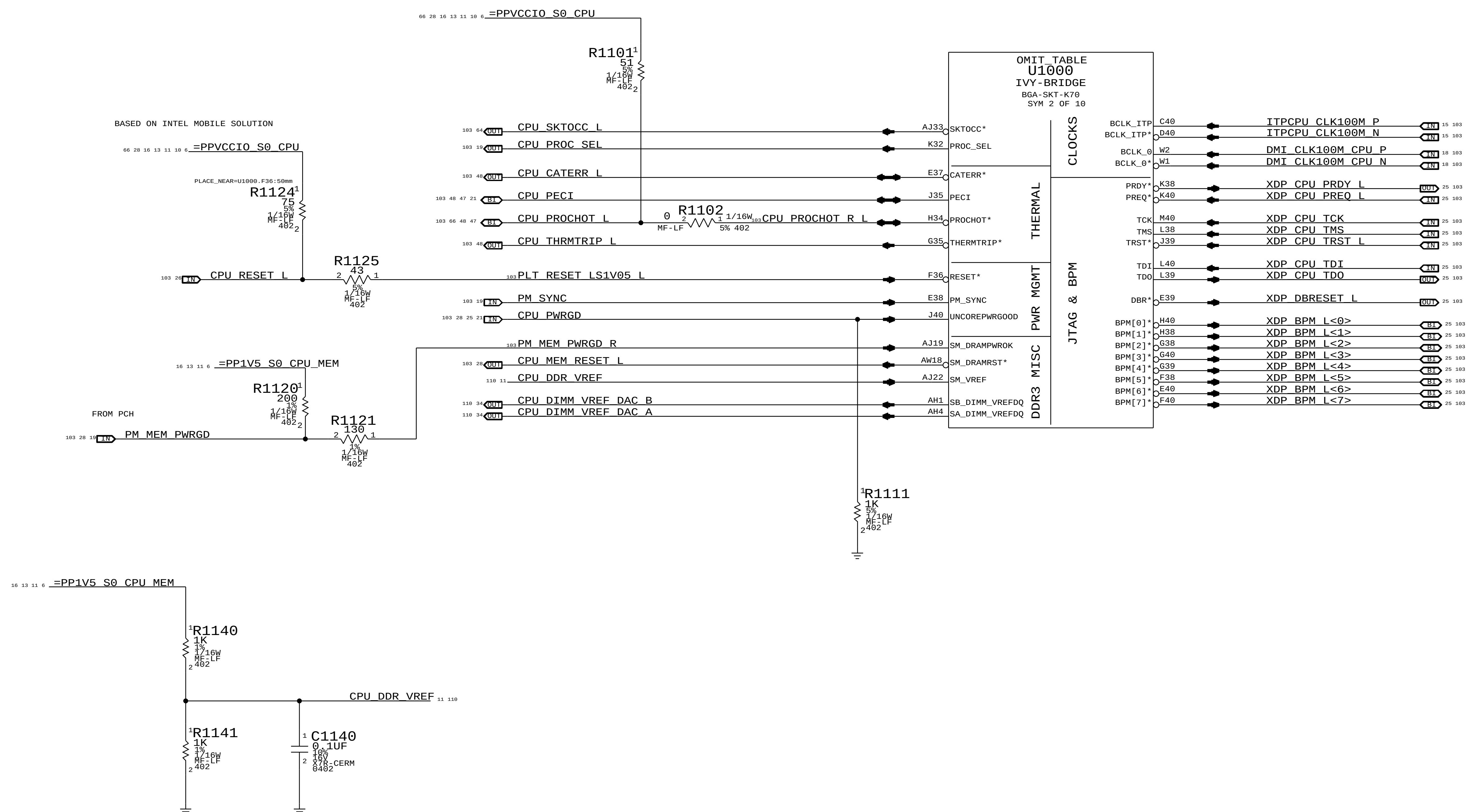
PAGE TITLE
CPU DMI/PEG/FDI/RSVD

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

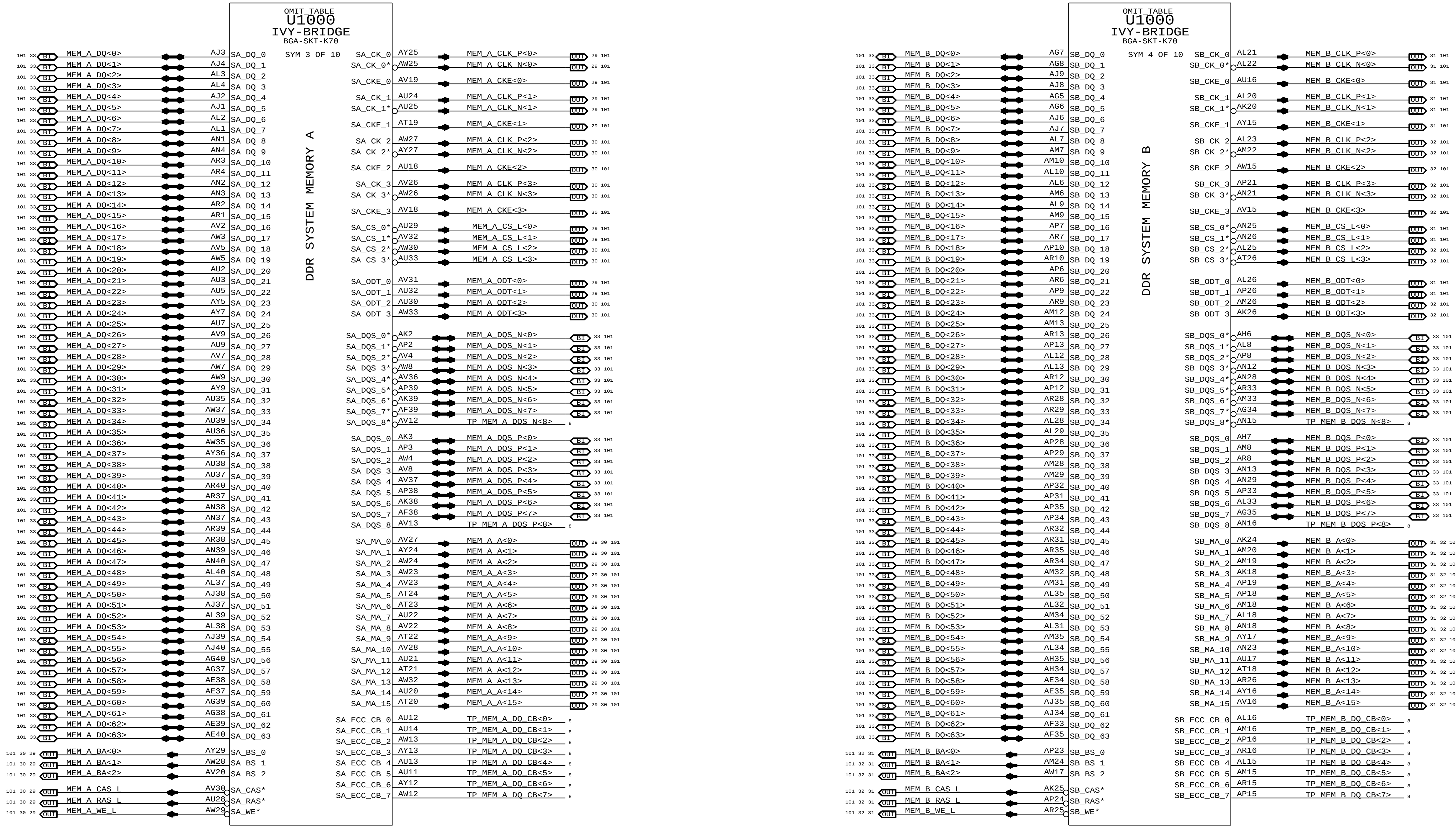
CPU CLOCK/MISC/JTAG

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

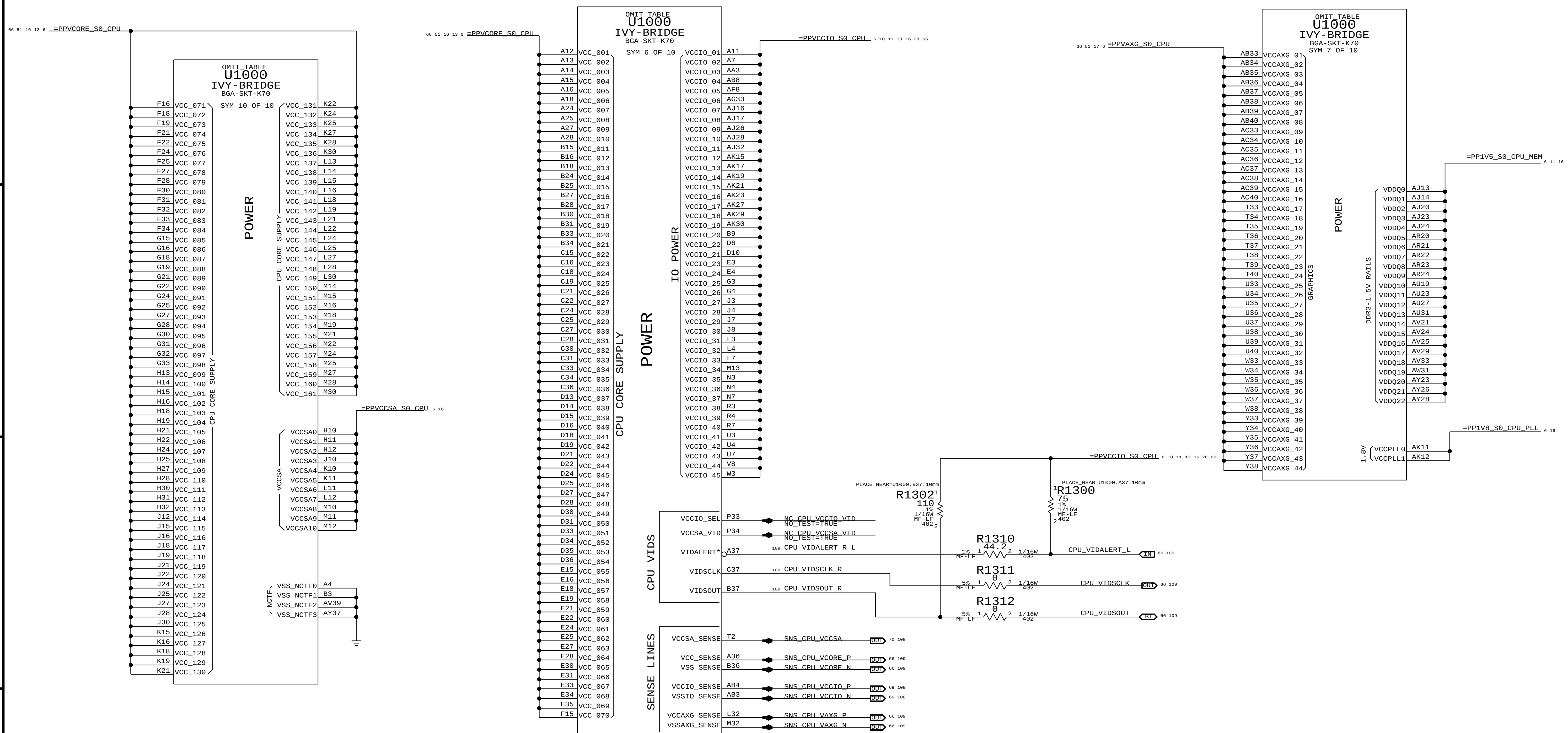
CPU DDR3 INTERFACES

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

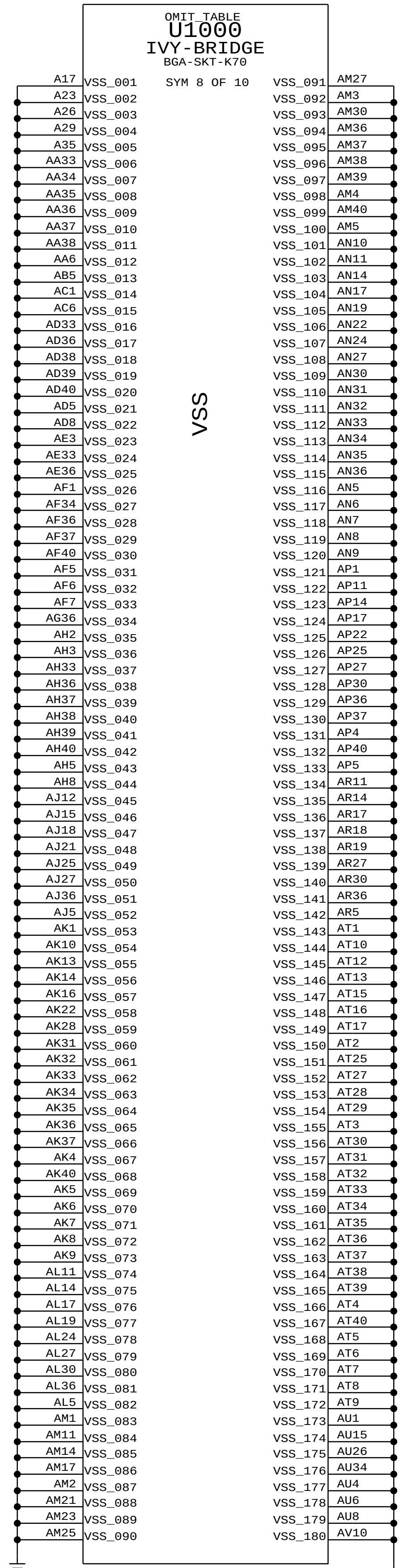
CPU POWER

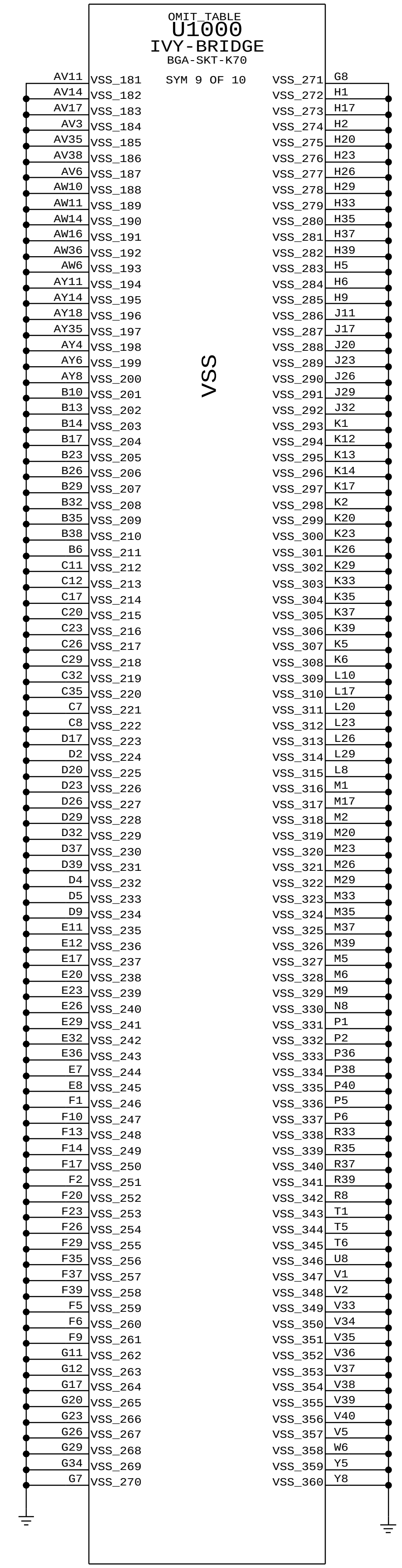
Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH





SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

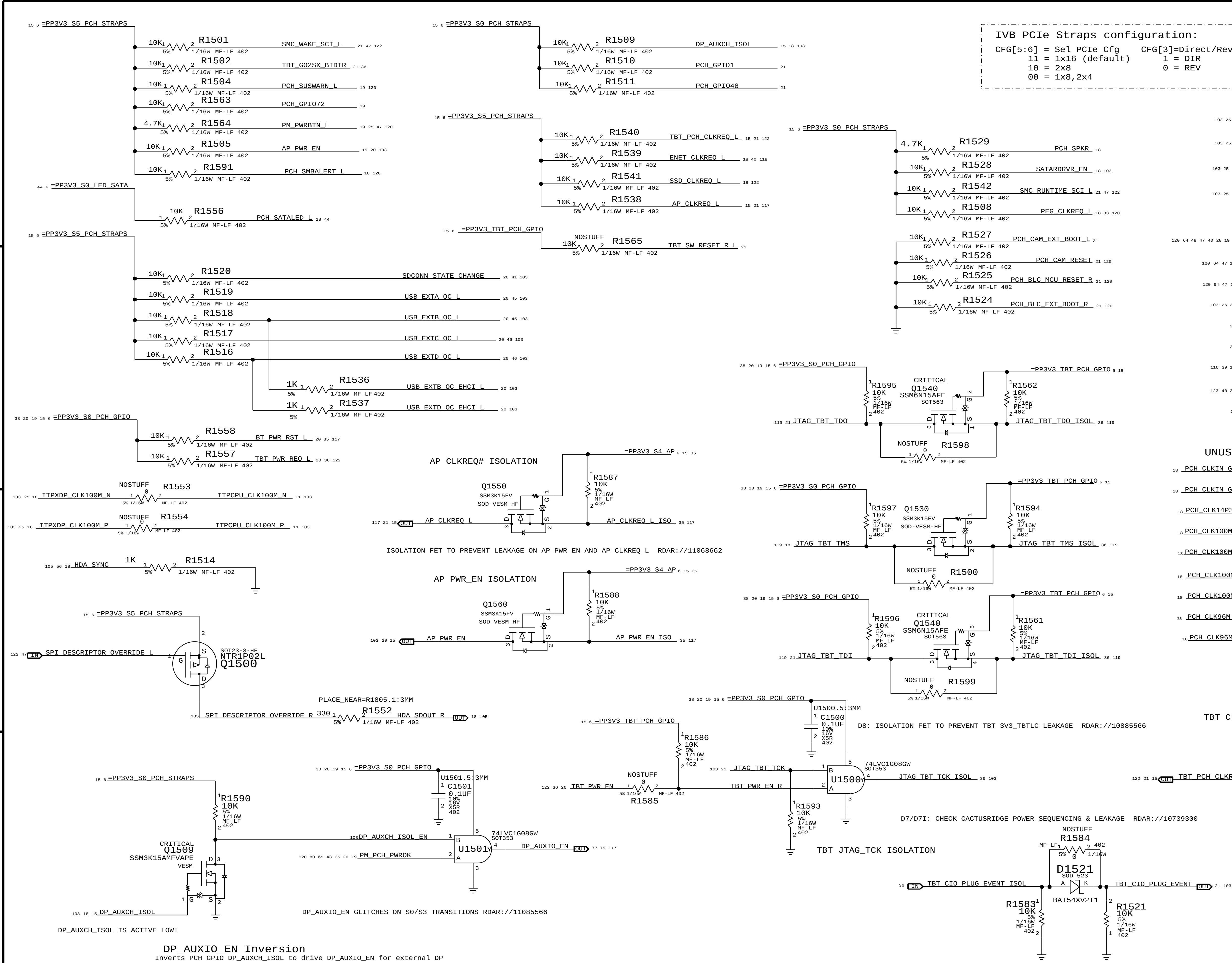
CPU GROUNDS

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

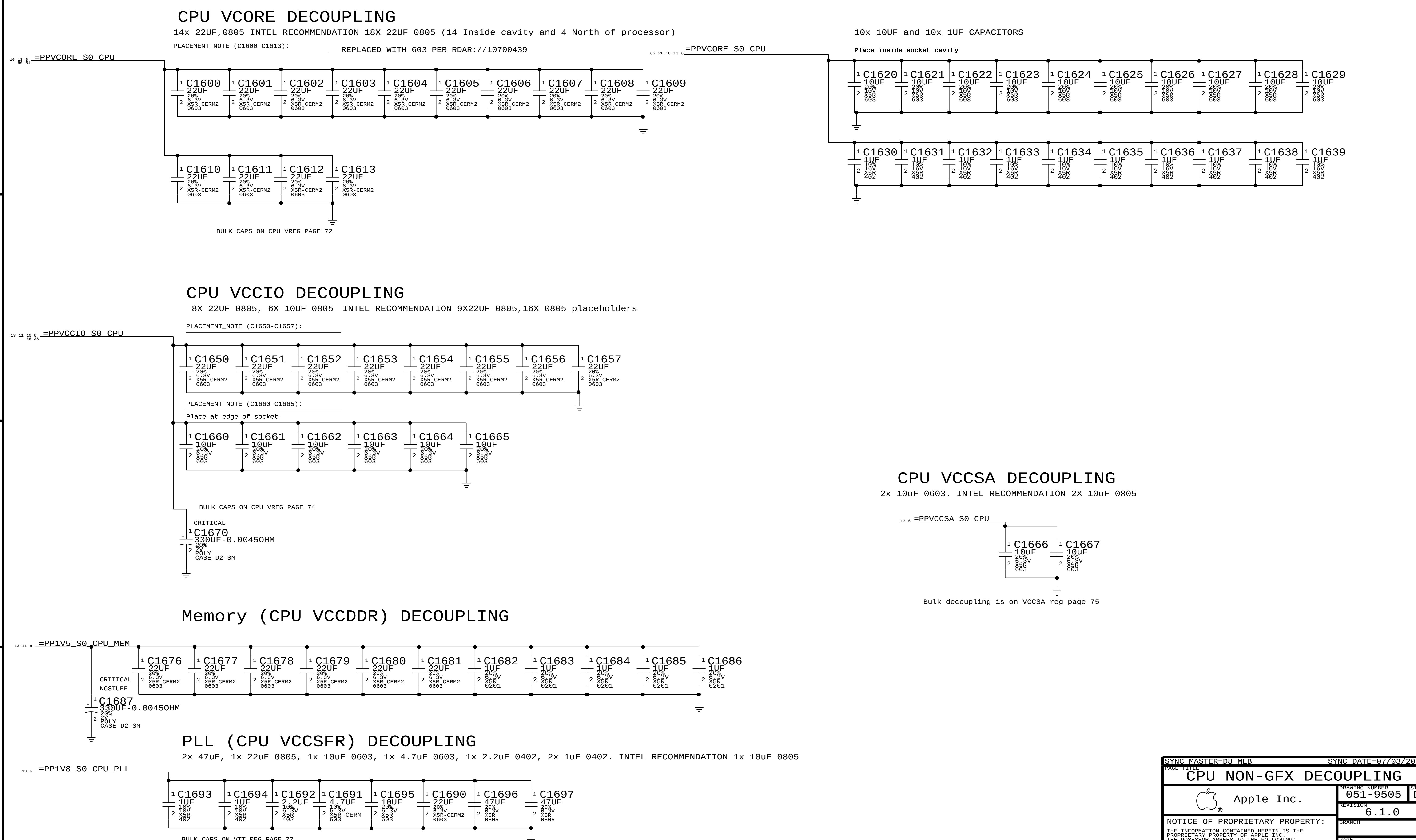
STRAPS,PULL UPS,PULL DOWNS FOR PCH AND CPU

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

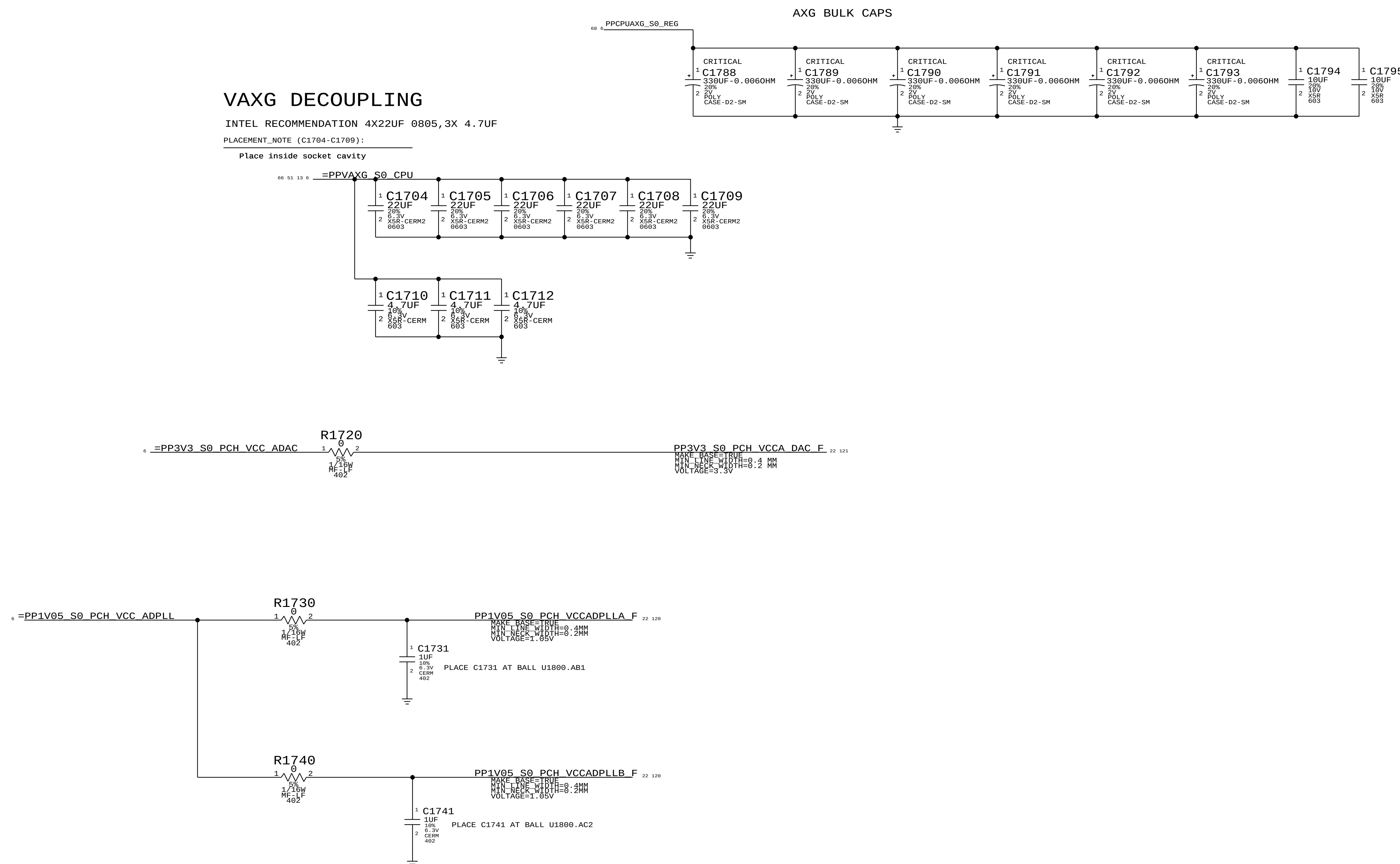
CPU NON-GFX DECOUPLING

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

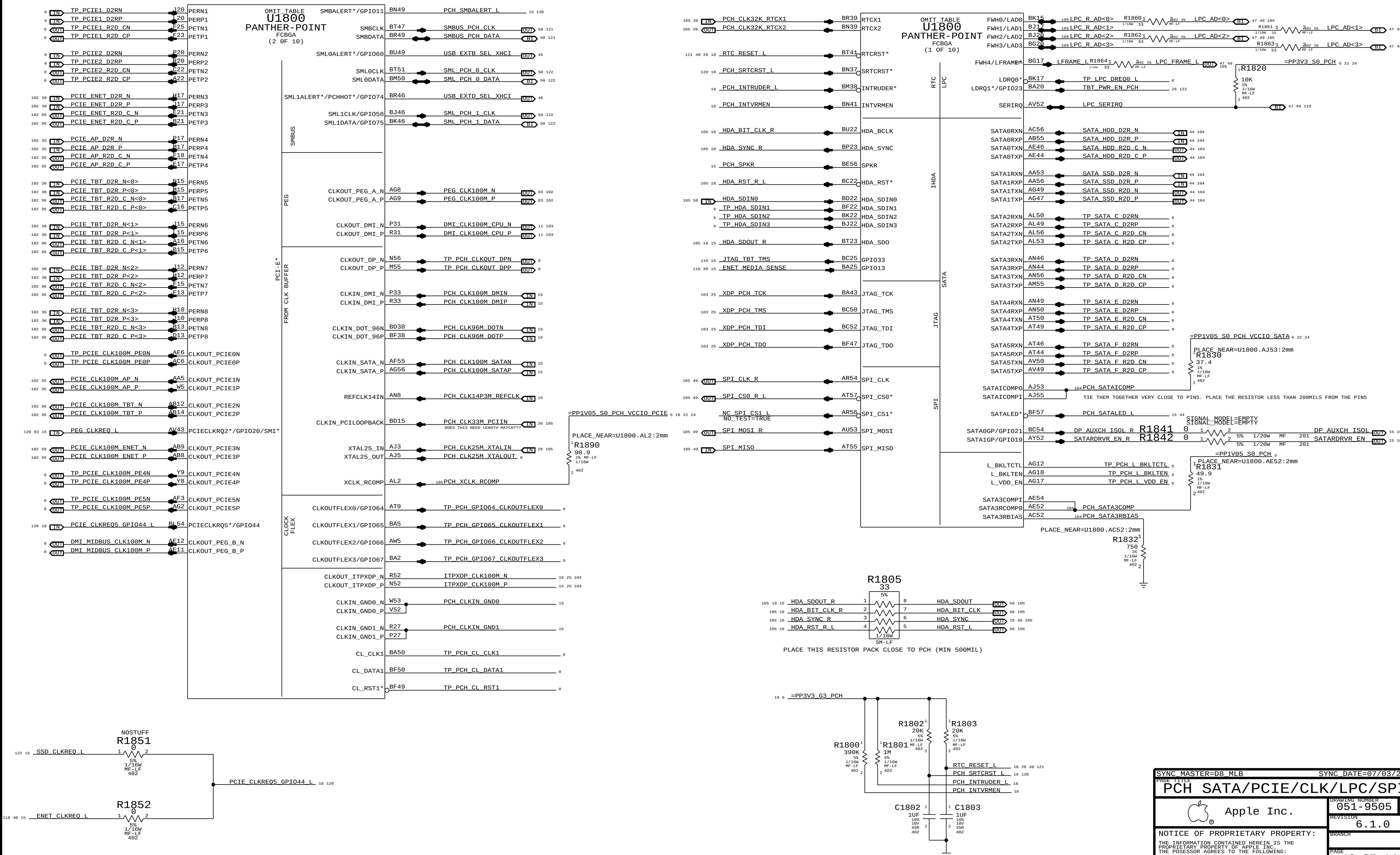


SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE
GFX DECOUPLING & PCH PWR ALIAS

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D
REVISION 6.1.0
BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE
PCH SATA/PCIE/CLK/LPC/SPI

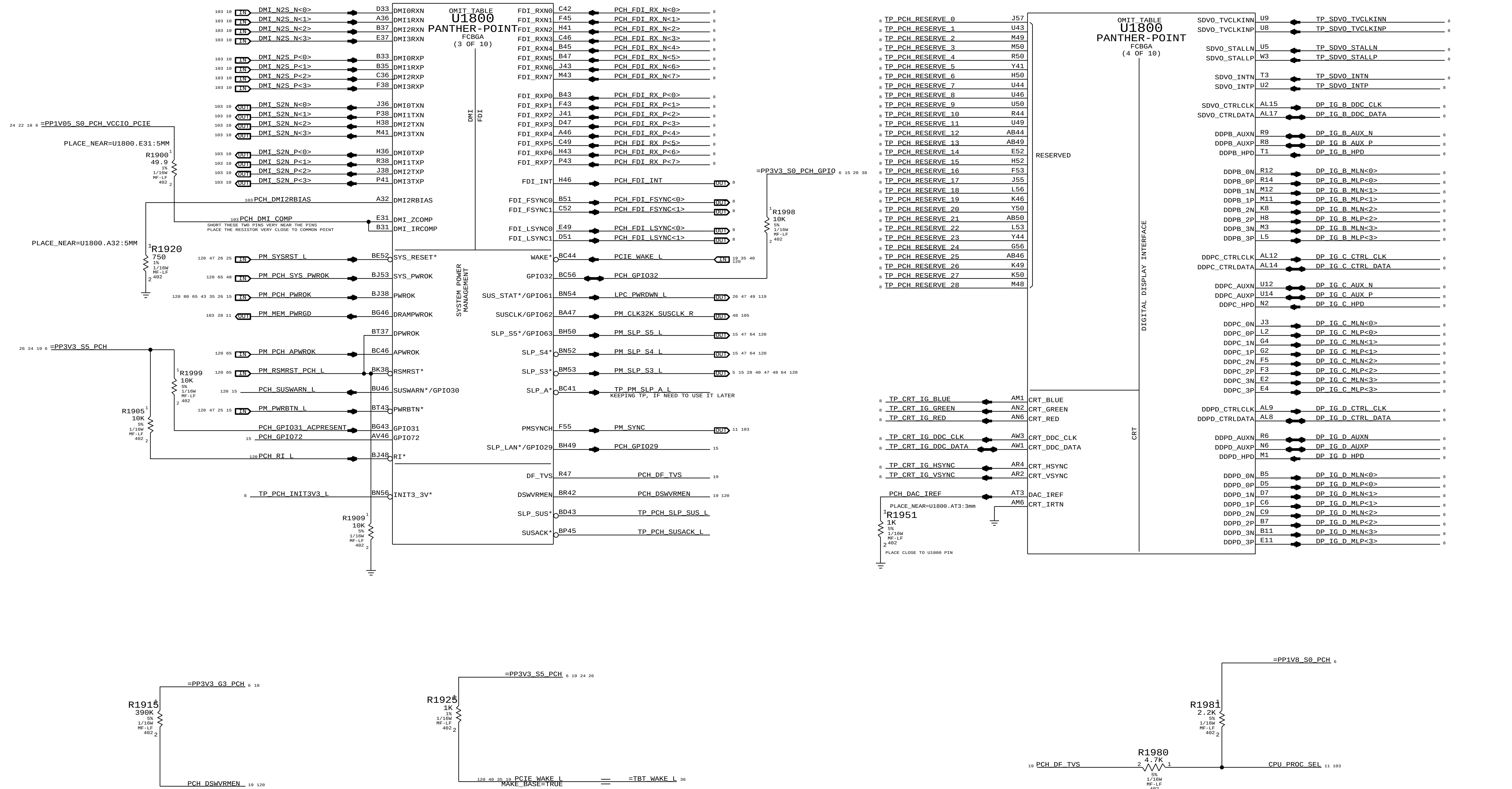
Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

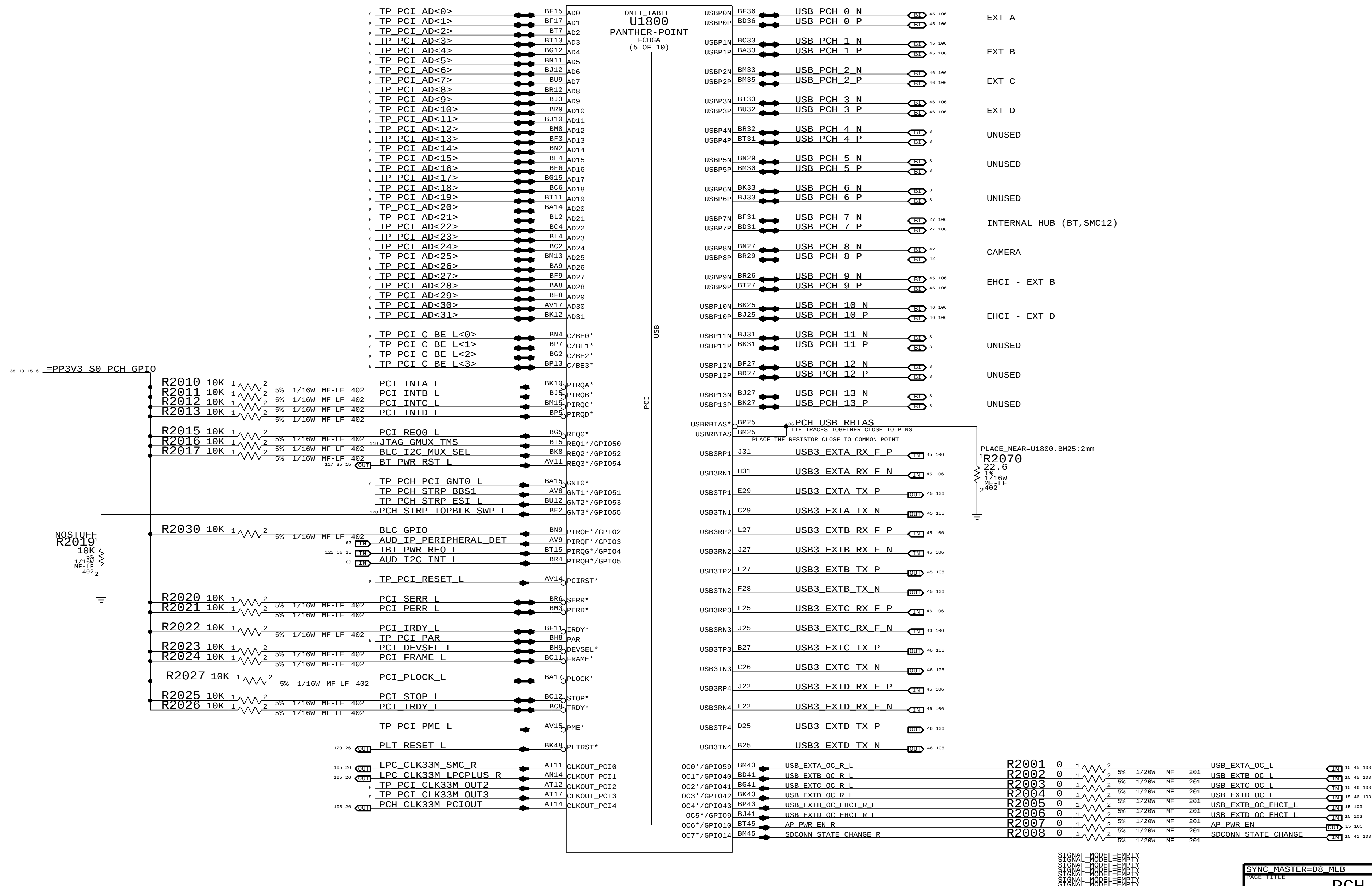
PCH DMI/FDI/GRAPHICS

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

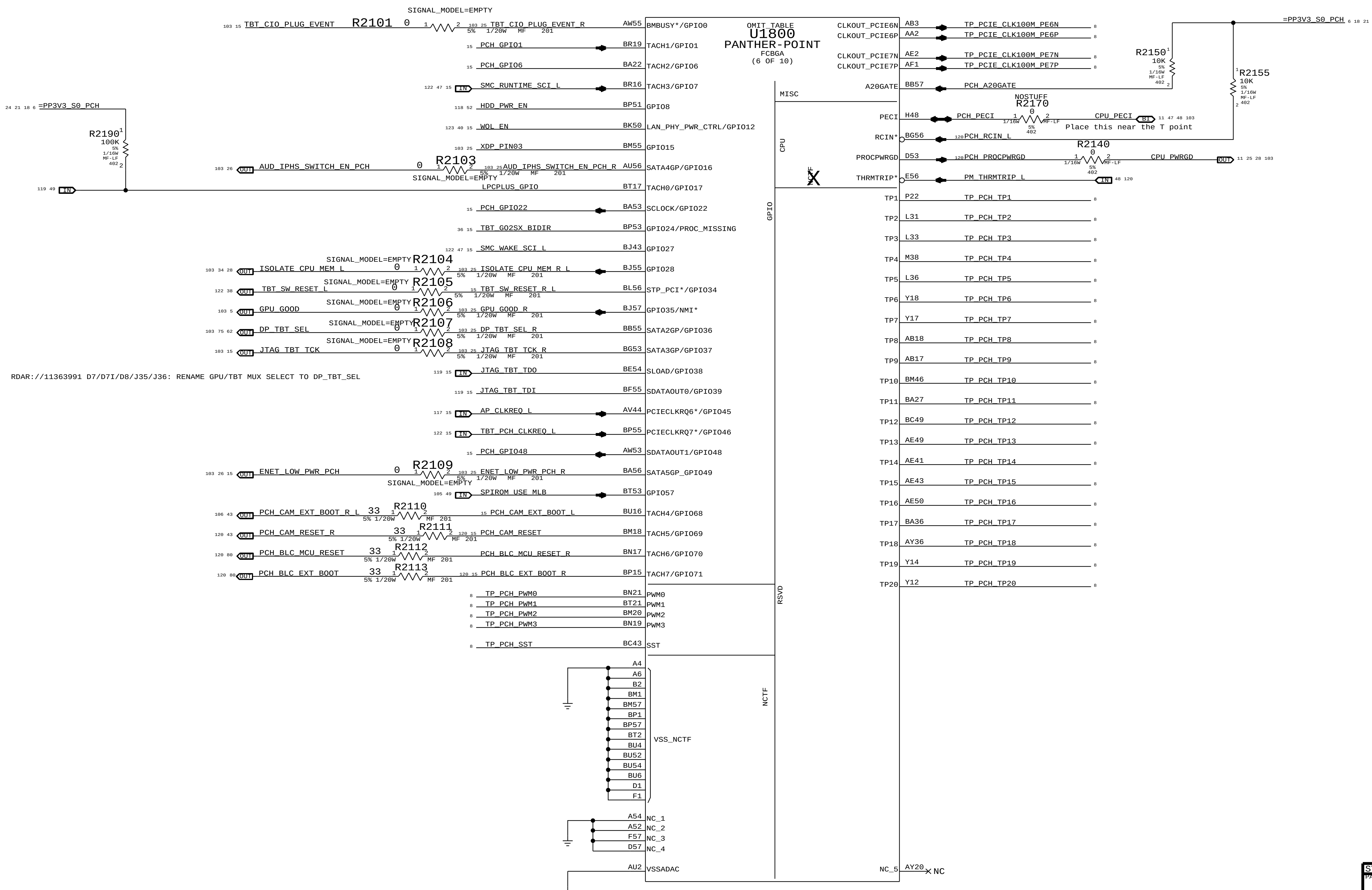
PCH PCI/USB

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

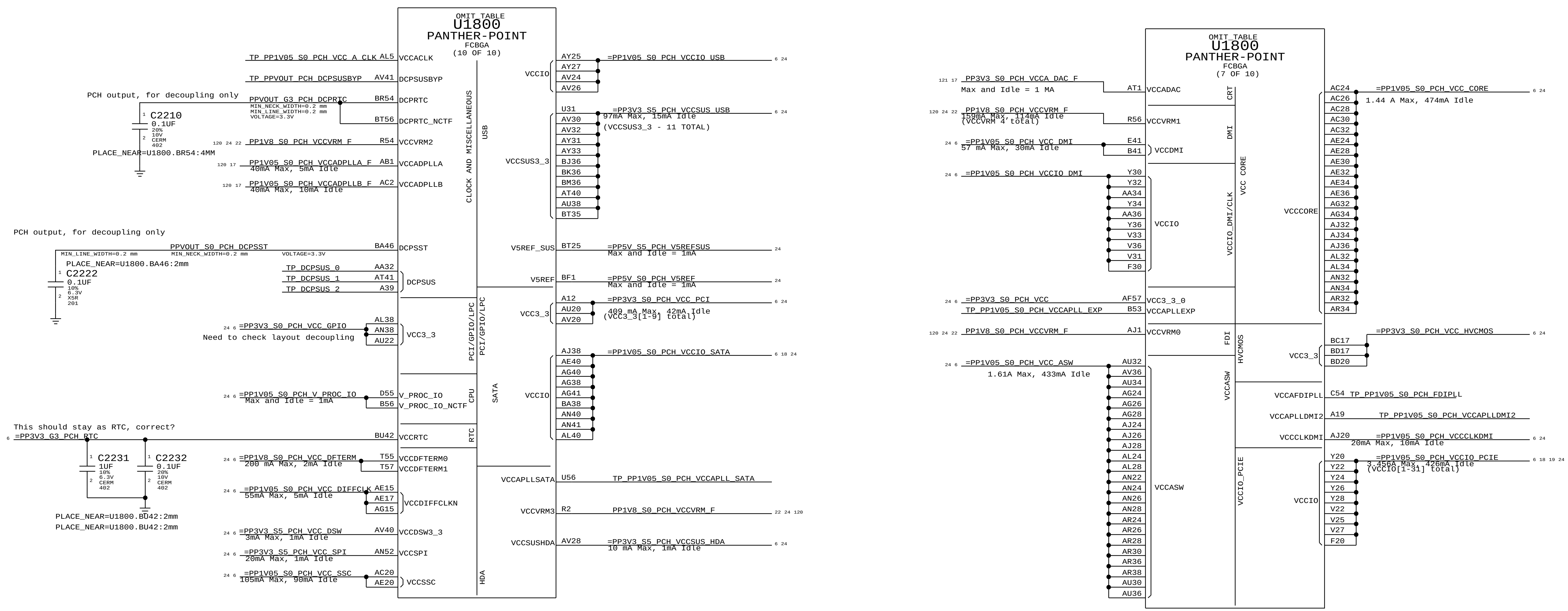
PCH MISC

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

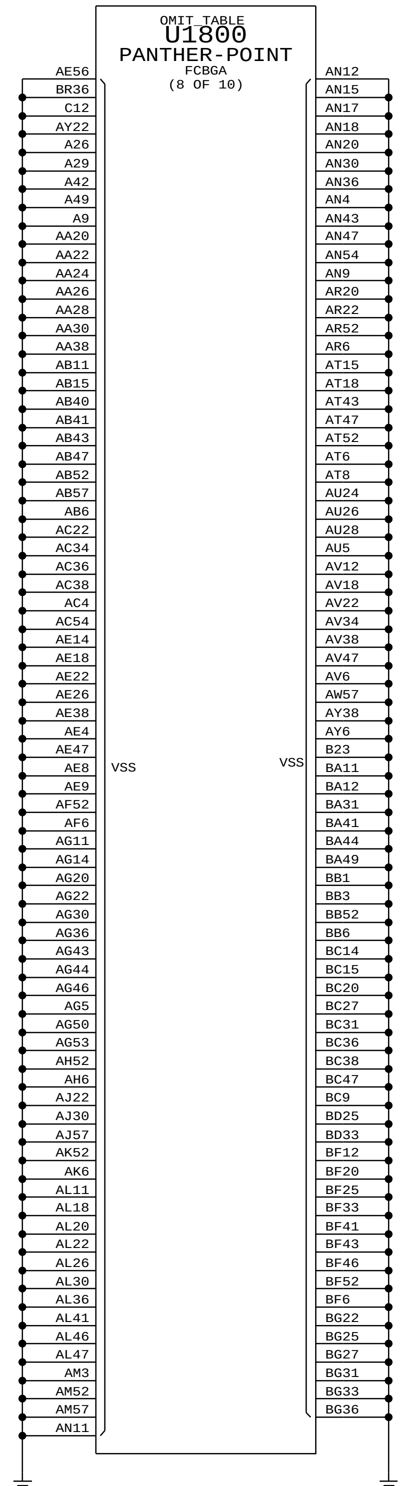
PAGE TITLE
PCH POWER

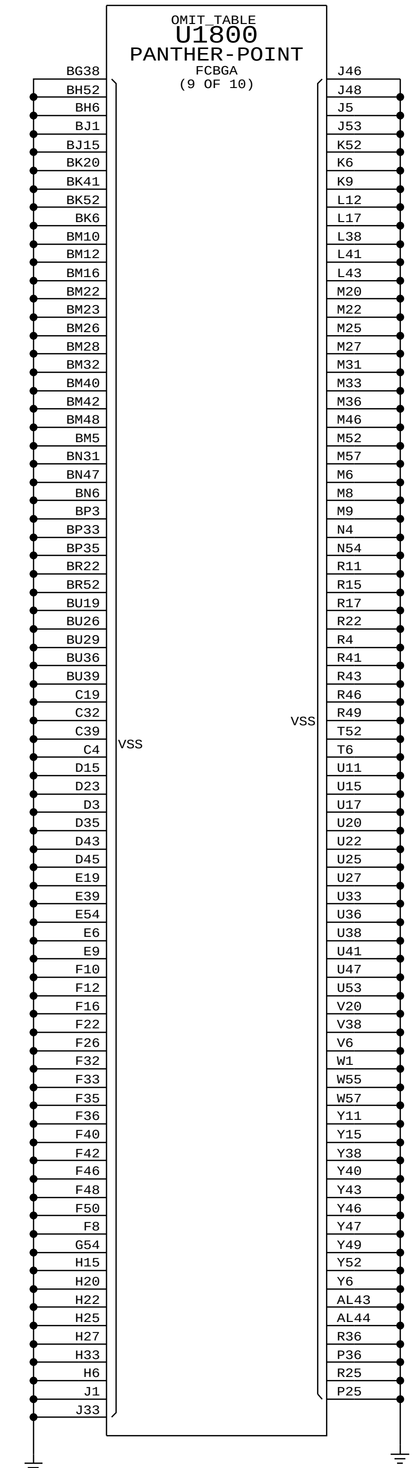
Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH





SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

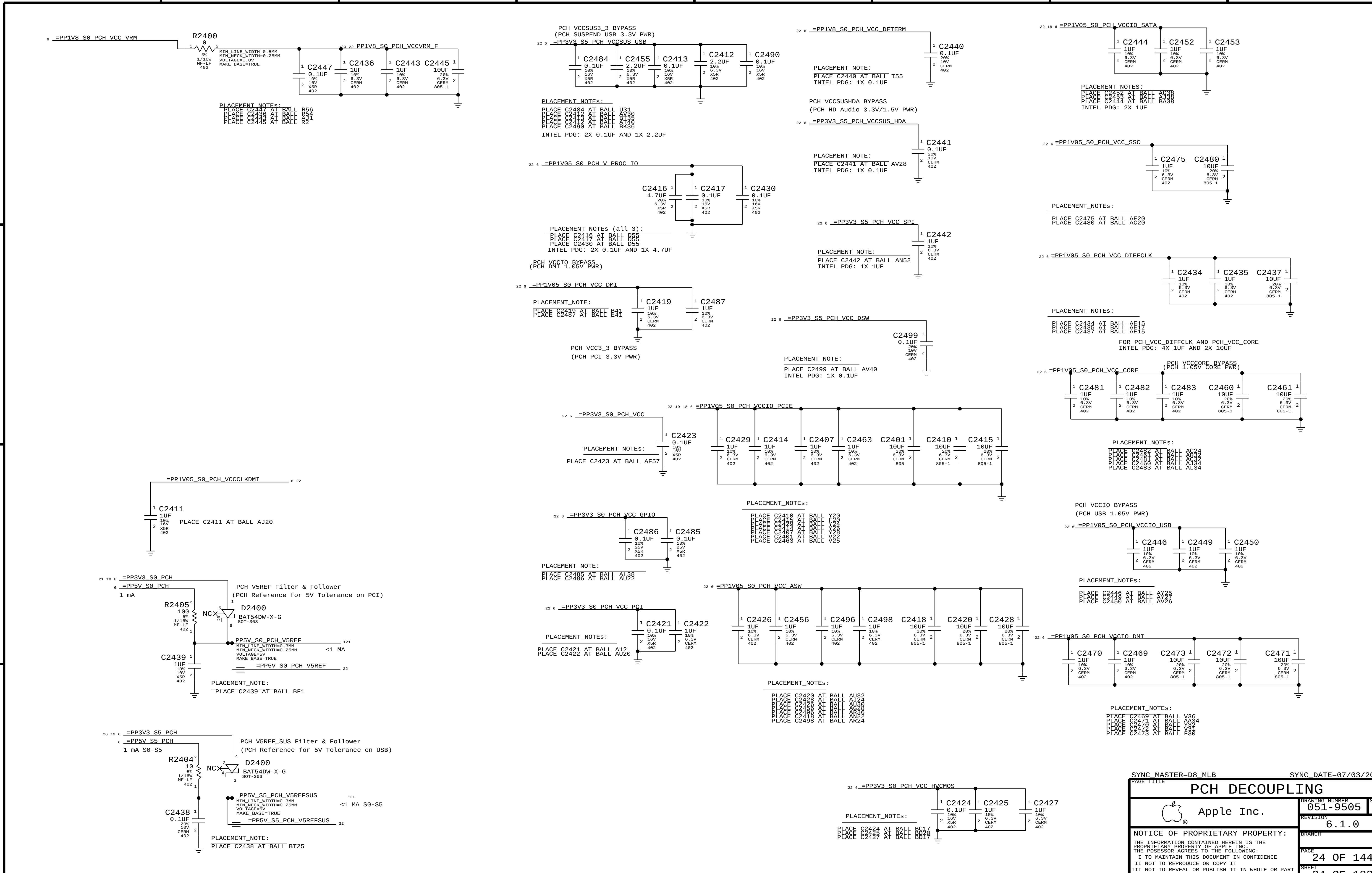
PCH GROUNDS

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

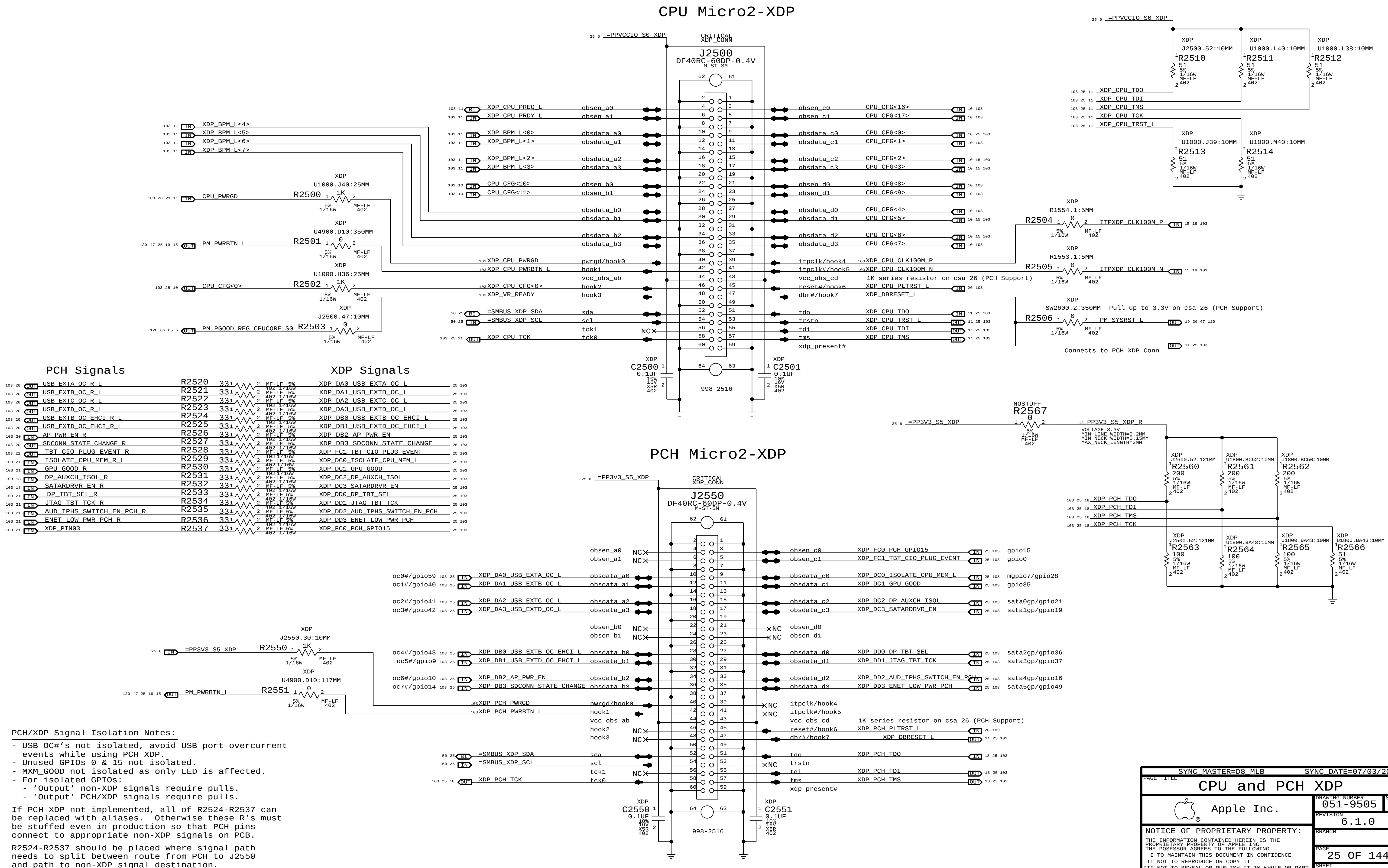
PCH DECOUPLING

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



PCH/XDP Signal Isolation Notes:

- USB OC#'s not isolated, avoid USB port overcurrent events while using PCH XDP.
- Unused GPIOs 0 & 15 not isolated.
- MXM_GOOD not isolated as only LED is affected.
- For isolated GPIOs:
  - 'Output' non-XDP signals require pulls.
  - 'Output' PCH/XDP signals require pulls.

If PCH XDP not implemented, all of R2524-R2537 can be replaced with aliases. Otherwise these R's must be stuffed even in production so that PCH pins connect to appropriate non-XDP signals on PCB.

R2524-R2537 should be placed where signal path needs to split between route from PCH to J2550 and path to non-XDP signal destination.

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

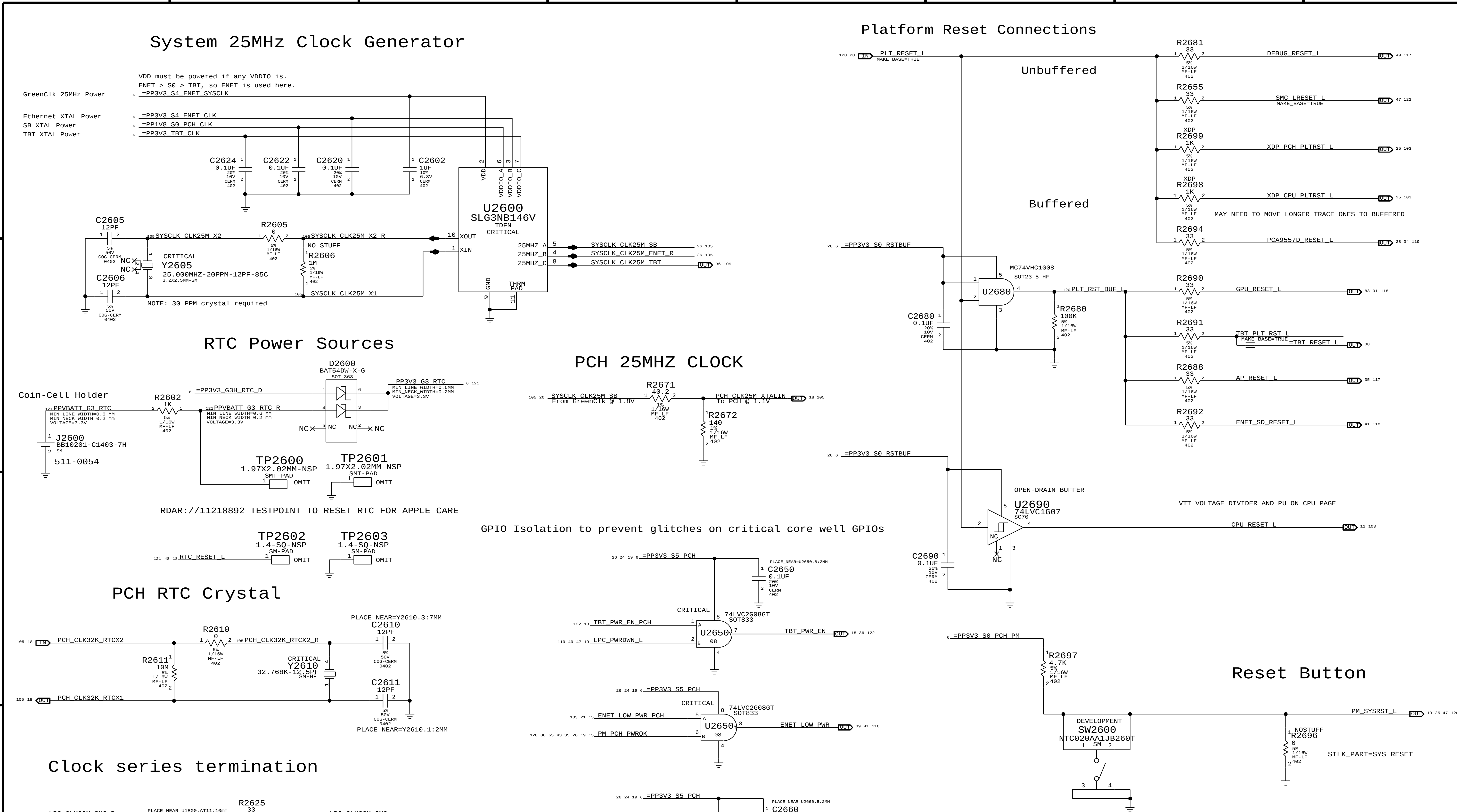
PAGE TITLE: CPU and PCH XDP

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

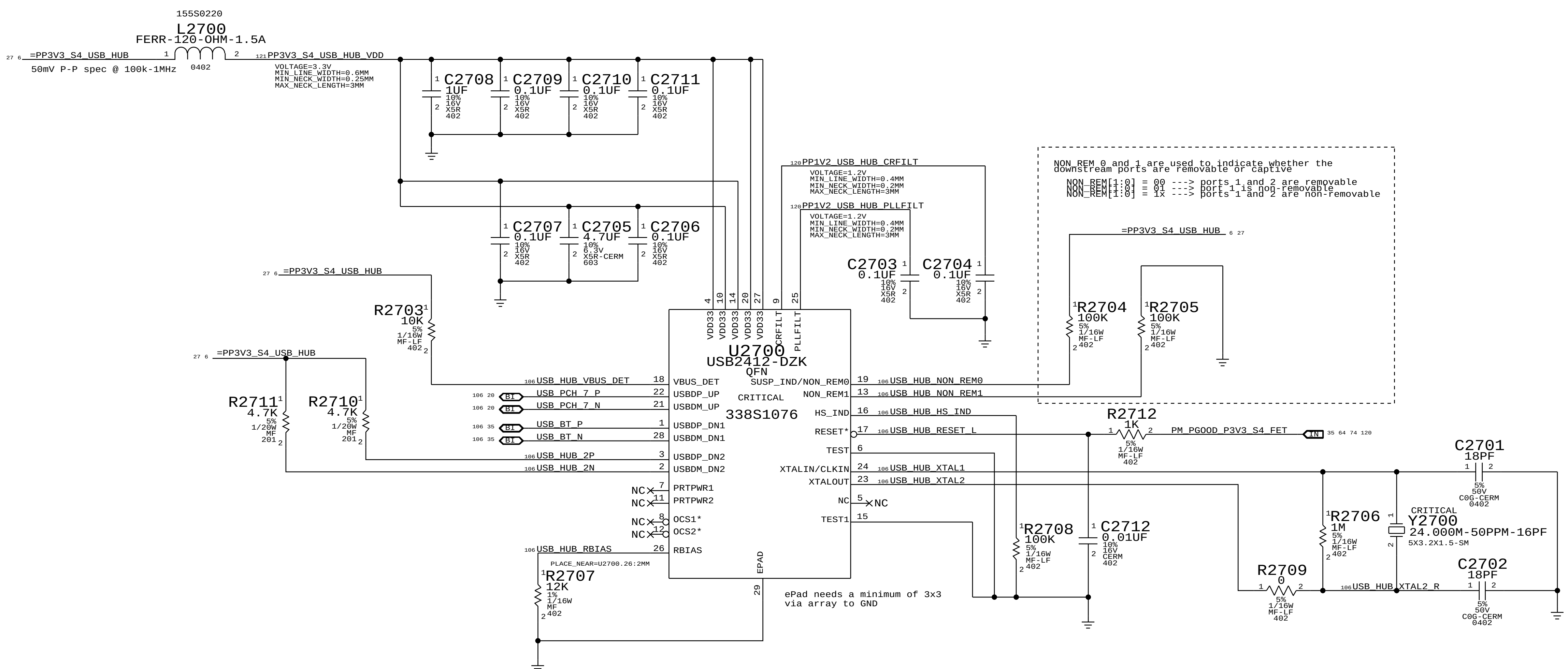
PAGE TITLE: CHIPSET SUPPORT

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

USB 2.0 HUB (BT/SMC)

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

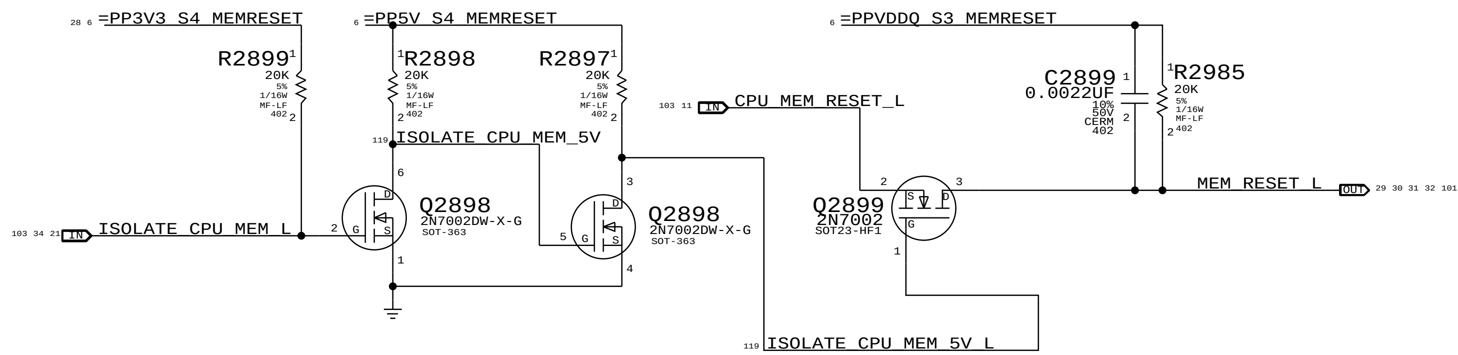
REVISION 6.1.0

BRANCH

MEM_RESET_L Generator

The circuits below handle MEMVTT power during S0->S3->S0 transitions, as well
as isolating the CPU's SM_DRAMRST# output from the SO-DIMMs when necessary.

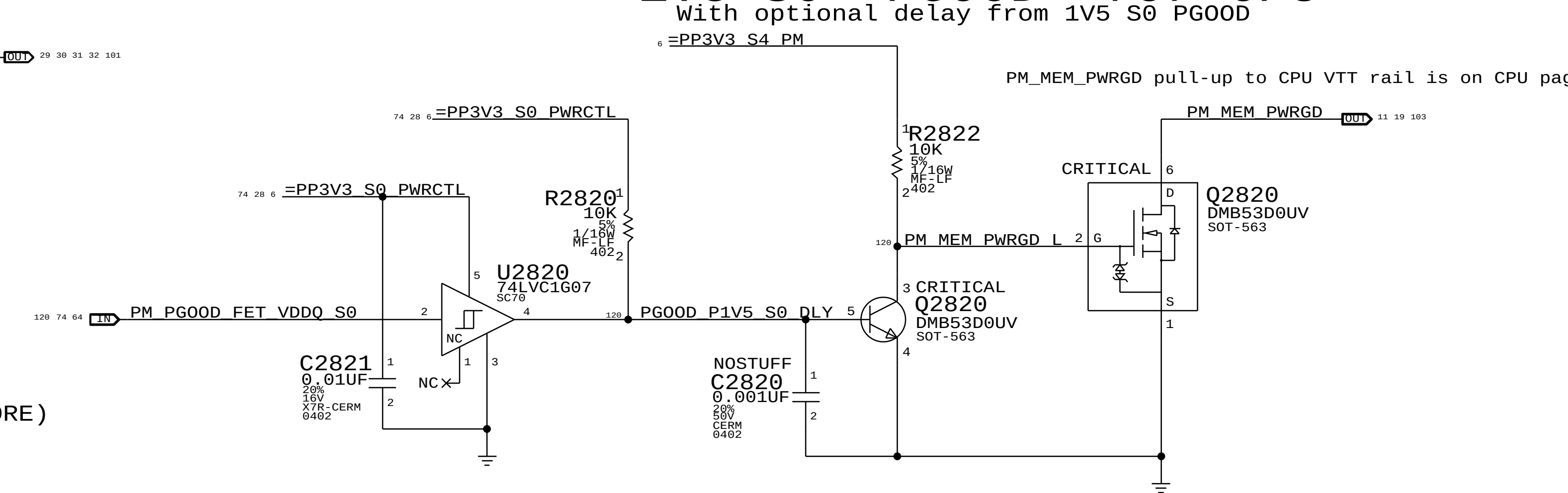
ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behaviour of signals.
WHEN HIGH: MEM_RESET_L NOT ISOLATED.
WHEN LOW: MEM_RESET_L IS ISOLATED.
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L (Block CPU from driving MEM_RESET_L in S3)



1V5 S0 "PGOOD" for CPU

With optional delay from 1V5 S0 PGOOD

PM_MEM_PWRGD pull-up to CPU VTT rail is on CPU page

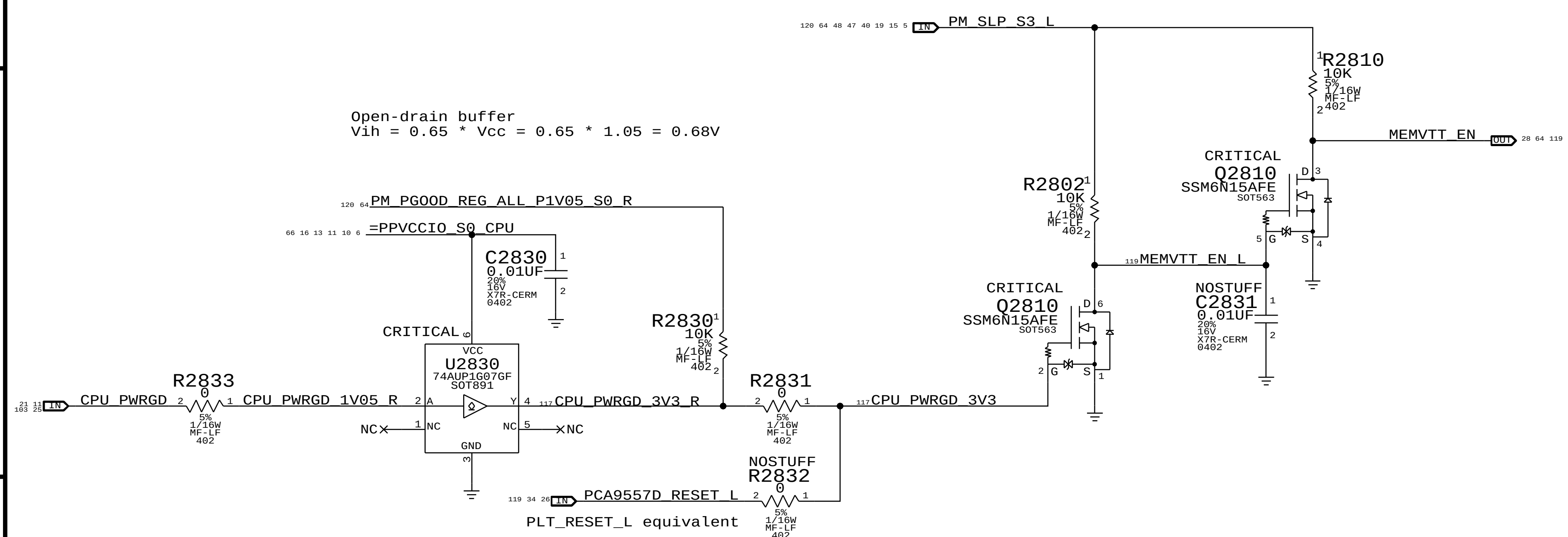


MEMVTT_EN Generator

rdar://11117167

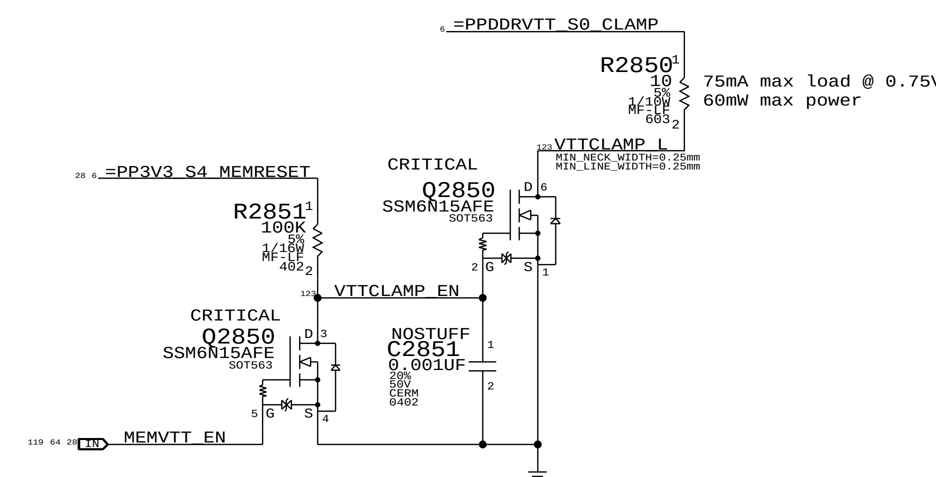
Enables MEMVTT when PCH drives CPU PWRGD.

CPU does not drive MEM_CKE until VCCORE activated but CPU 1V5 (VDDQ) leaks into it.
Clamping MEMVTT will keep the MEM_CKE low until CPU actively controls it.
MEMVTT Clamp actively holds MEMVTT rail low until MEMVTT is enabled.
MEMVTT_EN = CPU_PWRGD * PM_SLP_S3_L (VTT is enabled when PCH tells CPU to enable VCCORE)



MEMVTT Clamp

Ensures CKE signals are held low in S3 and in S0 before CPU PWRGD



S0 to S3 to S0

Step	ISOLATE_CPU_MEM_L	PM_SLP_S3_L	CPU_PWRGD	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
0	1	1	1	CPU_MEM_RESET_L	1	1
1	0	1	1	1	1	1
2	0	1	1	1	1	1
3	0	0	0	X	1	0
4	0	0	0	X	1	0
5	0	1	1	0 (*)	1	1
6	0	1	1	1	1	1
7	1	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: On a S5->S0 transition, ISOLATE_CPU_MEM_L will default low.
Rails will power-up as if from S3, but MEM_RESET_L now needs to be asserted in S0. Software must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE: CPU Memory S3 Support

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

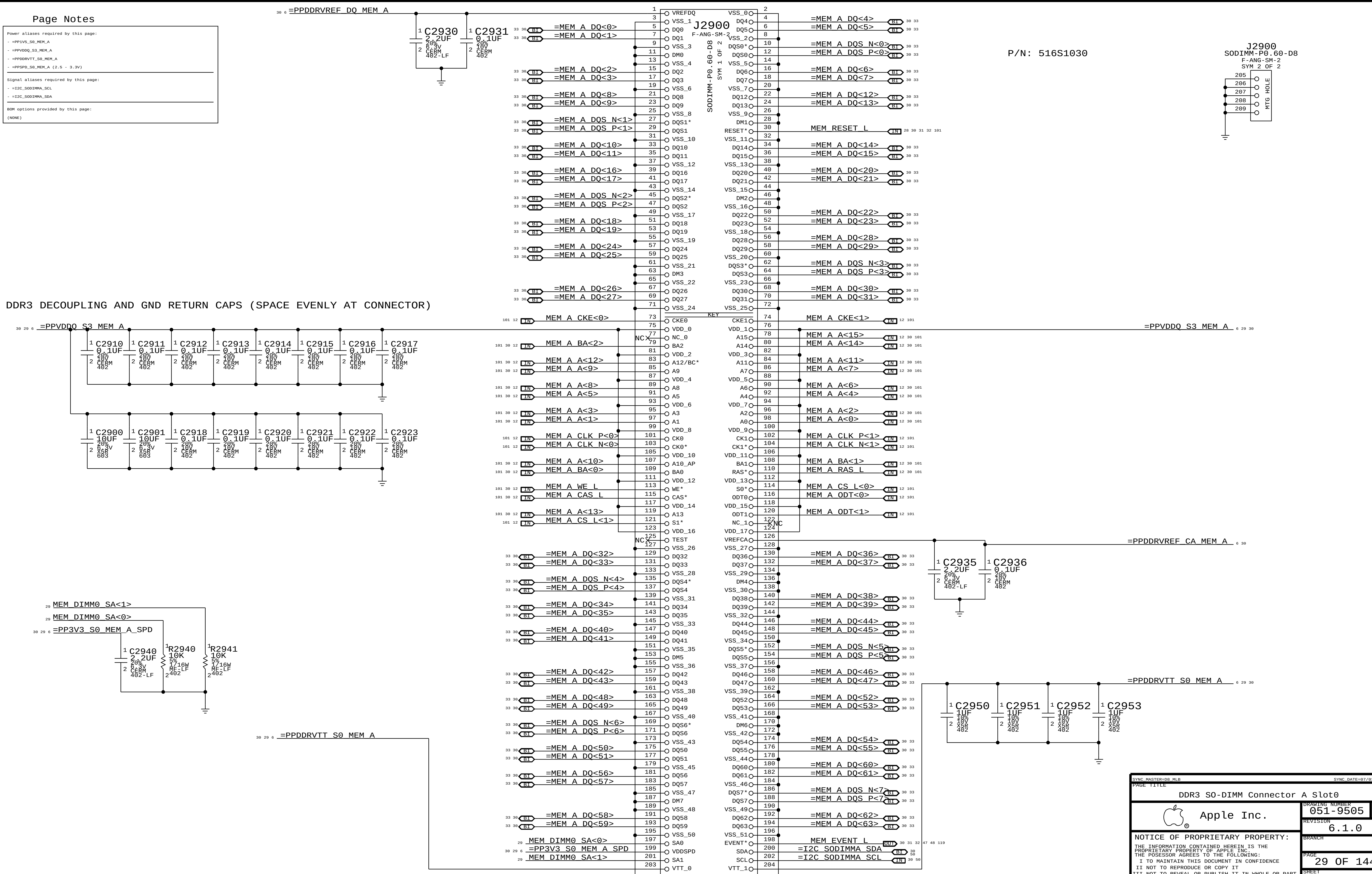
BRANCH

Page Notes

Power aliases required by this page:
- =PP1V5_S0_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVTT_S0_MEM_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:
(NONE)



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE: DDR3 SO-DIMM Connector A Slot0

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH

Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVTT_S0_MEM_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

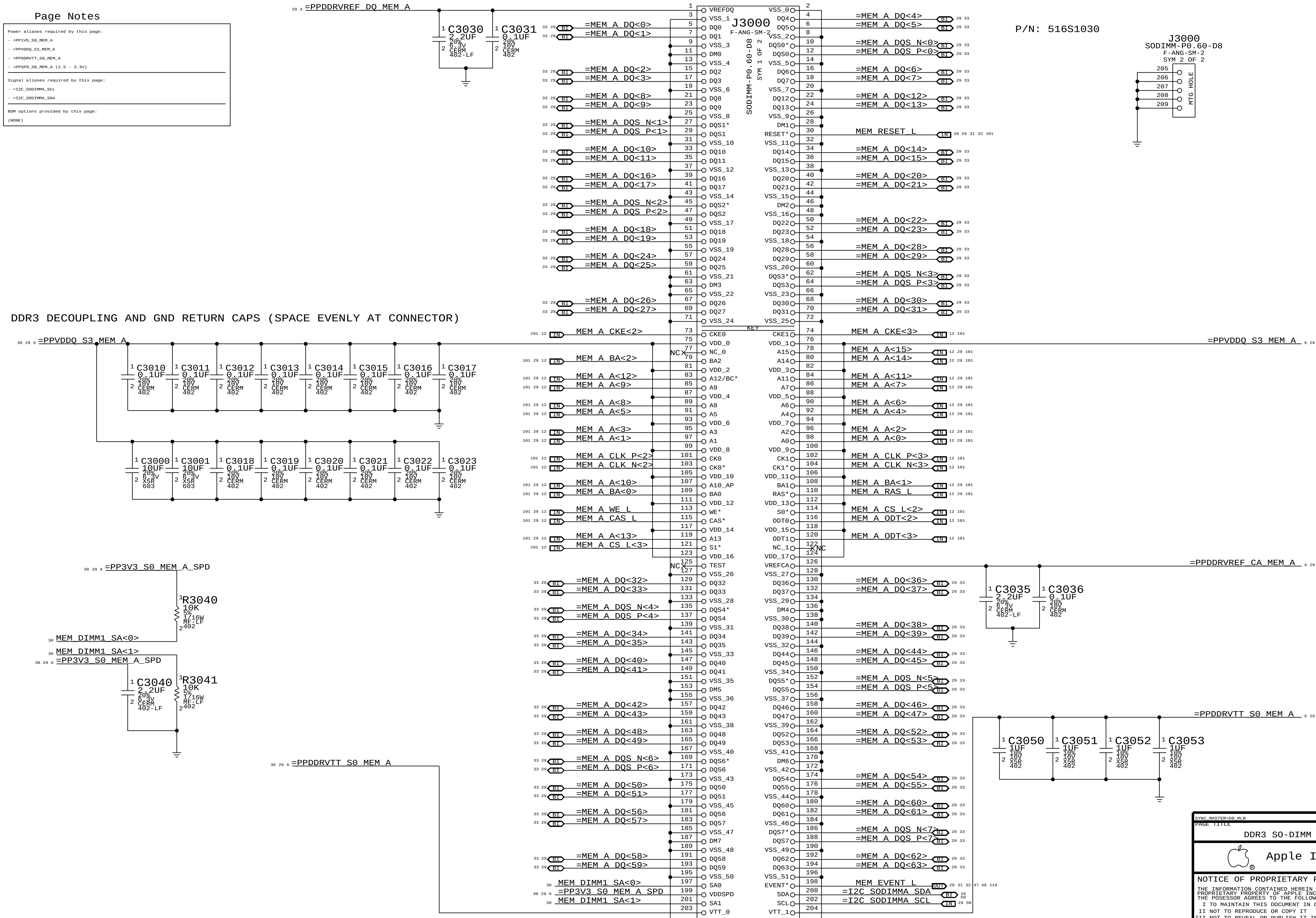
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)

P/N: 516S1030

DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

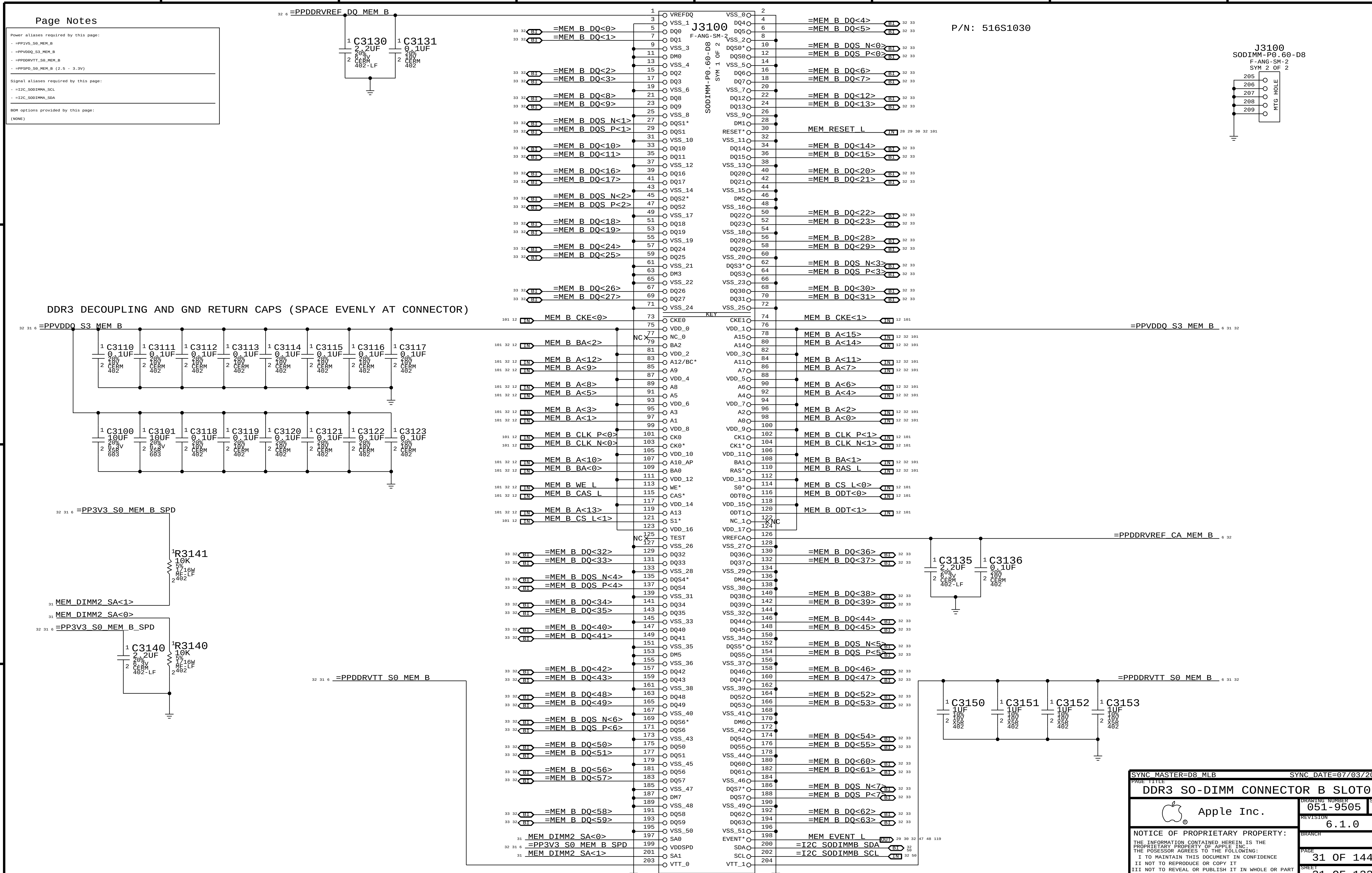
PAGE TITLE: DDR3 SO-DIMM Connector A Slot1

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

DDR3 SO-DIMM CONNECTOR B SLOT0

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

Page Notes

Power aliases required by this page:

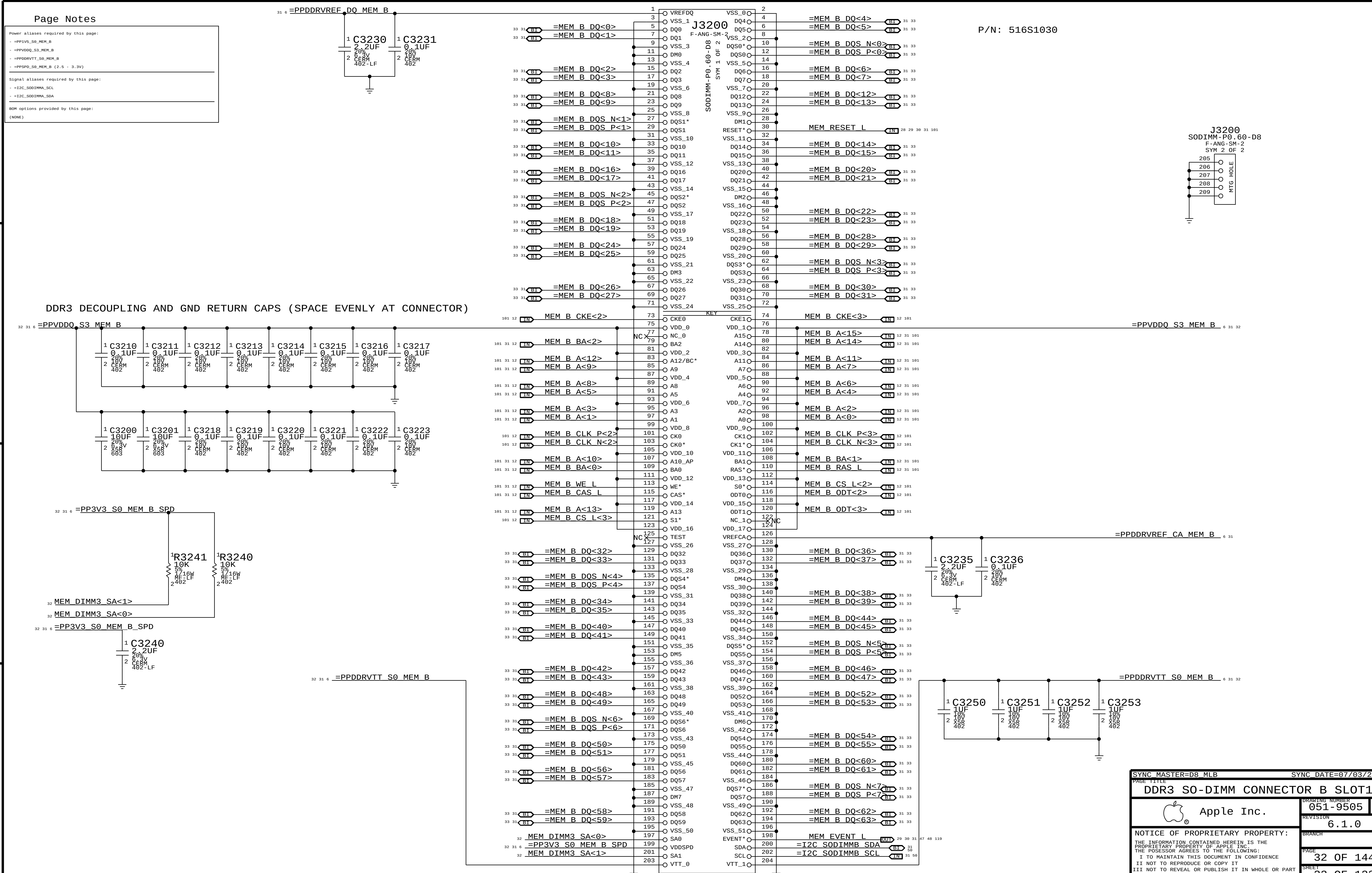
- =PP1V5_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVTT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

DDR3 SO-DIMM CONNECTOR B SLOT1

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

101 12	MEM_A_DQS_N<0>	MAKE_BASE=TRUE	=MEM_A_DQS_N<0>	29 30	101 12	MEM_B_DQS_N<0>	MAKE_BASE=TRUE	=MEM_B_DQS_N<0>	31 32
101 12	MEM_A_DQS_P<0>	MAKE_BASE=TRUE	=MEM_A_DQS_P<0>	29 30	101 12	MEM_B_DQS_P<0>	MAKE_BASE=TRUE	=MEM_B_DQS_P<0>	31 32
101 12	MEM_A_DQ<7>	MAKE_BASE=TRUE	=MEM_A_DQ<7>	29 30	101 12	MEM_B_DQ<7>	MAKE_BASE=TRUE	=MEM_B_DQ<3>	31 32
101 12	MEM_A_DQ<6>	MAKE_BASE=TRUE	=MEM_A_DQ<6>	29 30	101 12	MEM_B_DQ<6>	MAKE_BASE=TRUE	=MEM_B_DQ<6>	31 32
101 12	MEM_A_DQ<5>	MAKE_BASE=TRUE	=MEM_A_DQ<4>	29 30	101 12	MEM_B_DQ<5>	MAKE_BASE=TRUE	=MEM_B_DQ<5>	31 32
101 12	MEM_A_DQ<4>	MAKE_BASE=TRUE	=MEM_A_DQ<1>	29 30	101 12	MEM_B_DQ<4>	MAKE_BASE=TRUE	=MEM_B_DQ<4>	31 32
101 12	MEM_A_DQ<3>	MAKE_BASE=TRUE	=MEM_A_DQ<3>	29 30	101 12	MEM_B_DQ<3>	MAKE_BASE=TRUE	=MEM_B_DQ<7>	31 32
101 12	MEM_A_DQ<2>	MAKE_BASE=TRUE	=MEM_A_DQ<2>	29 30	101 12	MEM_B_DQ<2>	MAKE_BASE=TRUE	=MEM_B_DQ<2>	31 32
101 12	MEM_A_DQ<1>	MAKE_BASE=TRUE	=MEM_A_DQ<5>	29 30	101 12	MEM_B_DQ<1>	MAKE_BASE=TRUE	=MEM_B_DQ<1>	31 32
101 12	MEM_A_DQ<0>	MAKE_BASE=TRUE	=MEM_A_DQ<0>	29 30	101 12	MEM_B_DQ<0>	MAKE_BASE=TRUE	=MEM_B_DQ<0>	31 32
101 12	MEM_A_DQS_N<1>	MAKE_BASE=TRUE	=MEM_A_DQS_N<1>	29 30	101 12	MEM_B_DQS_N<1>	MAKE_BASE=TRUE	=MEM_B_DQS_N<1>	31 32
101 12	MEM_A_DQS_P<1>	MAKE_BASE=TRUE	=MEM_A_DQS_P<1>	29 30	101 12	MEM_B_DQS_P<1>	MAKE_BASE=TRUE	=MEM_B_DQS_P<1>	31 32
101 12	MEM_A_DQ<15>	MAKE_BASE=TRUE	=MEM_A_DQ<15>	29 30	101 12	MEM_B_DQ<15>	MAKE_BASE=TRUE	=MEM_B_DQ<11>	31 32
101 12	MEM_A_DQ<14>	MAKE_BASE=TRUE	=MEM_A_DQ<14>	29 30	101 12	MEM_B_DQ<14>	MAKE_BASE=TRUE	=MEM_B_DQ<10>	31 32
101 12	MEM_A_DQ<13>	MAKE_BASE=TRUE	=MEM_A_DQ<13>	29 30	101 12	MEM_B_DQ<13>	MAKE_BASE=TRUE	=MEM_B_DQ<13>	31 32
101 12	MEM_A_DQ<12>	MAKE_BASE=TRUE	=MEM_A_DQ<12>	29 30	101 12	MEM_B_DQ<12>	MAKE_BASE=TRUE	=MEM_B_DQ<12>	31 32
101 12	MEM_A_DQ<11>	MAKE_BASE=TRUE	=MEM_A_DQ<11>	29 30	101 12	MEM_B_DQ<11>	MAKE_BASE=TRUE	=MEM_B_DQ<15>	31 32
101 12	MEM_A_DQ<10>	MAKE_BASE=TRUE	=MEM_A_DQ<10>	29 30	101 12	MEM_B_DQ<10>	MAKE_BASE=TRUE	=MEM_B_DQ<14>	31 32
101 12	MEM_A_DQ<9>	MAKE_BASE=TRUE	=MEM_A_DQ<9>	29 30	101 12	MEM_B_DQ<9>	MAKE_BASE=TRUE	=MEM_B_DQ<8>	31 32
101 12	MEM_A_DQ<8>	MAKE_BASE=TRUE	=MEM_A_DQ<8>	29 30	101 12	MEM_B_DQ<8>	MAKE_BASE=TRUE	=MEM_B_DQ<9>	31 32
101 12	MEM_A_DQS_N<2>	MAKE_BASE=TRUE	=MEM_A_DQS_N<2>	29 30	101 12	MEM_B_DQS_N<2>	MAKE_BASE=TRUE	=MEM_B_DQS_N<2>	31 32
101 12	MEM_A_DQS_P<2>	MAKE_BASE=TRUE	=MEM_A_DQS_P<2>	29 30	101 12	MEM_B_DQS_P<2>	MAKE_BASE=TRUE	=MEM_B_DQS_P<2>	31 32
101 12	MEM_A_DQ<23>	MAKE_BASE=TRUE	=MEM_A_DQ<23>	29 30	101 12	MEM_B_DQ<23>	MAKE_BASE=TRUE	=MEM_B_DQ<22>	31 32
101 12	MEM_A_DQ<22>	MAKE_BASE=TRUE	=MEM_A_DQ<22>	29 30	101 12	MEM_B_DQ<22>	MAKE_BASE=TRUE	=MEM_B_DQ<18>	31 32
101 12	MEM_A_DQ<21>	MAKE_BASE=TRUE	=MEM_A_DQ<21>	29 30	101 12	MEM_B_DQ<21>	MAKE_BASE=TRUE	=MEM_B_DQ<21>	31 32
101 12	MEM_A_DQ<20>	MAKE_BASE=TRUE	=MEM_A_DQ<20>	29 30	101 12	MEM_B_DQ<20>	MAKE_BASE=TRUE	=MEM_B_DQ<20>	31 32
101 12	MEM_A_DQ<19>	MAKE_BASE=TRUE	=MEM_A_DQ<18>	29 30	101 12	MEM_B_DQ<19>	MAKE_BASE=TRUE	=MEM_B_DQ<23>	31 32
101 12	MEM_A_DQ<18>	MAKE_BASE=TRUE	=MEM_A_DQ<19>	29 30	101 12	MEM_B_DQ<18>	MAKE_BASE=TRUE	=MEM_B_DQ<19>	31 32
101 12	MEM_A_DQ<17>	MAKE_BASE=TRUE	=MEM_A_DQ<17>	29 30	101 12	MEM_B_DQ<17>	MAKE_BASE=TRUE	=MEM_B_DQ<17>	31 32
101 12	MEM_A_DQ<16>	MAKE_BASE=TRUE	=MEM_A_DQ<16>	29 30	101 12	MEM_B_DQ<16>	MAKE_BASE=TRUE	=MEM_B_DQ<16>	31 32
101 12	MEM_A_DQS_N<3>	MAKE_BASE=TRUE	=MEM_A_DQS_N<3>	29 30	101 12	MEM_B_DQS_N<3>	MAKE_BASE=TRUE	=MEM_B_DQS_N<3>	31 32
101 12	MEM_A_DQS_P<3>	MAKE_BASE=TRUE	=MEM_A_DQS_P<3>	29 30	101 12	MEM_B_DQS_P<3>	MAKE_BASE=TRUE	=MEM_B_DQS_P<3>	31 32
101 12	MEM_A_DQ<31>	MAKE_BASE=TRUE	=MEM_A_DQ<31>	29 30	101 12	MEM_B_DQ<31>	MAKE_BASE=TRUE	=MEM_B_DQ<31>	31 32
101 12	MEM_A_DQ<30>	MAKE_BASE=TRUE	=MEM_A_DQ<30>	29 30	101 12	MEM_B_DQ<30>	MAKE_BASE=TRUE	=MEM_B_DQ<30>	31 32
101 12	MEM_A_DQ<29>	MAKE_BASE=TRUE	=MEM_A_DQ<29>	29 30	101 12	MEM_B_DQ<29>	MAKE_BASE=TRUE	=MEM_B_DQ<29>	31 32
101 12	MEM_A_DQ<28>	MAKE_BASE=TRUE	=MEM_A_DQ<28>	29 30	101 12	MEM_B_DQ<28>	MAKE_BASE=TRUE	=MEM_B_DQ<28>	31 32
101 12	MEM_A_DQ<27>	MAKE_BASE=TRUE	=MEM_A_DQ<27>	29 30	101 12	MEM_B_DQ<27>	MAKE_BASE=TRUE	=MEM_B_DQ<27>	31 32
101 12	MEM_A_DQ<26>	MAKE_BASE=TRUE	=MEM_A_DQ<26>	29 30	101 12	MEM_B_DQ<26>	MAKE_BASE=TRUE	=MEM_B_DQ<26>	31 32
101 12	MEM_A_DQ<25>	MAKE_BASE=TRUE	=MEM_A_DQ<25>	29 30	101 12	MEM_B_DQ<25>	MAKE_BASE=TRUE	=MEM_B_DQ<25>	31 32
101 12	MEM_A_DQ<24>	MAKE_BASE=TRUE	=MEM_A_DQ<24>	29 30	101 12	MEM_B_DQ<24>	MAKE_BASE=TRUE	=MEM_B_DQ<24>	31 32
101 12	MEM_A_DQS_N<4>	MAKE_BASE=TRUE	=MEM_A_DQS_N<4>	29 30	101 12	MEM_B_DQS_N<4>	MAKE_BASE=TRUE	=MEM_B_DQS_N<4>	31 32
101 12	MEM_A_DQS_P<4>	MAKE_BASE=TRUE	=MEM_A_DQS_P<4>	29 30	101 12	MEM_B_DQS_P<4>	MAKE_BASE=TRUE	=MEM_B_DQS_P<4>	31 32
101 12	MEM_A_DQ<39>	MAKE_BASE=TRUE	=MEM_A_DQ<39>	29 30	101 12	MEM_B_DQ<39>	MAKE_BASE=TRUE	=MEM_B_DQ<39>	31 32
101 12	MEM_A_DQ<38>	MAKE_BASE=TRUE	=MEM_A_DQ<38>	29 30	101 12	MEM_B_DQ<38>	MAKE_BASE=TRUE	=MEM_B_DQ<38>	31 32
101 12	MEM_A_DQ<37>	MAKE_BASE=TRUE	=MEM_A_DQ<32>	29 30	101 12	MEM_B_DQ<37>	MAKE_BASE=TRUE	=MEM_B_DQ<37>	31 32
101 12	MEM_A_DQ<36>	MAKE_BASE=TRUE	=MEM_A_DQ<37>	29 30	101 12	MEM_B_DQ<36>	MAKE_BASE=TRUE	=MEM_B_DQ<36>	31 32
101 12	MEM_A_DQ<35>	MAKE_BASE=TRUE	=MEM_A_DQ<35>	29 30	101 12	MEM_B_DQ<35>	MAKE_BASE=TRUE	=MEM_B_DQ<35>	31 32
101 12	MEM_A_DQ<34>	MAKE_BASE=TRUE	=MEM_A_DQ<34>	29 30	101 12	MEM_B_DQ<34>	MAKE_BASE=TRUE	=MEM_B_DQ<34>	31 32
101 12	MEM_A_DQ<33>	MAKE_BASE=TRUE	=MEM_A_DQ<33>	29 30	101 12	MEM_B_DQ<33>	MAKE_BASE=TRUE	=MEM_B_DQ<33>	31 32
101 12	MEM_A_DQ<32>	MAKE_BASE=TRUE	=MEM_A_DQ<36>	29 30	101 12	MEM_B_DQ<32>	MAKE_BASE=TRUE	=MEM_B_DQ<32>	31 32
101 12	MEM_A_DQS_N<5>	MAKE_BASE=TRUE	=MEM_A_DQS_N<5>	29 30	101 12	MEM_B_DQS_N<5>	MAKE_BASE=TRUE	=MEM_B_DQS_N<5>	31 32
101 12	MEM_A_DQS_P<5>	MAKE_BASE=TRUE	=MEM_A_DQS_P<5>	29 30	101 12	MEM_B_DQS_P<5>	MAKE_BASE=TRUE	=MEM_B_DQS_P<5>	31 32
101 12	MEM_A_DQ<47>	MAKE_BASE=TRUE	=MEM_A_DQ<47>	29 30	101 12	MEM_B_DQ<47>	MAKE_BASE=TRUE	=MEM_B_DQ<47>	31 32
101 12	MEM_A_DQ<46>	MAKE_BASE=TRUE	=MEM_A_DQ<46>	29 30	101 12	MEM_B_DQ<46>	MAKE_BASE=TRUE	=MEM_B_DQ<46>	31 32
101 12	MEM_A_DQ<45>	MAKE_BASE=TRUE	=MEM_A_DQ<45>	29 30	101 12	MEM_B_DQ<45>	MAKE_BASE=TRUE	=MEM_B_DQ<45>	31 32
101 12	MEM_A_DQ<44>	MAKE_BASE=TRUE	=MEM_A_DQ<44>	29 30	101 12	MEM_B_DQ<44>	MAKE_BASE=TRUE	=MEM_B_DQ<44>	31 32
101 12	MEM_A_DQ<43>	MAKE_BASE=TRUE	=MEM_A_DQ<43>	29 30	101 12	MEM_B_DQ<43>	MAKE_BASE=TRUE	=MEM_B_DQ<43>	31 32
101 12	MEM_A_DQ<42>	MAKE_BASE=TRUE	=MEM_A_DQ<42>	29 30	101 12	MEM_B_DQ<42>	MAKE_BASE=TRUE	=MEM_B_DQ<42>	31 32
101 12	MEM_A_DQ<41>	MAKE_BASE=TRUE	=MEM_A_DQ<41>	29 30	101 12	MEM_B_DQ<41>	MAKE_BASE=TRUE	=MEM_B_DQ<41>	31 32
101 12	MEM_A_DQ<40>	MAKE_BASE=TRUE	=MEM_A_DQ<40>	29 30	101 12	MEM_B_DQ<40>	MAKE_BASE=TRUE	=MEM_B_DQ<40>	31 32
101 12	MEM_A_DQS_N<6>	MAKE_BASE=TRUE	=MEM_A_DQS_N<6>	29 30	101 12	MEM_B_DQS_N<6>	MAKE_BASE=TRUE	=MEM_B_DQS_N<6>	31 32
101 12	MEM_A_DQS_P<6>	MAKE_BASE=TRUE	=MEM_A_DQS_P<6>	29 30	101 12	MEM_B_DQS_P<6>	MAKE_BASE=TRUE	=MEM_B_DQS_P<6>	31 32
101 12	MEM_A_DQ<55>	MAKE_BASE=TRUE	=MEM_A_DQ<55>	29 30	101 12	MEM_B_DQ<55>	MAKE_BASE=TRUE	=MEM_B_DQ<55>	31 32
101 12	MEM_A_DQ<54>	MAKE_BASE=TRUE	=MEM_A_DQ<54>	29 30	101 12	MEM_B_DQ<54>	MAKE_BASE=TRUE	=MEM_B_DQ<54>	31 32
101 12	MEM_A_DQ<53>	MAKE_BASE=TRUE	=MEM_A_DQ<53>	29 30	101 12	MEM_B_DQ<53>	MAKE_BASE=TRUE	=MEM_B_DQ<52>	31 32
101 12	MEM_A_DQ<52>	MAKE_BASE=TRUE	=MEM_A_DQ<52>	29 30	101 12	MEM_B_DQ<52>	MAKE_BASE=TRUE	=MEM_B_DQ<53>	31 32
101 12	MEM_A_DQ<51>	MAKE_BASE=TRUE	=MEM_A_DQ<51>	29 30	101 12	MEM_B_DQ<51>	MAKE_BASE=TRUE	=MEM_B_DQ<51>	31 32
101 12	MEM_A_DQ<50>	MAKE_BASE=TRUE	=MEM_A_DQ<50>	29 30	101 12	MEM_B_DQ<50>	MAKE_BASE=TRUE	=MEM_B_DQ<50>	31 32
101 12	MEM_A_DQ<49>	MAKE_BASE=TRUE	=MEM_A_DQ<49>	29 30	101 12	MEM_B_DQ<49>	MAKE_BASE=TRUE	=MEM_B_DQ<48>	31 32
101 12	MEM_A_DQ<48>	MAKE_BASE=TRUE	=MEM_A_DQ<48>	29 30	101 12	MEM_B_DQ<48>	MAKE_BASE=TRUE	=MEM_B_DQ<49>	31 32
101 12	MEM_A_DQS_N<7>	MAKE_BASE=TRUE	=MEM_A_DQS_N<7>	29 30	101 12	MEM_B_DQS_N<7>	MAKE_BASE=TRUE	=MEM_B_DQS_N<7>	31 32
101 12	MEM_A_DQS_P<7>	MAKE_BASE=TRUE	=MEM_A_DQS_P<7>	29 30	101 12	MEM_B_DQS_P<7>	MAKE_BASE=TRUE	=MEM_B_DQS_P<7>	31 32
101 12	MEM_A_DQ<63>	MAKE_BASE=TRUE	=MEM_A_DQ<62>	29 30	101 12	MEM_B_DQ<63>	MAKE_BASE=TRUE	=MEM_B_DQ<63>	31 32
101 12	MEM_A_DQ<62>	MAKE_BASE=TRUE	=MEM_A_DQ<58>	29 30	101 12	MEM_B_DQ<62>	MAKE_BASE=TRUE	=MEM_B_DQ<58>	31 32
101 12	MEM_A_DQ<61>	MAKE_BASE=TRUE	=MEM_A_DQ<61>	29 30	101 12	MEM_B_DQ<61>	MAKE_BASE=TRUE	=MEM_B_DQ<60>	31 32
101 12	MEM_A_DQ<60>	MAKE_BASE=TRUE	=MEM_A_DQ<60>	29 30	101 12	MEM_B_DQ<60>	MAKE_BASE=TRUE	=MEM_B_DQ<61>	31 32
101 12	MEM_A_DQ<59>	MAKE_BASE=TRUE	=MEM_A_DQ<63>	29 30	101 12	MEM_B_DQ<59>	MAKE_BASE=TRUE	=MEM_B_DQ<59>	31 32
101 12	MEM_A_DQ<58>	MAKE_BASE=TRUE	=MEM_A_DQ<59>	29 30	101 12	MEM_B_DQ<58>	MAKE_BASE=TRUE	=MEM_B_DQ<62>	31 32
101 12	MEM_A_DQ<57>	MAKE_BASE=TRUE	=MEM_A_DQ<57>	29 30	101 12	MEM_B_DQ<57>	MAKE_BASE=TRUE	=MEM_B_DQ<56>	31 32
101 12	MEM_A_DQ<56>	MAKE_BASE=TRUE	=MEM_A_DQ<56>	29 30	101 12	MEM_B_DQ<56>	MAKE_BASE=TRUE	=MEM_B_DQ<57>	31 32

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

DDR3 ALIASES AND BITSWAPS

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

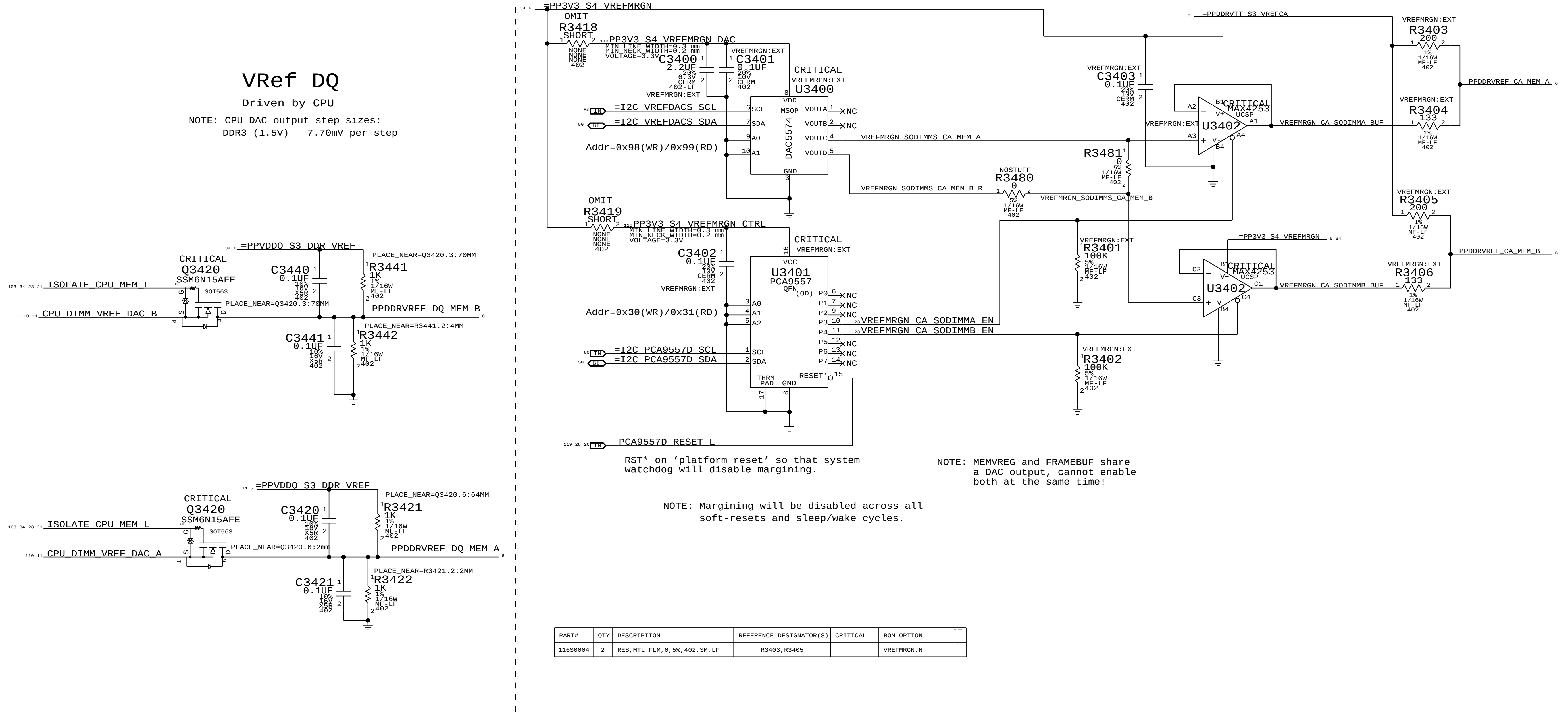
REVISION 6.1.0

BRANCH

VRef DQ

Driven by CPU

NOTE: CPU DAC output step sizes:
DDR3 (1.5V)   7.70mV per step



RST* on 'platform reset' so that system watchdog will disable margining.

NOTE: MEMVREG and FRAMEBUF share a DAC output, cannot enable both at the same time!

NOTE: Margining will be disabled across all soft-resets and sleep/wake cycles.

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
116S0004	2	RES,MTL FLM,0,5%,402,SM,LF	R3403,R3405		VREFMRGN:N

	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% VRef)
DAC Channel:	C	C	D	D
PCA9557D Pin:	3	4	5	6
Nominal value	0.75V (DAC: 0x3A)		1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:	0.300V - 1.200V (+/- 450mV)		1.000V - 2.000V (+/- 500mV)	1.056V - 1.442V (+/- 180mV)
DAC range:	0.000V - 1.501V (0x00 - 0x74)		0.000V - 3.000V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:	+3.4mA - -3.4mA (- = sourced)		+61uA - -61uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:	7.69mV / step @ output		8.59mV / step @ output	1.51mV / step @ output

SYNC_MASTER=D8_MLB   SYNC_DATE=07/03/2012

PAGE TITLE
DDR3/FRAMEBUF VREF MARGINING

Apple Inc.

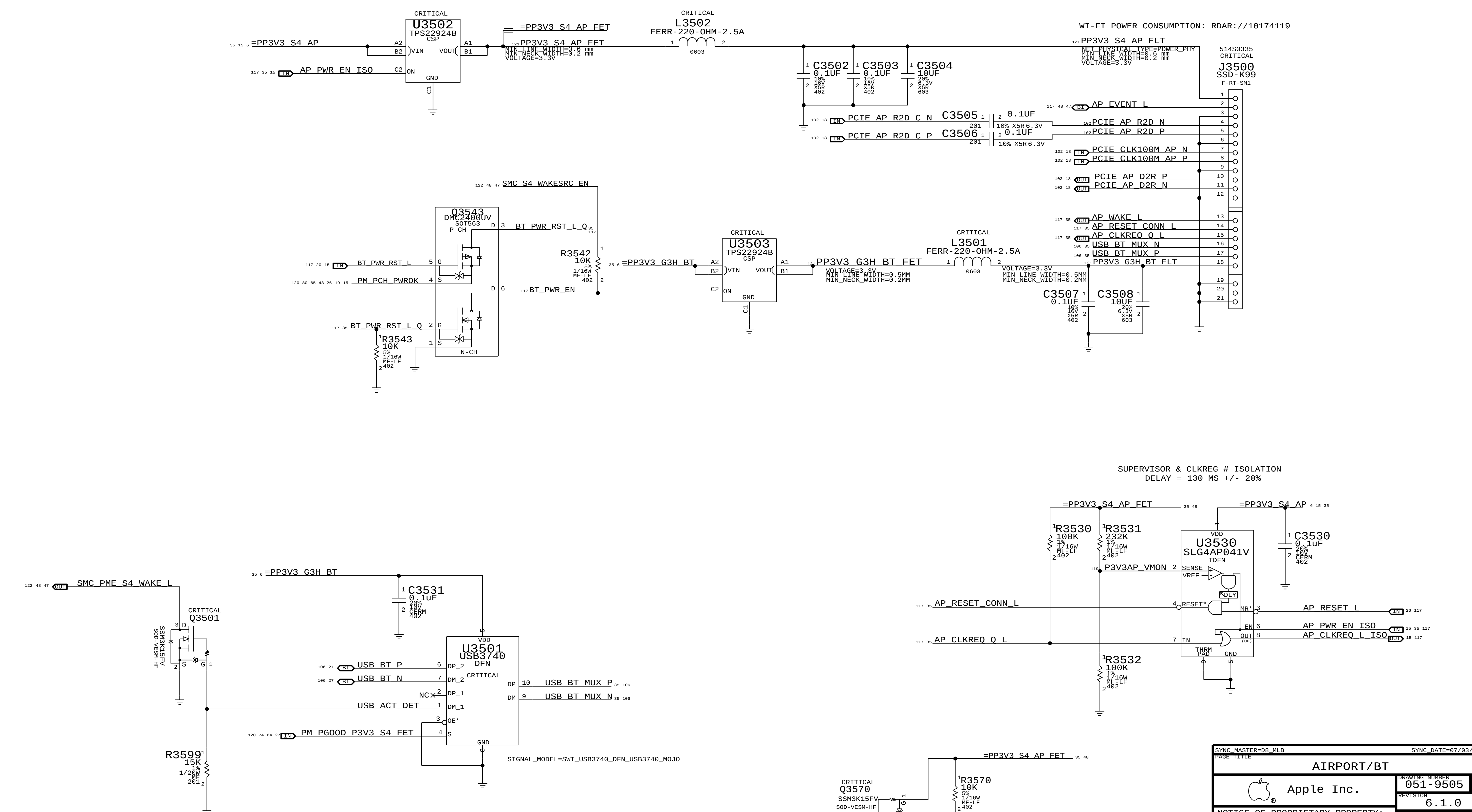
DRAWING NUMBER 051-9505   SIZE D

REVISION 6.1.0

BRANCH

AIRPORT
BLUETOOTH

WI-FI POWER CONSUMPTION: RDAR://10174119



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

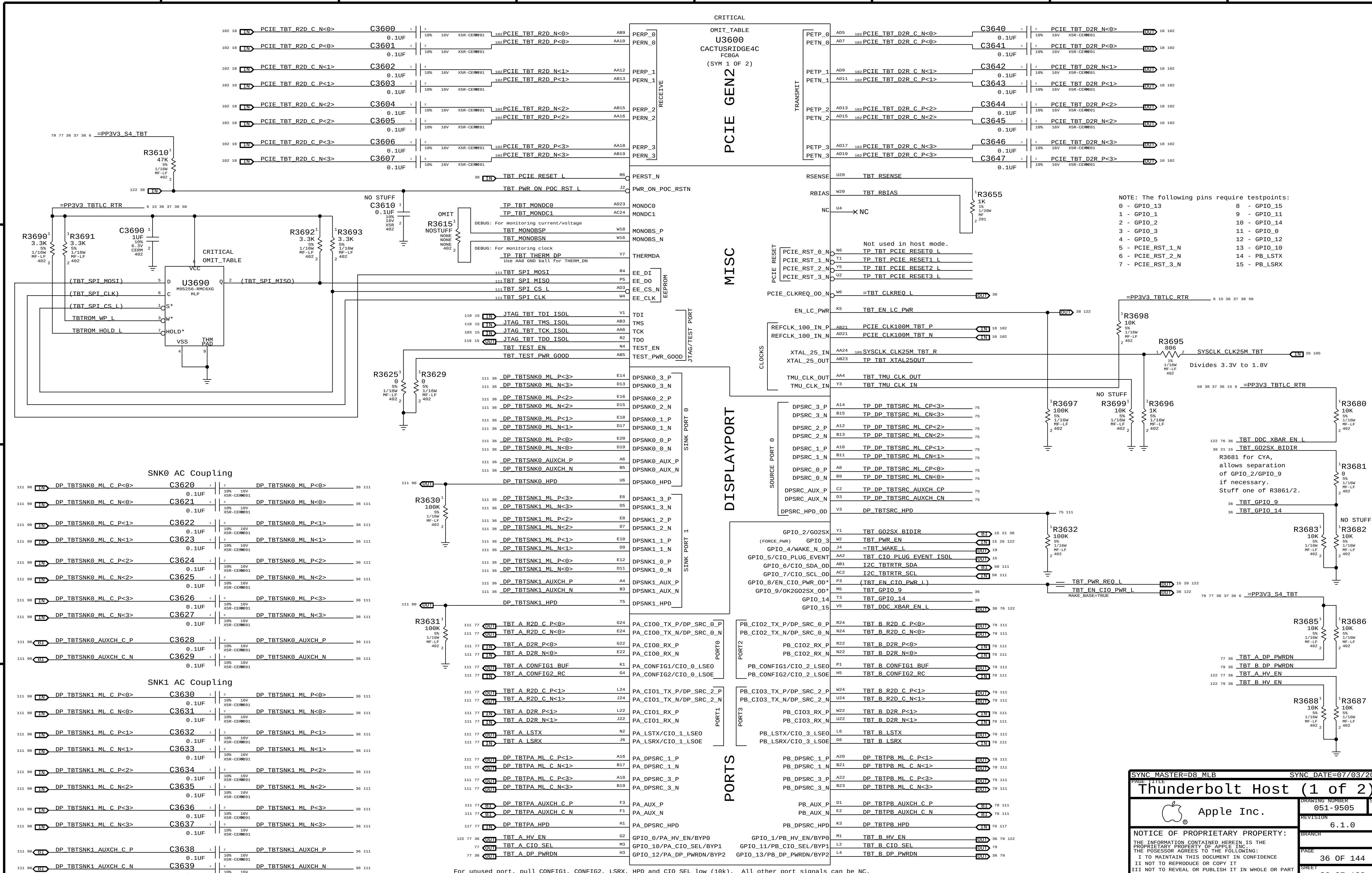
PAGE TITLE: AIRPORT/BT

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

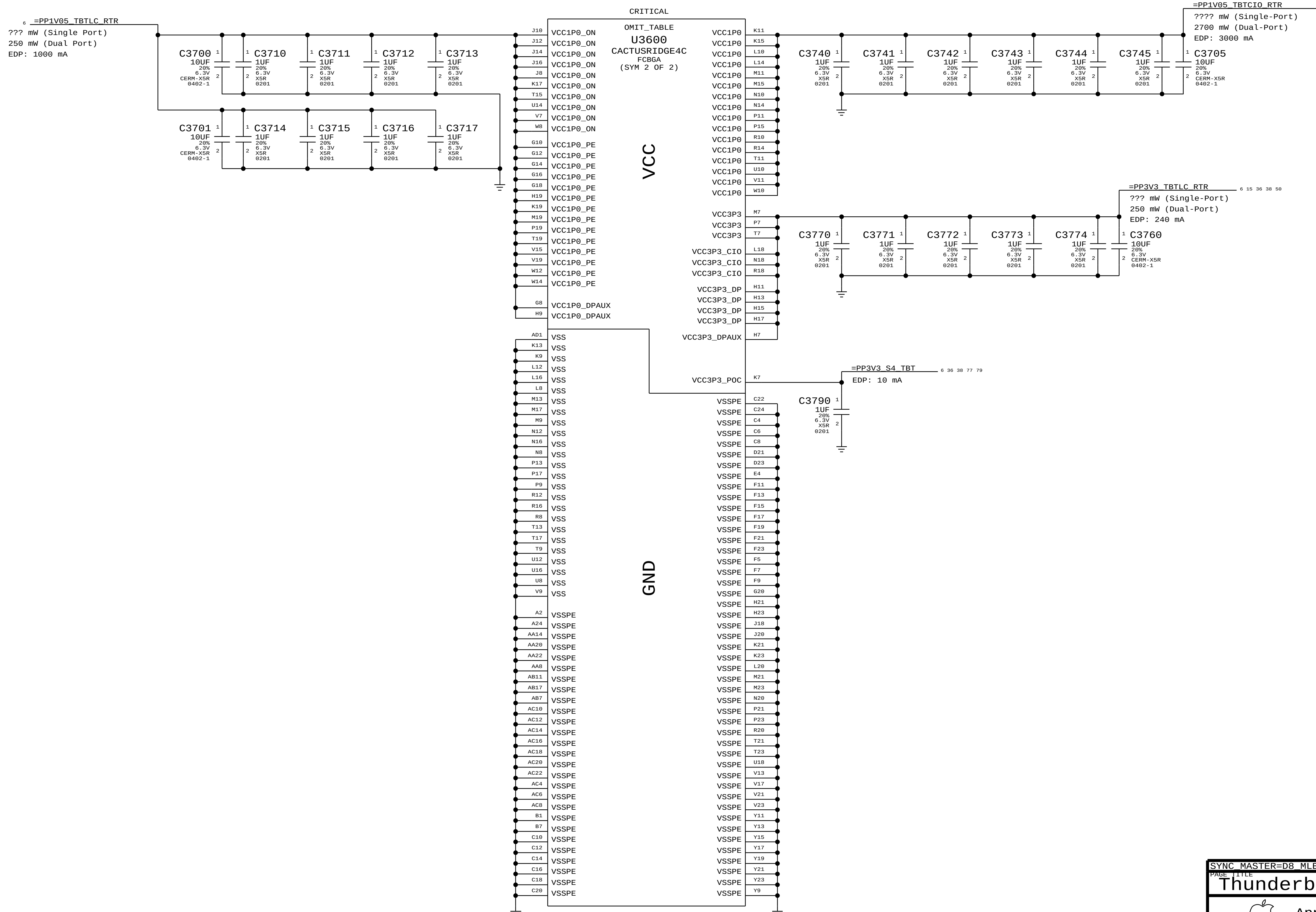
Thunderbolt Host (1 of 2)

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



EDP current / power consumption figures from CR DG v0.57, IBL doc #472455.

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE: Thunderbolt Host (2 of 2)

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH

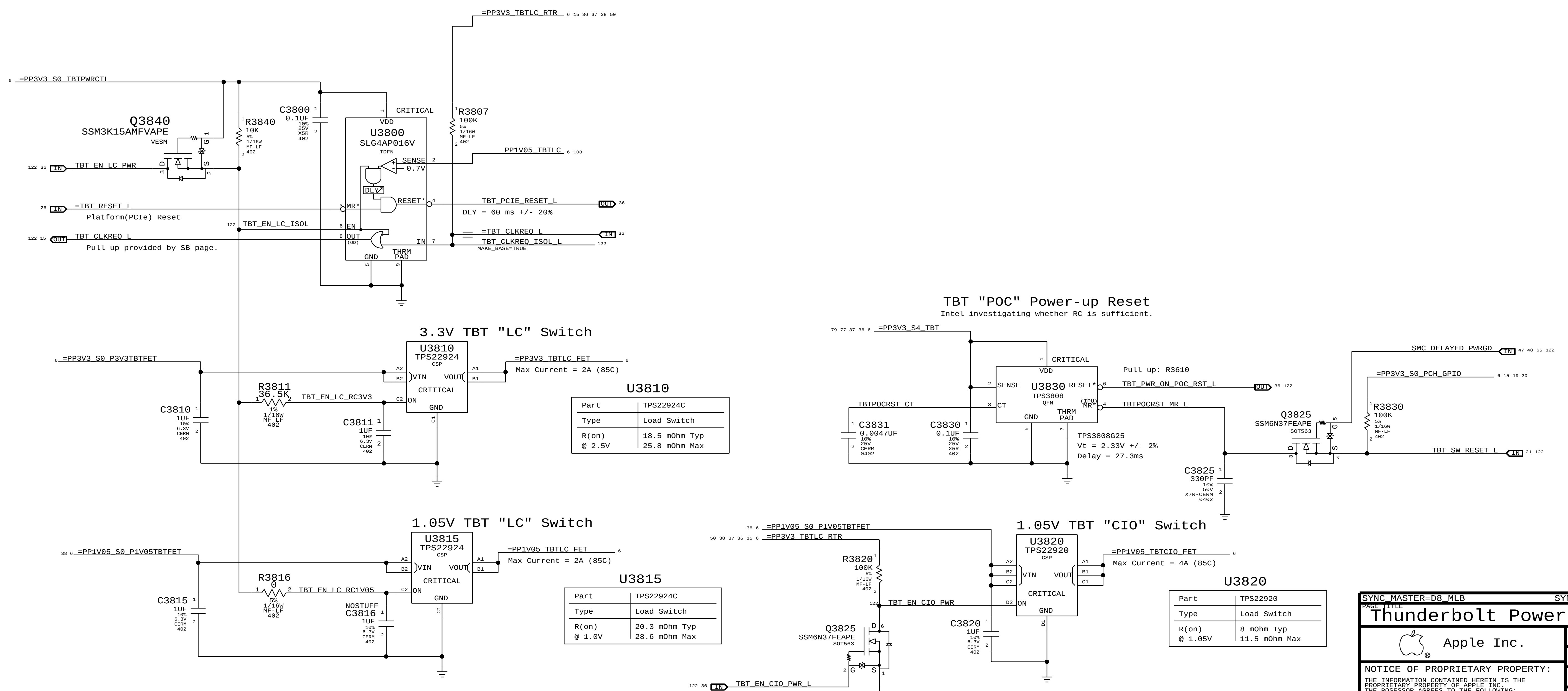
Page Notes

Power aliases required by this page:
- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_TBT_P3V3TBTFET (3.3V FET Input)
- =PP3V3_TBT_FET (3.3V FET Output)
- =PP3V3_S0_TBTPWRCTL
- =PP1V05_TBT_P1V05TBTFET (1.05V FET Input)
- =PP1V05_TBT_FET (1.05V FET Output)

Signal aliases required by this page:
- =TBT_CLKREQ_L
- =TBT_RESET_L

BOM options provided by this page:
TBTBST:Y - Stuffs 15V boost circuitry.

Supervisor & CLKREQ# Isolation



U3810

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max

U3815

Part	TPS22924C
Type	Load Switch
R(on) @ 1.0V	20.3 mOhm Typ 28.6 mOhm Max

U3820

Part	TPS22920
Type	Load Switch
R(on) @ 1.05V	8 mOhm Typ 11.5 mOhm Max

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE: Thunderbolt Power Support

Apple Inc.

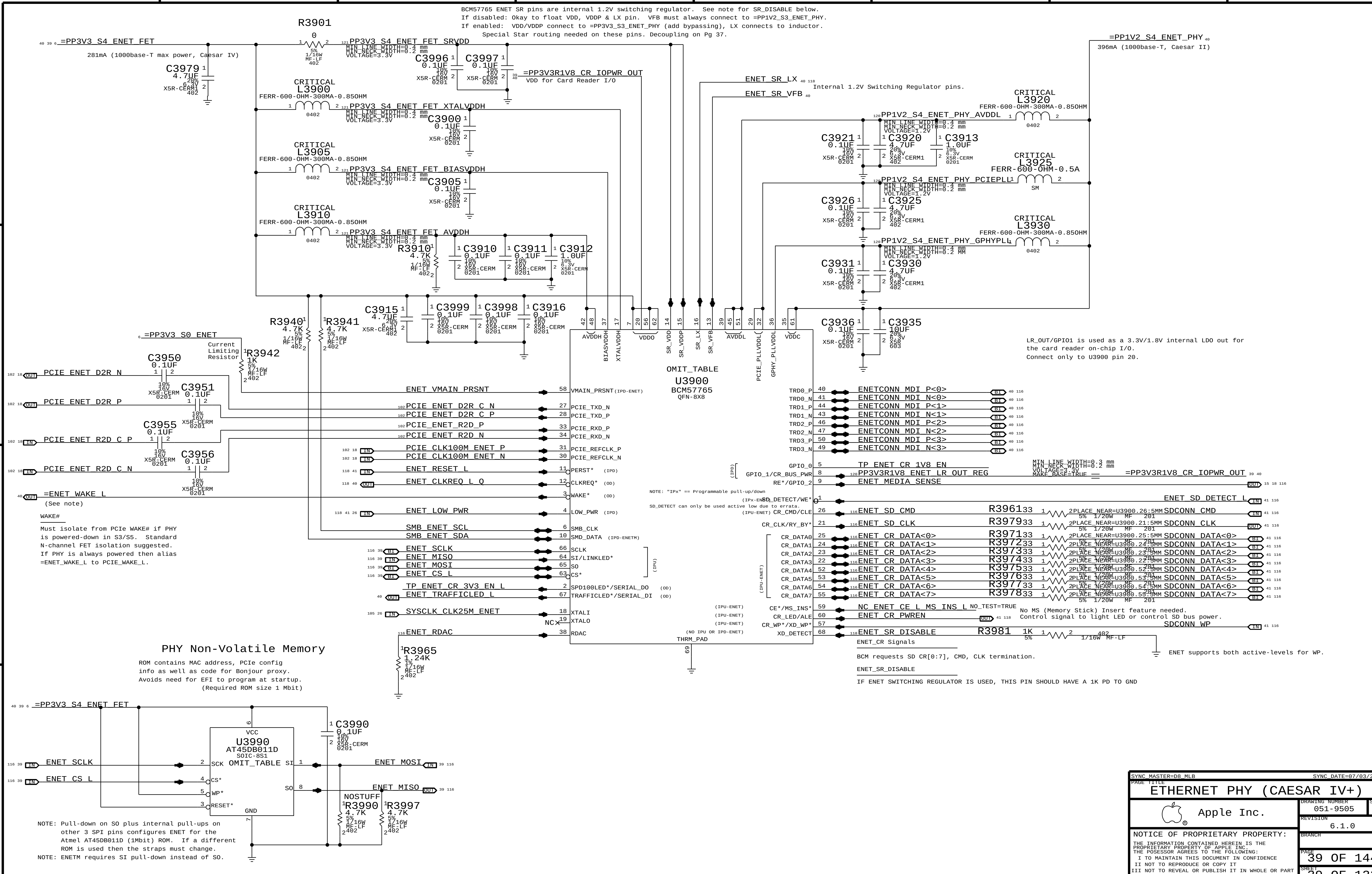
DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH

PAGE: 38 OF 144

SHEET: 38 OF 123



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

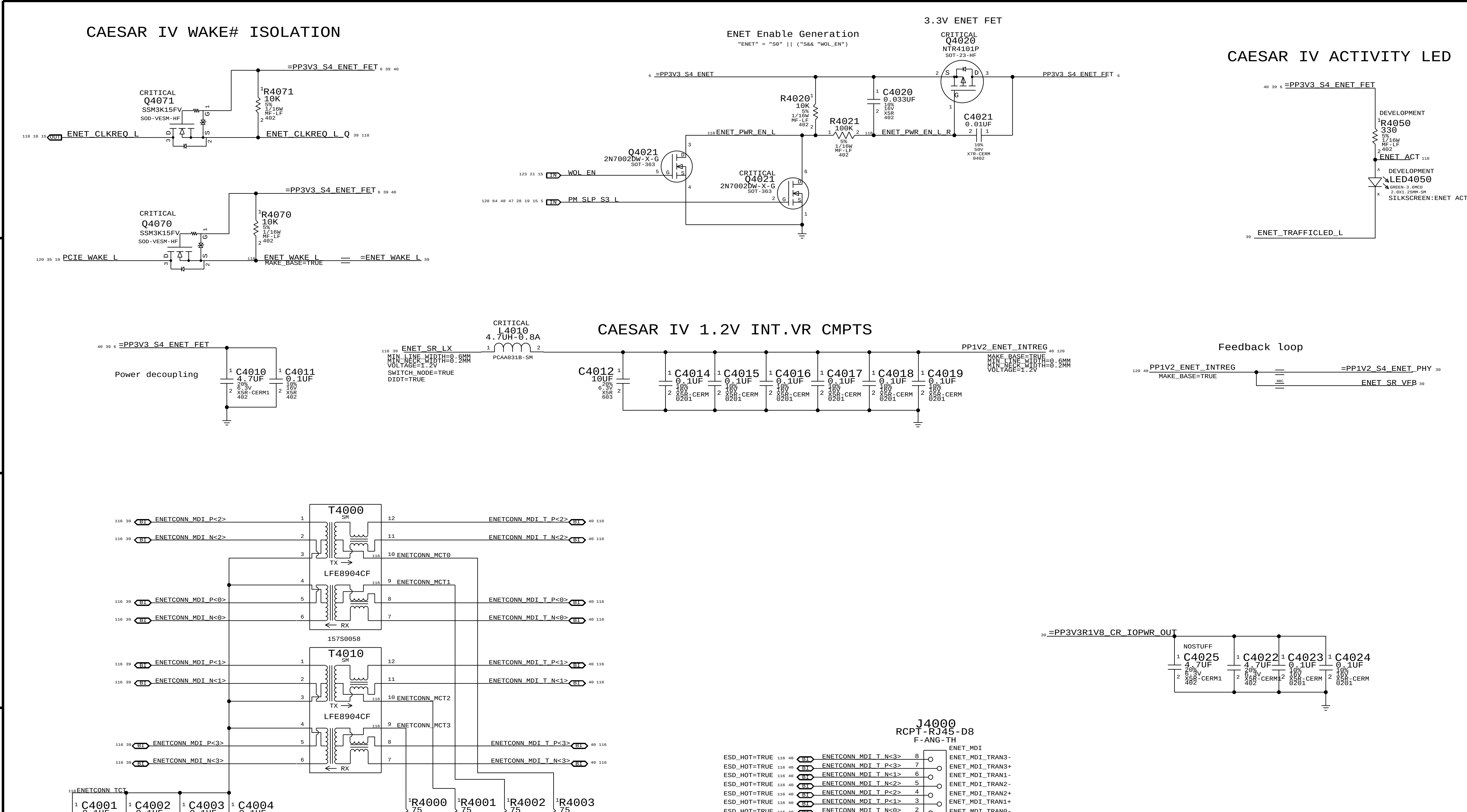
ETHERNET PHY (CAESAR IV+)

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE: Ethernet Support & Connector

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

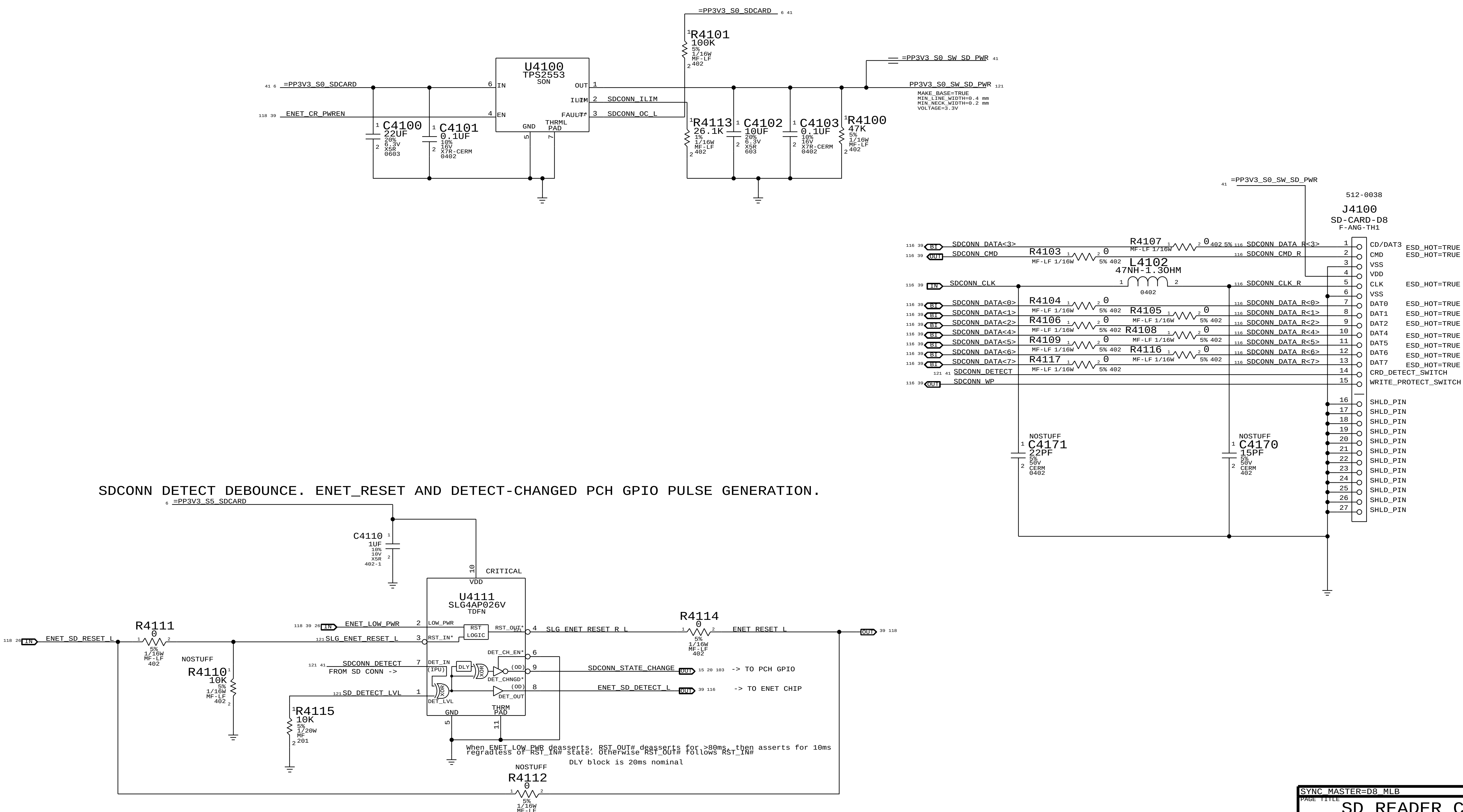
REVISION: 6.1.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:

SD CARD 3.3V OVERCURRENT PROTECTION

SDCONN DETECT DEBOUNCE. ENET_RESET AND DETECT-CHANGED PCH GPIO PULSE GENERATION.



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE: SD READER CONNECTOR

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

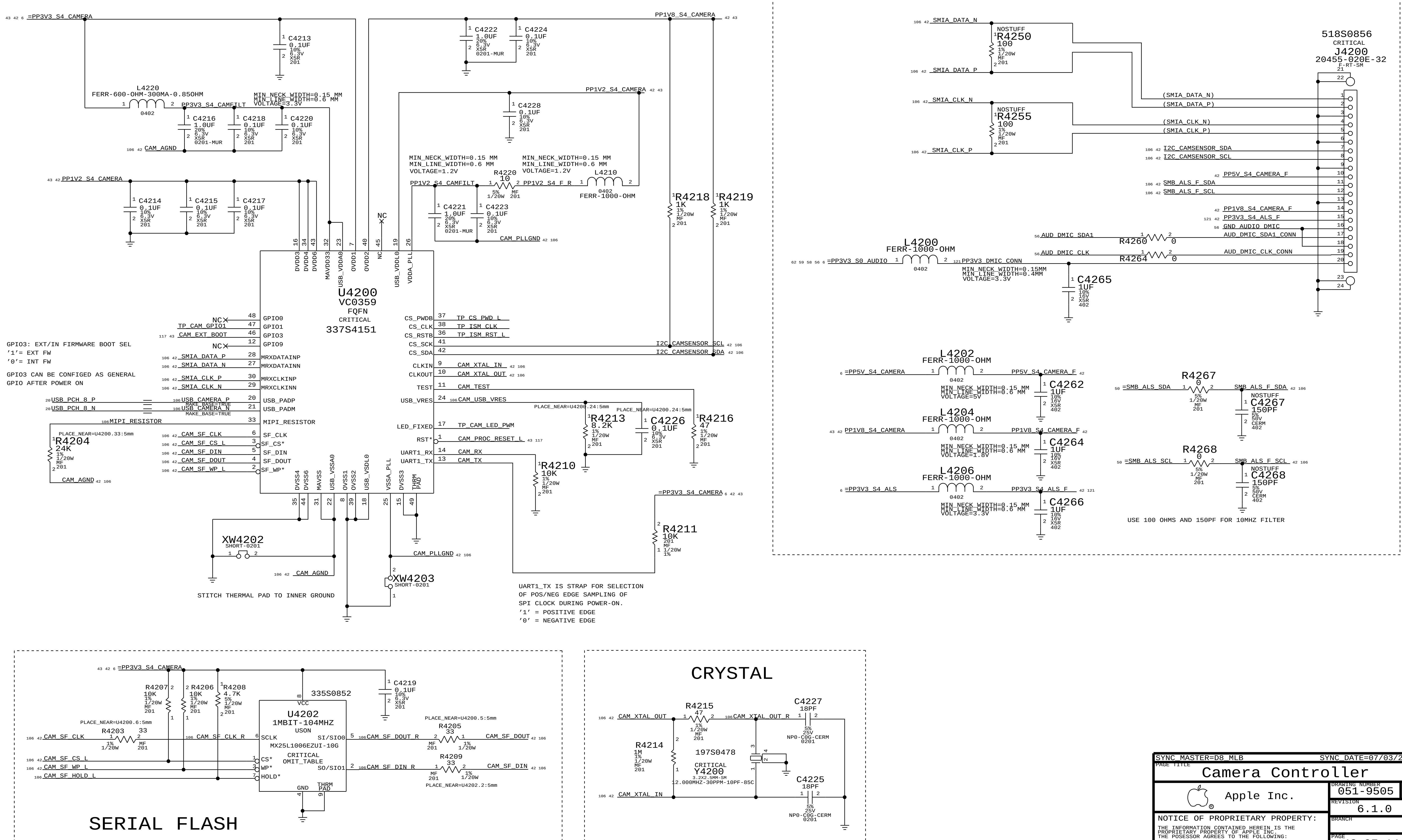
REVISION: 6.1.0

BRANCH

PAGE: 41 OF 144

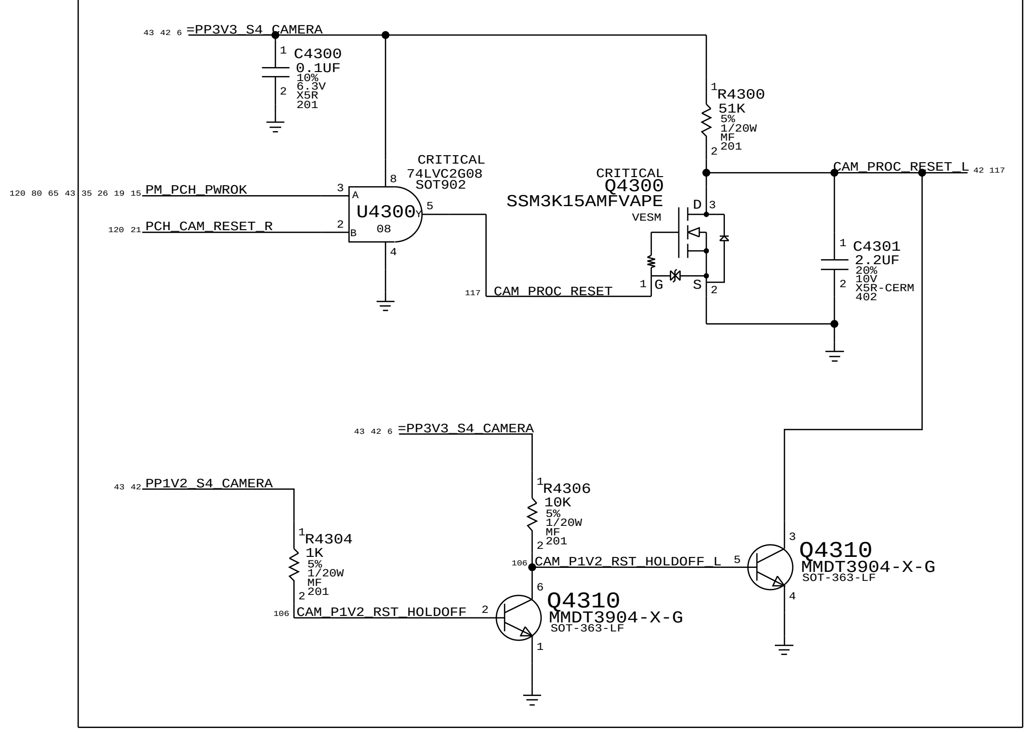
SHEET: 41 OF 123

USB CAMERA CONTROLLER

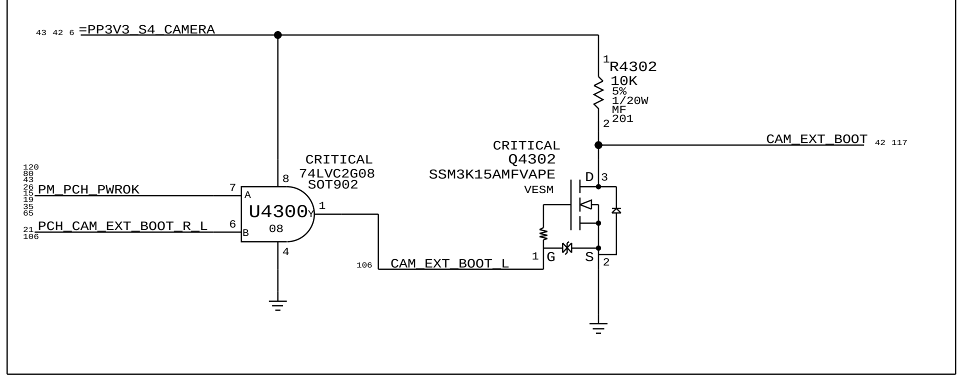


SYNC_MASTER=D8_MLB		SYNC_DATE=07/03/2012
PAGE TITLE: Camera Controller		
Apple Inc.	DRAWING NUMBER: 051-9505	SIZE: D
	REVISION: 6.1.0	
NOTICE OF PROPRIETARY PROPERTY:	BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE; II NOT TO REPRODUCE OR COPY IT; III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART; IV ALL RIGHTS RESERVED	PAGE: 42 OF 144	
	SHEET: 42 OF 123	

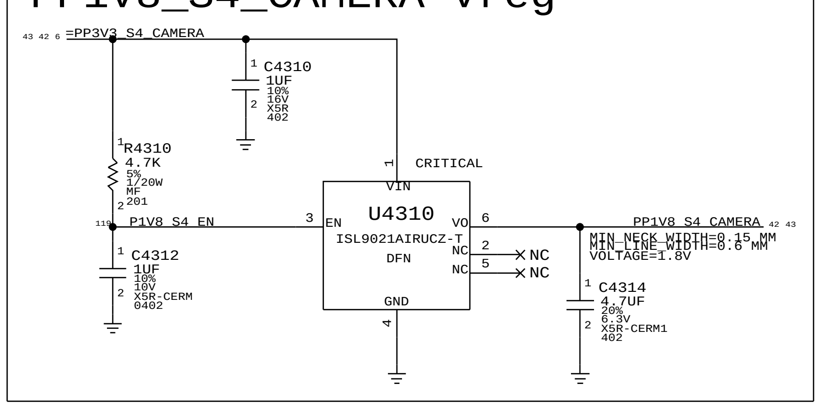
Camera Processor Reset



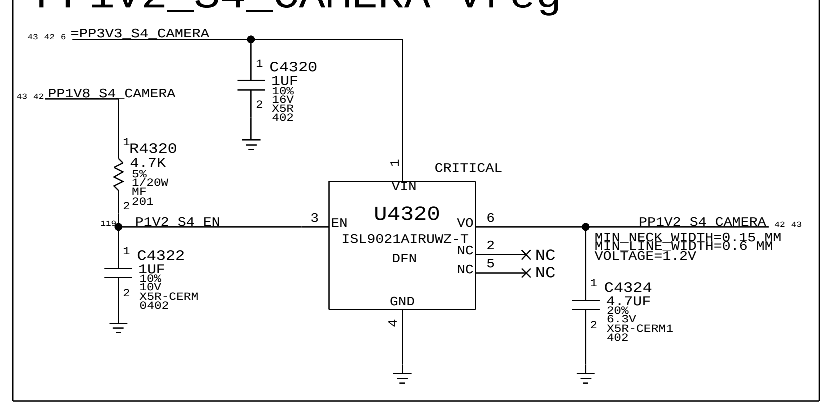
Camera Processor ExtBoot Cntl



PP1V8_S4_CAMERA Vreg



PP1V2_S4_CAMERA Vreg



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

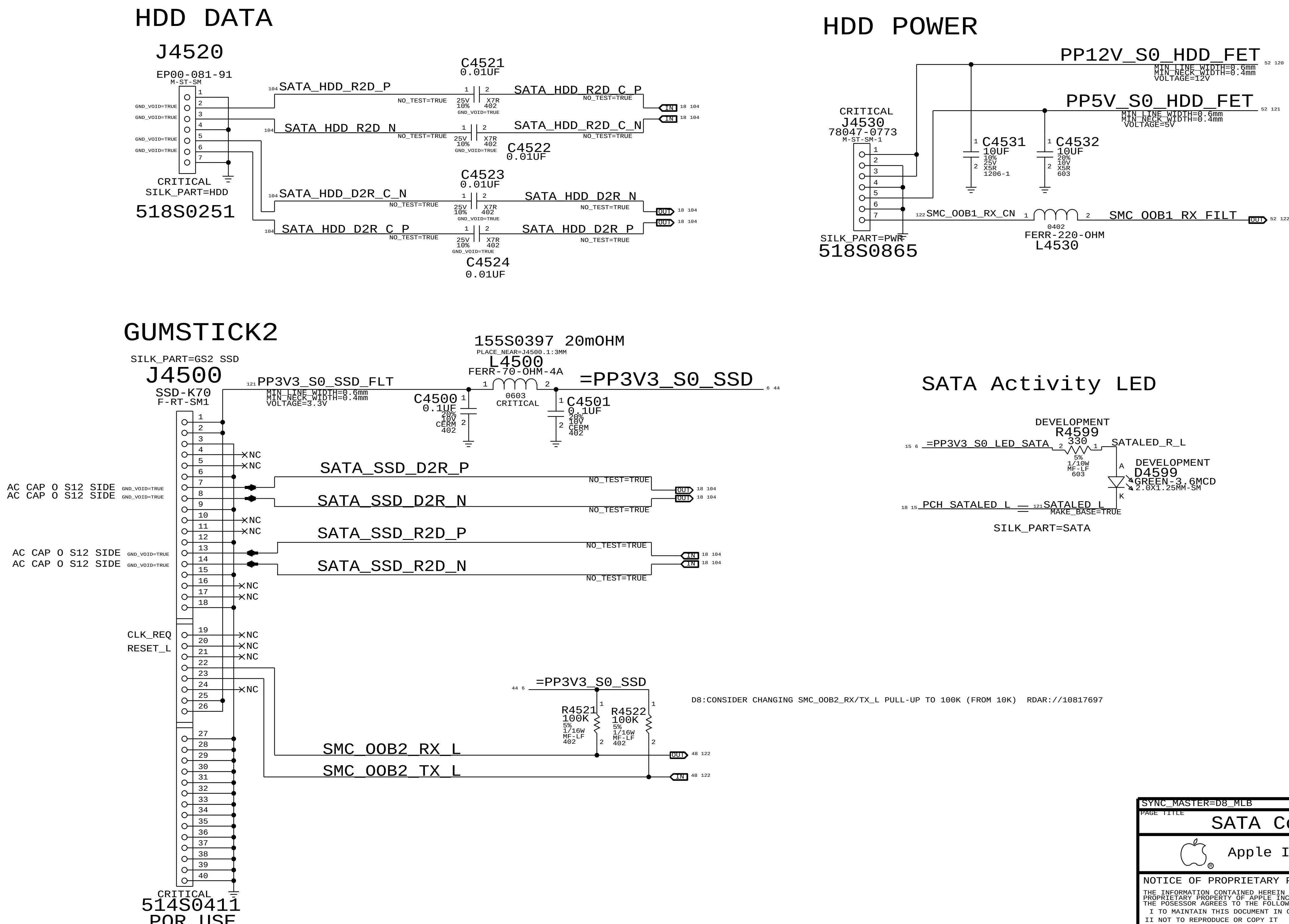
Camera Controller Support

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

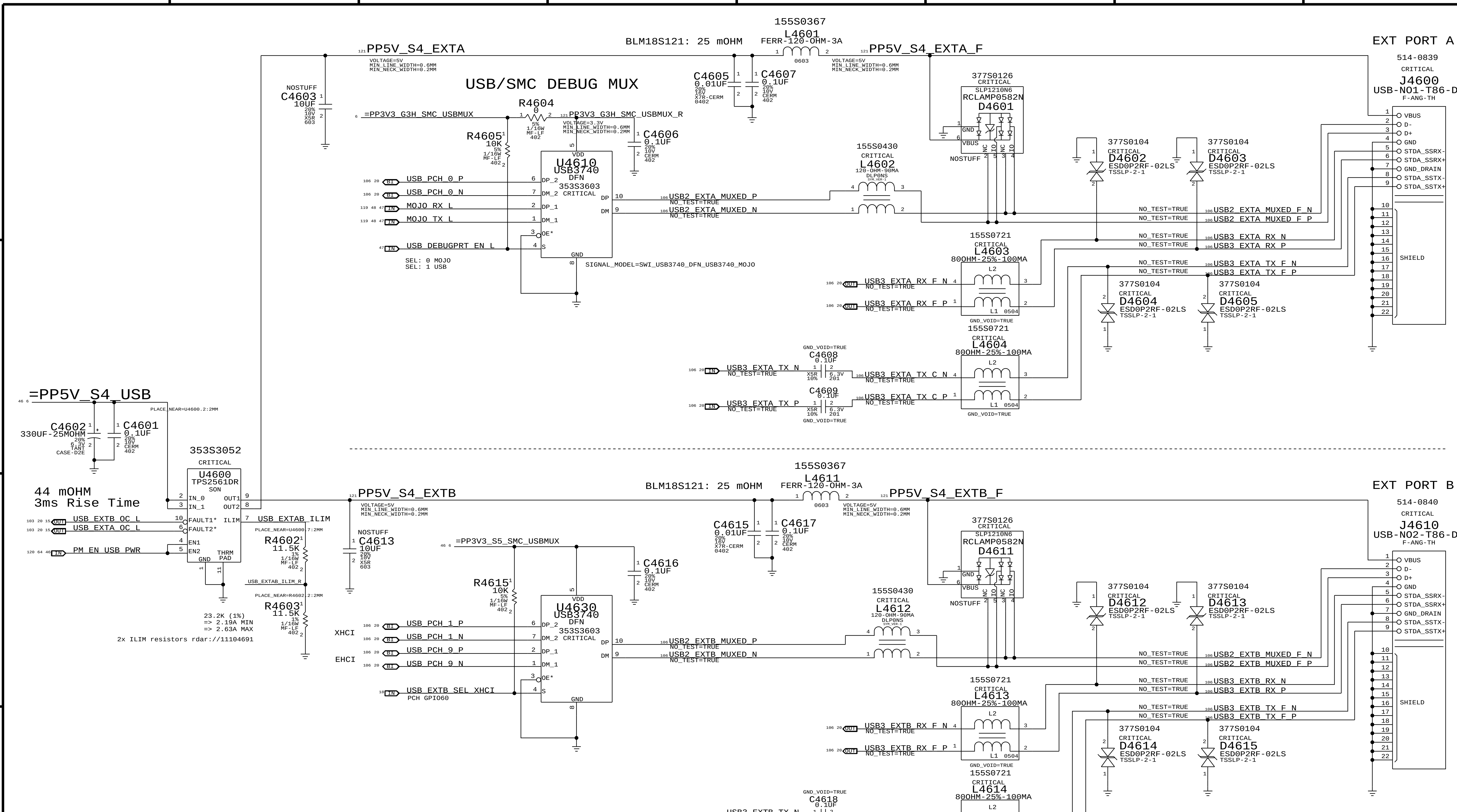
SATA Connectors

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

EXTERNAL USB PORTS A & B

DRAWING NUMBER 051-9505 SIZE D

Apple Inc.

REVISION 6.1.0

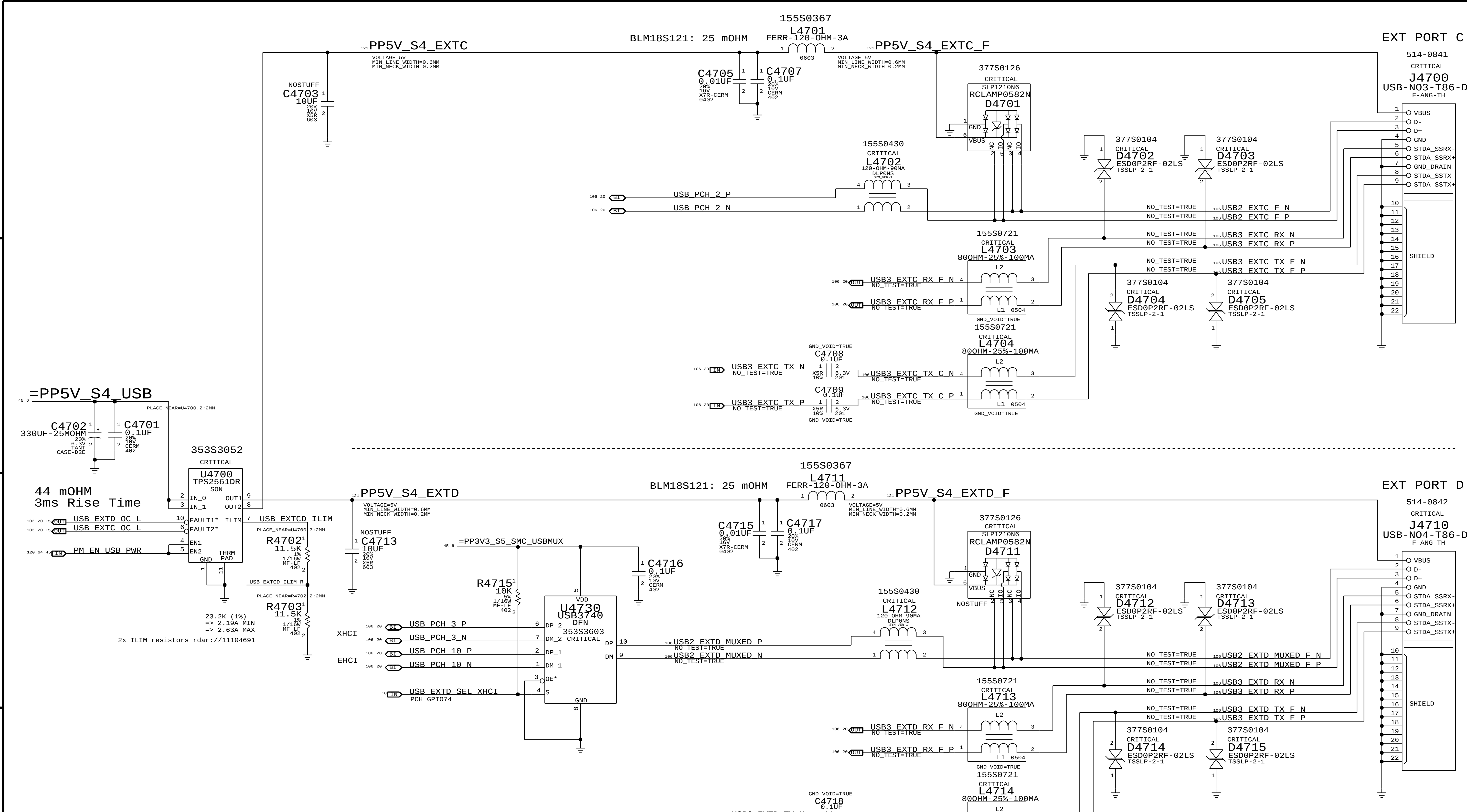
BRANCH

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

PAGE 46 OF 144

SHEET 45 OF 123



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE
EXTERNAL USB PORTS C & D

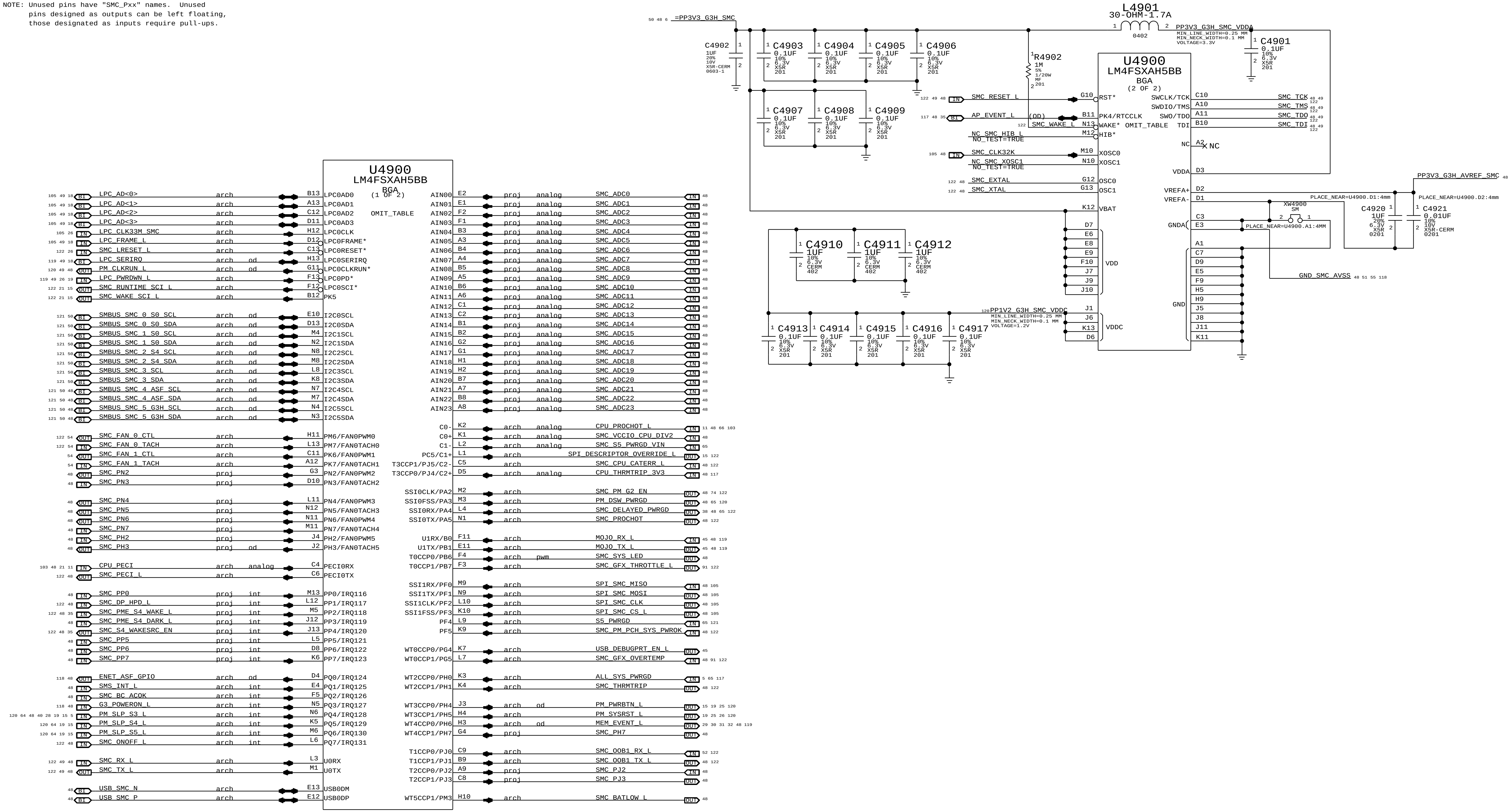
Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

NOTE: Unused pins have "SMC_Pxx" names. Unused
pins designed as outputs can be left floating,
those designated as inputs require pull-ups.



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

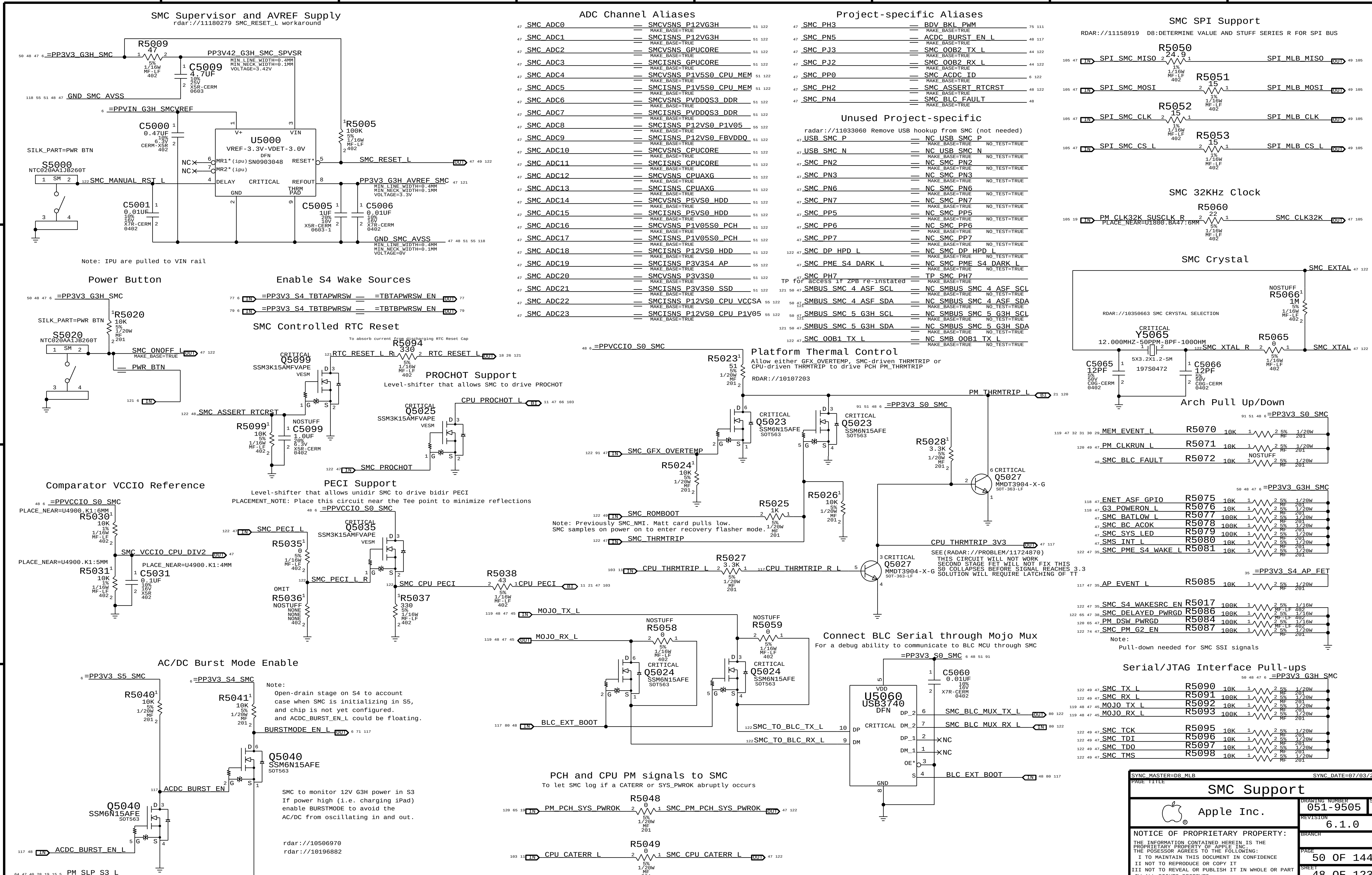
SMC

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

SMC Support

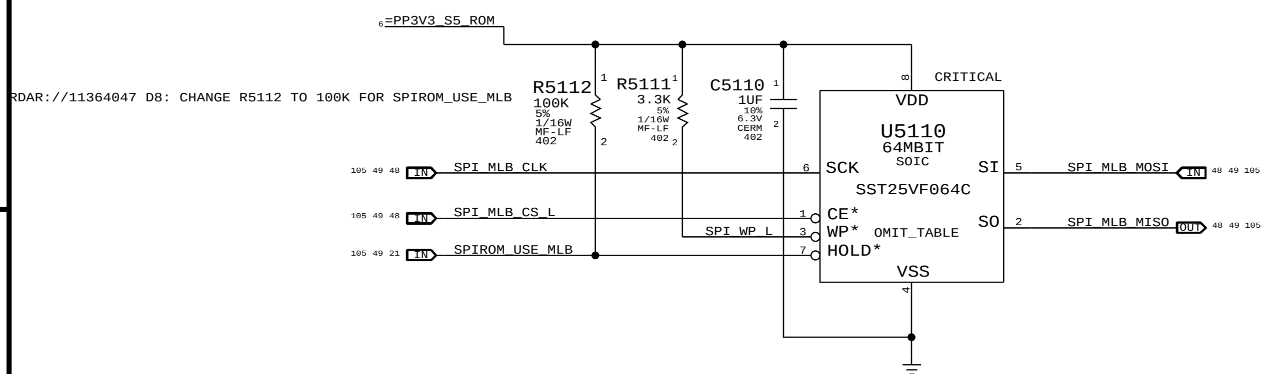
Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

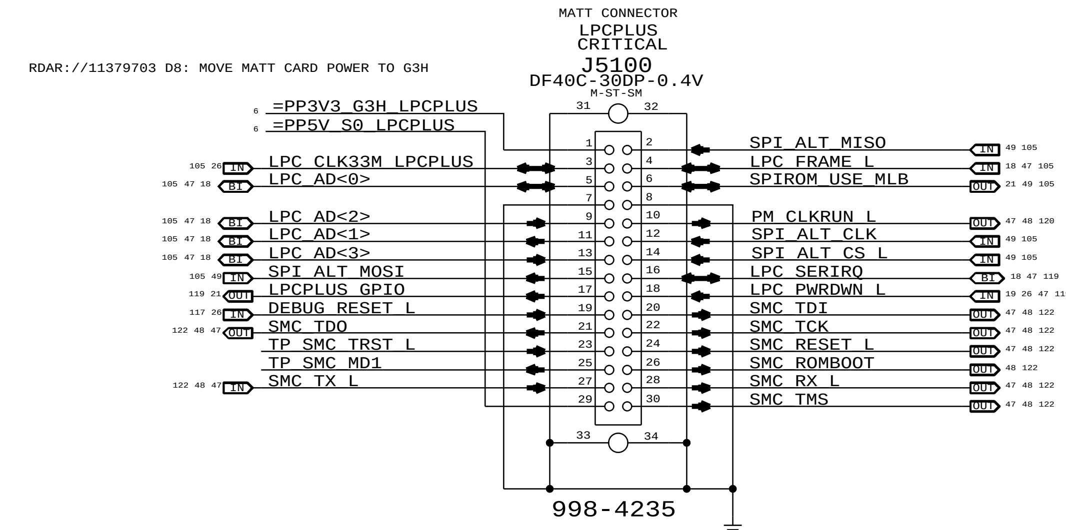
REVISION 6.1.0

BRANCH

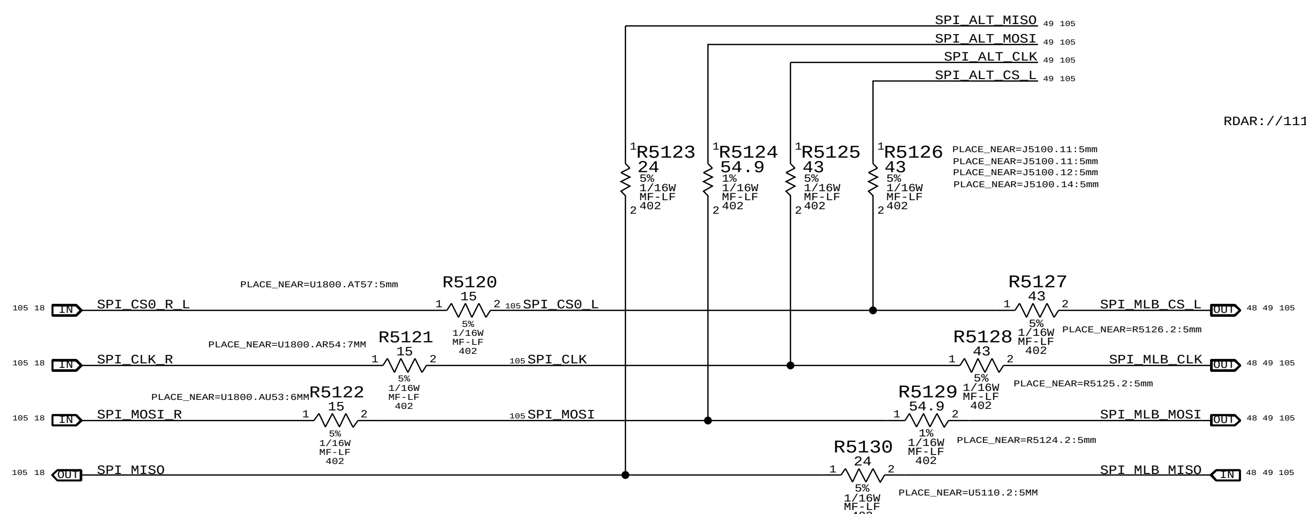
SPI BootROM



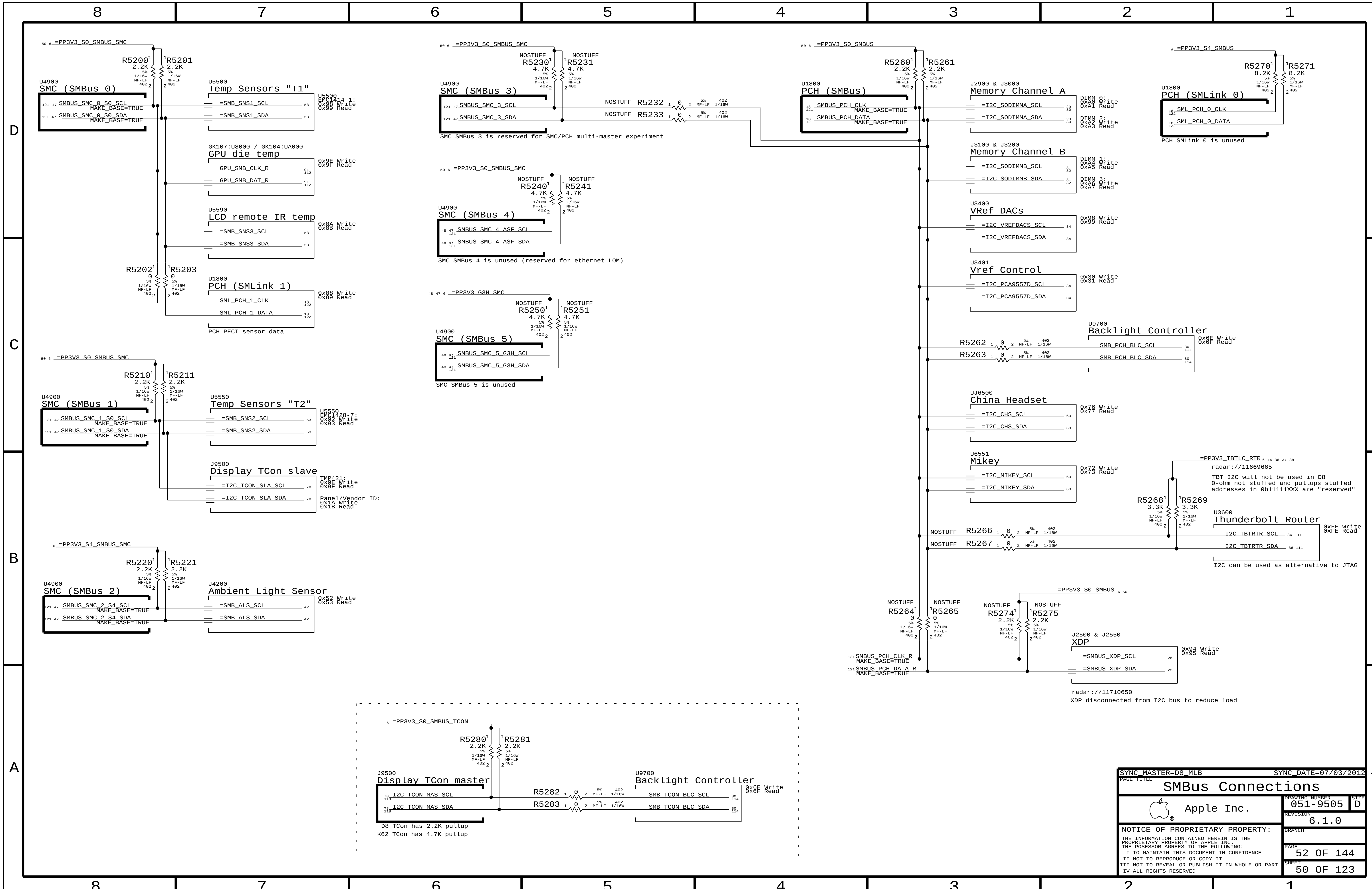
LPC+SPI Connector



SPI Series Termination



SYNC_MASTER=D8_MLB	SYNC_DATE=07/03/2012
PAGE TITLE	SPI and Debug Connector
Apple Inc.	DRAWING NUMBER 051-9505 SIZE D
	REVISION 6.1.0
NOTICE OF PROPRIETARY PROPERTY:	BRANCH
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	PAGE 51 OF 144
	SHEET 49 OF 123



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

SMBus Connections

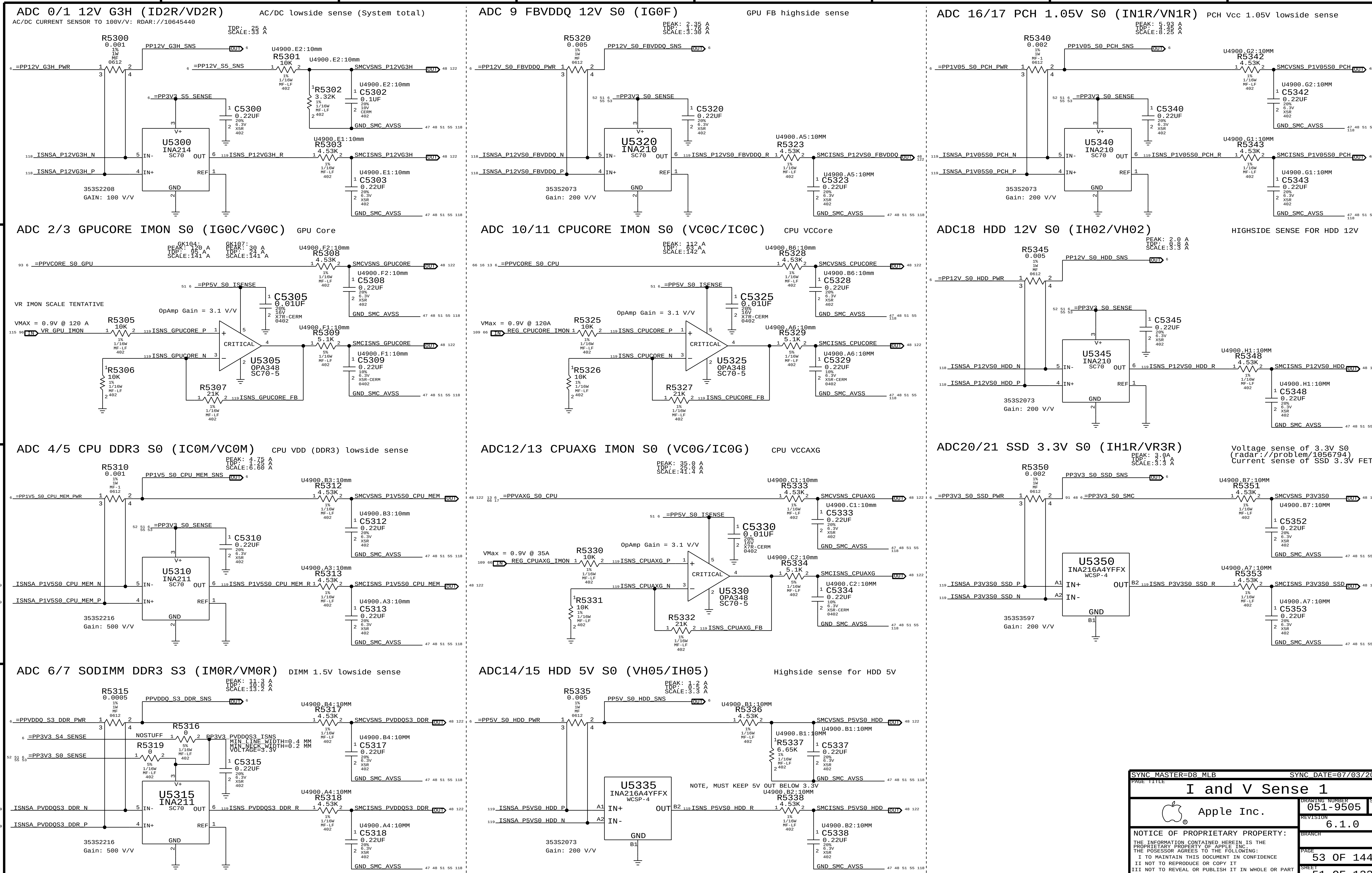
Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

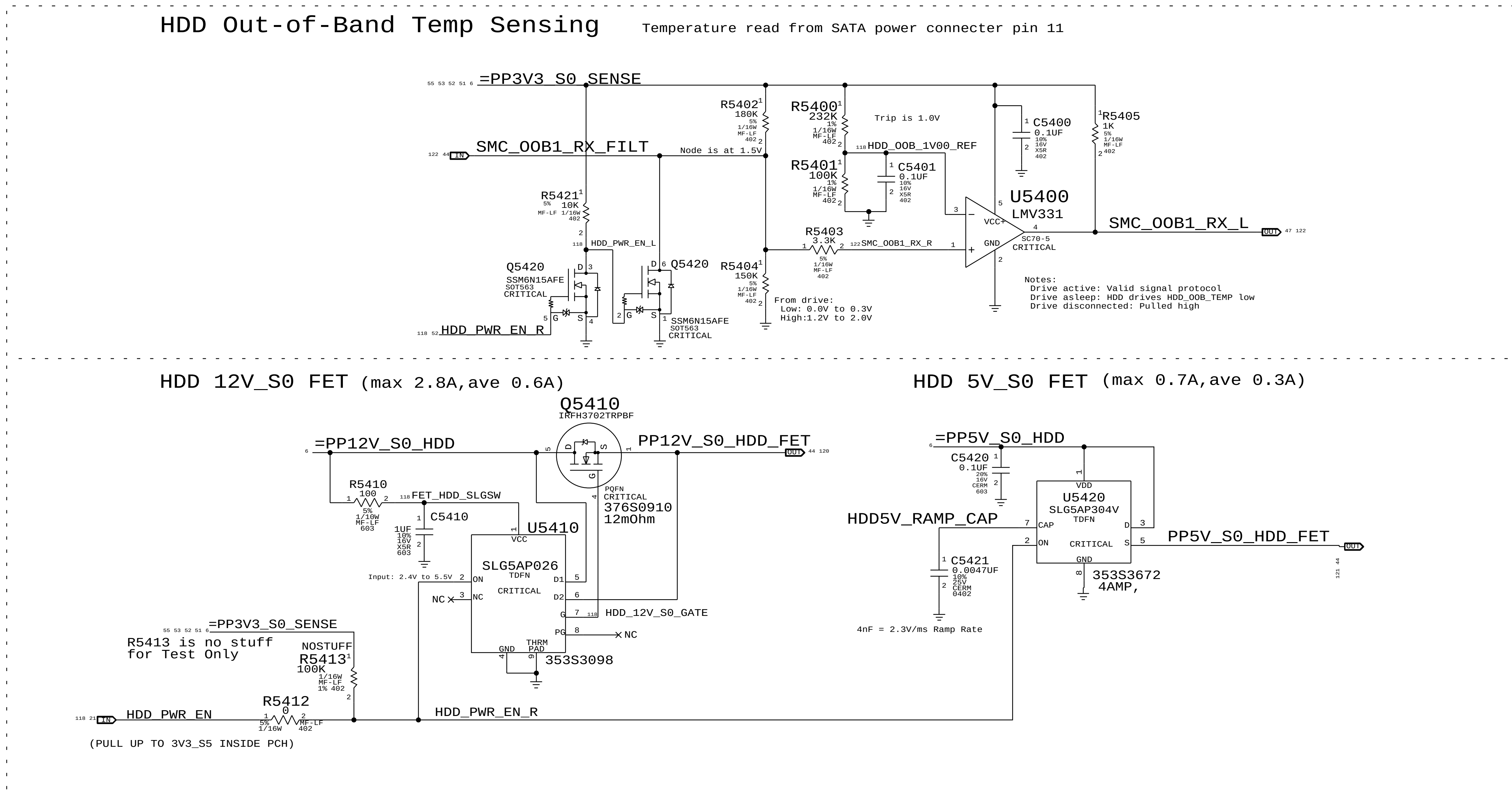
I and V Sense 1

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



REMOVE R5413 AND SHORT R5412 AFTER HDD_PWR_EN WORKS

SYNC-MASTER=D8-MLB SYNC-DATE=07/03/2012

PAGE TITLE
HDD/SSD Temp Sense

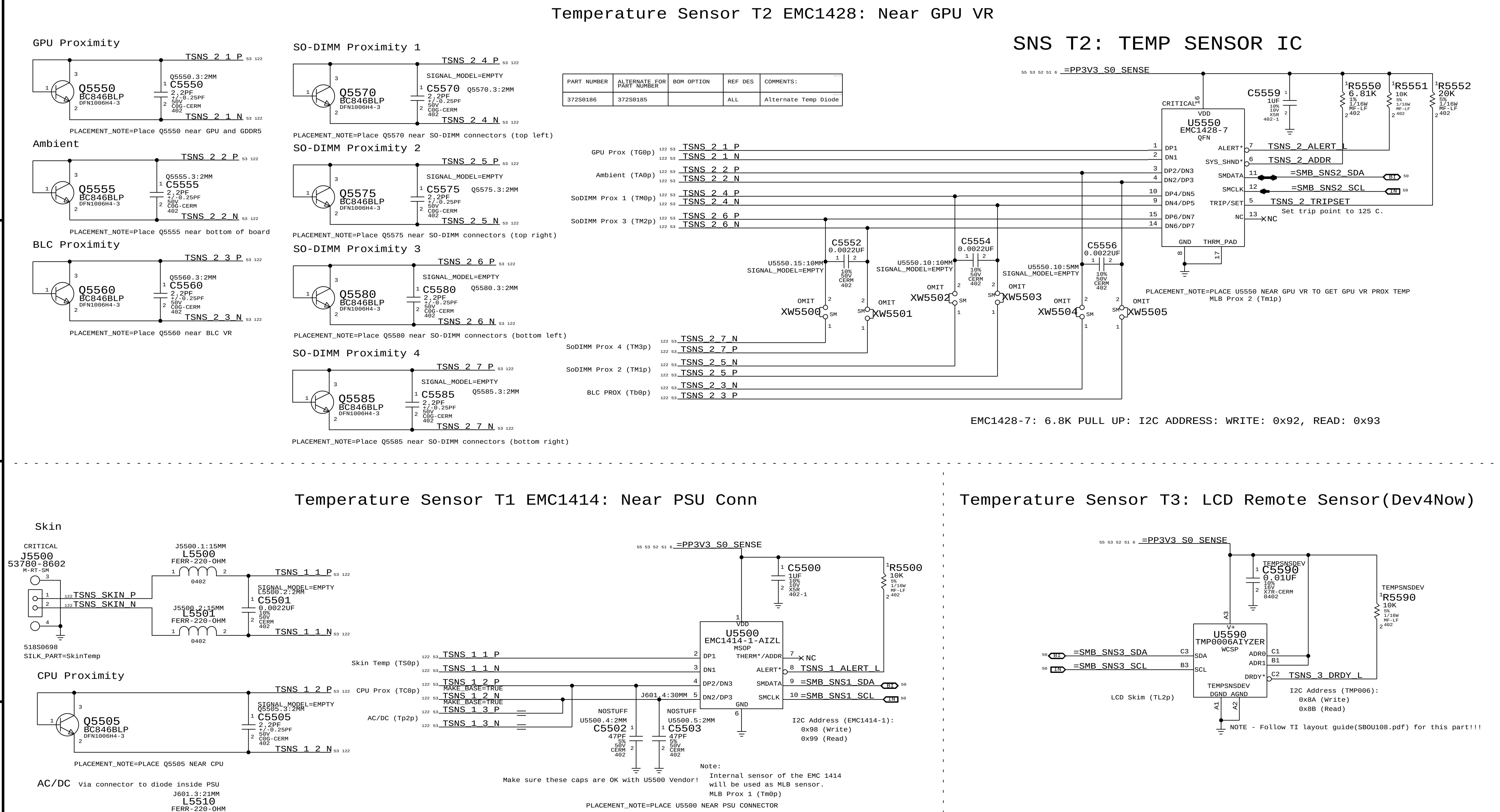
Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

Temperature Sensors

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

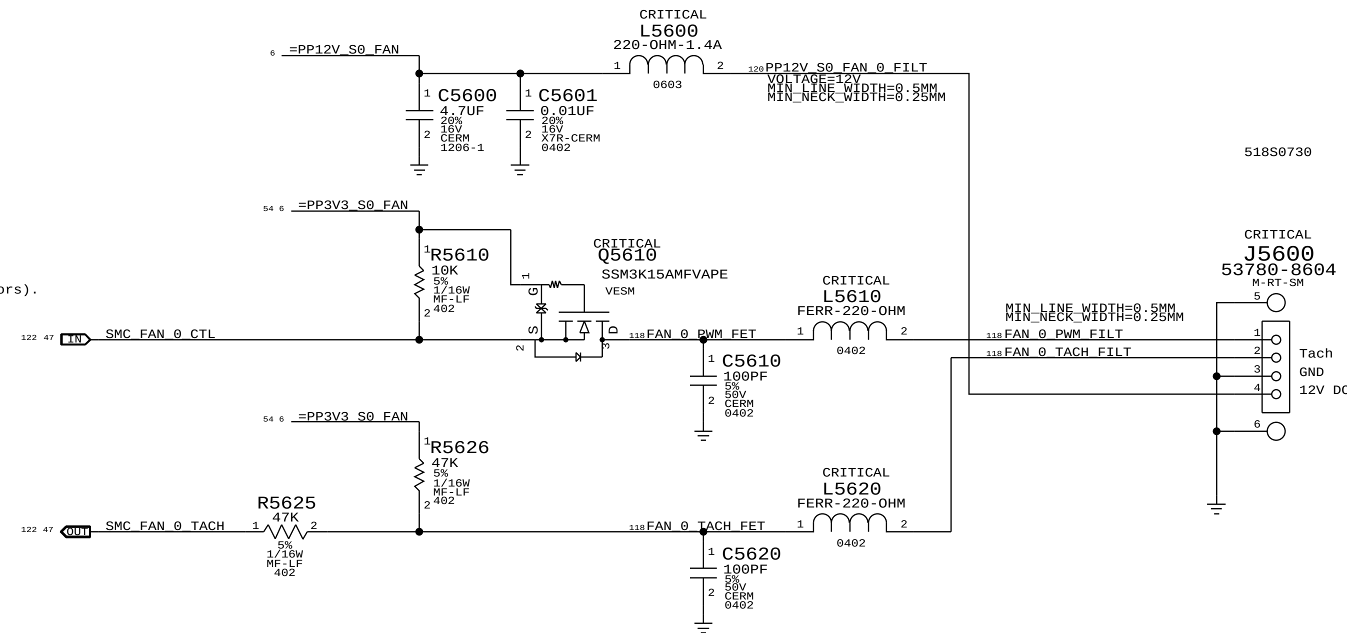
BRANCH

SMC Fan 0 (System)

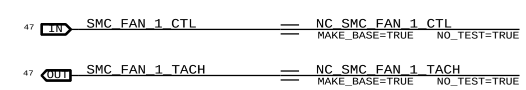
Note:
The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q5610 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q5610 is at common and the SMC sinks current when Q5610 is on.

This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

Otherwise, this is simply a pass-FET.
See RADAR: 10565825- D7: Need scematic and PCB file of fan(All Vendors).



SMC Fan 1 (Unused)



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

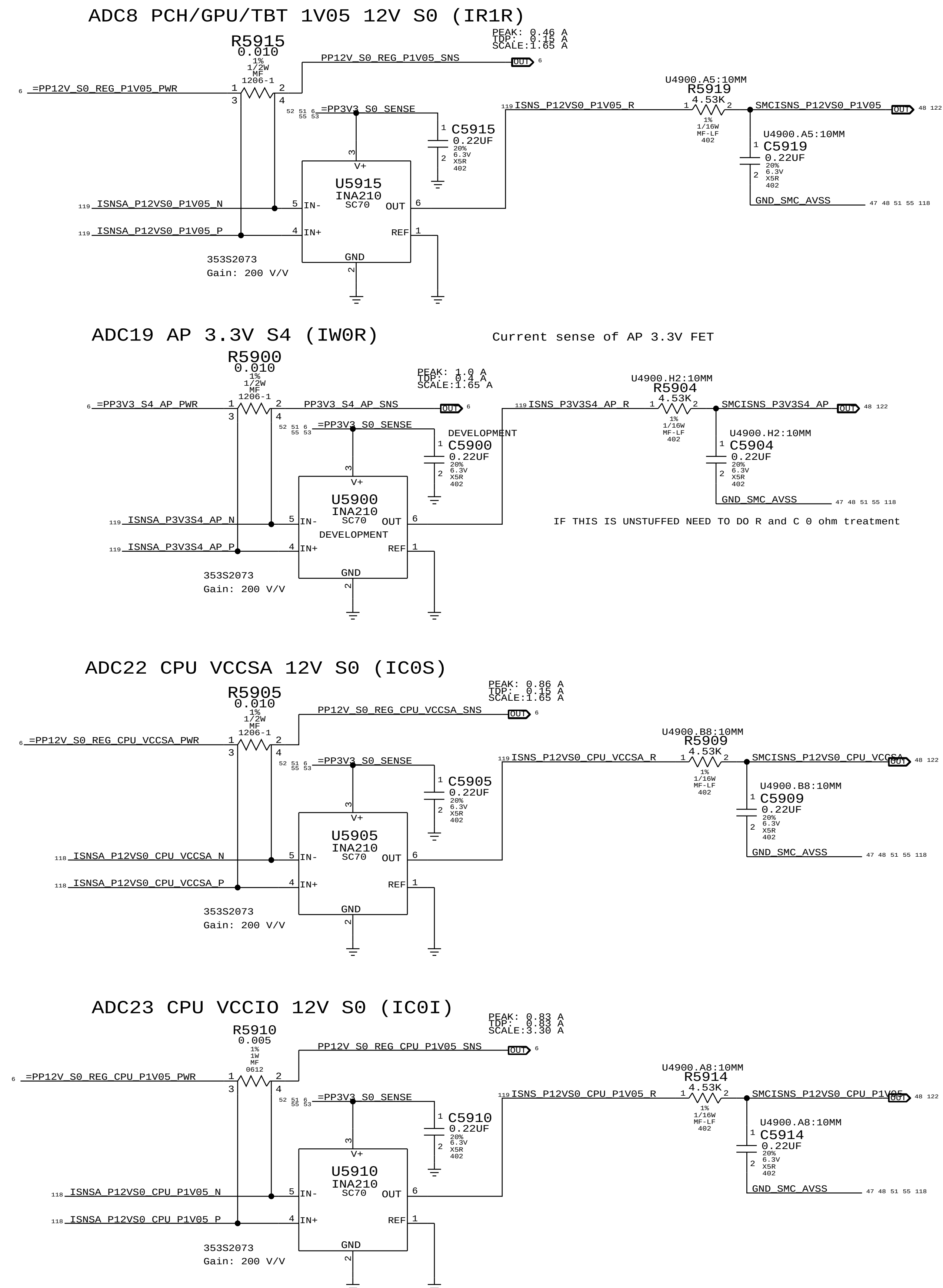
PAGE TITLE: System Fan

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

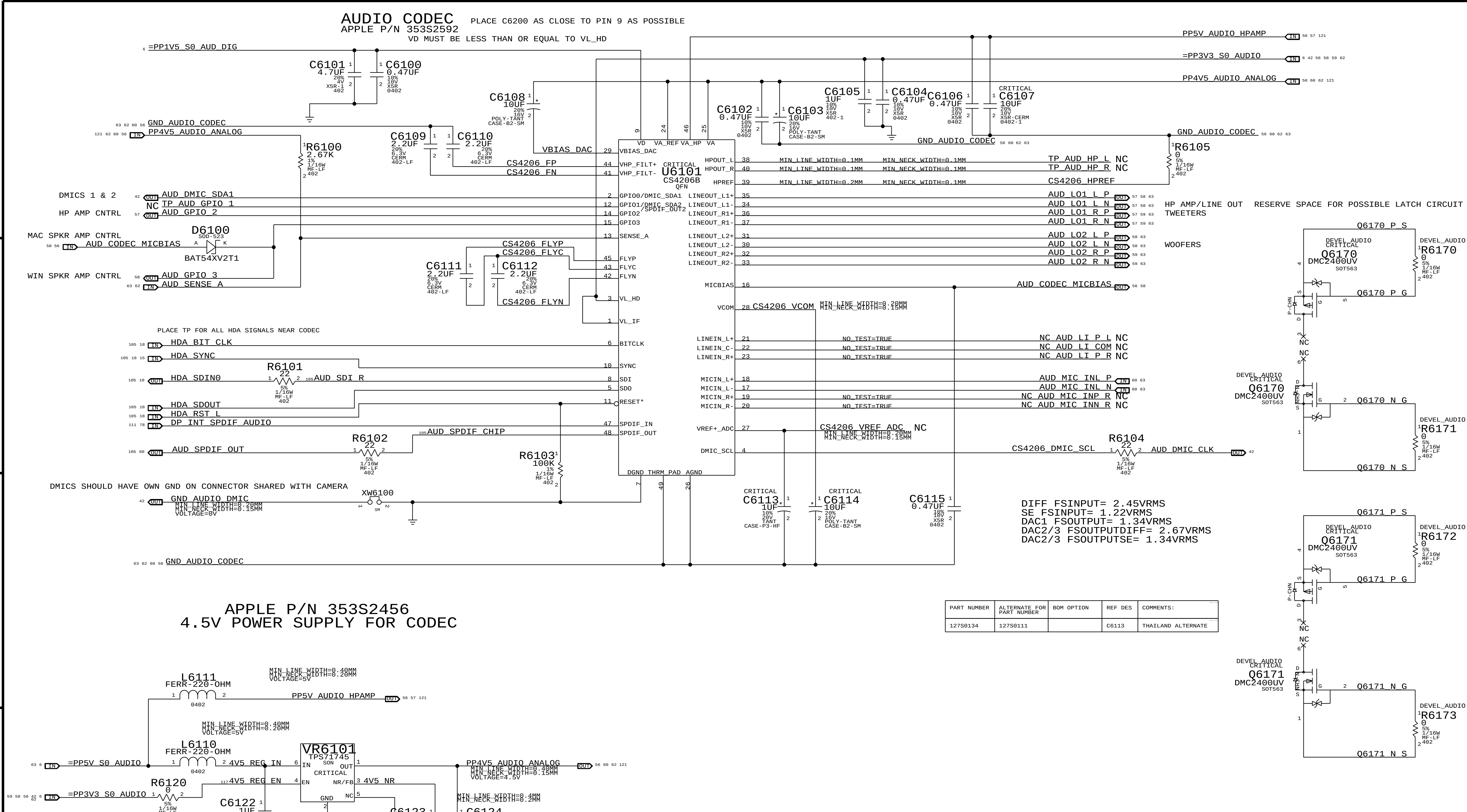
I and V Sense 2

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
127S0134	127S0111		C6113	THAILAND ALTERNATE

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

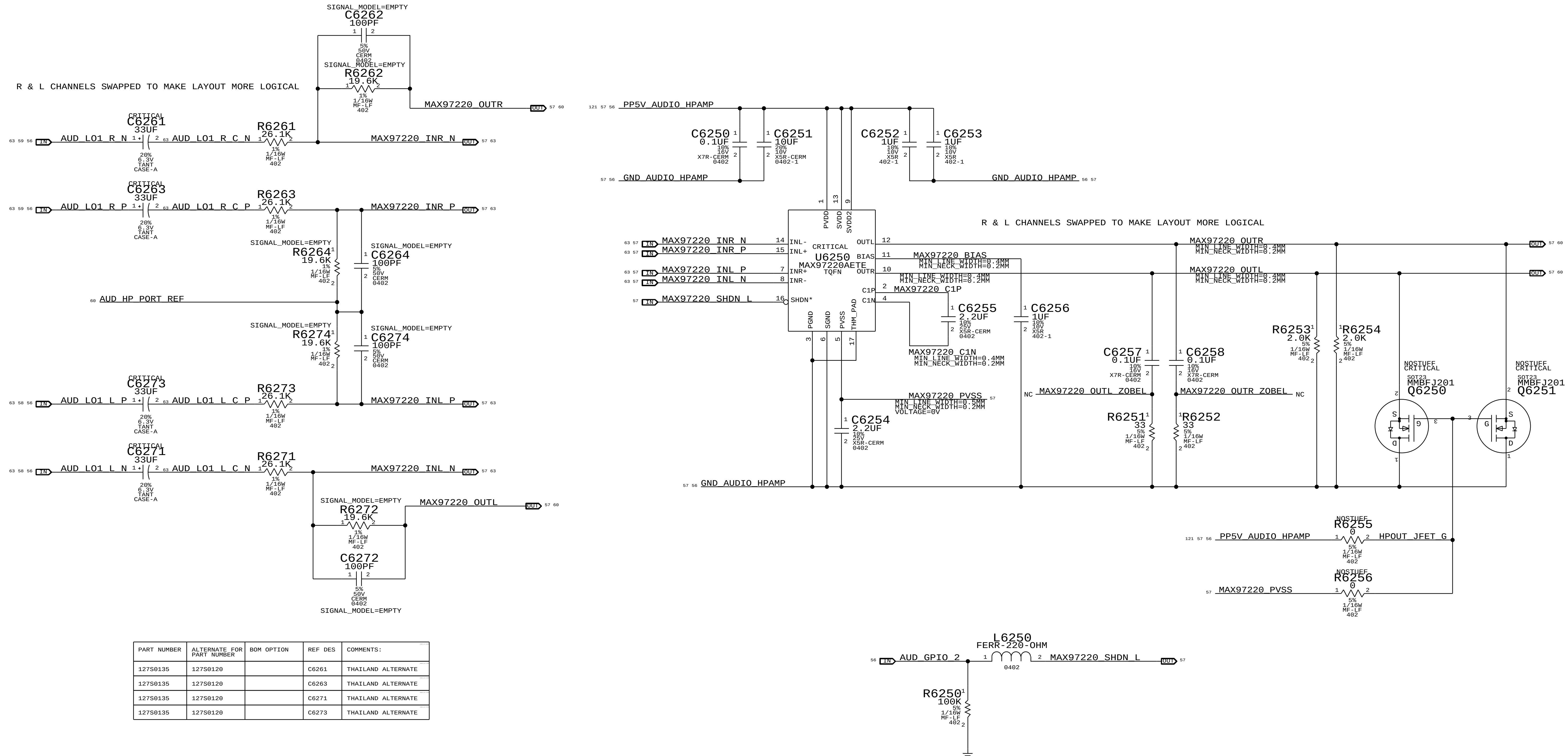
AUDIO: CODEC/REGULATORS

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
127S0135	127S0120		C6261	THAILAND ALTERNATE
127S0135	127S0120		C6263	THAILAND ALTERNATE
127S0135	127S0120		C6271	THAILAND ALTERNATE
127S0135	127S0120		C6273	THAILAND ALTERNATE

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

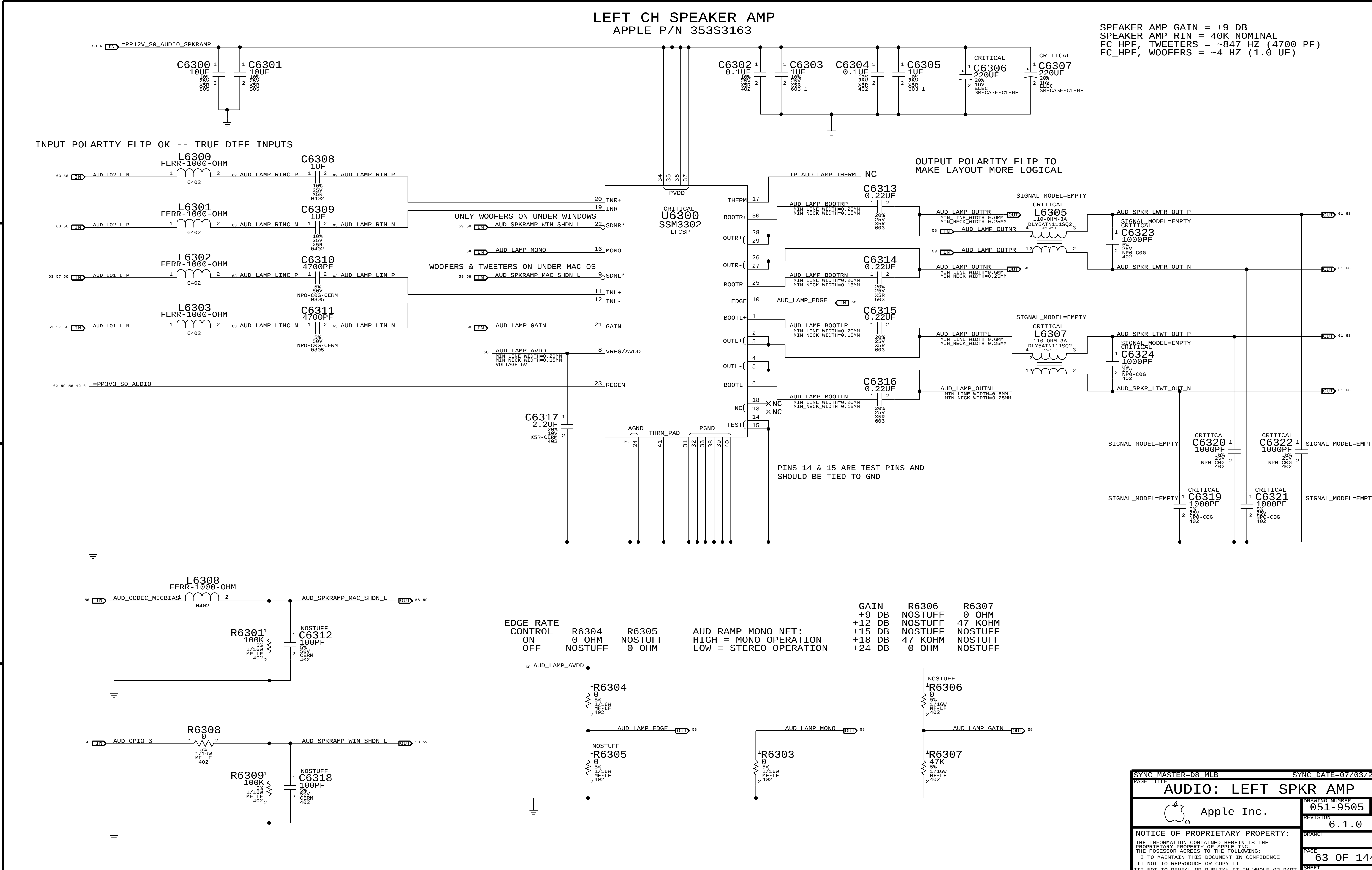
AUDIO: HEADPHONE AMP

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

AUDIO: LEFT SPKR AMP

Apple Inc.

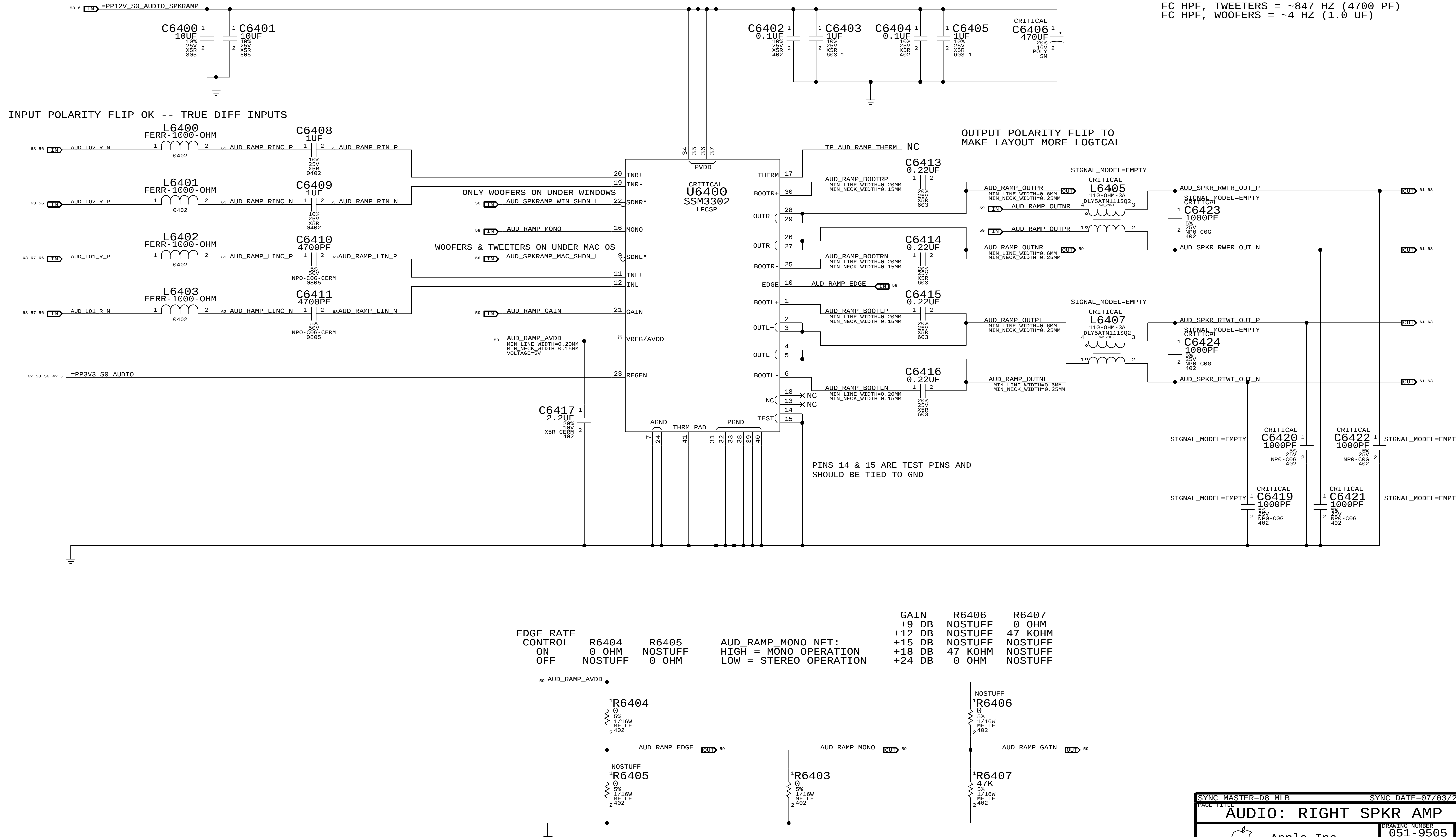
DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

RIGHT CH SPEAKER AMP
APPLE P/N 353S3163

SPEAKER AMP GAIN = +9 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~4 HZ (1.0 UF)



EDGE RATE CONTROL	R6404	R6405
ON	0 OHM	NOSTUFF
OFF	NOSTUFF	0 OHM

AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN	R6406	R6407
+9 DB	NOSTUFF	0 OHM
+12 DB	NOSTUFF	47 KOHM
+15 DB	NOSTUFF	NOSTUFF
+18 DB	47 KOHM	NOSTUFF
+24 DB	0 OHM	NOSTUFF

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE
AUDIO: RIGHT SPKR AMP

Apple Inc.

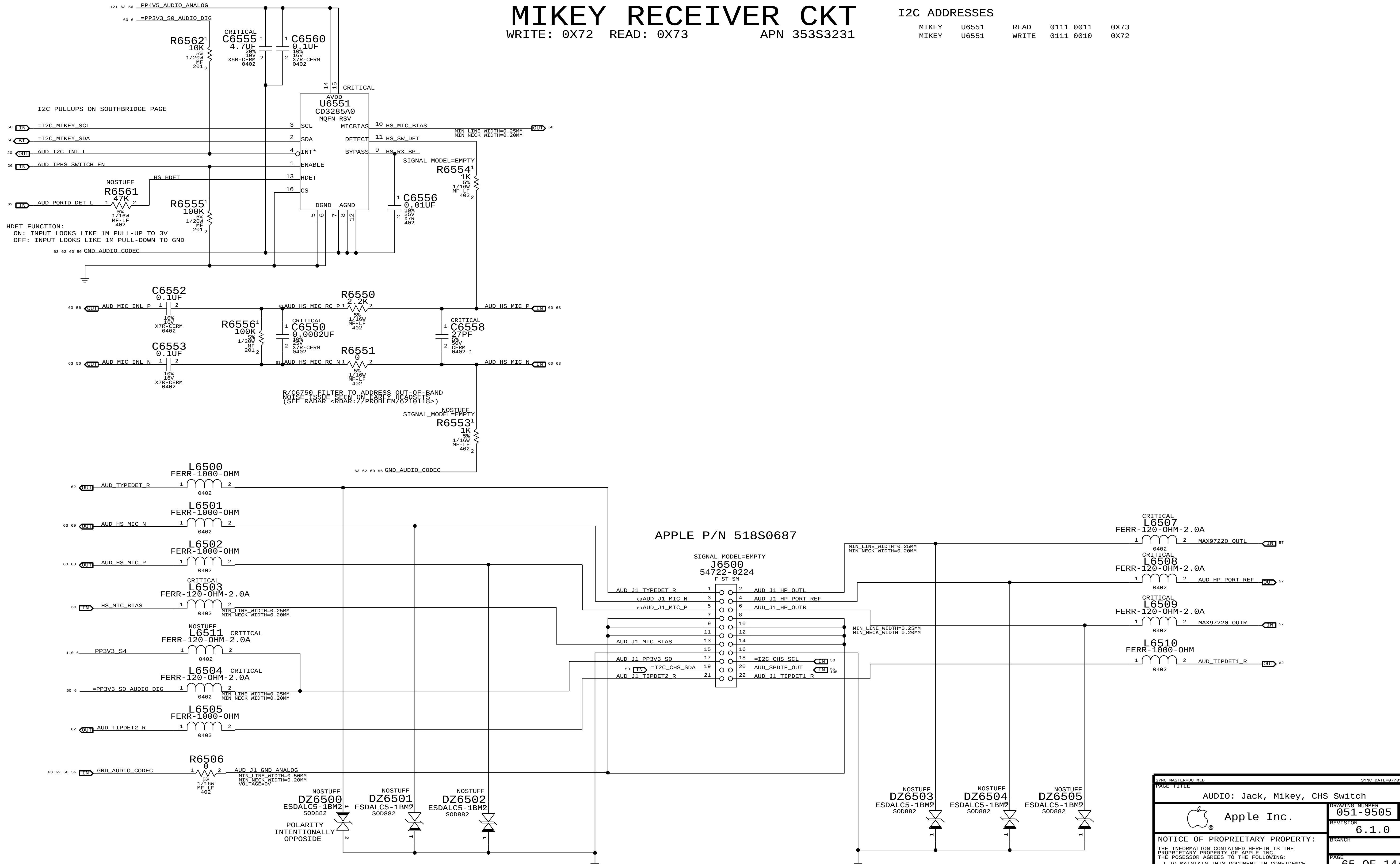
DRAWING NUMBER 051-9505 SIZE D
REVISION 6.1.0
BRANCH
PAGE 64 OF 144
SHEET 59 OF 123

MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S3231

I2C ADDRESSES

MIKEY	U6551	READ	0111 0011	0X73
MIKEY	U6551	WRITE	0111 0010	0X72



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE: AUDIO: Jack, Mikey, CHS Switch

Apple Inc.

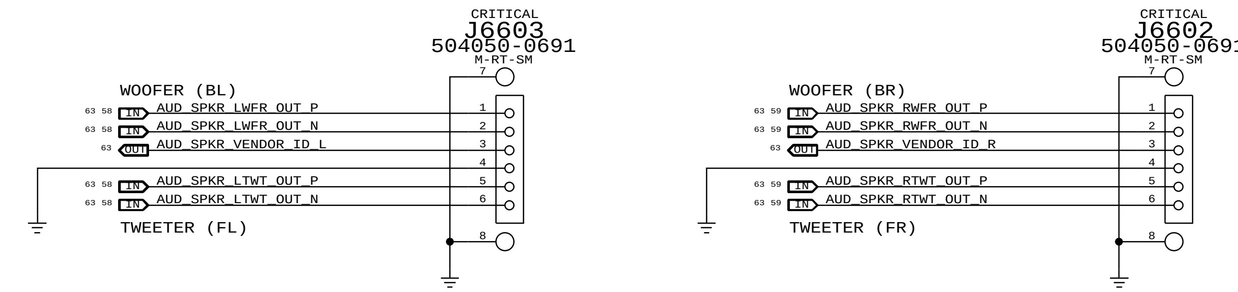
DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

SPEAKER CABLE CONNECTORS

APPLE P/N 518S0862



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

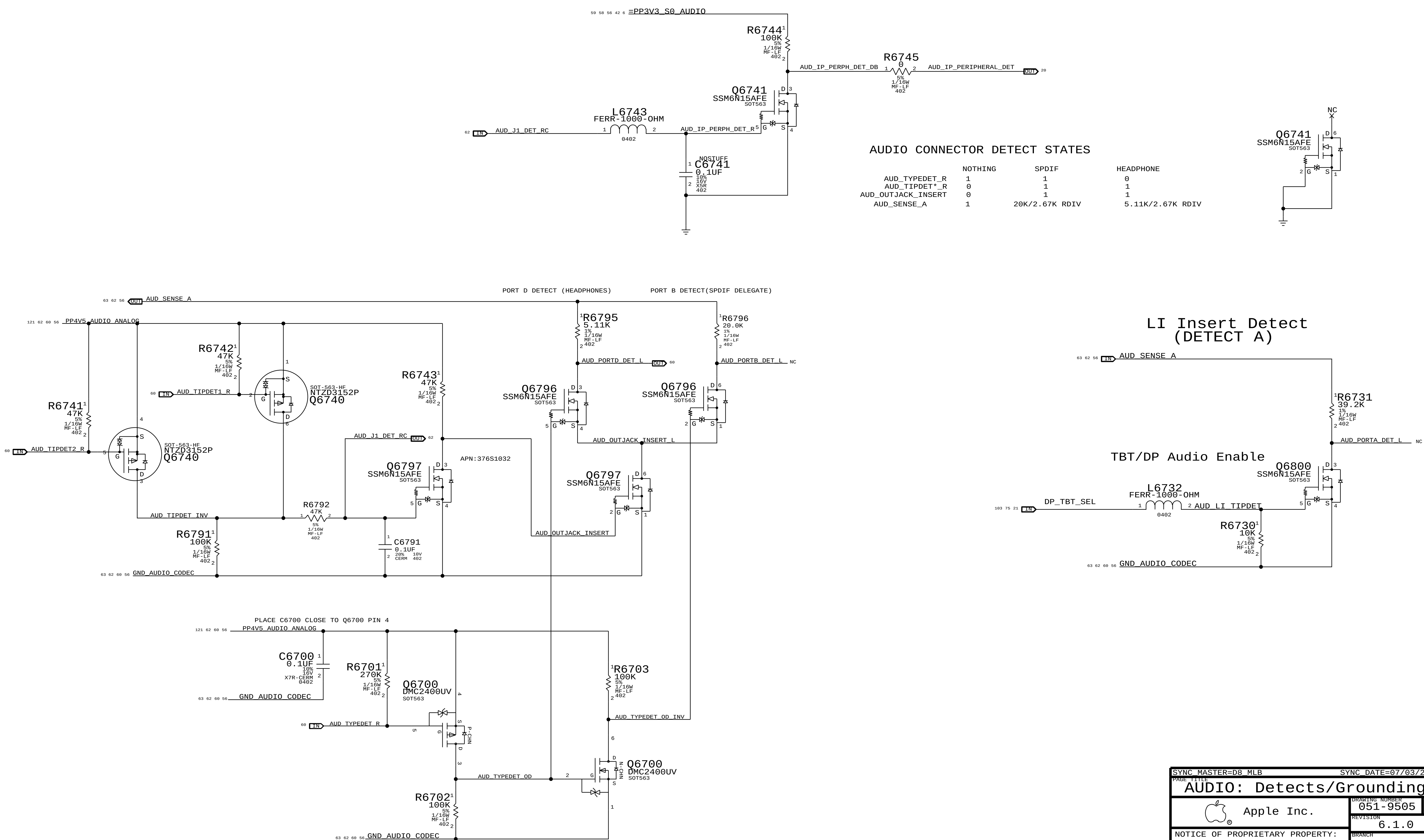
PAGE TITLE

Audio: Spkr/Mic Conn.

Apple Inc.

DRAWING NUMBER	051-9505
SIZE	D
REVISION	6.1.0
BRANCH	
PAGE	66 OF 144
SHEET	61 OF 123

IPHS HS Detect Debounce CKT



AUDIO CONNECTOR DETECT STATES

	NOTHING	SPDIF	HEADPHONE
AUD_TYPEDET_R	1	1	0
AUD_TIPDET*_R	0	1	1
AUD_OUTJACK_INSERT	0	1	1
AUD_SENSE_A	1	20K/2.67K RDIV	5.11K/2.67K RDIV

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

AUDIO: Detects/Grounding

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	MAC SHDN	WIN SHDN	DET ASSIGNMENT
HP/LINE OUT	0X03 (3)	0X03 (3)	0X0A (10,D)	GPIO_2	GPIO_2	0X0A (DET D)
PRIMARY SPKRS (WFR)	0X04 (4)	0X04 (4)	0X0B (11)	MICBIAS	GPIO_3	N/A
SECONDARY SPKRS (TWT)	0X03 (3)	0X03 (3)	0X0A (10,V24)	MICBIAS	N/A	N/A
SPDIF OUT	N/A	0X08 (8)	0x10 (16)	N/A	N/A	0X0D (DET B)

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
SPDIF IN	0X07 (7)	0X0F (15)	N/A	0X09 (DET A)
INTERNAL MIC ARRAY	0X06 (6) 0X05 (5)	0X0E (14,LEFT & RIGHT) 0X12 (18,LEFT)	N/A	N/A
EXTERNAL MIC	0X06 (6)	0X0D (13,V22,B,LEFT)	PANTHER POINT GPIO 16	PANTHER POINT GPIO 5 (RCVR_INT) PANTHER POINT GPIO 3 (PERIPH DET)

OTHER DETECT

FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
MULTIPLE SPKR VENDORS	N/A	N/A	N/A	0X0C (DET C)

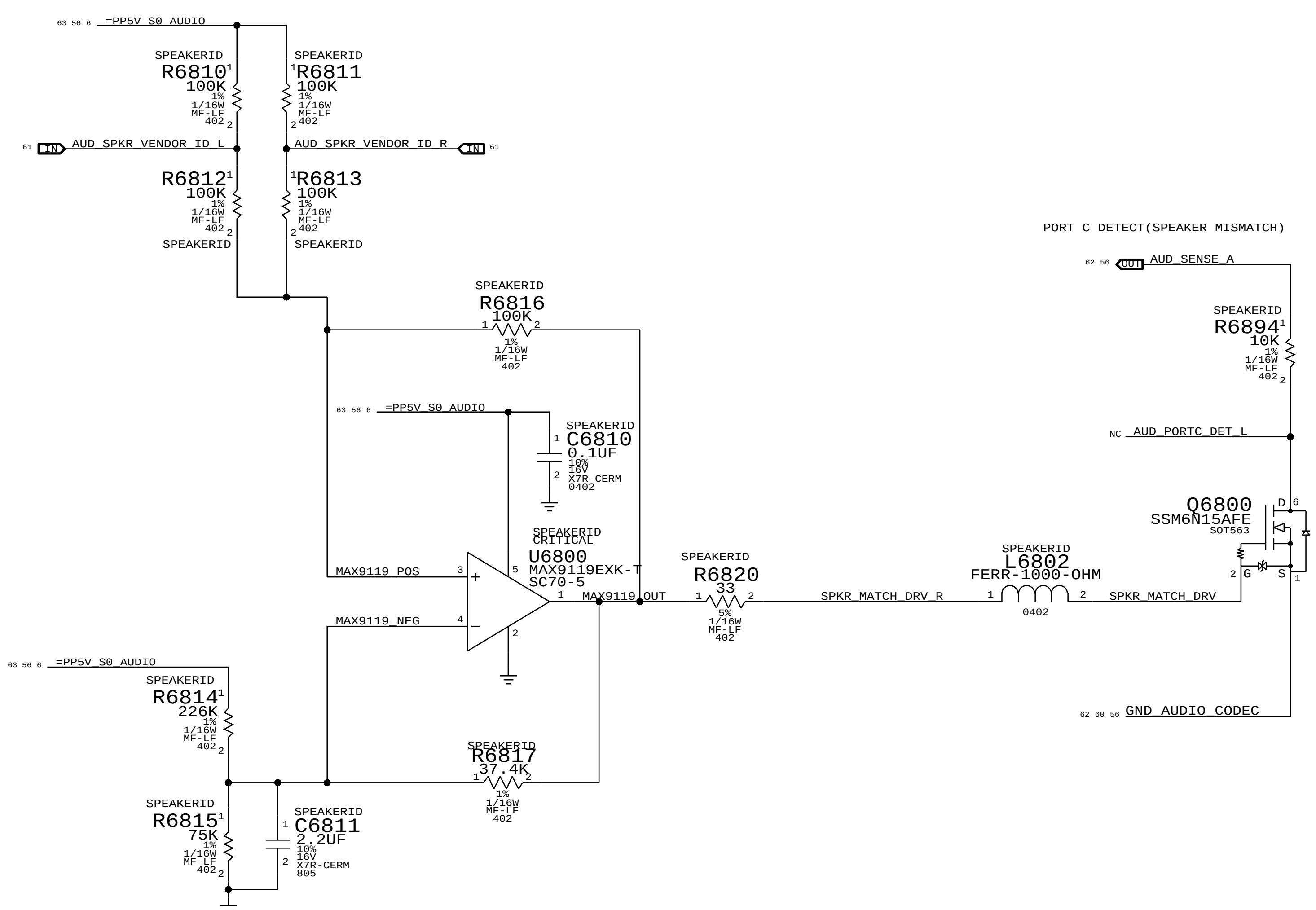
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
AUDIO	*	0.1 MM	?
SPKROUT	*	0.2 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
AUDIODIFF	*	Y	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
SPKROUTDIFF	*	Y	0.6 MM	0.25 MM	10 MM	0.2 MM	0.2 MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
AUDIODIFF	*	AUDIODIFF
SPKROUTDIFF	*	SPKROUTDIFF

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I215	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_P	56 57 58
I216	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_N	56 57 58
I217	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_C_P	57
I218	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_C_N	57
I211	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_P	56 57 59
I212	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_N	56 57 59
I210	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_C_P	57
I209	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_C_N	57
I208	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_P	56 58
I206	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_N	56 58
I207	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_P	56 59
I204	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_N	56 59
I205	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LINC_P	59
I203	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LINC_N	59
I220	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RINC_P	59
I219	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RINC_N	59
I222	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LIN_P	59
I221	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LIN_N	59
I224	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RIN_P	59
I223	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RIN_N	59
I226	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LINC_P	58
I225	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LINC_N	58
I227	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RINC_P	58
I229	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RINC_N	58
I228	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LIN_P	58
I230	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LIN_N	58
I264	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RIN_P	58
I263	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RIN_N	58
I236	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INL_P	57
I238	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INL_N	57
I237	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INR_P	57
I239	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INR_N	57
I240	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RWFR_OUT_P	59 61
I241	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RWFR_OUT_N	59 61
I242	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RTWT_OUT_P	59 61
I243	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RTWT_OUT_N	59 61
I244	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LWFR_OUT_P	58 61
I246	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LWFR_OUT_N	58 61
I245	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LTWT_OUT_P	58 61
I247	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LTWT_OUT_N	58 61
I326	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_INL_P	56 60
I327	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_INL_N	56 60
I328	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_RC_P	60
I329	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_RC_N	60
I254	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_P	60
I255	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_N	60
I324	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_J1_MIC_P	60
I325	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_J1_MIC_N	60

CIRCUIT THEORY OF OPERATION AVAILABLE IN <RDAR://PROBLEM/9776522>



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

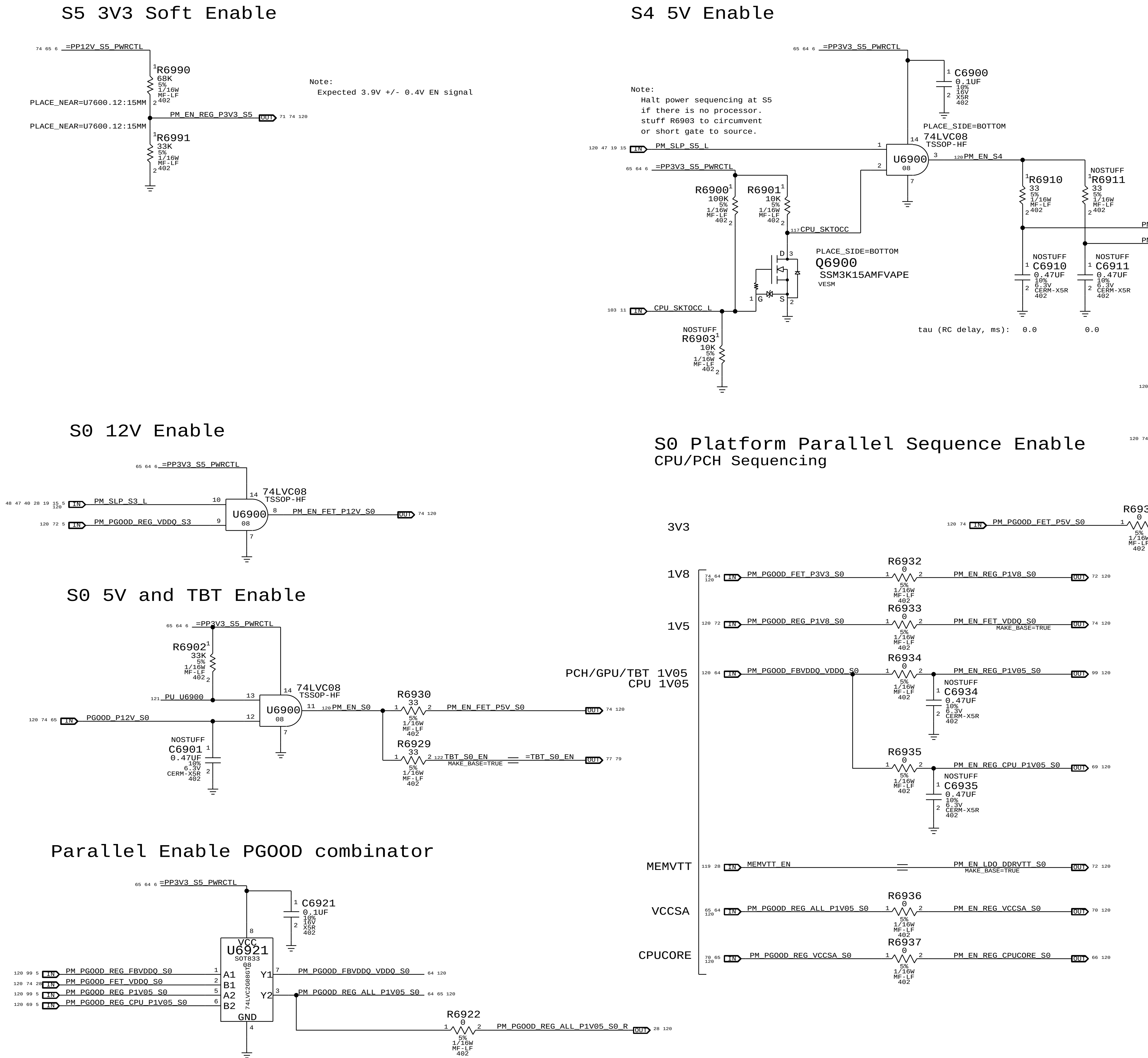
AUDIO: Speaker ID

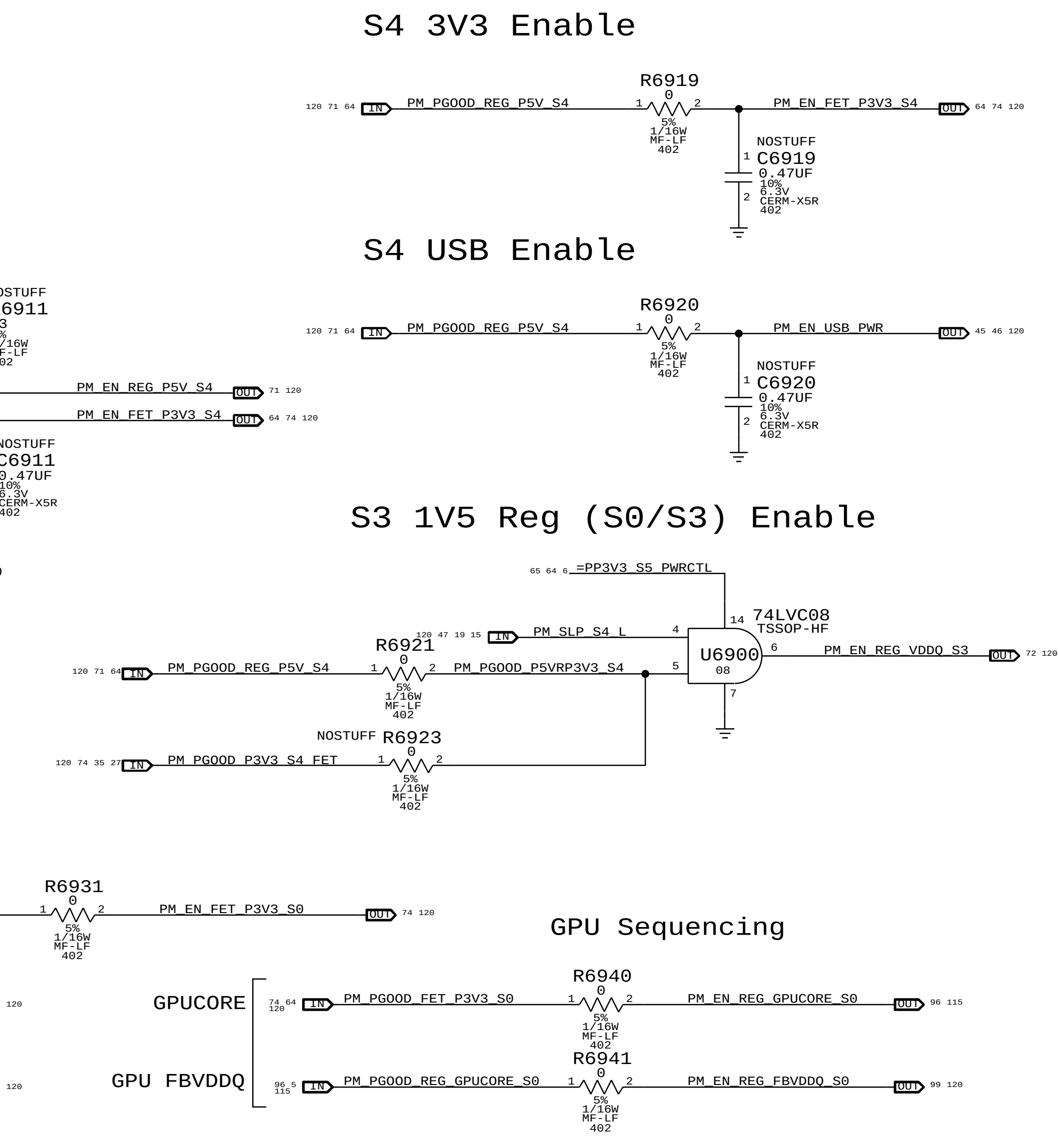
Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH





Rail definitions

Platform: All processor non-Core and non-Graphics (5 V, 3.3 V, 1.8 V 1.5 V, PCH Core/PLL/VRM)
Uncore: VccSA, VDDQ, VccA (1.8 V), VccIO (VccSA, VccA, and VccIO must ramp within 50 ms of each other)

Notes on sequencing requirements

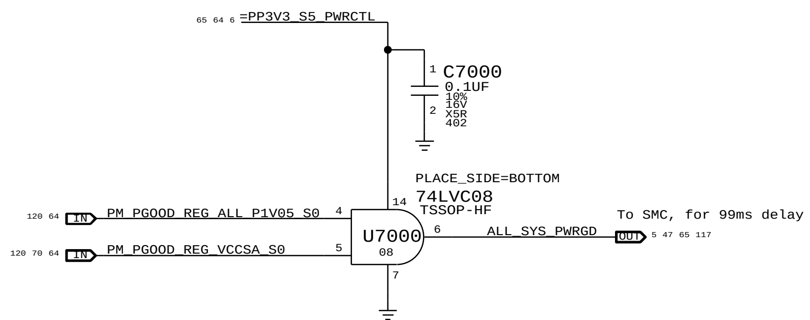
Intel:
1. No hard specification on platform rails
2. SMC guarantees timing on PCH DPWROK and PWROK
3. CPU VDDQ must ramp before MEMVTT and vice versa on power down. It has no relationship to any other rails.

NVIDIA GFX:
1. VDD33 (our 3V3_S0)
2. IFPA/B_IOVDD (1.8 V) with or after 3V3_S0 (unused in our implementation)
3. NVVDD (our GPUCORE) after IFPA/B_IOVDD
4. FBVDDQ (our GPU_VDDQ) after NVVDD
5. PEX_VDD (our GPU_1V05) after FBVDDQ
6. IFPC/D/E/F_IOVDD (1.05V) wit or after PEX_VDD
7. The ramp time for any rail must be more than 40 uS
8. All rails must be powered off within 10 ms from first rail powering off

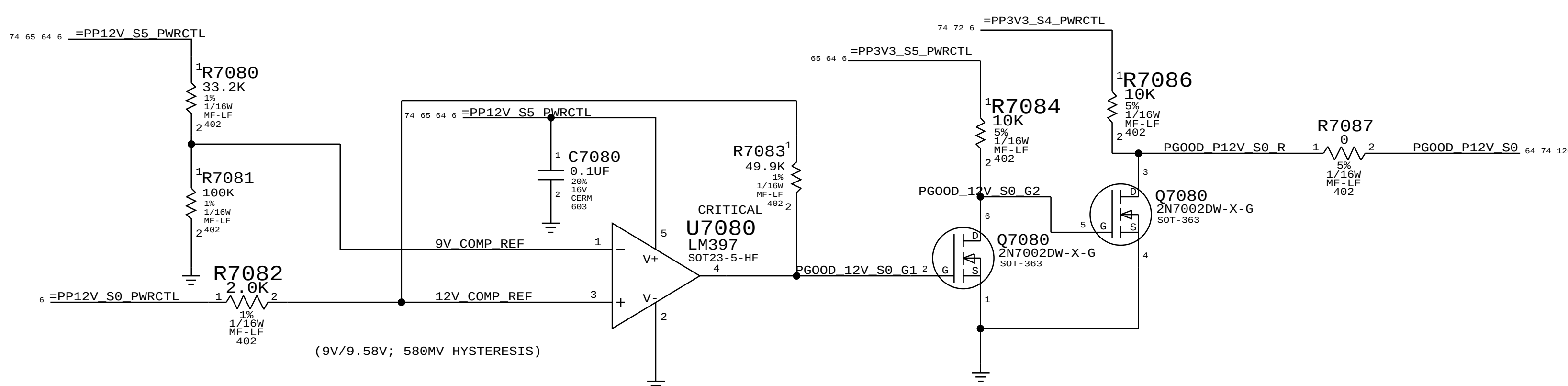
SYNC_MASTER=D8_MLB	SYNC_DATE=07/03/2012
PAGE TITLE	PM Regulator Enables
Apple Inc.	DRAWING NUMBER 051-9505 SIZE D
	REVISION 6.1.0
	BRANCH
	PAGE 69 OF 144
	SHEET 64 OF 123

Platform Power Good Derive SMC ALL_SYS_PWRGD

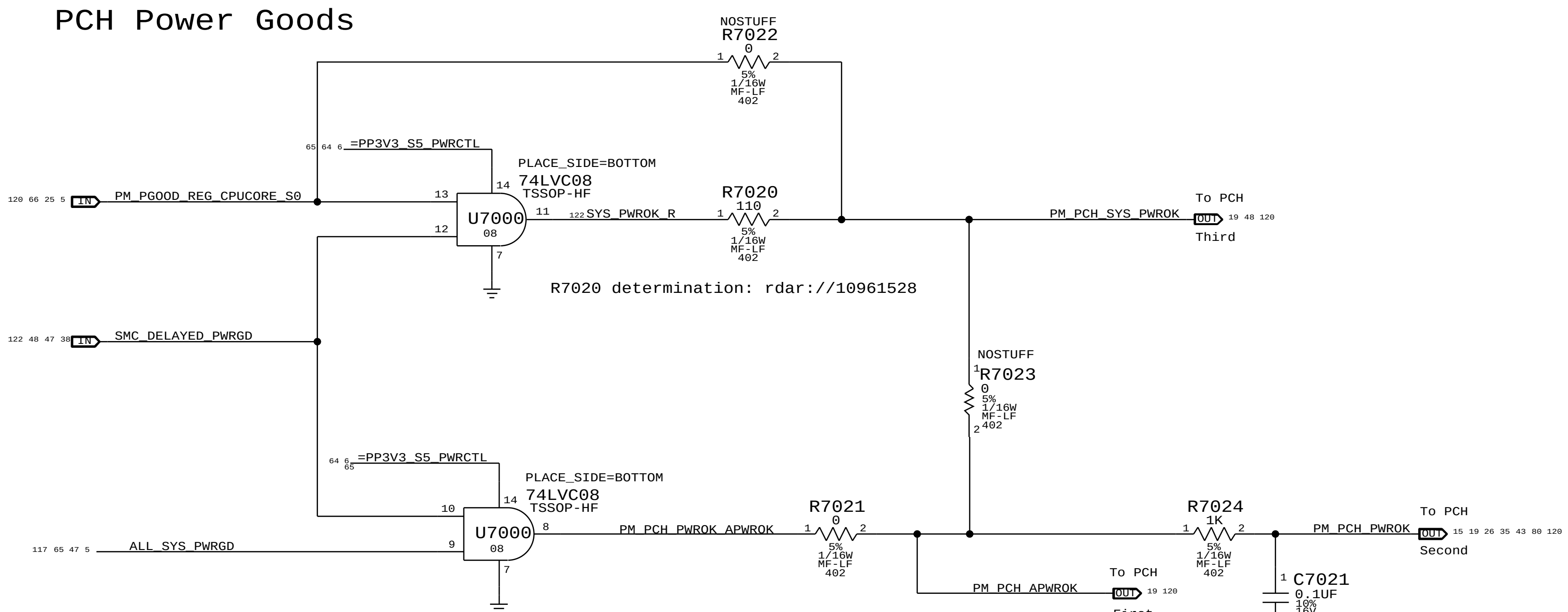
The end of the power sequence for S0 rails except CPU CORE.



PGOOD COMPARATORS FOR PP12V_S0



PCH Power Goods



radar://11043352 Need AND Gate to deassert PM_PCH_PWROK to PCH when unexpected power loss happens

Resume Reset

Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

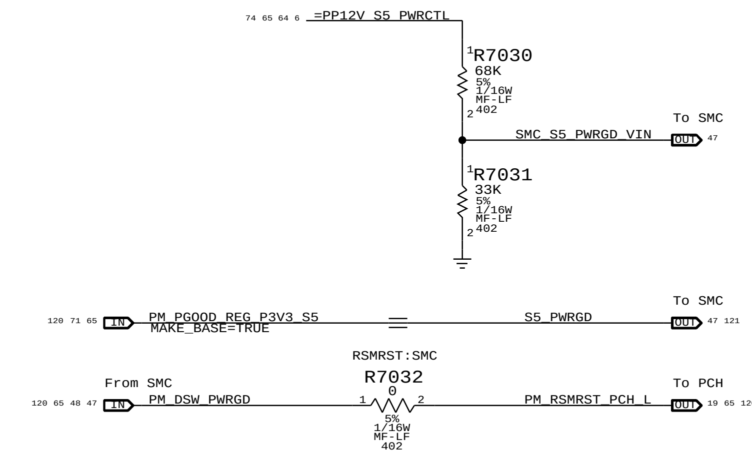
Note:
The iMac K70K72 designs does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

Requirements:
Power on:
Asserted at least 10 ms after all suspend well power is valid
Power off or loss of AC:
Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V to allow PCH to switch suspend well to battery without excessive loading

Primary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation.

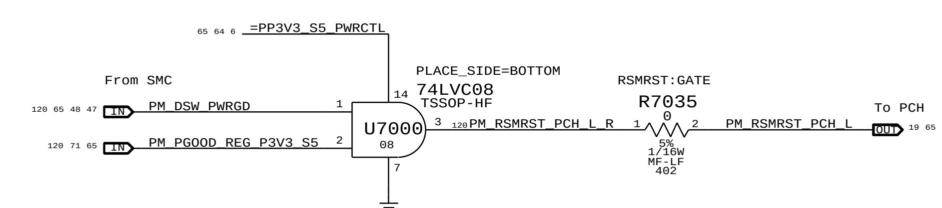
SMC de-asserts RSMRST# (PM_DSW_PWRGD) when S5_PWRGD input is asserted and SMC_S5_PWRGD_VIN input is above comparator input level of 1.5 V.

SMC asserts RSMRST# (PM_DSW_PWRGD) when SMC_S5_PWRGD_VIN input drops from 1.8 V to 1.5 V (as implemented) when 12 V S5 rail drops to 10 V.



Secondary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.

RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

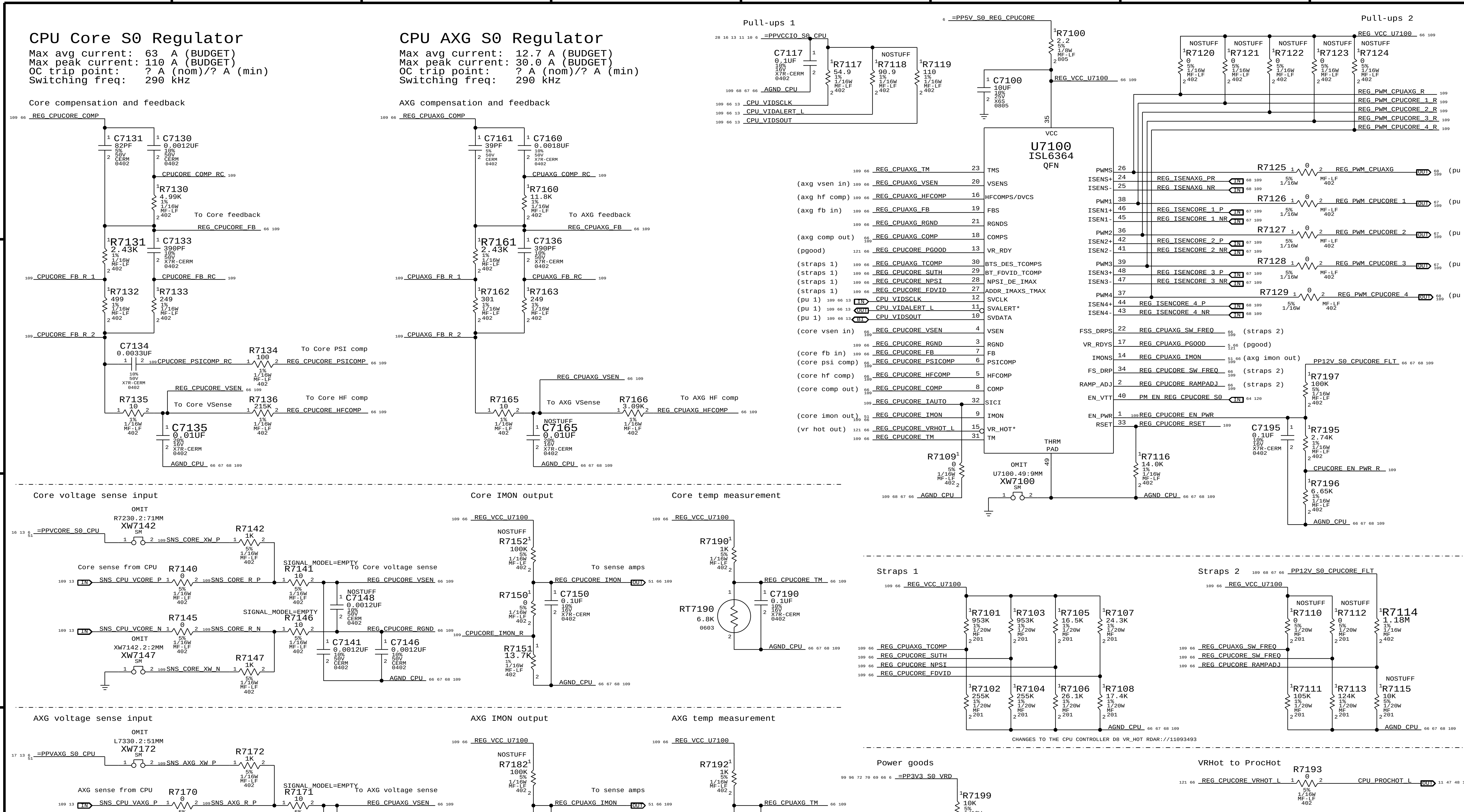
PAGE TITLE: PM Power Good

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

VReg CPU Core/AXG Cntl

Apple Inc.

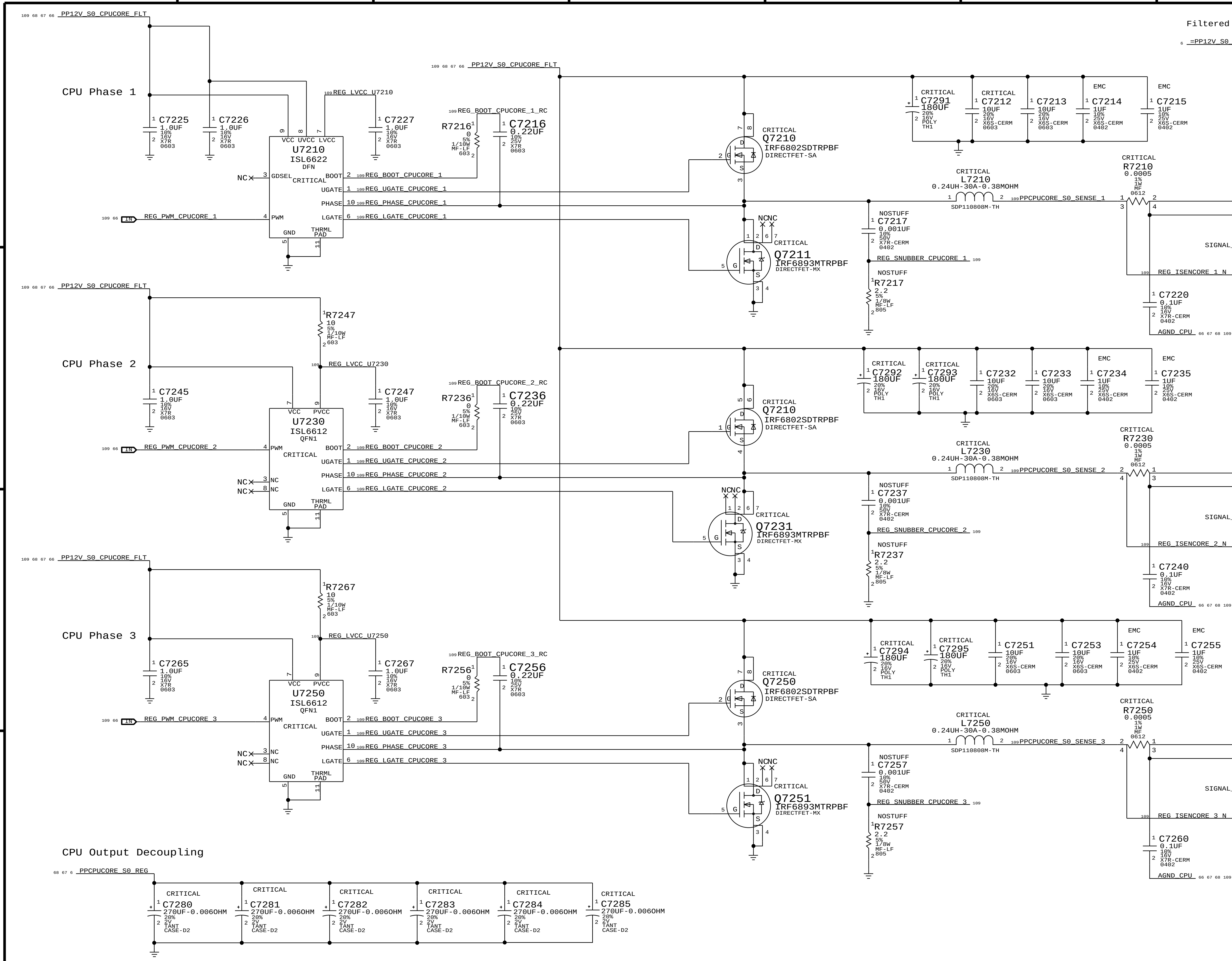
DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

PAGE 71 OF 144

SHEET 66 OF 123



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

VReg CPU Core Phases

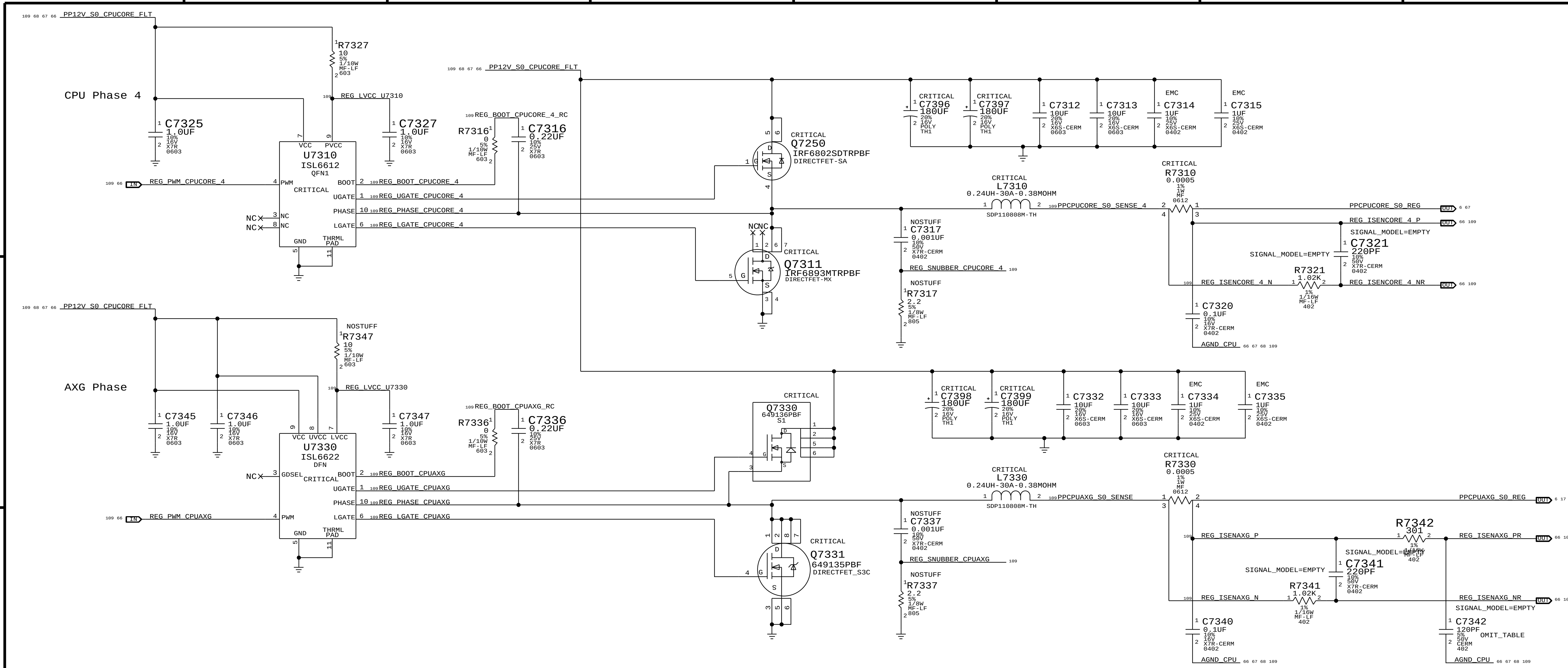
Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
131S0513	1	120PF,5%,50V,0402,SMD	C7342		

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

VReg CPU AXG Phases

Apple Inc.

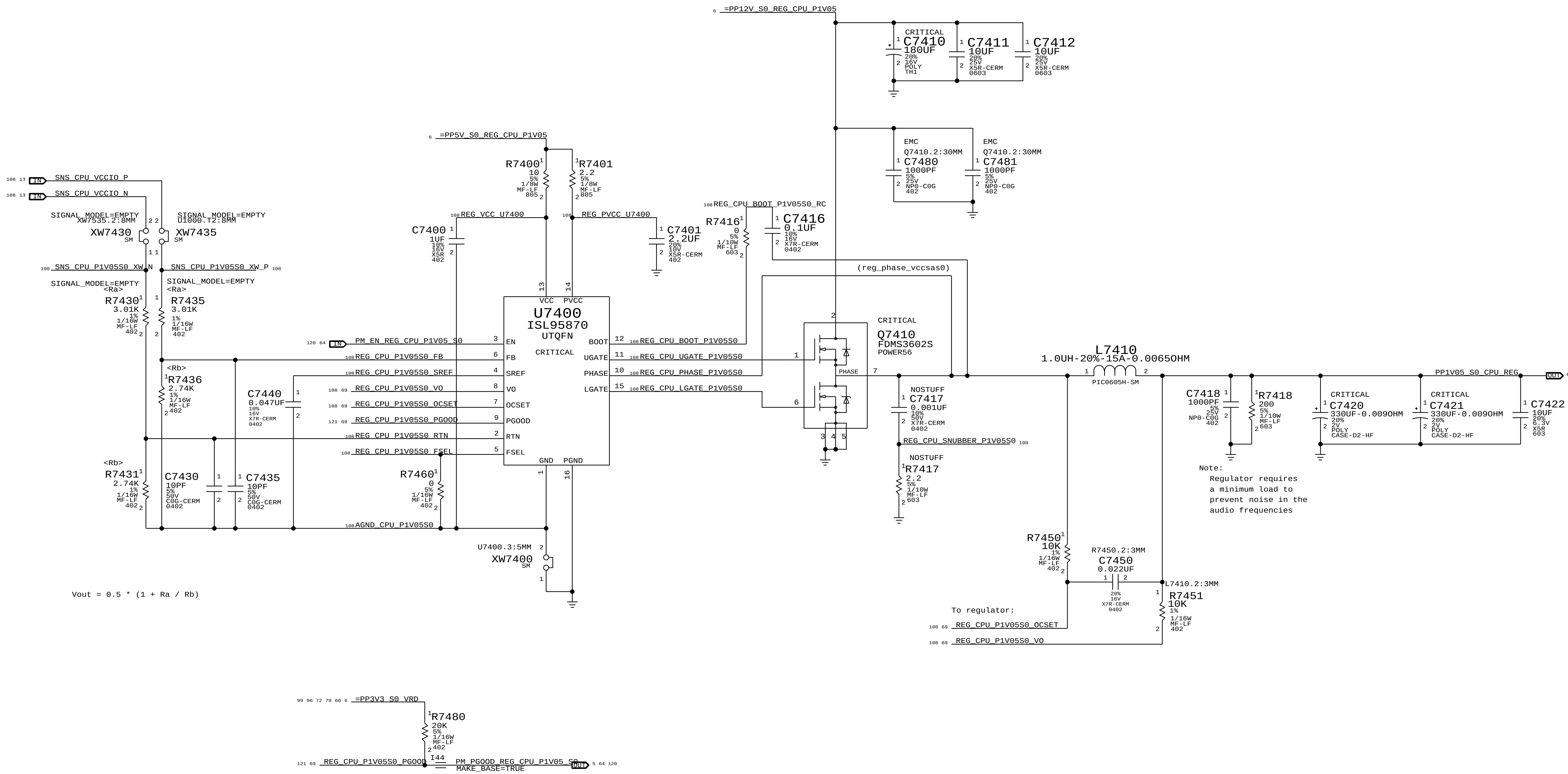
DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

CPU VccIO (1.05V) S0 Regulator

Max avg current: 8.10 A (BUDGET)
Max peak current: 8.50 A (BUDGET)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 kHz



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE: VReg CPU 1.05V S0

Apple Inc.

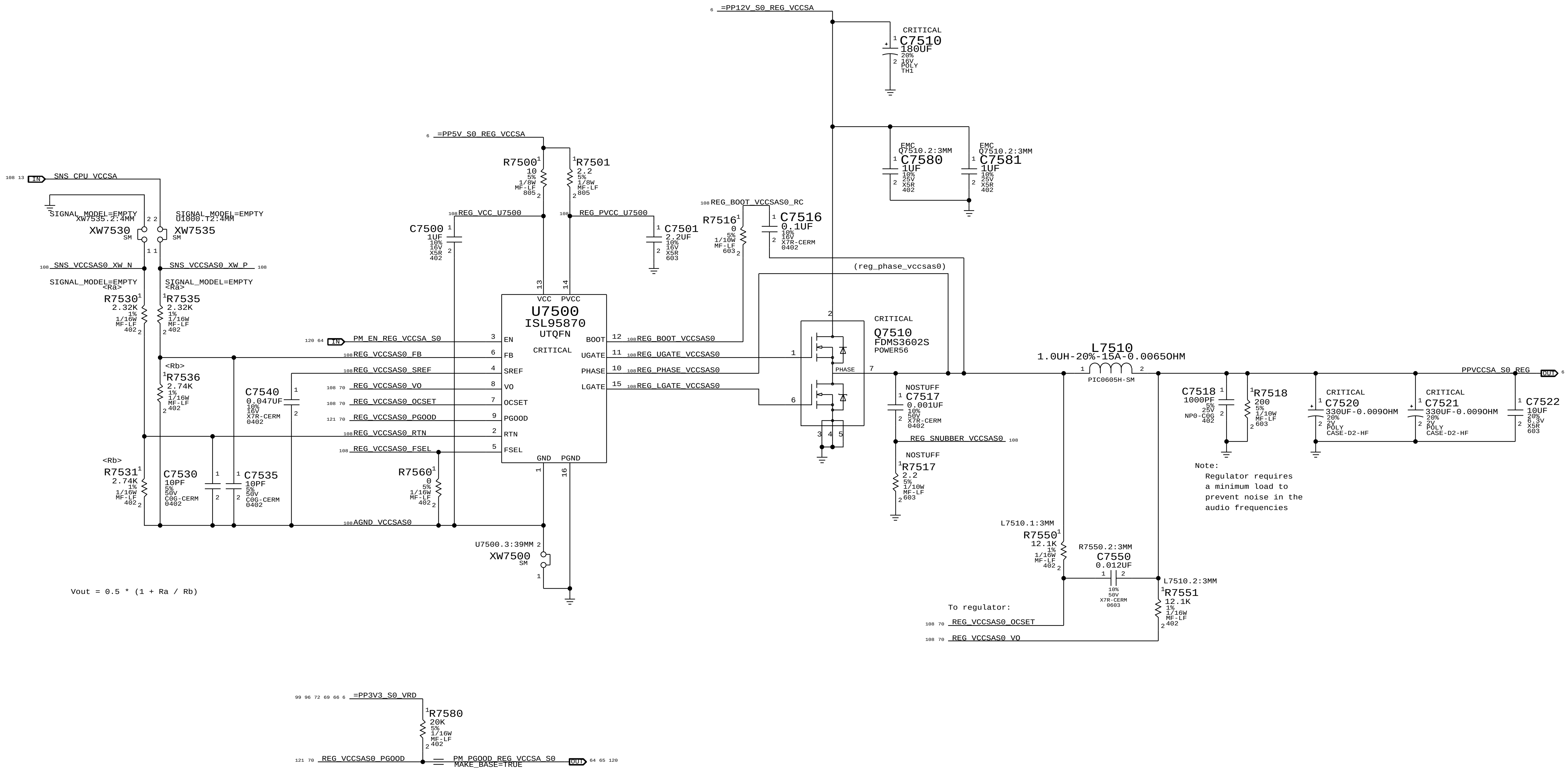
DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH

CPU VccSA (0.925V) S0 Regulator

Max avg current: 12.07 A (BUDGET)
Max peak current: 30 A (BUDGET)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 kHz



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

VReg CPU VccSA S0

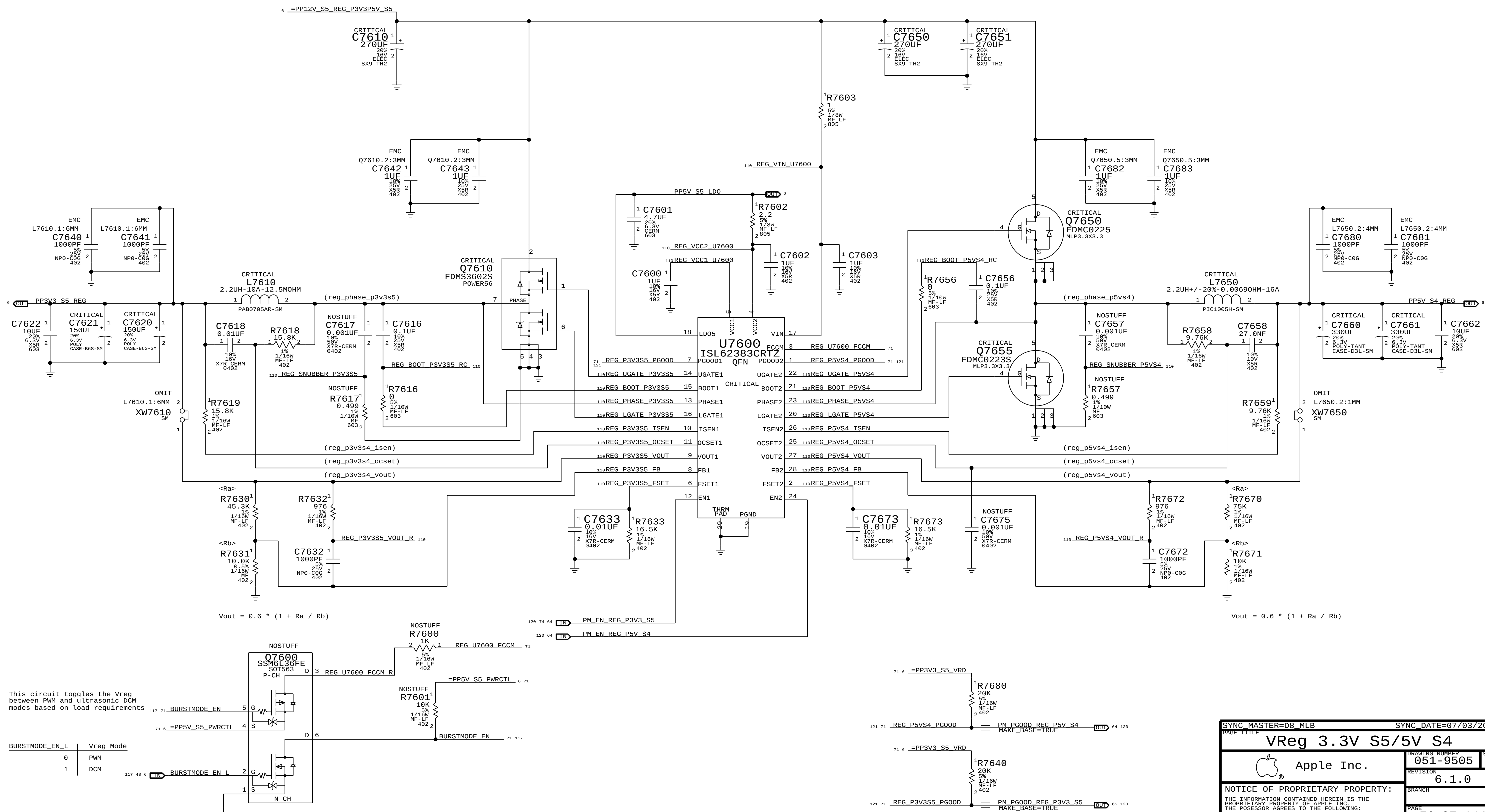
Apple Inc.	DRAWING NUMBER 051-9505 SIZE D
	REVISION 6.1.0
NOTICE OF PROPRIETARY PROPERTY:	BRANCH
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	PAGE 75 OF 144
	SHEET 70 OF 123

3.3V S5 Regulator

Max avg current: 6 A (design)/ 4.85 A (budget)
Max peak current: ? A (design)/ 6.6 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 350 kHz

5V S4 Regulator

Max avg current: 10 A (design)/ 6.08 A (budget)
Max peak current: ? A (design)/ 6.9 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 350 kHz



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

VReg 3.3V S5/5V S4

Apple Inc.

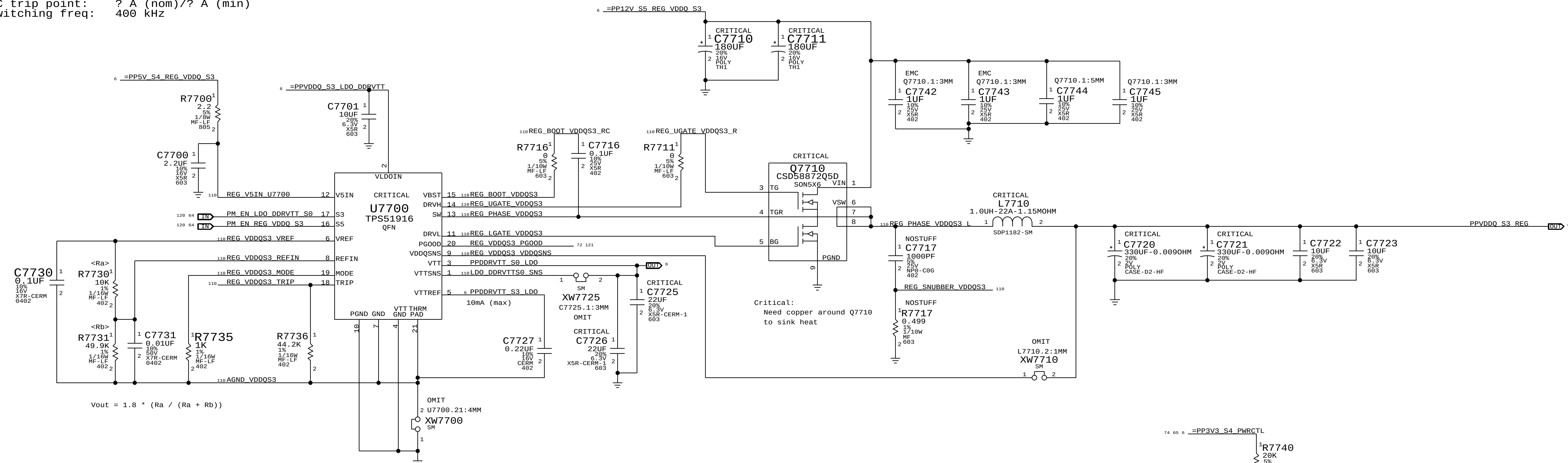
DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

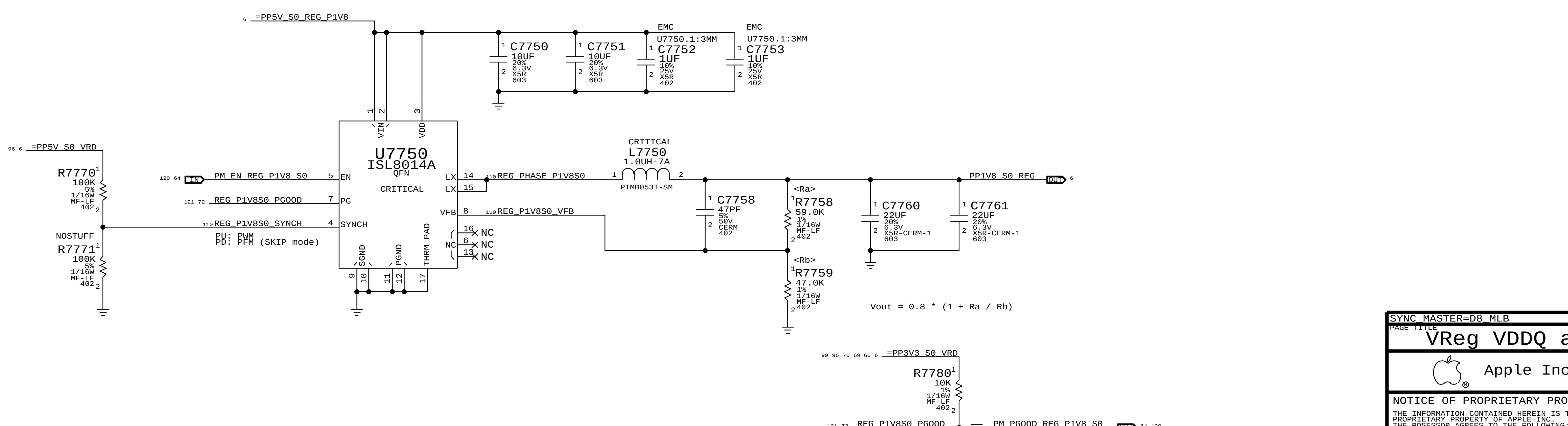
VDDQ (1.5V) S3 Regulator

Max avg current: 9.0 A (BUDGET)
Max peak current: 11.3 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: 400 kHz



1.8V S0 Regulator

Max avg current: 0.6 A (BUDGET)
Max peak current: 1.7 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: ? kHz



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

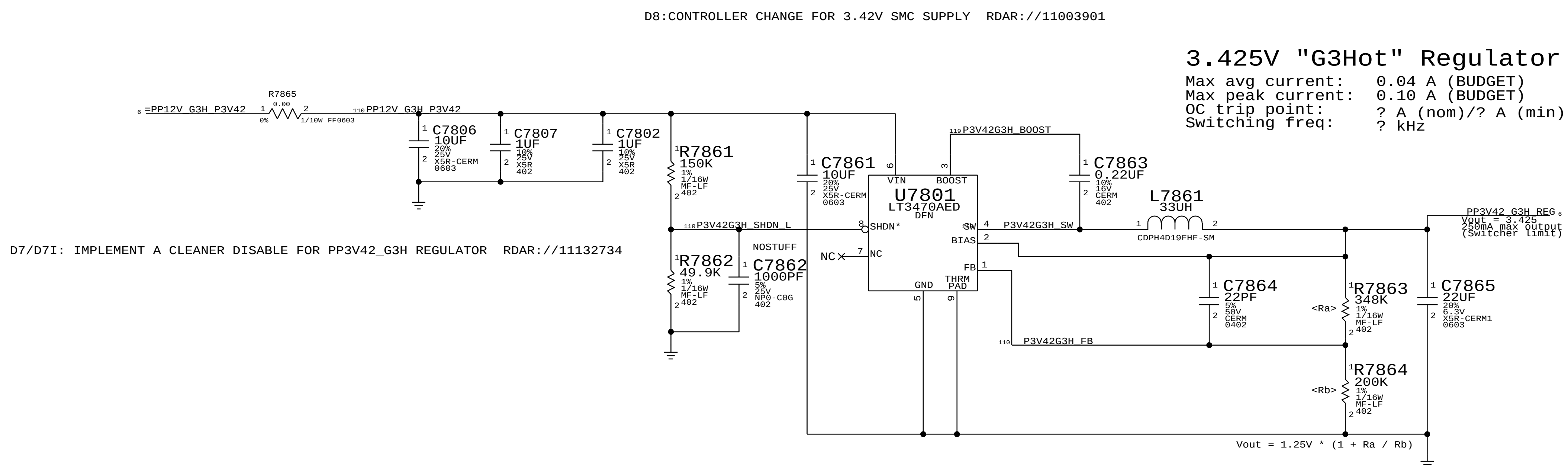
PAGE TITLE: VReg VDDQ and 1.8V S0

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

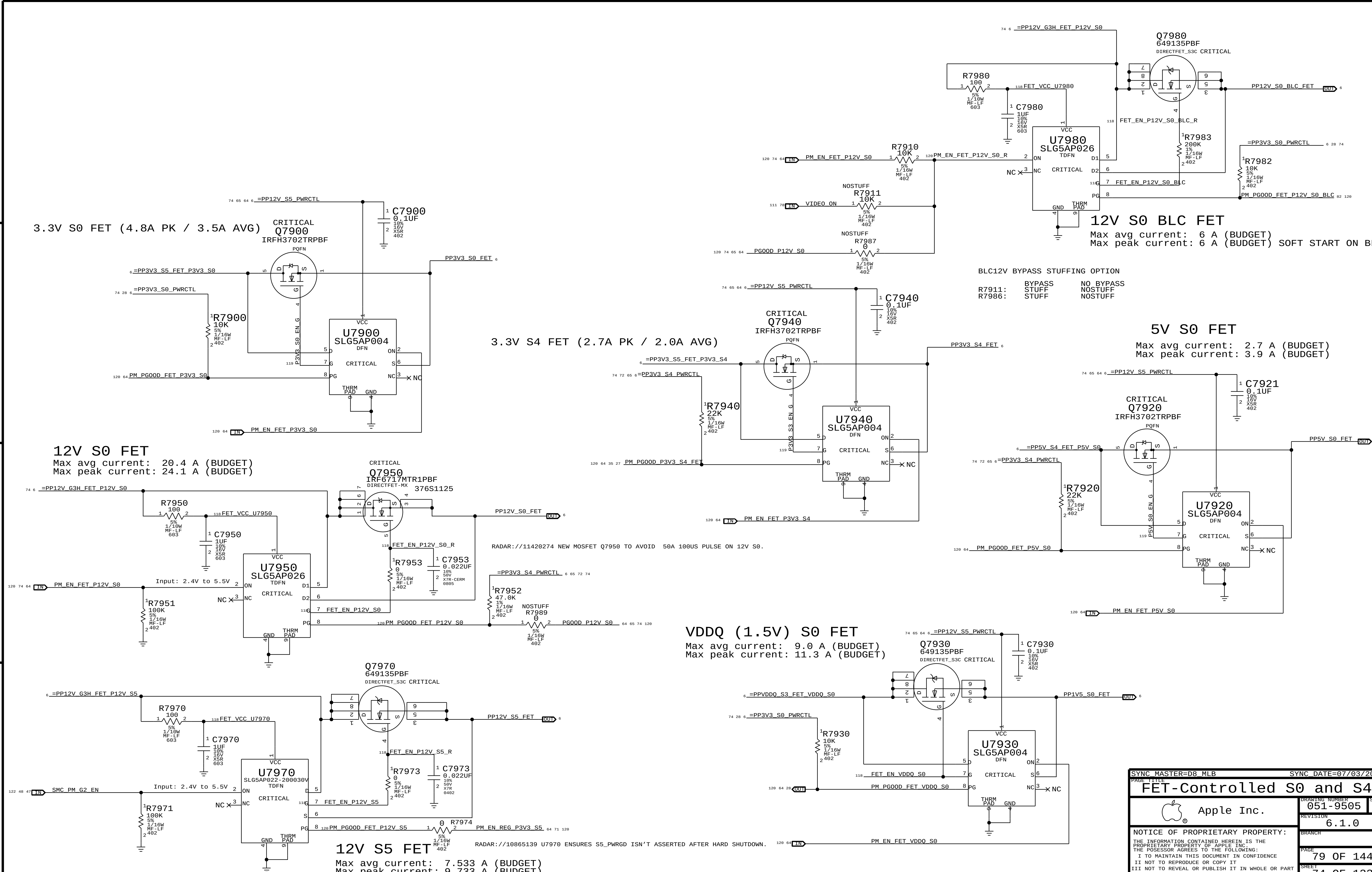
VREG 3.42V G3HOT

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

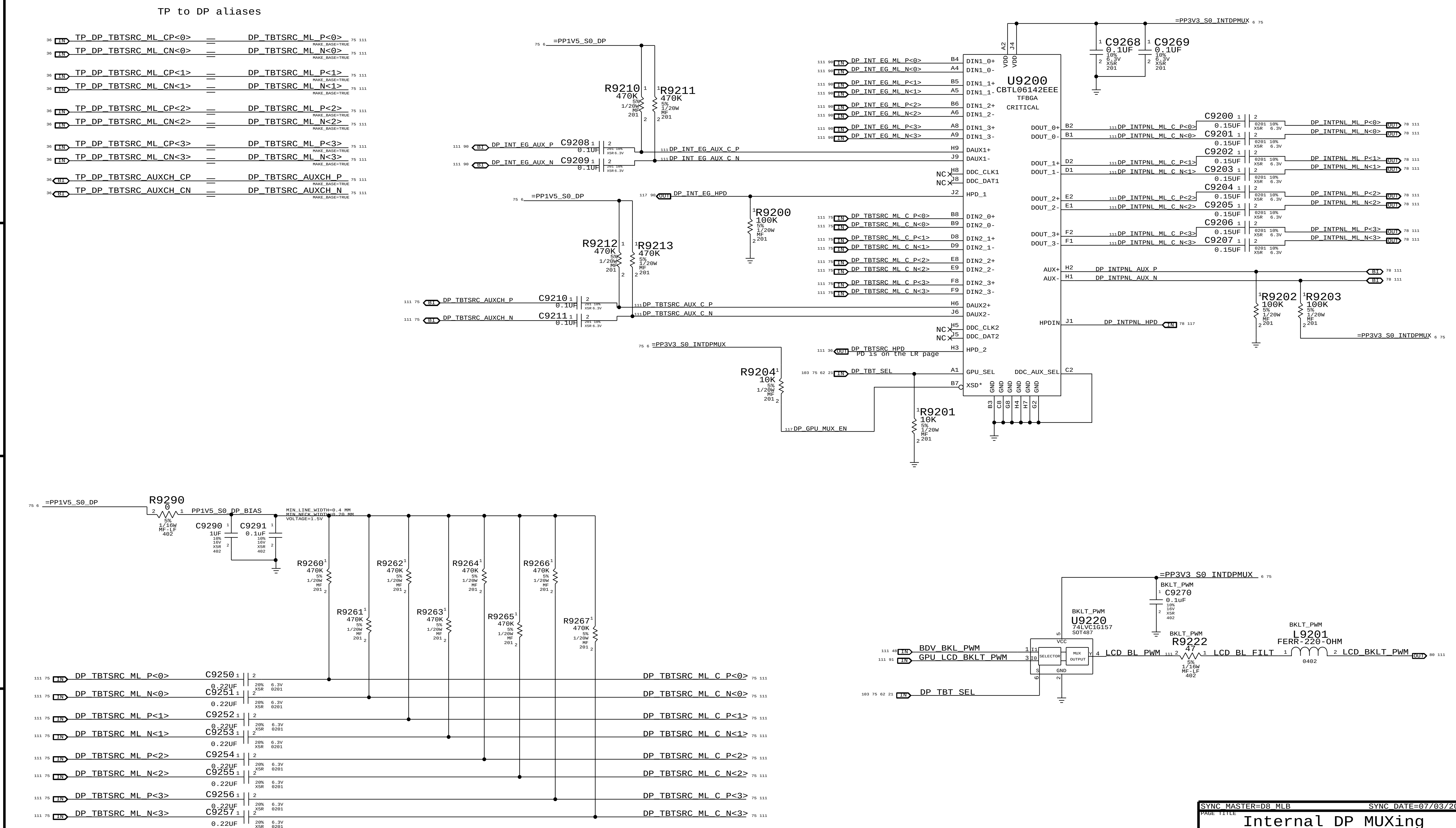
FET-Controlled S0 and S4

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

Internal DP MUXing

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

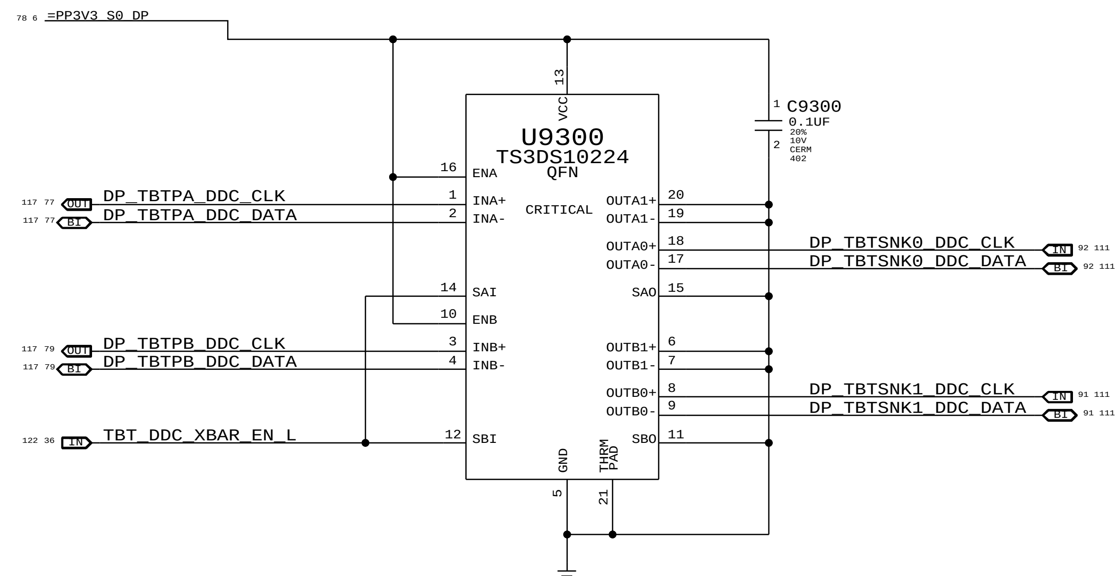
REVISION 6.1.0

BRANCH

PAGE 92 OF 144

SHEET 75 OF 123

Dual-Port Host DDC Crossbar



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

TBT DDC Crossbar

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

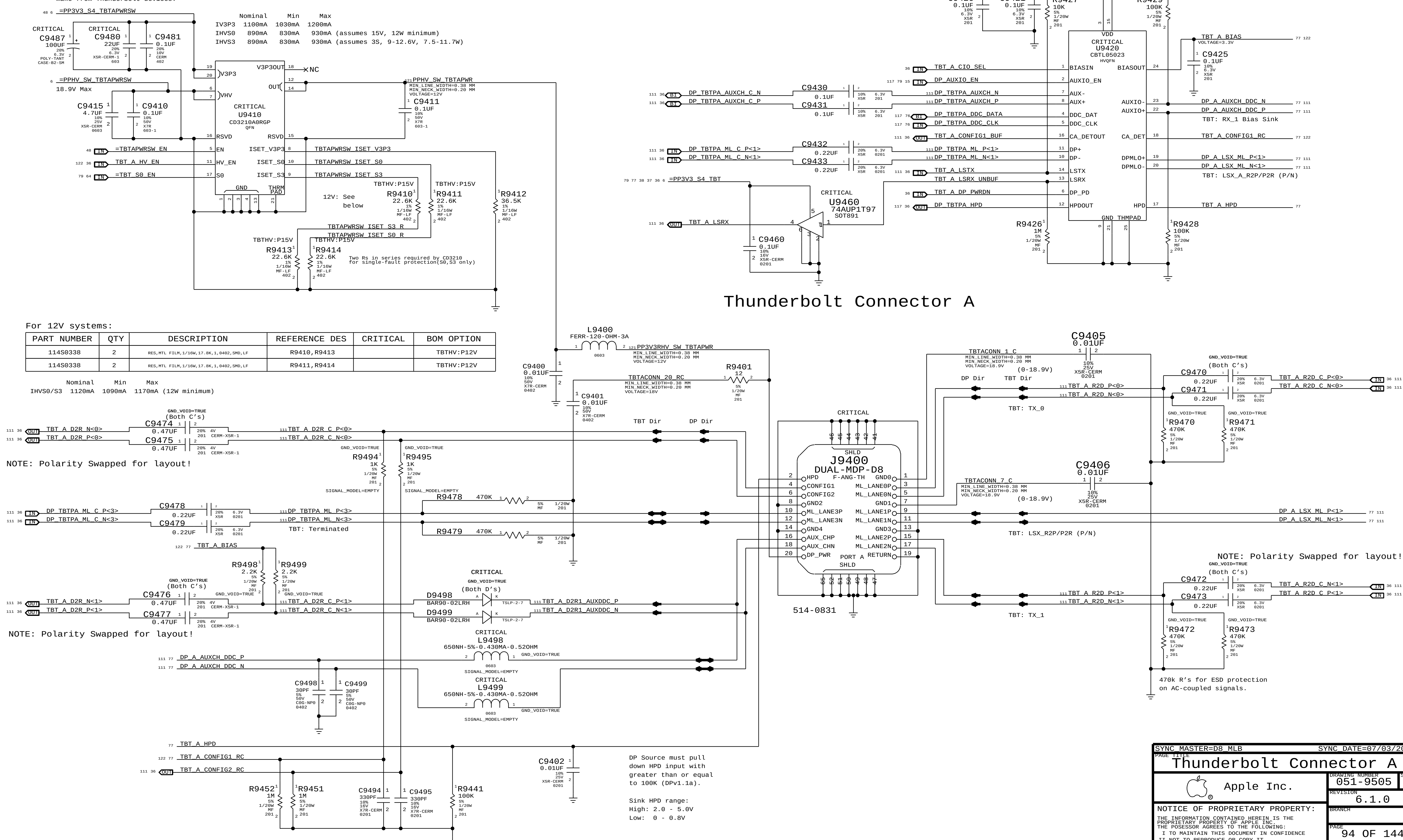
PAGE 93 OF 144

SHEET 76 OF 123

3.3V/HV Power MUX

V3P3 must be S4 to support
wake from Thunderbolt devices.

	Nominal	Min	Max	
IV3P3	1100mA	1030mA	1200mA	
IHVS0	890mA	830mA	930mA	(assumes 15V, 12W minimum)
IHVS3	890mA	830mA	930mA	(assumes 3S, 9-12.6V, 7.5-11.7W)



For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9410,R9413		TBTHV:P12V
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9411,R9414		TBTHV:P12V

	Nominal	Min	Max
IHVS0/S3	1120mA	1090mA	1170mA (12W minimum)

Thunderbolt Connector A

DP Source must pull
down HPD input with
greater than or equal
to 100K (DPv1.1a).

Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE: Thunderbolt Connector A

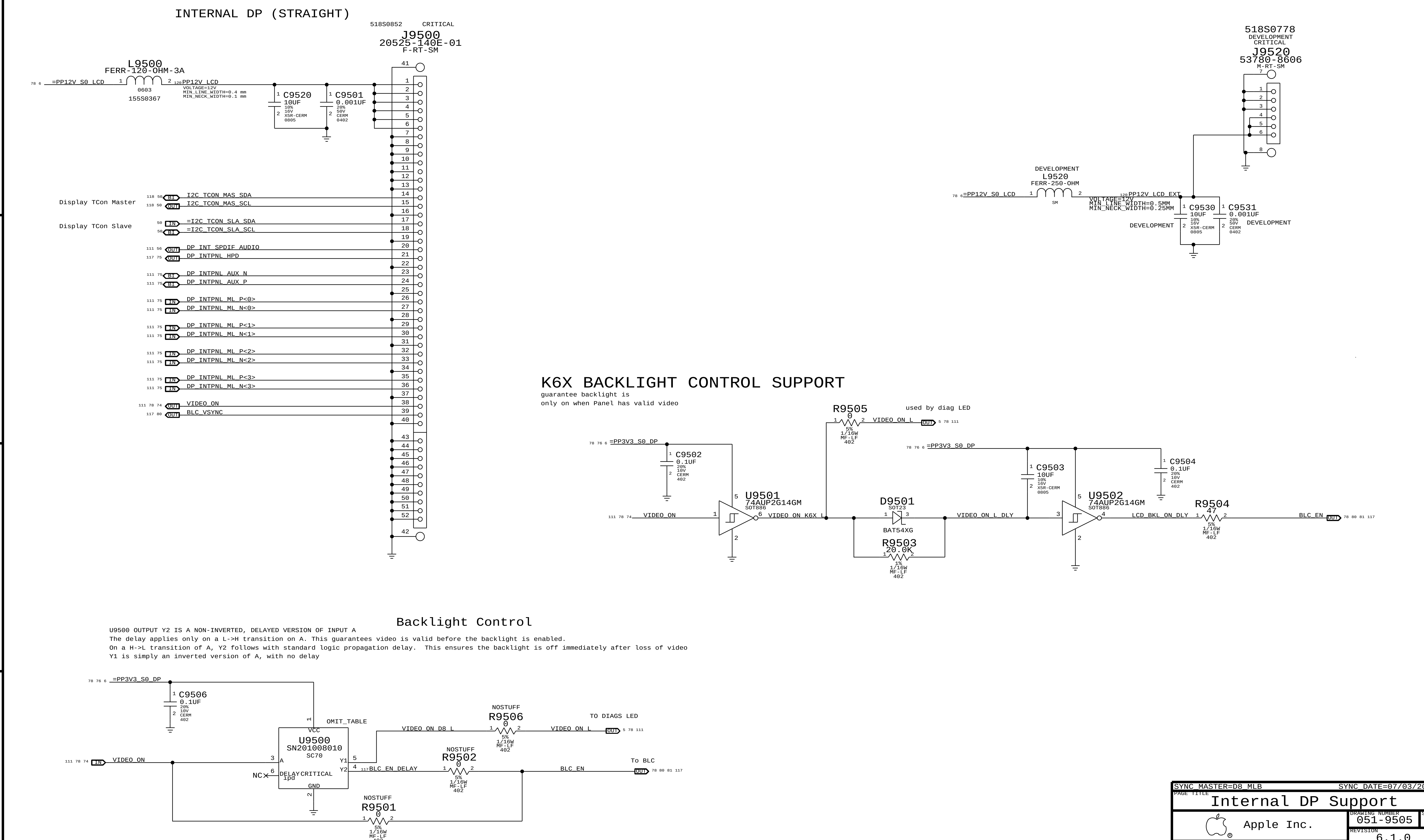
Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH

INTERNAL DP (STRAIGHT)



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
311S0566	1	ONE BIT SELECTABLE DELAY	U9500	Y	

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE
Internal DP Support

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

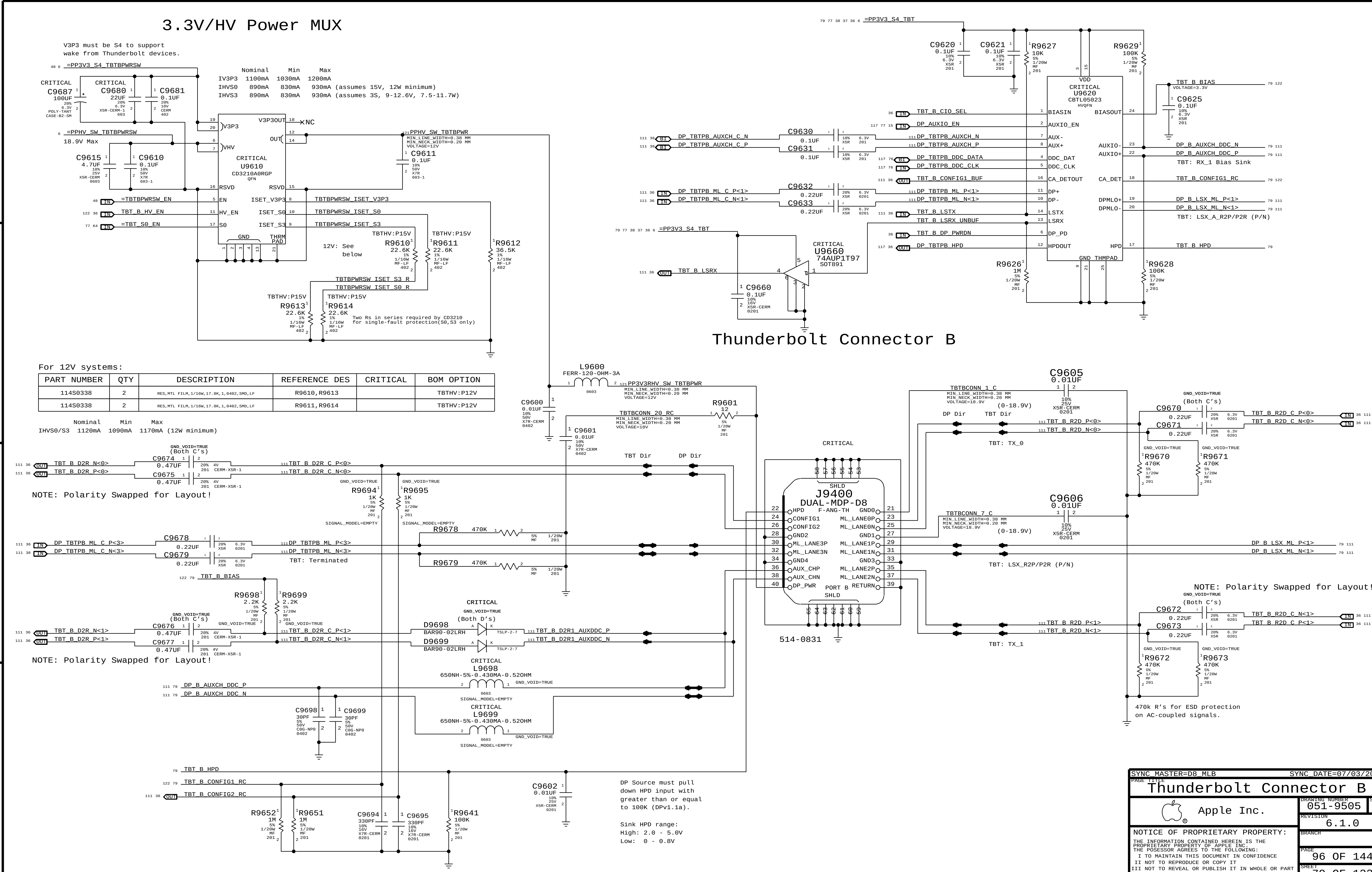
REVISION 6.1.0

BRANCH

3.3V/HV Power MUX

V3P3 must be S4 to support
wake from Thunderbolt devices.

	Nominal	Min	Max	
IV3P3	1100mA	1030mA	1200mA	
IHVS0	890mA	830mA	930mA	(assumes 15V, 12W minimum)
IHVS3	890mA	830mA	930mA	(assumes 3S, 9-12.6V, 7.5-11.7W)



For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9610,R9613		TBTHV:P12V
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9611,R9614		TBTHV:P12V

	Nominal	Min	Max	
IHVS0/S3	1120mA	1090mA	1170mA	(12W minimum)

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

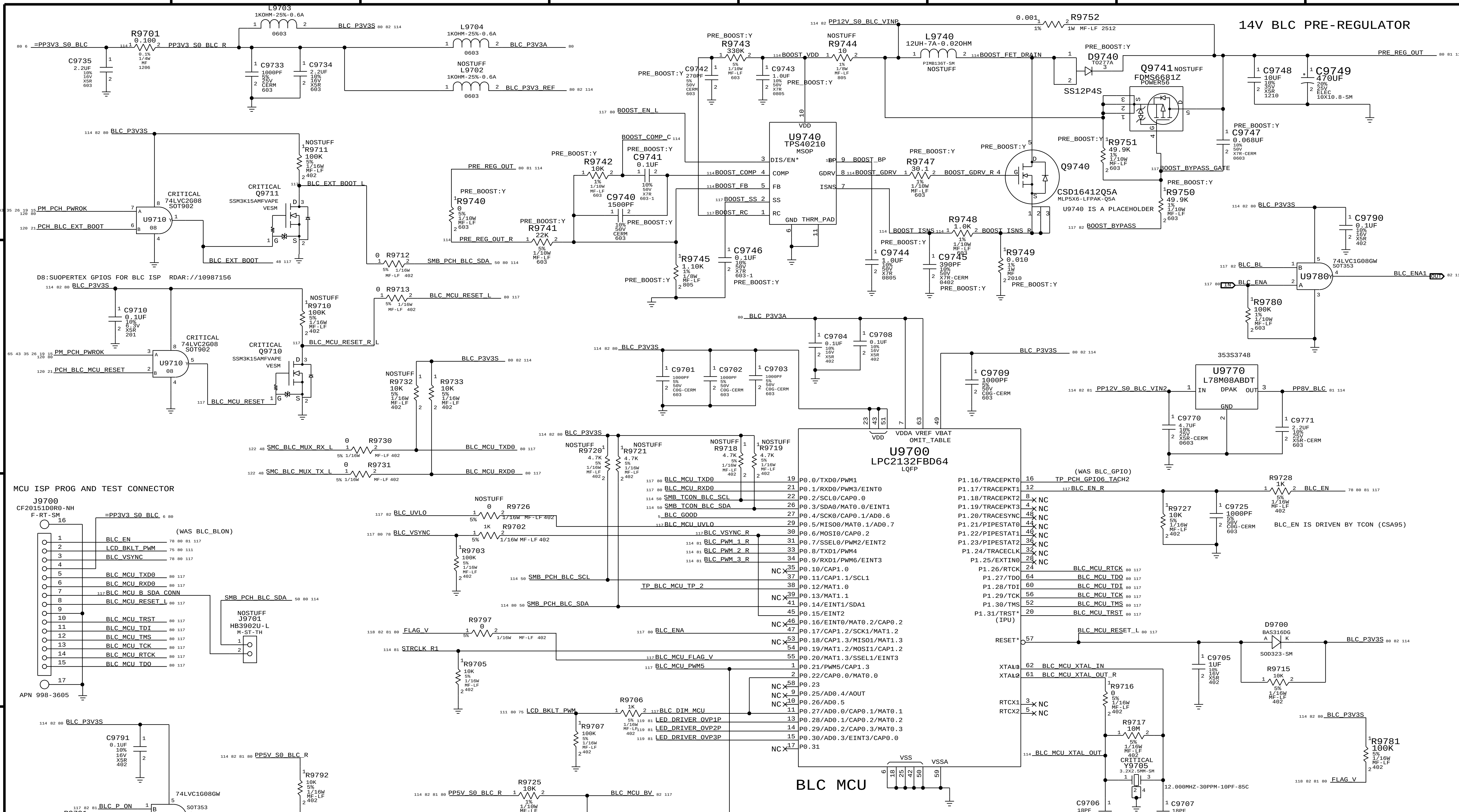
PAGE TITLE: Thunderbolt Connector B

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

Backlight Controller MCU

Apple Inc.

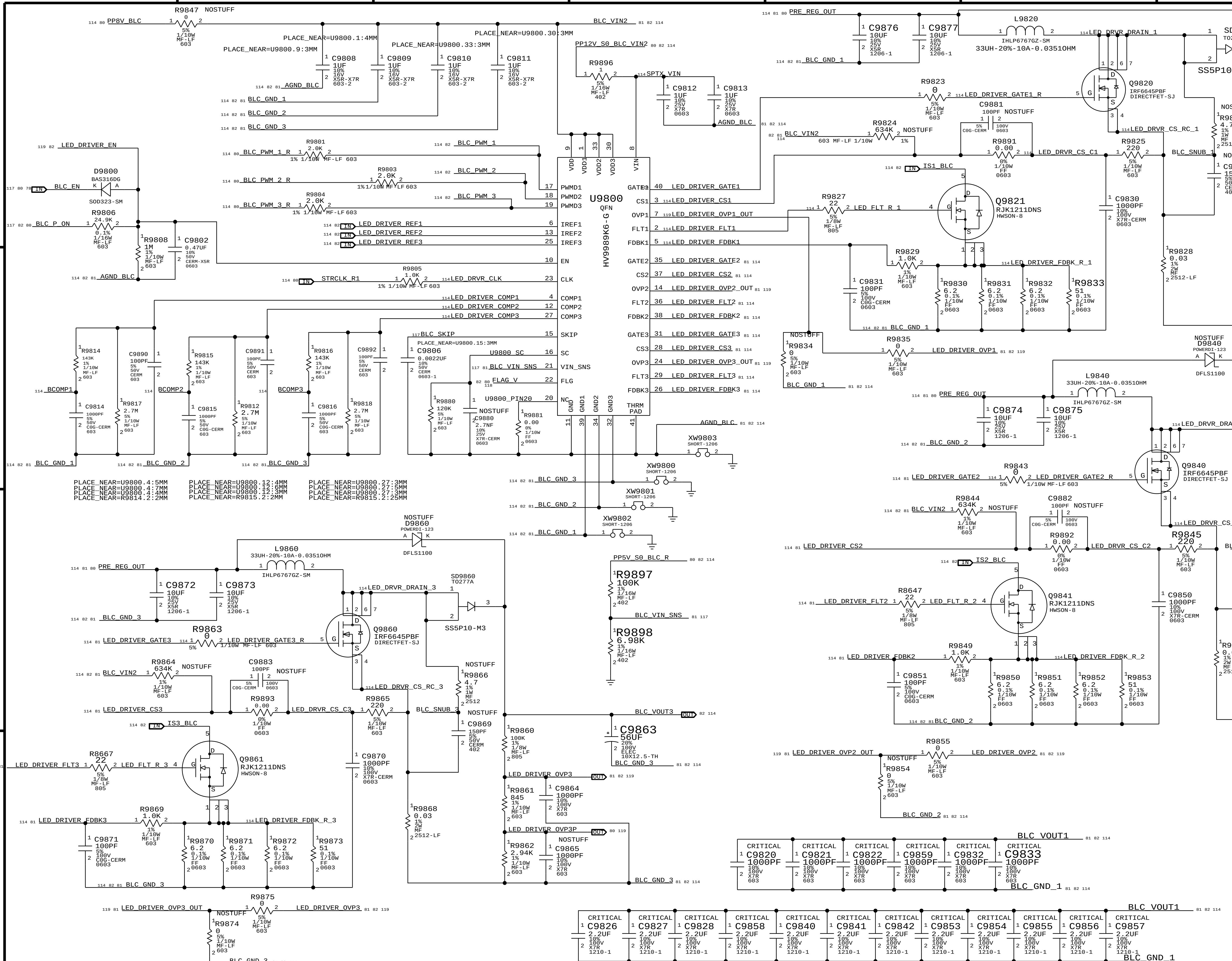
DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

PAGE 97 OF 144

SHEET 80 OF 123



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

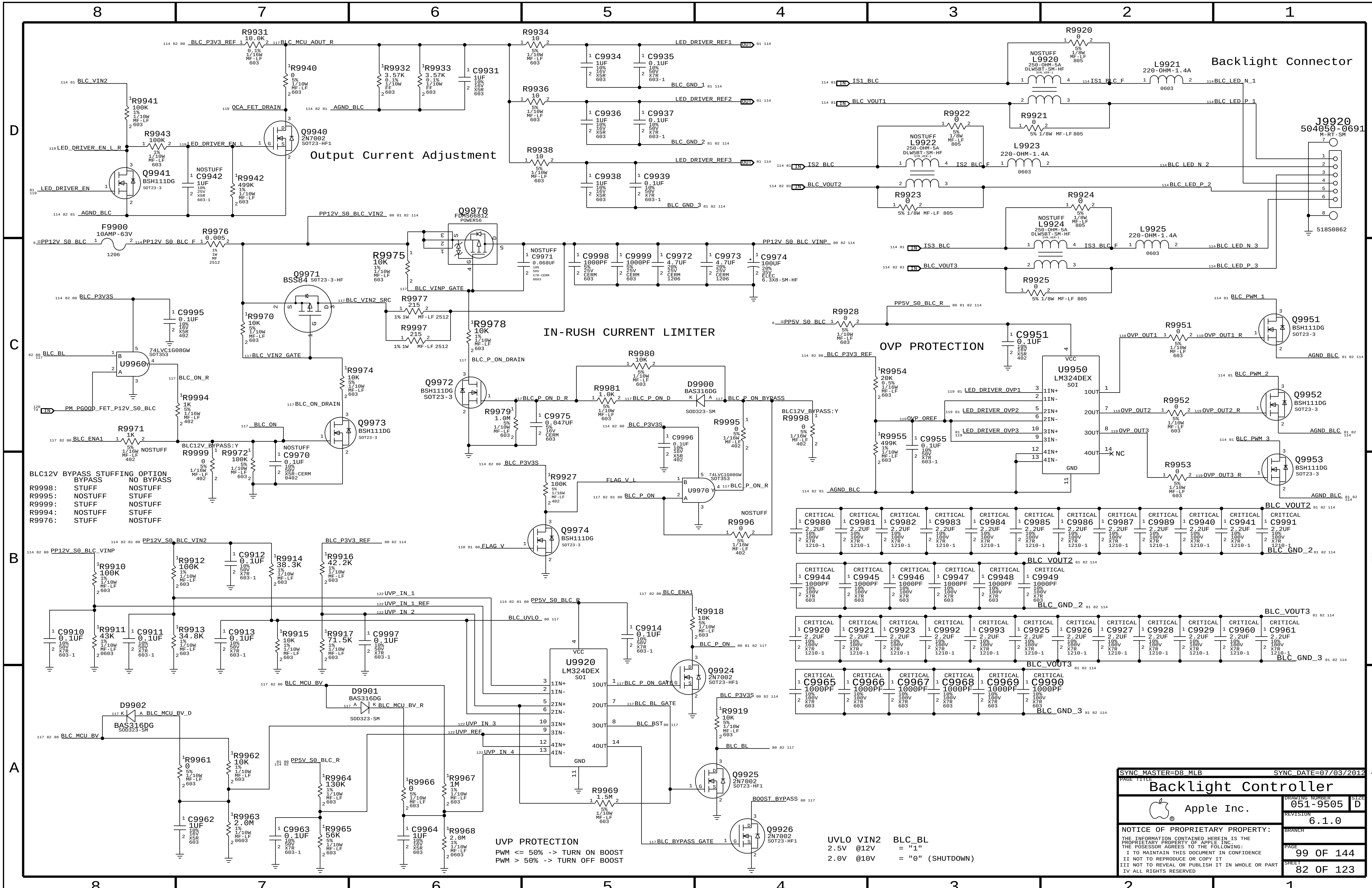
Backlight LED Driver

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

Backlight Controller

Apple Inc.

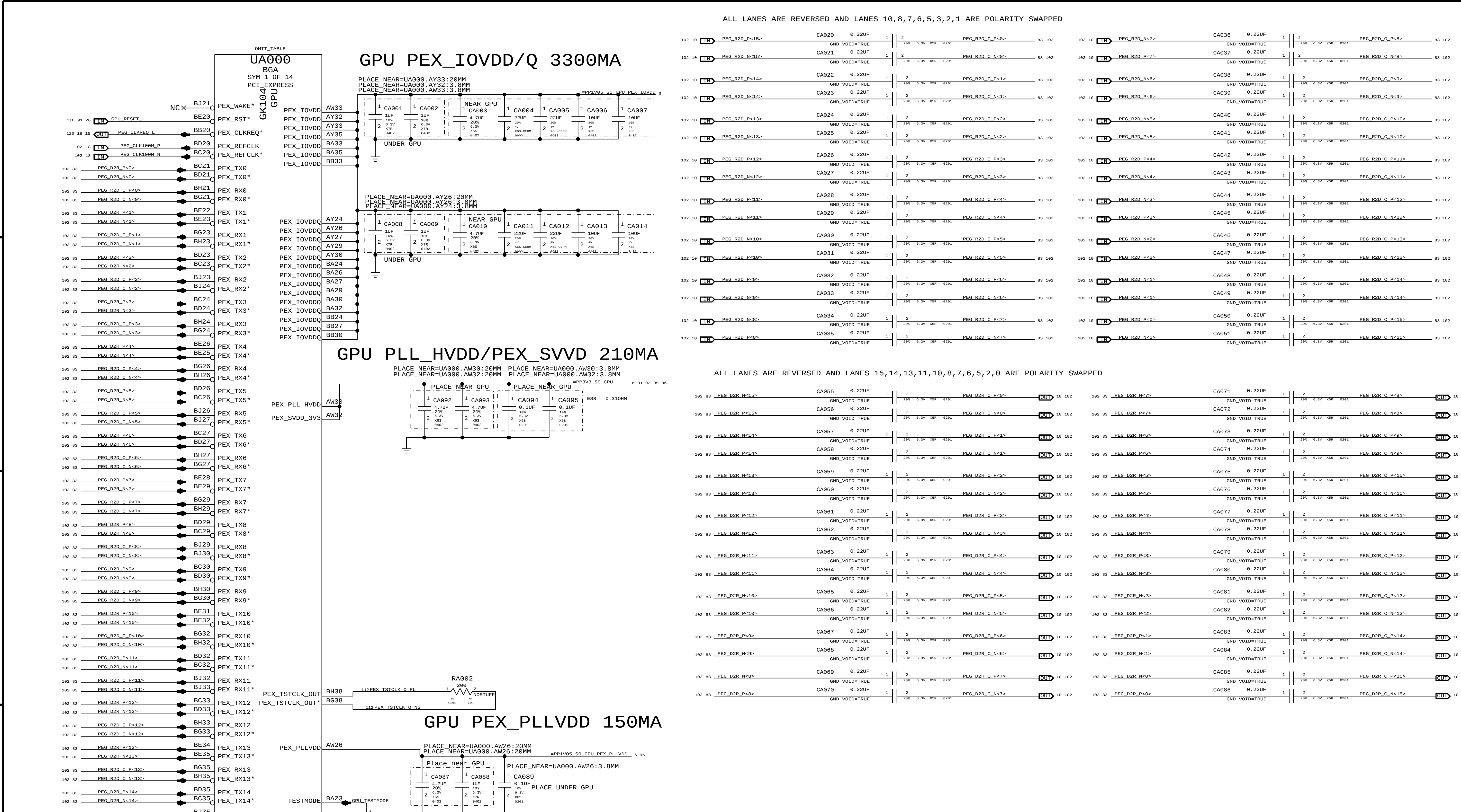
DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

PAGE 99 OF 144

SHEET 82 OF 123



Page Notes

Power aliases required by this page:

- =PP1V05_S0_GPU_PEX_IOVDD
- =PP3V3_S0_GPU
- =PP1V05_S0_GPU_PEX_PLLVDD

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

(NONE)

SYNC_MASTER=D8_YAN SYNC_DATE=04/09/2012

PAGE TITLE

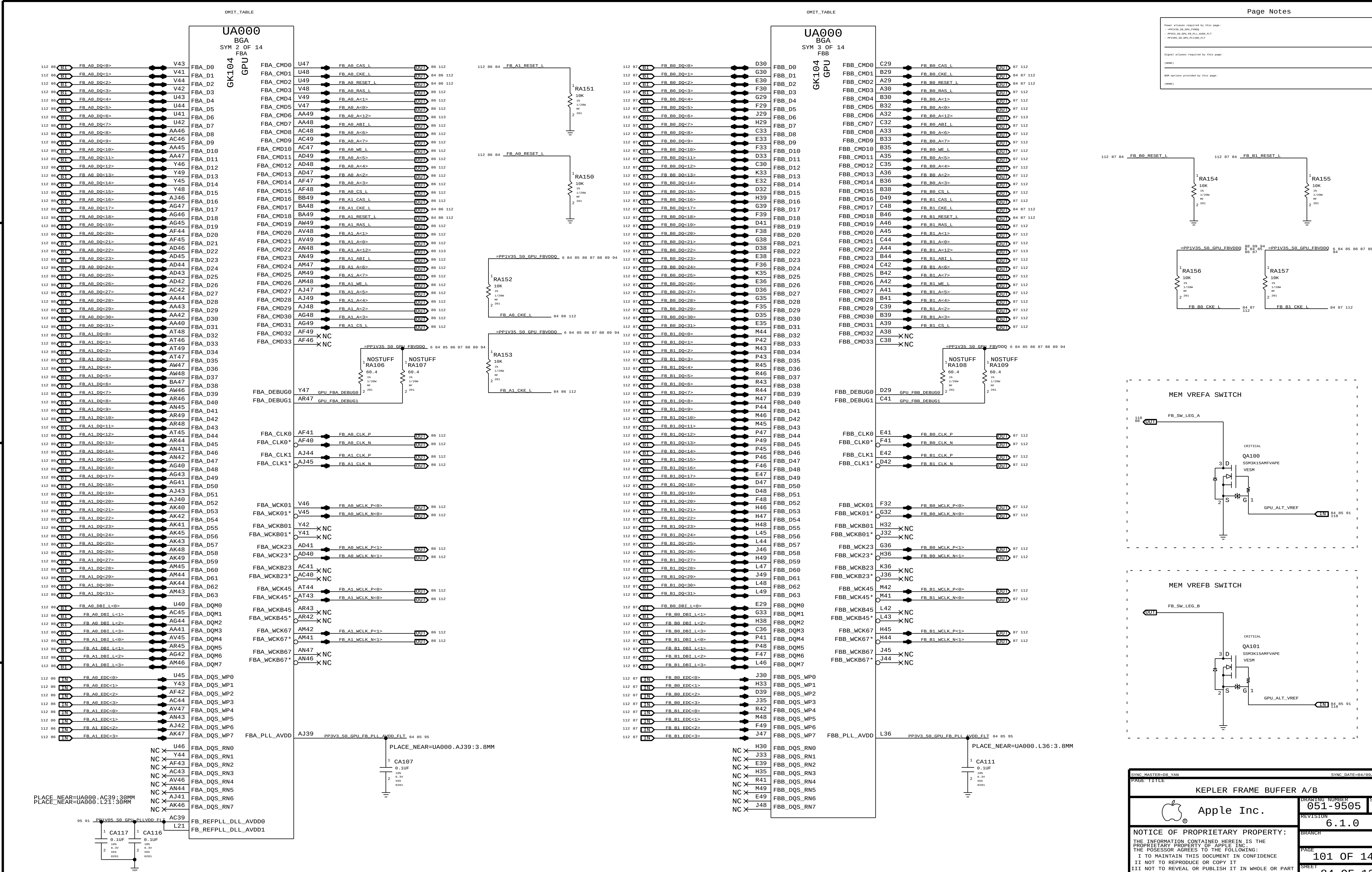
KEPLER PCI-E

Apple Inc.

DRAWING NUMBER	SIZE
051-9505	D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_YAN SYNC_DATE=04/09/2012

PAGE TITLE

KEPLER FRAME BUFFER A/B

Apple Inc.

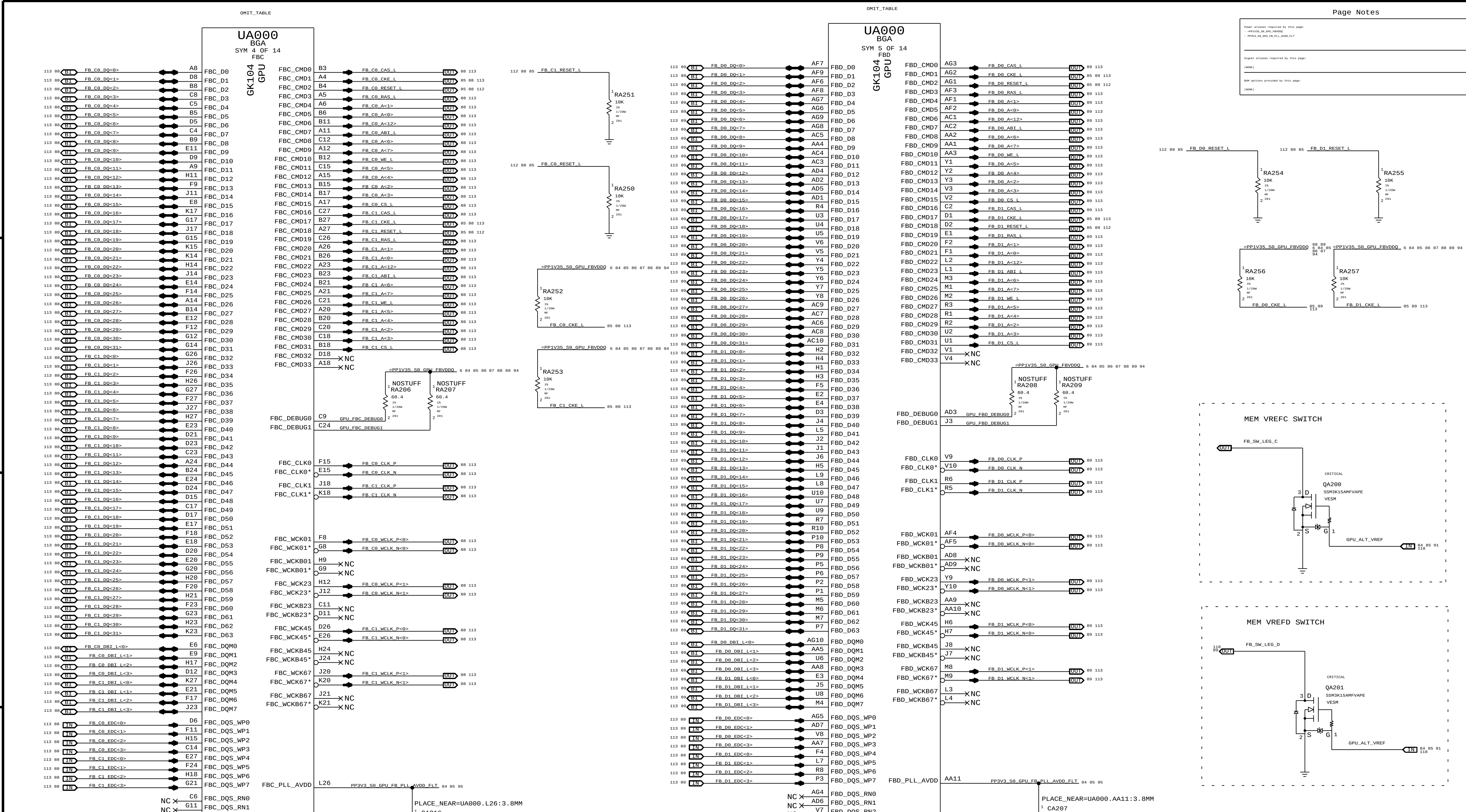
DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

PAGE 101 OF 144

SHEET 84 OF 123



SYNC_MASTER=D8_VAN SYNC_DATE=04/09/2012

PAGE TITLE

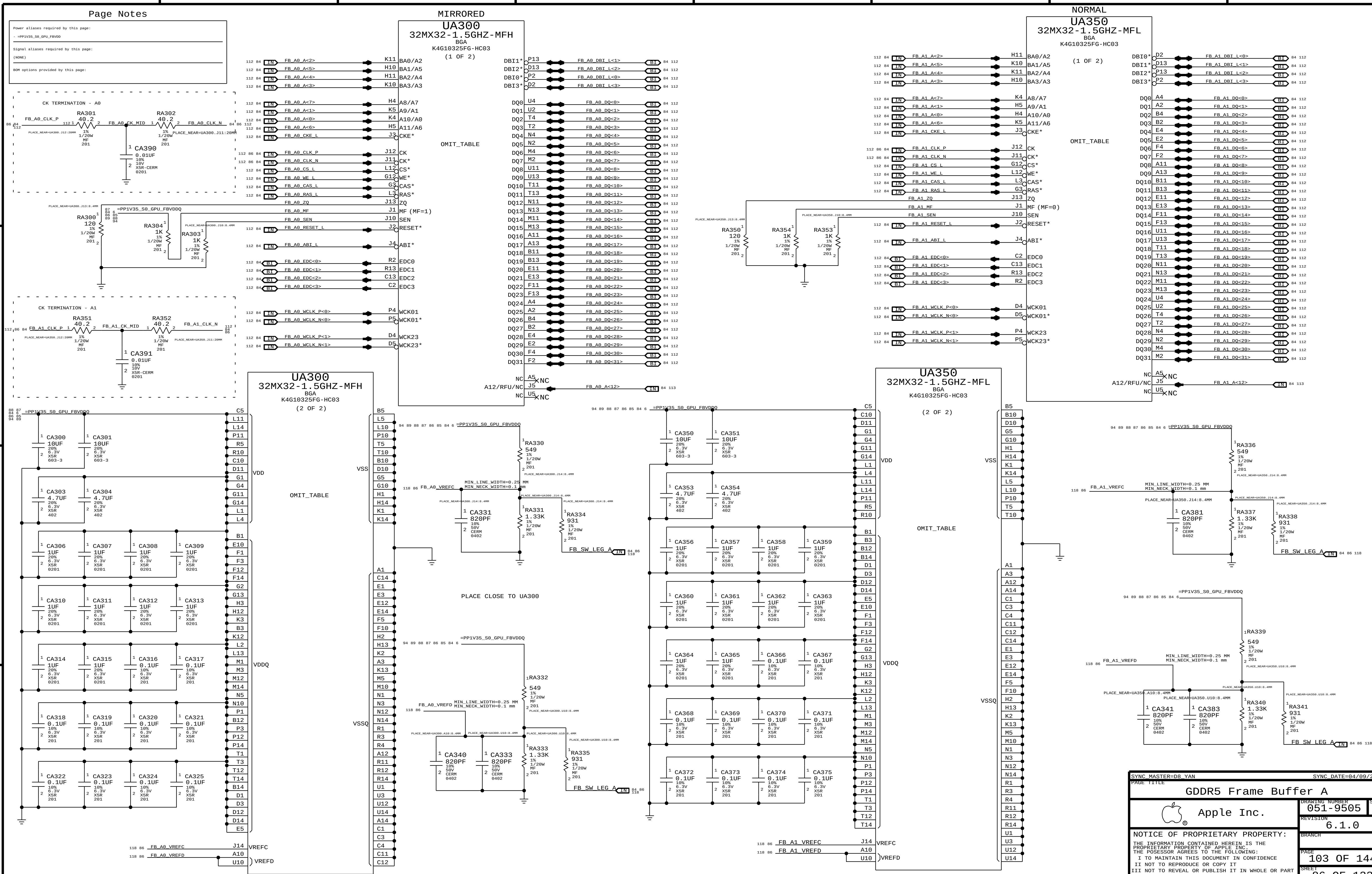
KEPLER FRAME BUFFER C/D

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_YAN SYNC_DATE=04/09/2012

PAGE TITLE

GDDR5 Frame Buffer A

Apple Inc.

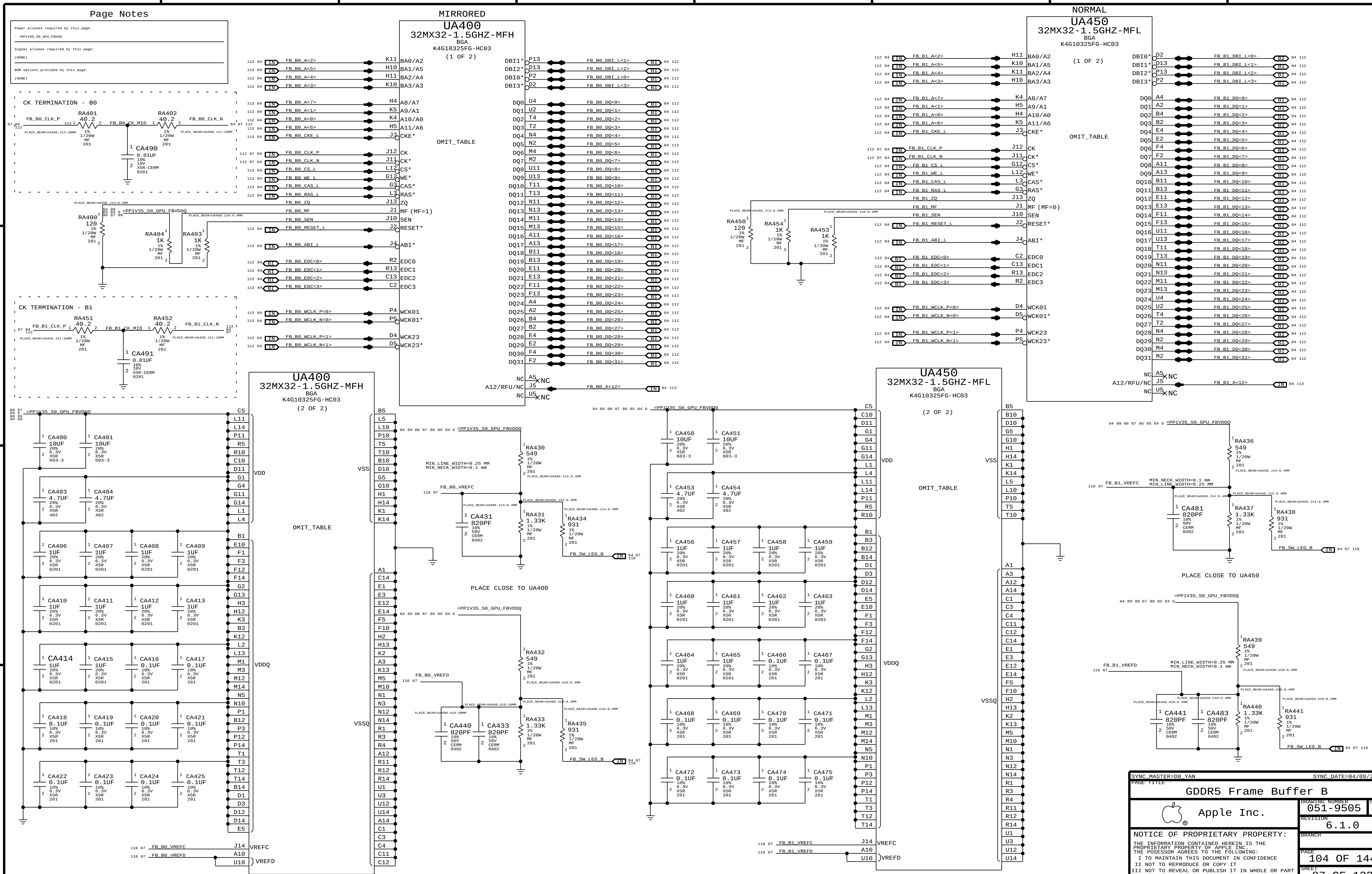
DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

PAGE 103 OF 144

SHEET 86 OF 123



SYNC_MASTER=D8_YAN SYNC_DATE=04/09/2012

PAGE TITLE

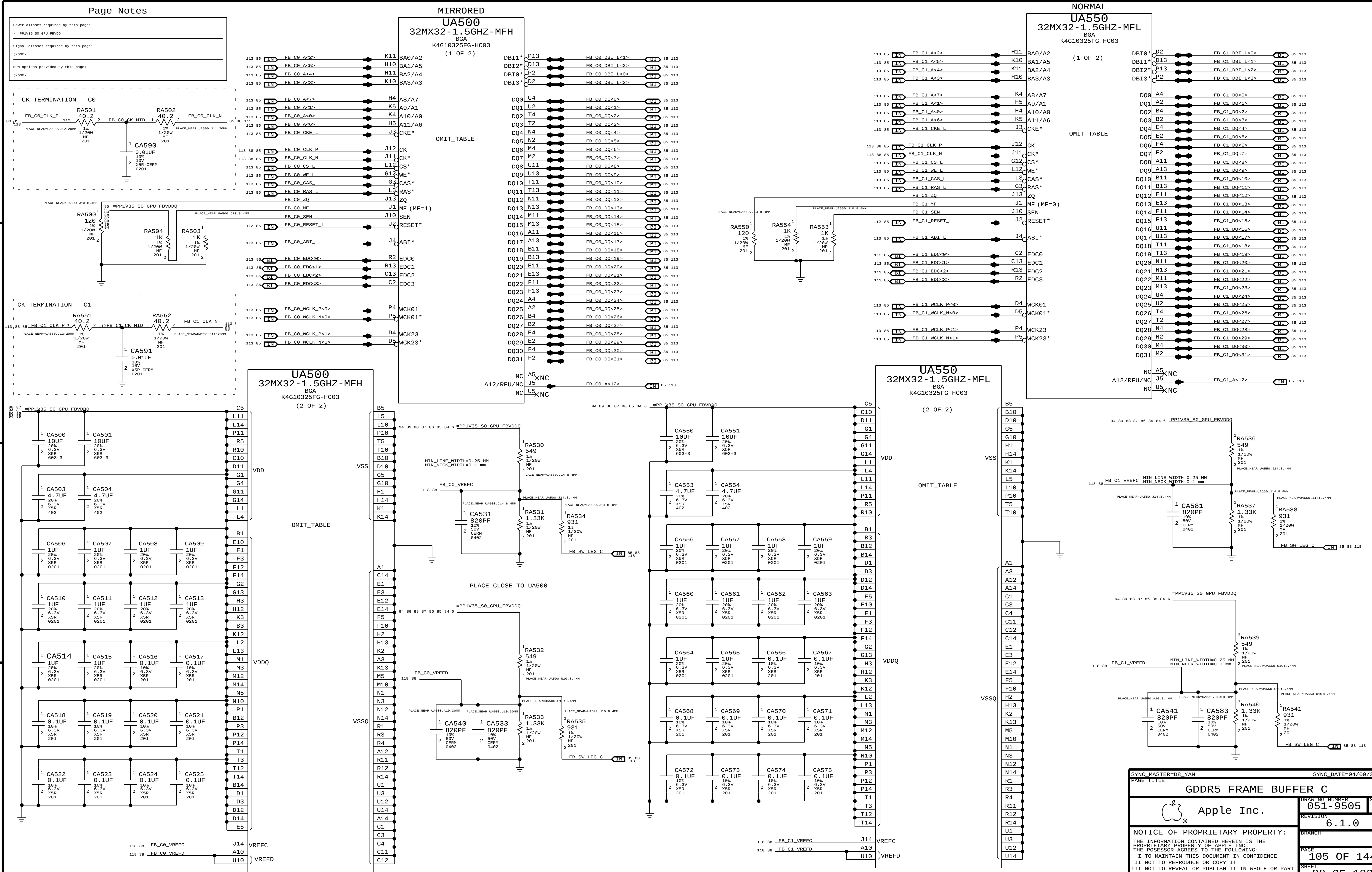
GDDR5 Frame Buffer B

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_YAN SYNC_DATE=04/09/2012

PAGE TITLE

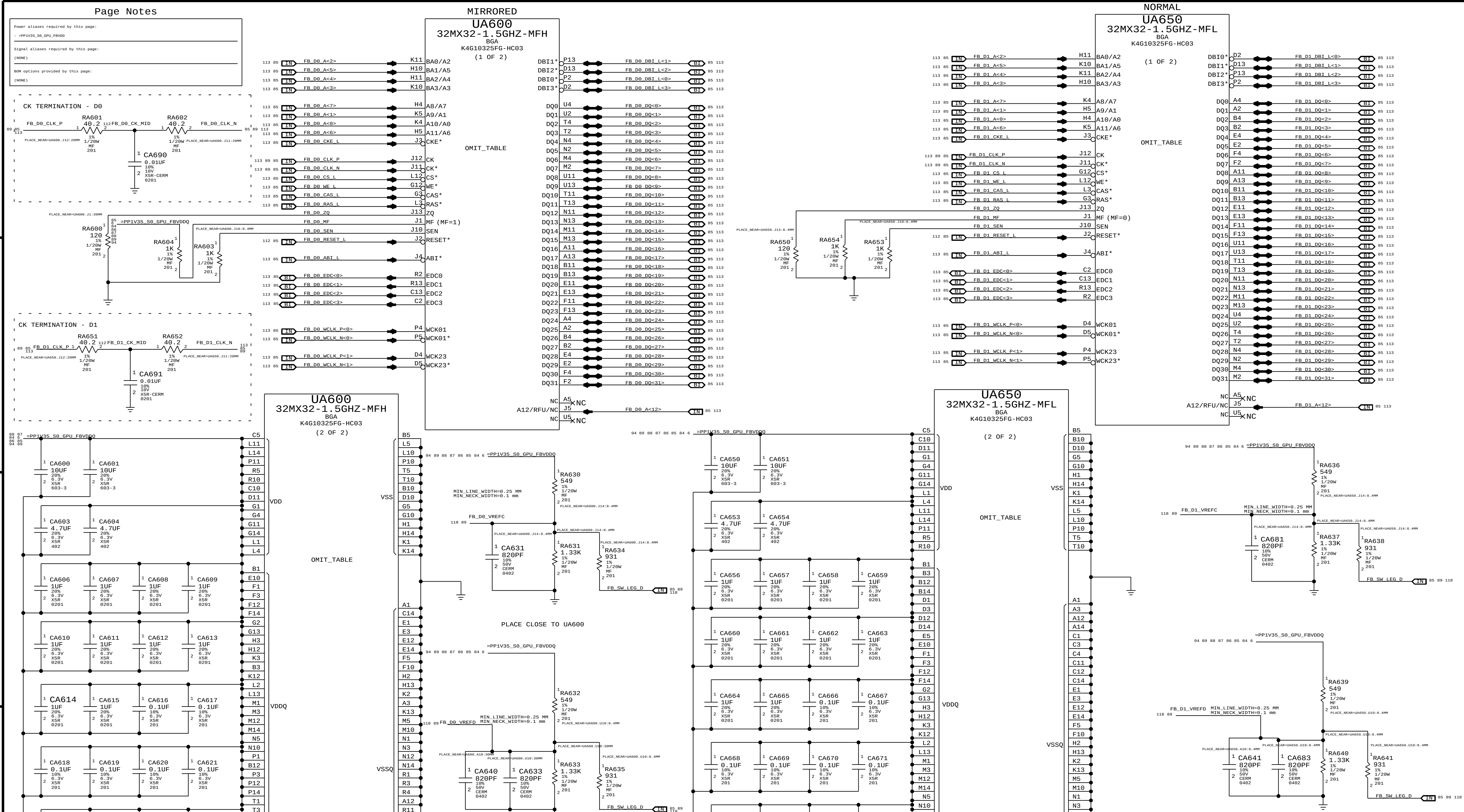
GDDR5 FRAME BUFFER C

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_YAN SYNC_DATE=04/09/2012

PAGE TITLE

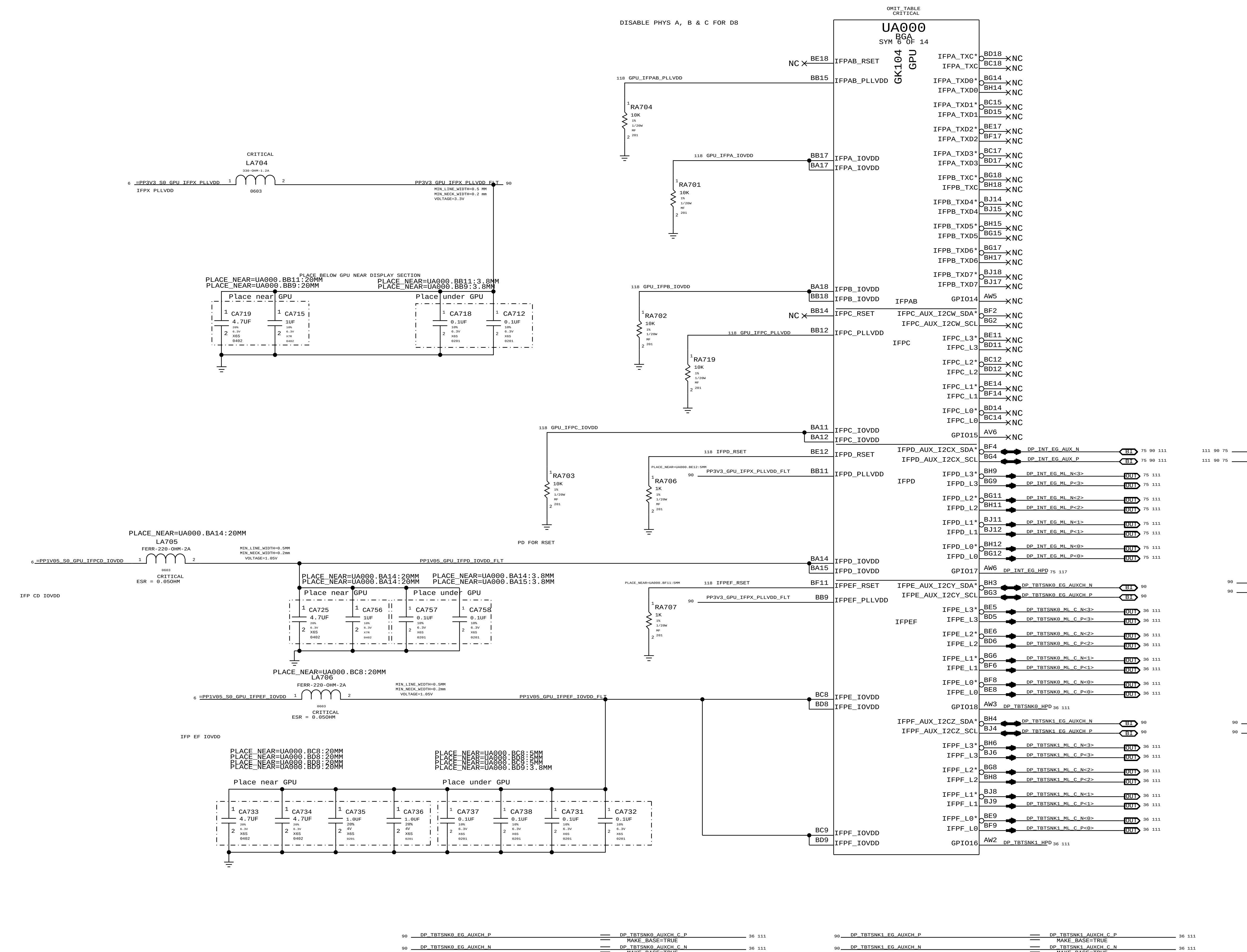
GDDR5 FRAME BUFFER D

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

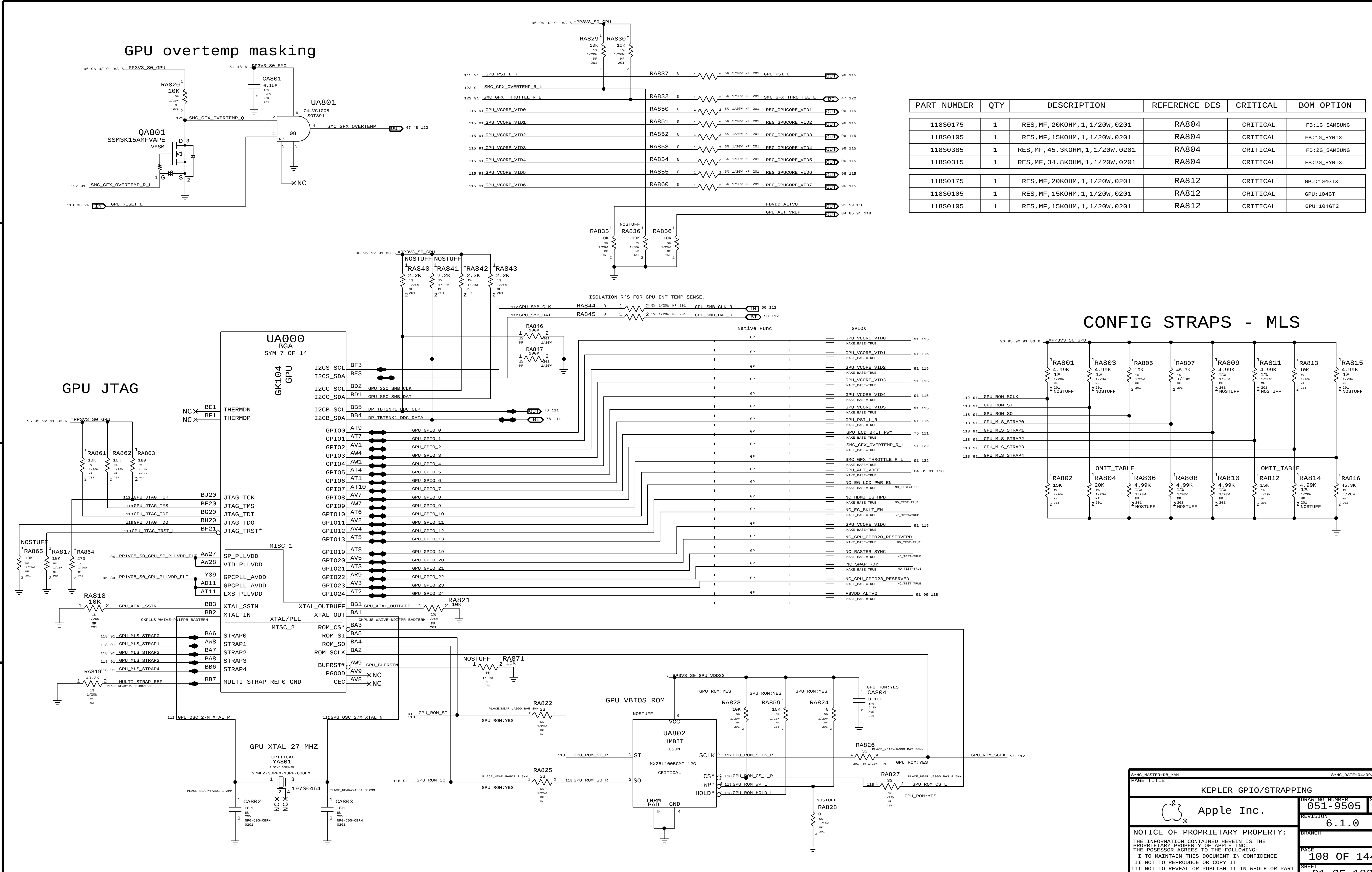


SYNC_MASTER=D8_YAN
SYNC_DATE=04/09/2012

PAGE TITLE
KEPLER EDP/DP/GPIO

Apple Inc.

DRAWING NUMBER 051-9505
SIZE D
REVISION 6.1.0
BRANCH
PAGE 107 OF 144
SHEET 90 OF 123



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0175	1	RES,MF,20KOHM,1,1/20W,0201	RA804	CRITICAL	FB:1G_SAMSUNG
118S0105	1	RES,MF,15KOHM,1,1/20W,0201	RA804	CRITICAL	FB:1G_HYNIX
118S0385	1	RES,MF,45.3KOHM,1,1/20W,0201	RA804	CRITICAL	FB:2G_SAMSUNG
118S0315	1	RES,MF,34.8KOHM,1,1/20W,0201	RA804	CRITICAL	FB:2G_HYNIX
118S0175	1	RES,MF,20KOHM,1,1/20W,0201	RA812	CRITICAL	GPU:104GTX
118S0105	1	RES,MF,15KOHM,1,1/20W,0201	RA812	CRITICAL	GPU:104GT
118S0105	1	RES,MF,15KOHM,1,1/20W,0201	RA812	CRITICAL	GPU:104GT2

SYNC_MASTER=D8_YAN SYNC_DATE=04/09/2012

PAGE TITLE

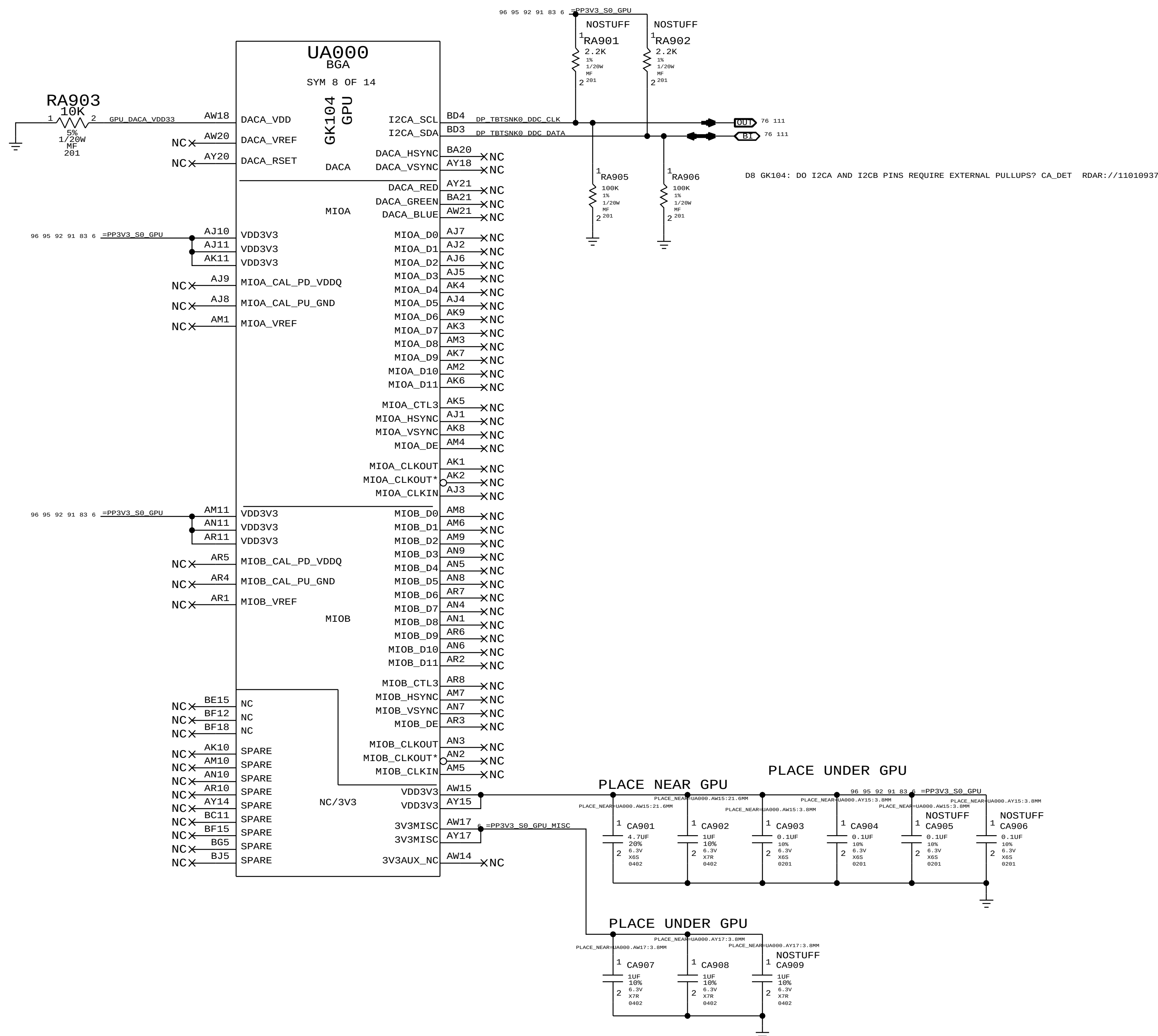
KEPLER GPIO/STRAPPING

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH



SYNC_MASTER=D8_YAN SYNC_DATE=04/09/2012

PAGE TITLE

KEPLER MISC

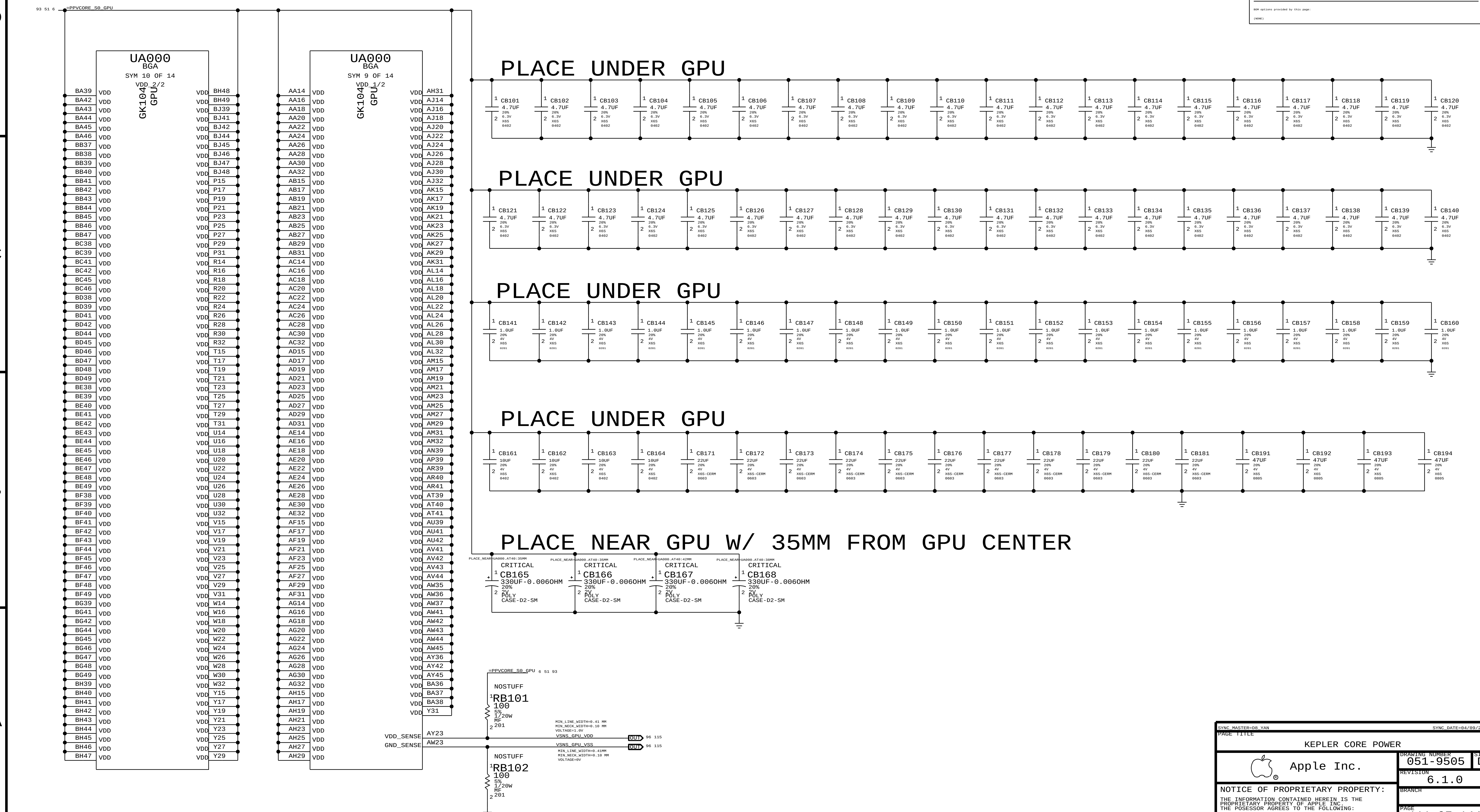
Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

GPU NVVDD DECOUPLING I(EDP)=95A



Page Notes

Power aliases required by this page:
- =PPVCORE_S0_GPU

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

SYNC_MASTER=D8_YAN SYNC_DATE=04/09/2012

PAGE TITLE
KEPLER CORE POWER

Apple Inc.

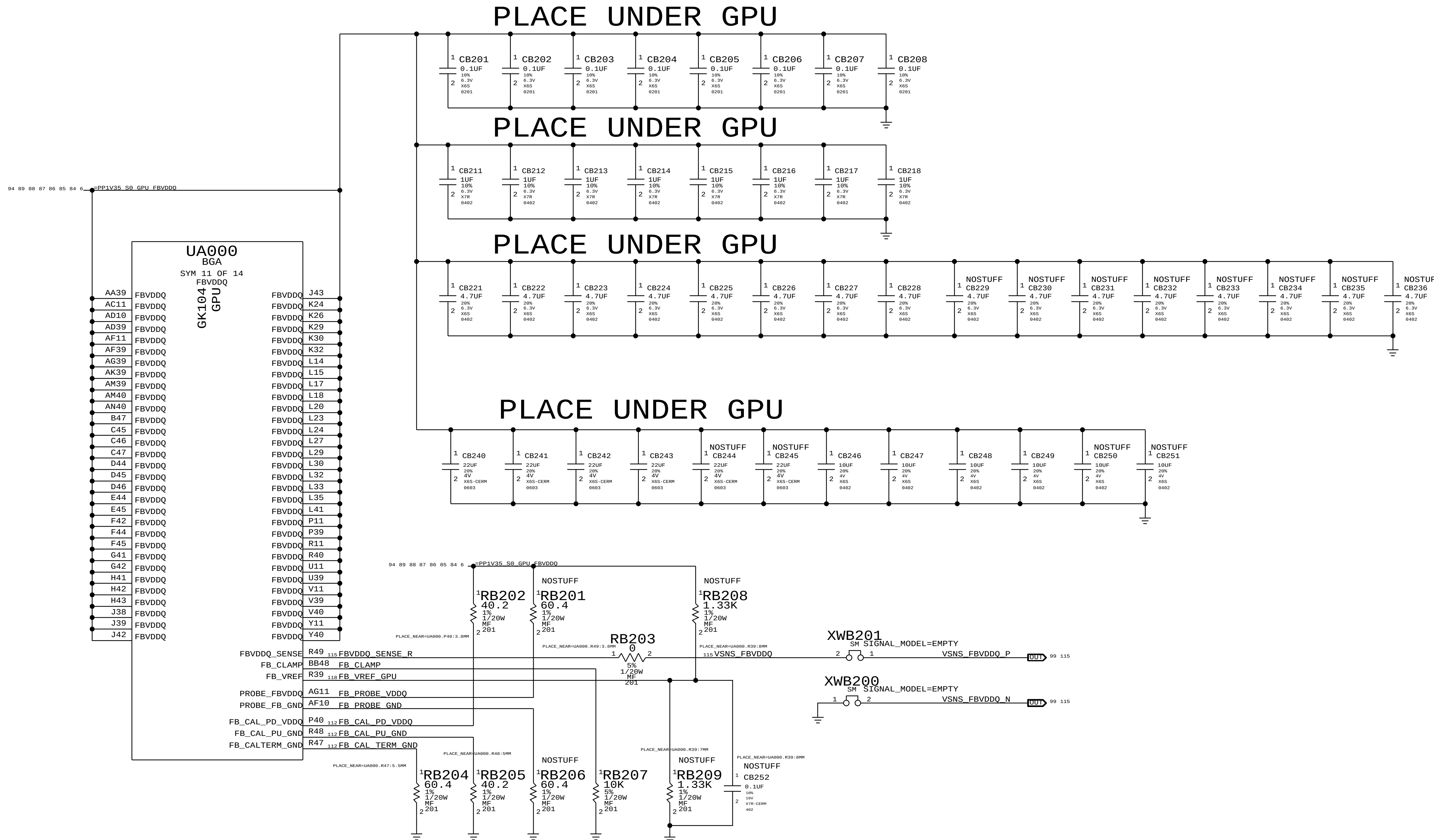
DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

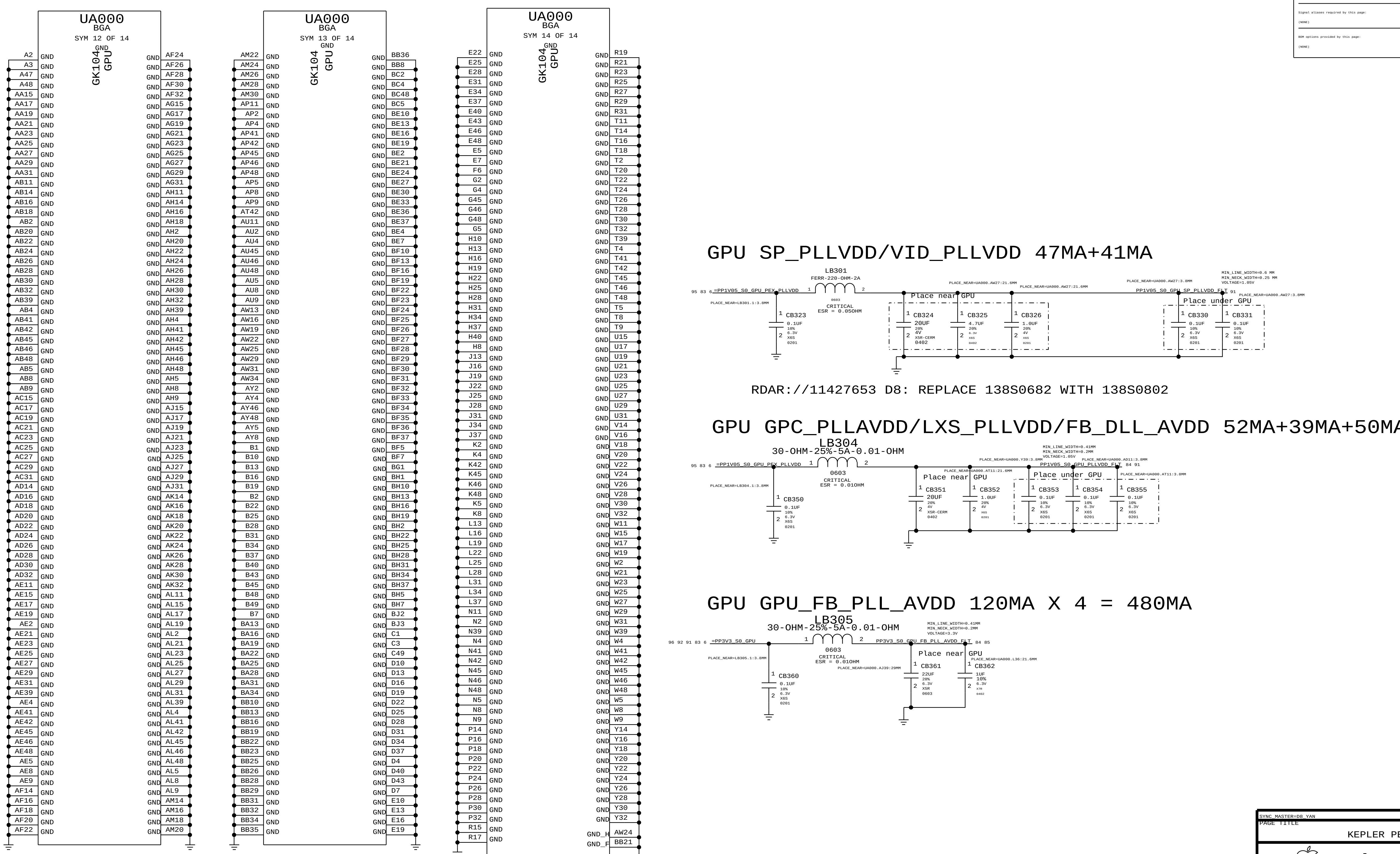
GPU FBVDD/Q DECOUPLING I(EDP PEAK)=26A

Page Notes



SYNC_MASTER=D8_YAN	SYNC_DATE=04/09/2012
PAGE TITLE	KEPLER FBVDD/Q POWER
Apple Inc.	DRAWING NUMBER 051-9505 SIZE D
	REVISION 6.1.0
	BRANCH
	PAGE 112 OF 144
	SHEET 94 OF 123

Page Notes



SYNC_MASTER=D8_YAN SYNC_DATE=04/09/2012

PAGE TITLE

KEPLER PEX PWR/GNDS

Apple Inc.

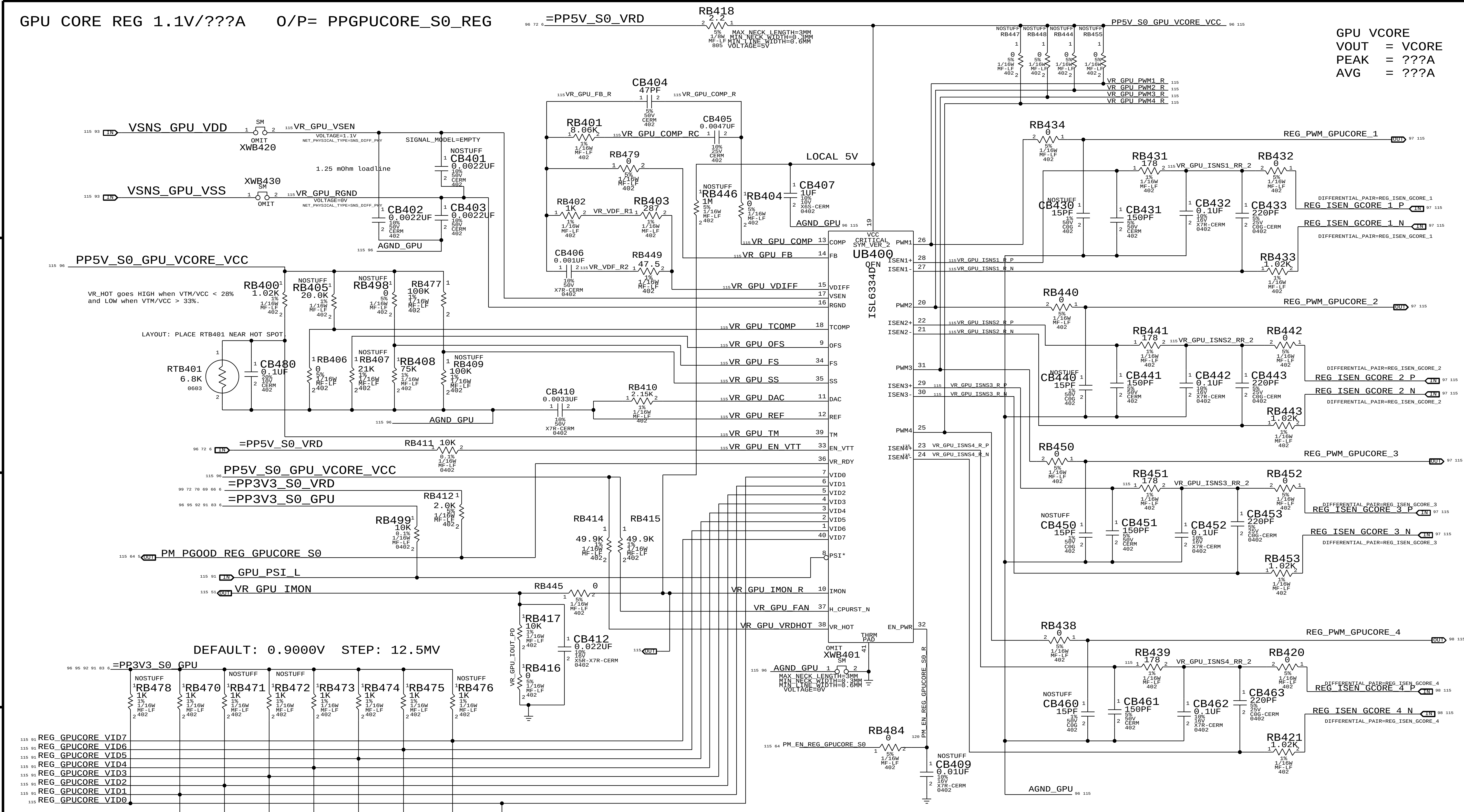
DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

PAGE 113 OF 144

SHEET 95 OF 123



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

VReg GPU Core Phases

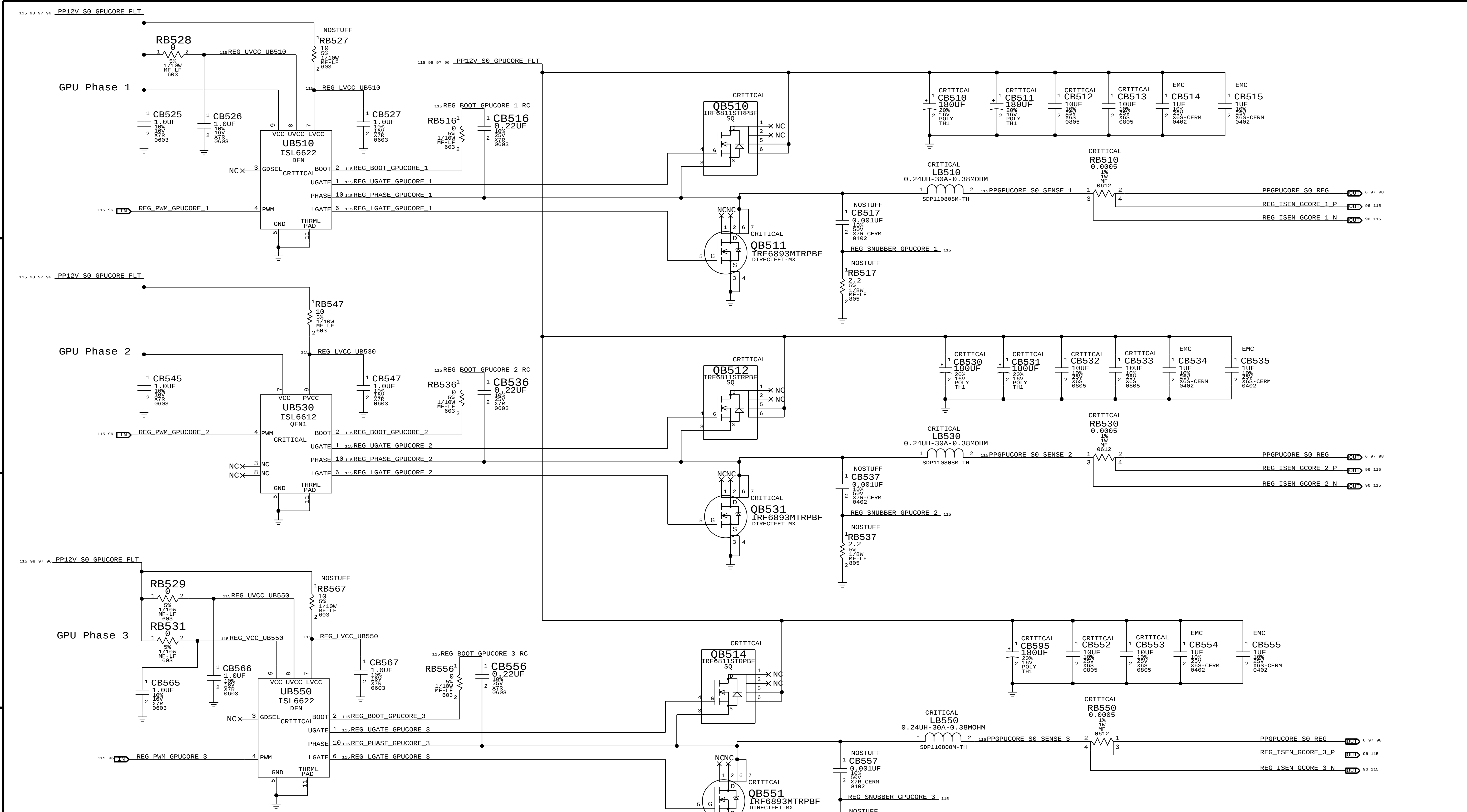
Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

VReg GPU Core Phases

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

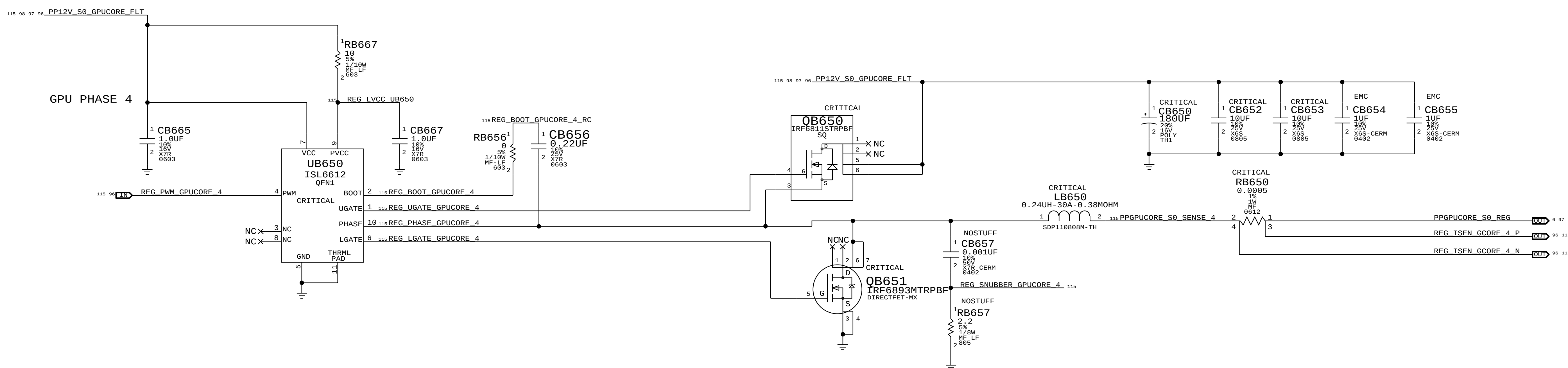
BRANCH

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.
THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

PAGE 115 OF 144

SHEET 97 OF 123



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE
VREG GPU CORE PHASE 4

Apple Inc.

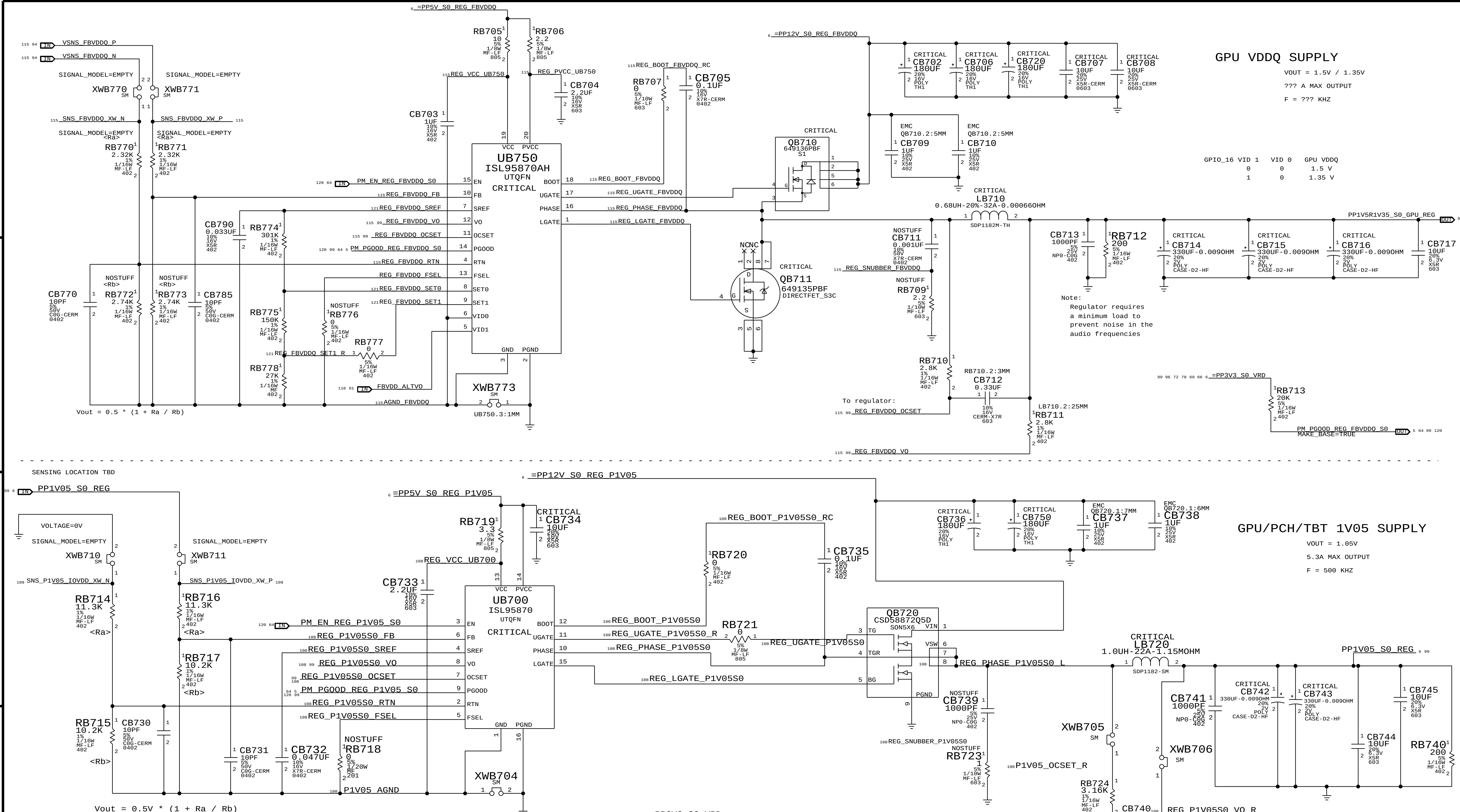
DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

PAGE 116 OF 144

SHEET 98 OF 123



SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE

GPU VDDQ AND 1V05 GPU/PCH/TBT VREGS

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

D8 BOARD SPECIFIC PHYSICAL AND SPACING CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,ISL8,ISL9,ISL10,ISL11,BOTTOM	NO_TYPE,BGA,BGA_TBT	MM	16.2

General Physical Rule Definitions

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	0.1 MM	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	*	Y	0.185 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	ISL5,ISL8	Y	0.205 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	TOP,BOTTOM	Y	0.220 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	*	Y	0.150 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	ISL5,ISL8	Y	0.165 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	TOP,BOTTOM	Y	0.175 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	*	Y	0.130 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	ISL5,ISL8	Y	0.145 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	TOP,BOTTOM	Y	0.155 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	*	Y	0.115 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	ISL5,ISL8	Y	0.126 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	TOP,BOTTOM	Y	0.135 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	*	Y	0.090 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	ISL5,ISL8	Y	0.100 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	TOP,BOTTOM	Y	0.105 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	*	Y	0.075 MM	0.075 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	ISL5,ISL8	Y	0.080 MM	0.080 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	TOP,BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	Y	0.171 MM	0.085 MM	=STANDARD	0.130 MM	0.1 MM
68_OHM_DIFF	TOP,BOTTOM	Y	0.185 MM	0.085 MM	=STANDARD	0.150 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	0.136 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.141 MM	0.085 MM	=STANDARD	0.185 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	Y	0.121 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	0.109 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.111 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	0.086 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.090 MM	0.085 MM	=STANDARD	0.230 MM	0.1 MM

Compensation Physical Rule Definition

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
COMP_SE	*	Y	0.305 MM	0.105 MM	3 MM	=STANDARD	=STANDARD

NOTE: line width based on 12 mil recommendation
NOTE: neck width based on 4 mil recommendation

BGA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=STANDARD	?

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_ISO	*	=STANDARD	8000
GND_P2MM	*	=2:1_SPACING	1000
PWR_P2MM	*	=2:1_SPACING	1100

GENERIC

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GENERIC_ISO	*	=1:1_SPACING	?
PM_ISO	*	=1:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PM	*	*	PM_ISO
PM	GND	*	DEFAULT

BGA Area Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1MM

General Spacing Definitions

Default

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

Fixed and Dielectric

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?
1X_DIELECTRIC	TOP,BOTTOM	0.071 MM	?
1X_DIELECTRIC	ISL3,ISL10	0.101 MM	?
1X_DIELECTRIC	*	0.076 MM	?

Board Stack-up

FINISHED BOARD THICKNESS: 1.94 MM

Top	Signal	0.5 oz (Cu plated)
	Prepreg	0.071 MM
2	Plane	1 oz
	Core	0.101 MM
3	Signal	0.5 oz
	Prepreg	0.115 MM
4	Plane	1 oz
	Core	0.076 MM
5	Signal	0.5 oz
	Prepreg	0.380 MM
6	Plane	1 oz
	Core	0.076 MM
7	Plane	1 oz
	Prepreg	0.380 MM
8	Signal	0.5 oz
	Core	0.076 MM
9	Plane	1 oz
	Prepreg	0.115 MM
10	Signal	0.5 oz
	Core	0.101 MM
11	Plane	1 oz
	Prepreg	0.071 MM
Btm	Signal	0.5 oz (Cu plated)

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE
D8 RULE DEFINITIONS

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

DDR3

DDR3-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DDR_34S	*	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=STANDARD	=STANDARD
DDR_34S	BOTTOM	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	2.0 MM	=STANDARD	=STANDARD
DDR_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
DDR_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
DDR_42S	BOTTOM	=42_OHM_SE	=55_OHM_SE	=55_OHM_SE	2.0 MM	=STANDARD	=STANDARD
DDR_42S	ISL5,ISL8	=42_OHM_SE	=55_OHM_SE	=55_OHM_SE	2.0 MM	=STANDARD	=STANDARD
DDR_42S_D	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	0.1016 MM	0.1016 MM
DDR_42S_D	BOTTOM	=42_OHM_SE	=55_OHM_SE	=55_OHM_SE	2.0 MM	0.1016 MM	0.1016 MM
DDR_42S_D	ISL5,ISL8	=42_OHM_SE	=55_OHM_SE	=55_OHM_SE	2.0 MM	0.1016 MM	0.1016 MM
DDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
DDR_68D	*	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF

Minimum diff spacing is 4 mil
Table 3-5, Intel Doc# 473718

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
POWER_DDR_P4MM	*	Y	0.400 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_DDR_PHY	*	POWER_DDR_P4MM
DDR_CLK_PHY	*	DDR_68D
DDR_CTRL_PHY	*	DDR_39S
DDR_CMD_PHY	*	DDR_34S
DDR_DQ_PHY	*	DDR_42S
DDR_DQS_PHY	*	DDR_42S_D

DDR3-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DDR_CLK_ISO	TOP,BOTTOM	=5.5X_DIELECTRIC	?
DDR_CLK_ISO	ISL3,ISL10	=4.9X_DIELECTRIC	?
DDR_CLK_ISO	*	=6.5X_DIELECTRIC	?
DDR_CTRL_ISO	TOP,BOTTOM	=4.5X_DIELECTRIC	?
DDR_CTRL_ISO	ISL3,ISL10	=4.0X_DIELECTRIC	?
DDR_CTRL_ISO	*	=5.3X_DIELECTRIC	?
DDR_CTRL2CTRL	TOP,BOTTOM	=3.0X_DIELECTRIC	?
DDR_CTRL2CTRL	ISL3,ISL10	=2.6X_DIELECTRIC	?
DDR_CTRL2CTRL	*	=3.5X_DIELECTRIC	?
DDR_CMD_ISO	TOP,BOTTOM	=4.5X_DIELECTRIC	?
DDR_CMD_ISO	ISL3,ISL10	=4.0X_DIELECTRIC	?
DDR_CMD_ISO	*	=5.3X_DIELECTRIC	?
DDR_CMD2CMD	TOP,BOTTOM	=2.3X_DIELECTRIC	?
DDR_CMD2CMD	ISL3,ISL10	=2.0X_DIELECTRIC	?
DDR_CMD2CMD	*	=2.7X_DIELECTRIC	?
DDR_DATA_ISO	TOP,BOTTOM	=4.5X_DIELECTRIC	?
DDR_DATA_ISO	ISL3,ISL10	=4.0X_DIELECTRIC	?
DDR_DATA_ISO	*	=5.3X_DIELECTRIC	?
DDR_DQ2DQ	TOP,BOTTOM	=3.2X_DIELECTRIC	900
DDR_DQ2DQ	ISL3,ISL10	=2.8X_DIELECTRIC	900
DDR_DQ2DQ	*	=3.8X_DIELECTRIC	900
DDR_DQ2DQS	TOP,BOTTOM	=3.7X_DIELECTRIC	?
DDR_DQ2DQS	ISL3,ISL10	=3.3X_DIELECTRIC	?
DDR_DQ2DQS	*	=4.4X_DIELECTRIC	?
DDR_BL2BL	TOP,BOTTOM	=4.5X_DIELECTRIC	?
DDR_BL2BL	ISL3,ISL10	=4.0X_DIELECTRIC	?
DDR_BL2BL	*	=5.3X_DIELECTRIC	?
DDR_CH2CH	TOP,BOTTOM	=9.1X_DIELECTRIC	?
DDR_CH2CH	ISL3,ISL10	=8.2X_DIELECTRIC	?
DDR_CH2CH	*	=10.9X_DIELECTRIC	?
CMD2DATA_ISO	TOP,BOTTOM	=7X_DIELECTRIC	?
CMD2DATA_ISO	ISL3,ISL10	=5X_DIELECTRIC	?
CMD2DATA_ISO	*	=5X_DIELECTRIC	?

DDR3 Power-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_DDR_ISO	*	=4.3X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER_DDR	*	*	POWER_DDR_ISO

Constraints

Clocks: CK[3:0], CK#[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CLK	*	*	DDR_CLK_ISO

Control: CS#[3:0], CKE[3:0], ODT[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CTRL	*	*	DDR_CTRL_ISO
DDR_CTRL	DDR_CTRL	*	DDR_CTRL2CTRL

Command: MA[15:0], RAS#, CAS#, WE# BS[2:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CMD	*	*	DDR_CMD_ISO
DDR_CMD	DDR_CMD	*	DDR_CMD2CMD
DDR_CMD	DDR_A_DQ_BYTE*	*	CMD2DATA_ISO
DDR_CMD	DDR_A_DQS*	*	CMD2DATA_ISO
DDR_CMD	DDR_B_DQ_BYTE*	*	CMD2DATA_ISO
DDR_CMD	DDR_B_DQS*	*	CMD2DATA_ISO

Data: DQS[7:0], DQS#[7:0], DQ[63:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	
DDR_A_DQ_BYTE*	*	*	DDR_DATA_ISO	
DDR_A_DQS*	*	*	DDR_DATA_ISO	
DDR_B_DQ_BYTE*	*	*	DDR_DATA_ISO	
DDR_B_DQS*	*	*	DDR_DATA_ISO	
DDR_*_DQ_BYTE*	=SAME	*	DDR_DQ2DQ	See Note (3)
DDR_A_DQ_BYTE*	DDR_A_DQS*	*	DDR_DQ2DQS	See Note (1)
DDR_A_DQ_BYTE*	DDR_A_DQ_BYTE*	*	DDR_BL2BL	See Note (3)
DDR_B_DQ_BYTE*	DDR_B_DQS*	*	DDR_DQ2DQS	See Note (1)
DDR_B_DQ_BYTE*	DDR_B_DQ_BYTE*	*	DDR_BL2BL	
DDR_A_*	DDR_B_*	*	DDR_CH2CH	See Note (2)

Note (1):
Deliberately set DQ to DQS spacing to 3:1 to avoid adding complexity to contraints, even though it can be less. Only one rule per channel is needed by trading off a little space.

Note (2):
Intel suggests 25 mil (0.65 mm) spacing for via to channel, and via to pad to two different channels. DDR3 draws about 20 mA per trace with edge rates in the 100s of ps. The main coupling mechanism is capacitive. A 0.65 mm spacing is used for power nets, which draw far more current (inductive coupling however). These rules are far too conservative. To meet these rules, the spacing must be applied to the net.

Note (3):
In order for the constraints DDR_*_DQ_BYTE* to =SAME to win out over DDR_{A,B}_DQ_BYTE* to DDR_{A,B}_DQ_BYTE* so that the small intra-bytelane spacing is used, the spacing rule DDR_DQ2DQ must have a weight greater than DDR_BL2BL.

DDR3

	Electrical Contraint Set	Physical	Spacing		
Channel A					
I178	DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_P<1..0>	12 29
I179	DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_N<1..0>	12 29
I255	DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_P<3..2>	12 30
I256	DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_N<3..2>	12 30
I180	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CKE<1..0>	12 29
I181	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CS_L<1..0>	12 29
I182	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_ODT<1..0>	12 29
I247	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CKE<3..2>	12 30
I248	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CS_L<3..2>	12 30
I249	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_ODT<3..2>	12 30
I183	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_A<15..0>	12 29 30
I184	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_BA<2..0>	12 29 30
I185	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_RAS_L	12 29 30
I186	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_CAS_L	12 29 30
I187	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_WE_L	12 29 30
I188	DDR_A_DQ_BYTE0	DDR_DQ_PHY	DDR_A_DQ_BYTE0	MEM_A_DQ<7..0>	12 33
I189	DDR_A_DQ_BYTE1	DDR_DQ_PHY	DDR_A_DQ_BYTE1	MEM_A_DQ<15..8>	12 33
I191	DDR_A_DQ_BYTE2	DDR_DQ_PHY	DDR_A_DQ_BYTE2	MEM_A_DQ<23..16>	12 33
I190	DDR_A_DQ_BYTE3	DDR_DQ_PHY	DDR_A_DQ_BYTE3	MEM_A_DQ<31..24>	12 33
I192	DDR_A_DQ_BYTE4	DDR_DQ_PHY	DDR_A_DQ_BYTE4	MEM_A_DQ<39..32>	12 33
I193	DDR_A_DQ_BYTE5	DDR_DQ_PHY	DDR_A_DQ_BYTE5	MEM_A_DQ<47..40>	12 33
I194	DDR_A_DQ_BYTE6	DDR_DQ_PHY	DDR_A_DQ_BYTE6	MEM_A_DQ<55..48>	12 33
I195	DDR_A_DQ_BYTE7	DDR_DQ_PHY	DDR_A_DQ_BYTE7	MEM_A_DQ<63..56>	12 33
I196	DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS_P<0>	12 33
I197	DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS_N<0>	12 33
I198	DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS_P<1>	12 33
I199	DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS_N<1>	12 33
I200	DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS_P<2>	12 33
I201	DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS_N<2>	12 33
I203	DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS_P<3>	12 33
I202	DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS_N<3>	12 33
I204	DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS_P<4>	12 33
I205	DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS_N<4>	12 33
I206	DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS_P<5>	12 33
I207	DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS_N<5>	12 33
I208	DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS_P<6>	12 33
I209	DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS_N<6>	12 33
I210	DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS_P<7>	12 33
I211	DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS_N<7>	12 33
Channel B					
I212	DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_P<1..0>	12 31
I213	DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_N<1..0>	12 31
I254	DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_P<3..2>	12 32
I253	DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_N<3..2>	12 32
I214	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CKE<1..0>	12 31
I215	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CS_L<1..0>	12 31
I216	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_ODT<1..0>	12 31
I250	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CKE<3..2>	12 32
I251	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CS_L<3..2>	12 32
I252	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_ODT<3..2>	12 32
I217	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_A<15..0>	12 31 32
I218	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_BA<2..0>	12 31 32
I219	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_RAS_L	12 31 32
I220	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_CAS_L	12 31 32
I221	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_WE_L	12 31 32
I222	DDR_B_DQ_BYTE0	DDR_DQ_PHY	DDR_B_DQ_BYTE0	MEM_B_DQ<7..0>	12 33
I223	DDR_B_DQ_BYTE1	DDR_DQ_PHY	DDR_B_DQ_BYTE1	MEM_B_DQ<15..8>	12 33
I225	DDR_B_DQ_BYTE2	DDR_DQ_PHY	DDR_B_DQ_BYTE2	MEM_B_DQ<23..16>	12 33
I224	DDR_B_DQ_BYTE3	DDR_DQ_PHY	DDR_B_DQ_BYTE3	MEM_B_DQ<31..24>	12 33
I226	DDR_B_DQ_BYTE4	DDR_DQ_PHY	DDR_B_DQ_BYTE4	MEM_B_DQ<39..32>	12 33
I227	DDR_B_DQ_BYTE5	DDR_DQ_PHY	DDR_B_DQ_BYTE5	MEM_B_DQ<47..40>	12 33
I228	DDR_B_DQ_BYTE6	DDR_DQ_PHY	DDR_B_DQ_BYTE6	MEM_B_DQ<55..48>	12 33
I229	DDR_B_DQ_BYTE7	DDR_DQ_PHY	DDR_B_DQ_BYTE7	MEM_B_DQ<63..56>	12 33
I230	DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS_P<0>	12 33
I231	DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS_N<0>	12 33
I232	DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS_P<1>	12 33
I233	DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS_N<1>	12 33
I234	DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS_P<2>	12 33
I235	DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS_N<2>	12 33
I237	DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS_P<3>	12 33
I236	DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS_N<3>	12 33
I238	DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS_P<4>	12 33
I239	DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS_N<4>	12 33
I240	DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS_P<5>	12 33
I241	DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS_N<5>	12 33
I242	DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS_P<6>	12 33
I243	DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS_N<6>	12 33
I244	DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS_P<7>	12 33
I245	DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS_N<7>	12 33
Reset					
I246		DDR_50S	CPU	MEM_RESET_L	28 29 30 31 32

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE
DDR3 Constraints

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

PCI EXPRESS

PCIe-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE_PHY	*	PCIE_80D
CLK_PCIE_PHY	*	PCIE_90D
COMP_PCIE_PHY	*	COMP_SE

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_ISO	TOP,BOTTOM	=5X_DIELECTRIC	?
PCIE_ISO	*	=4X_DIELECTRIC	?
CLK_PCIE_ISO	*	=5:1_SPACING	?
COMP_PCIE_ISO	*	=4:1_SPACING	?

PCIe Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
4.2.1	80	85	16	15.75	PCIe Gen2, PCIe Gen3, DMI

PCIe and DMI Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
4-5	50	50	15	15.75	PCIe. Impedance inferred from Table 4-7.
4-7	50	50	8	15.75	DMI. Numbers based on Intel stack-up.

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE	*	*	PCIE_ISO
CLK_PCIE	*	*	CLK_PCIE_ISO
COMP_PCIE	*	*	COMP_PCIE_ISO

PCIe (PCH)

	Electrical Contraint Set	Physical	Spacing	Net		
	x1 AirPort					
I599	PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_R2D_P	NO_TEST=TRUE	35
I600	PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_R2D_N	NO_TEST=TRUE	35
I601		PCIE_PHY	PCIE	PCIE_AP_R2D_C_P	NO_TEST=TRUE	18 35
I603		PCIE_PHY	PCIE	PCIE_AP_R2D_C_N	NO_TEST=TRUE	18 35
I602	PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_D2R_P	NO_TEST=TRUE	18 35
I604	PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_D2R_N	NO_TEST=TRUE	18 35
I607	PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_AP_P		18 35
I608	PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_AP_N		18 35
	x1 Caesar IV					
I610	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE_ENET_R2D_P	NO_TEST=TRUE	39
I609	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE_ENET_R2D_N	NO_TEST=TRUE	39
I611		PCIE_PHY	PCIE	PCIE_ENET_R2D_C_P	NO_TEST=TRUE	18 39
I613		PCIE_PHY	PCIE	PCIE_ENET_R2D_C_N	NO_TEST=TRUE	18 39
I612	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE_ENET_D2R_P	NO_TEST=TRUE	18 39
I614	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE_ENET_D2R_N	NO_TEST=TRUE	18 39
I615		PCIE_PHY	PCIE	PCIE_ENET_D2R_C_P	NO_TEST=TRUE	39
I616		PCIE_PHY	PCIE	PCIE_ENET_D2R_C_N	NO_TEST=TRUE	39
I617	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_ENET_P		18 39
I618	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_ENET_N		18 39

PCIE (PCH - TBT)

	Electrical Contraint Set	Physical	Spacing	Net		
	x4 Thunderbolt					
I571	PCIE_GEN2_R2D_PINV	PCIE_PHY	PCIE	PCIE_TBT_R2D_P<2..0>	NO_TEST=TRUE	36
I572	PCIE_GEN2_R2D_PINV	PCIE_PHY	PCIE	PCIE_TBT_R2D_N<2..0>	NO_TEST=TRUE	36
I620	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE_TBT_R2D_P<3>	NO_TEST=TRUE	36
I619	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE_TBT_R2D_N<3>	NO_TEST=TRUE	36
I573		PCIE_PHY	PCIE	PCIE_TBT_R2D_C_P<3..0>	NO_TEST=TRUE	18 36
I574		PCIE_PHY	PCIE	PCIE_TBT_R2D_C_N<3..0>	NO_TEST=TRUE	18 36
I575		PCIE_PHY	PCIE	PCIE_TBT_D2R_P<3..0>	NO_TEST=TRUE	18 36
I576		PCIE_PHY	PCIE	PCIE_TBT_D2R_N<3..0>	NO_TEST=TRUE	18 36
I577	PCIE_GEN2_D2R_PINV	PCIE_PHY	PCIE	PCIE_TBT_D2R_C_P<1..0>	NO_TEST=TRUE	36
I578	PCIE_GEN2_D2R_PINV	PCIE_PHY	PCIE	PCIE_TBT_D2R_C_N<1..0>	NO_TEST=TRUE	36
I622	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE_TBT_D2R_C_P<2>	NO_TEST=TRUE	36
I621	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE_TBT_D2R_C_N<2>	NO_TEST=TRUE	36
I623	PCIE_GEN2_D2R_PINV	PCIE_PHY	PCIE	PCIE_TBT_D2R_C_P<3>	NO_TEST=TRUE	36
I624	PCIE_GEN2_D2R_PINV	PCIE_PHY	PCIE	PCIE_TBT_D2R_C_N<3>	NO_TEST=TRUE	36
I579	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_TBT_P		18 36
I580	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_TBT_N		18 36

PCIe (CPU)

	Electrical Contraint Set	Physical	Spacing	Net		
	x16 Graphics					
I642	PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_P<15>	NO_TEST=TRUE	10 83
I643	PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_N<15>	NO_TEST=TRUE	10 83
I644		PCIE_PHY	PCIE	PEG_R2D_C_P<15>	NO_TEST=TRUE	83
I645		PCIE_PHY	PCIE	PEG_R2D_C_N<15>	NO_TEST=TRUE	83
I646	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_P<15>	NO_TEST=TRUE	83
I647	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_N<15>	NO_TEST=TRUE	83
I648		PCIE_PHY	PCIE	PEG_D2R_C_P<15>	NO_TEST=TRUE	10 83
I649		PCIE_PHY	PCIE	PEG_D2R_C_N<15>	NO_TEST=TRUE	10 83
I651	PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_P<14>	NO_TEST=TRUE	10 83
I650	PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_N<14>	NO_TEST=TRUE	10 83
I652		PCIE_PHY	PCIE	PEG_R2D_C_P<14>	NO_TEST=TRUE	83
I653		PCIE_PHY	PCIE	PEG_R2D_C_N<14>	NO_TEST=TRUE	83
I654	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_P<14>	NO_TEST=TRUE	83
I655	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_N<14>	NO_TEST=TRUE	83
I656		PCIE_PHY	PCIE	PEG_D2R_C_P<14>	NO_TEST=TRUE	10 83
I657		PCIE_PHY	PCIE	PEG_D2R_C_N<14>	NO_TEST=TRUE	10 83
I658	PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_P<13>	NO_TEST=TRUE	10 83
I709	PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_N<13>	NO_TEST=TRUE	10 83
I710		PCIE_PHY	PCIE	PEG_R2D_C_P<13>	NO_TEST=TRUE	83
I711		PCIE_PHY	PCIE	PEG_R2D_C_N<13>	NO_TEST=TRUE	83
I712	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_P<13>	NO_TEST=TRUE	83
I713	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_N<13>	NO_TEST=TRUE	83
I715		PCIE_PHY	PCIE	PEG_D2R_C_P<13>	NO_TEST=TRUE	10 83
I714		PCIE_PHY	PCIE	PEG_D2R_C_N<13>	NO_TEST=TRUE	10 83
I716	PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_P<12>	NO_TEST=TRUE	10 83
I717	PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_N<12>	NO_TEST=TRUE	10 83
I718		PCIE_PHY	PCIE	PEG_R2D_C_P<12>	NO_TEST=TRUE	83
I719		PCIE_PHY	PCIE	PEG_R2D_C_N<12>	NO_TEST=TRUE	83
I720	PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_P<12>	NO_TEST=TRUE	83
I721	PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_N<12>	NO_TEST=TRUE	83
I722		PCIE_PHY	PCIE	PEG_D2R_C_P<12>	NO_TEST=TRUE	10 83
I723		PCIE_PHY	PCIE	PEG_D2R_C_N<12>	NO_TEST=TRUE	10 83
I724	PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_P<11>	NO_TEST=TRUE	10 83
I725	PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_N<11>	NO_TEST=TRUE	10 83
I726		PCIE_PHY	PCIE	PEG_R2D_C_P<11>	NO_TEST=TRUE	83
I727		PCIE_PHY	PCIE	PEG_R2D_C_N<11>	NO_TEST=TRUE	83
I728	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_P<11>	NO_TEST=TRUE	83
I729	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_N<11>	NO_TEST=TRUE	83
I730		PCIE_PHY	PCIE	PEG_D2R_C_P<11>	NO_TEST=TRUE	10 83
I731		PCIE_PHY	PCIE	PEG_D2R_C_N<11>	NO_TEST=TRUE	10 83
I732	PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_P<10>	NO_TEST=TRUE	10 83
I734	PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_N<10>	NO_TEST=TRUE	10 83
I733		PCIE_PHY	PCIE	PEG_R2D_C_P<10>	NO_TEST=TRUE	83
I735		PCIE_PHY	PCIE	PEG_R2D_C_N<10>	NO_TEST=TRUE	83
I736	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_P<10>	NO_TEST=TRUE	83
I737	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_N<10>	NO_TEST=TRUE	83
I738		PCIE_PHY	PCIE	PEG_D2R_C_P<10>	NO_TEST=TRUE	10 83
I739		PCIE_PHY	PCIE	PEG_D2R_C_N<10>	NO_TEST=TRUE	10 83
I740	PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_P<9>	NO_TEST=TRUE	10 83
I741	PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_N<9>	NO_TEST=TRUE	10 83
I742		PCIE_PHY	PCIE	PEG_R2D_C_P<9>	NO_TEST=TRUE	83
I743		PCIE_PHY	PCIE	PEG_R2D_C_N<9>	NO_TEST=TRUE	83
I744	PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_P<9>	NO_TEST=TRUE	83
I745	PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_N<9>	NO_TEST=TRUE	83
I746		PCIE_PHY	PCIE	PEG_D2R_C_P<9>	NO_TEST=TRUE	10 83
I747		PCIE_PHY	PCIE	PEG_D2R_C_N<9>	NO_TEST=TRUE	10 83
I748	PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_P<8>	NO_TEST=TRUE	10 83
I749	PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_N<8>	NO_TEST=TRUE	10 83
I750		PCIE_PHY	PCIE	PEG_R2D_C_P<8>	NO_TEST=TRUE	83
I751		PCIE_PHY	PCIE	PEG_R2D_C_N<8>	NO_TEST=TRUE	83
I752	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_P<8>	NO_TEST=TRUE	83
I753	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_N<8>	NO_TEST=TRUE	83
I754		PCIE_PHY	PCIE	PEG_D2R_C_P<8>	NO_TEST=TRUE	10 83
I755		PCIE_PHY	PCIE	PEG_D2R_C_N<8>	NO_TEST=TRUE	10 83

PCIe (CPU)

	Electrical Contraint Set	Physical	Spacing	Net		
	x16 Graphics					
I625	PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_P<7>	NO_TEST=TRUE	10 83
I626	PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_N<7>	NO_TEST=TRUE	10 83
I627		PCIE_PHY	PCIE	PEG_R2D_C_P<7>	NO_TEST=TRUE	83
I629		PCIE_PHY	PCIE	PEG_R2D_C_N<7>	NO_TEST=TRUE	83
I628	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_P<7>	NO_TEST=TRUE	83
I630	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_N<7>	NO_TEST=TRUE	83
I632		PCIE_PHY	PCIE	PEG_D2R_C_P<7>	NO_TEST=TRUE	10 83
I631		PCIE_PHY	PCIE	PEG_D2R_C_N<7>	NO_TEST=TRUE	10 83
I634	PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_P<6>	NO_TEST=TRUE	10 83
I633	PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_N<6>	NO_TEST=TRUE	10 83
I636		PCIE_PHY	PCIE	PEG_R2D_C_P<6>	NO_TEST=TRUE	83
I635		PCIE_PHY	PCIE	PEG_R2D_C_N<6>	NO_TEST=TRUE	83
I637	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_P<6>	NO_TEST=TRUE	83
I639	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_N<6>	NO_TEST=TRUE	83
I638		PCIE_PHY	PCIE	PEG_D2R_C_P<6>	NO_TEST=TRUE	10 83
I640		PCIE_PHY	PCIE	PEG_D2R_C_N<6>	NO_TEST=TRUE	10 83
I641	PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_P<5>	NO_TEST=TRUE	10 83
I659	PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_N<5>	NO_TEST=TRUE	10 83
I660		PCIE_PHY	PCIE	PEG_R2D_C_P<5>	NO_TEST=TRUE	83
I661		PCIE_PHY	PCIE	PEG_R2D_C_N<5>	NO_TEST=TRUE	83
I662	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_P<5>	NO_TEST=TRUE	83
I663	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_N<5>	NO_TEST=TRUE	83
I664		PCIE_PHY	PCIE	PEG_D2R_C_P<5>	NO_TEST=TRUE	10 83
I665		PCIE_PHY	PCIE	PEG_D2R_C_N<5>	NO_TEST=TRUE	10 83
I666	PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_P<4>	NO_TEST=TRUE	10 83
I667	PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_N<4>	NO_TEST=TRUE	10 83
I668		PCIE_PHY	PCIE	PEG_R2D_C_P<4>	NO_TEST=TRUE	83
I669		PCIE_PHY	PCIE	PEG_R2D_C_N<4>	NO_TEST=TRUE	83
I670	PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_P<4>	NO_TEST=TRUE	83
I671	PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_N<4>	NO_TEST=TRUE	83
I672		PCIE_PHY	PCIE	PEG_D2R_C_P<4>	NO_TEST=TRUE	10 83
I673		PCIE_PHY	PCIE	PEG_D2R_C_N<4>	NO_TEST=TRUE	10 83
I674	PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_P<3>	NO_TEST=TRUE	10 83
I675	PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_N<3>	NO_TEST=TRUE	10 83
I676		PCIE_PHY	PCIE	PEG_R2D_C_P<3>	NO_TEST=TRUE	83
I677		PCIE_PHY	PCIE	PEG_R2D_C_N<3>	NO_TEST=TRUE	83
I678	PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_P<3>	NO_TEST=TRUE	83
I679	PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_N<3>	NO_TEST=TRUE	83
I680		PCIE_PHY	PCIE	PEG_D2R_C_P<3>	NO_TEST=TRUE	10 83
I681		PCIE_PHY	PCIE	PEG_D2R_C_N<3>	NO_TEST=TRUE	10 83
I682	PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_P<2>	NO_TEST=TRUE	10 83
I684	PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_N<2>	NO_TEST=TRUE	10 83
I683		PCIE_PHY	PCIE	PEG_R2D_C_P<2>	NO_TEST=TRUE	83
I685		PCIE_PHY	PCIE	PEG_R2D_C_N<2>	NO_TEST=TRUE	83
I686	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_P<2>	NO_TEST=TRUE	83
I687	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_N<2>	NO_TEST=TRUE	83
I688		PCIE_PHY	PCIE	PEG_D2R_C_P<2>	NO_TEST=TRUE	10 83
I689		PCIE_PHY	PCIE	PEG_D2R_C_N<2>	NO_TEST=TRUE	10 83
I690	PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_P<1>	NO_TEST=TRUE	10 83
I692	PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_N<1>	NO_TEST=TRUE	10 83
I691		PCIE_PHY	PCIE	PEG_R2D_C_P<1>	NO_TEST=TRUE	83
I693		PCIE_PHY	PCIE	PEG_R2D_C_N<1>	NO_TEST=TRUE	83
I694	PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_P<1>	NO_TEST=TRUE	83
I695	PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_N<1>	NO_TEST=TRUE	83
I697		PCIE_PHY	PCIE	PEG_D2R_C_P<1>	NO_TEST=TRUE	10 83
I696		PCIE_PHY	PCIE	PEG_D2R_C_N<1>	NO_TEST=TRUE	10 83
I698	PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_P<0>	NO_TEST=TRUE	10 83
I699	PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_N<0>	NO_TEST=TRUE	10 83
I700		PCIE_PHY	PCIE	PEG_R2D_C_P<0>	NO_TEST=TRUE	83
I701		PCIE_PHY	PCIE	PEG_R2D_C_N<0>	NO_TEST=TRUE	83
I702	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_P<0>	NO_TEST=TRUE	83
I703	PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_N<0>	NO_TEST=TRUE	83
I704		PCIE_PHY	PCIE	PEG_D2R_C_P<0>	NO_TEST=TRUE	10 83
I705		PCIE_PHY	PCIE	PEG_D2R_C_N<0>	NO_TEST=TRUE	10 83
	CPU PCIe Clocks					
I706	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG_CLK100M_P	NO_TEST=TRUE	18 83
I707	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG_CLK100M_N	NO_TEST=TRUE	18 83
	CPU PCIe Compensation					
I708		COMP_PCIE_PHY	COMP_PCIE	CPU_PEG_COMP		10

SYNC_MASTER=D8_MLB SYNC_DATE=02/06/2012

PAGE TITLE
CPU PCIe Constraints

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

DMI

DMI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DMI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
DMI_PHY	*	DMI_85D

CPU Misc / FDI

CPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
COMP_FDI_SE	*	Y	0.3 MM	0.25 MM	3 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
CPU_PHY	*	CPU_50S
COMP_FDI_PHY	*	COMP_FDI_SE

CPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_ISO	*	=3:1_SPACING	?
COMP_FDI_ISO	*	=4:1_SPACING	?

FDI Compensation Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Table	Trace	Design	Iso	Design	Comments
6-4	10	11.81	-	15.75	Using PCIe guidelines

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU	*	*	CPU_ISO
COMP_FDI	*	*	COMP_FDI_ISO

XDP

XDP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
XDP_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
XDP_PHY	*	XDP_50S

XDP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XDP_ISO	*	=2:1_SPACING	?
XDP_CLK_ISO	*	=4:1_SPACING	?

Desktop Debug Design Guide (Intel Doc# 430883)

Section	Imp	Design	Iso	Design	Comments
1.5	45-65	50	-	15.75	Isolation is for JTAG clocks. All signals default are 50 Ohm SE.

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XDP	*	*	XDP_ISO
CLK_JTAG	*	*	XDP_CLK_ISO
OBS_STROBE	*	*	XDP_CLK_ISO

DMI

	Electrical Contraint Set	Physical	Spacing	Net		Pages
	DMI					
I166	DMI_N2S	DMI_PHY	PCIE	DMI_N2S_P<3..0>	NO_TEST=TRUE	10 19
I165	DMI_N2S	DMI_PHY	PCIE	DMI_N2S_N<3..0>	NO_TEST=TRUE	10 19
I167	DMI_S2N	DMI_PHY	PCIE	DMI_S2N_P<3..0>	NO_TEST=TRUE	10 19
I168	DMI_S2N	DMI_PHY	PCIE	DMI_S2N_N<3..0>	NO_TEST=TRUE	10 19
I283	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	DMI_CLK100M_CPU_P		11 18
I284	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	DMI_CLK100M_CPU_N		11 18
	DMI Compensation					
I171		COMP_PCIE_PHY	COMP_PCIE	PCH_DMI_COMP		19
I272		COMP_PCIE_PHY	COMP_PCIE	PCH_DMI2RBIAS		19

CPU Misc.

	Electrical Contraint Set	Physical	Spacing	Net	Pages
	CPU Misc.				
I245		CPU_PHY	CPU	CPU_SKTOCC_L	11 64
I244		CPU_PHY	CPU	CPU_PROC_SEL	11 19
I246		CPU_PHY	CPU	CPU_CATERR_L	11 48
I251		CPU_PHY	CPU	CPU_PECI	11 21 47 48
I250		CPU_PHY	CPU	CPU_PROCHOT_L	11 47 48 66
I267		CPU_PHY	CPU	CPU_PROCHOT_R_L	11
I249		CPU_PHY	CPU	CPU_THRMTRIP_L	11 48
I248		CPU_PHY	CPU	CPU_RESET_L	11 26
I247		CPU_PHY	CPU	PLT_RESET_LS1V05_L	11
I256		CPU_PHY	CPU	PM_SYNC	11 19
I255		CPU_PHY	CPU	CPU_PWRGD	11 21 25 28
I260		CPU_PHY	CPU	PM_MEM_PWRGD	11 19 28
I254		CPU_PHY	CPU	PM_MEM_PWRGD_R	11
I253		CPU_PHY	CPU	CPU_MEM_RESET_L	11 28

FDI

	Electrical Contraint Set	Physical	Spacing	Net	Pages
	FDI Compensation				
I261		COMP_FDI_PHY	COMP_FDI	CPU_FDI_COMPIO	10

XDP

	Electrical Contraint Set	Physical	Spacing	Net	Pages
	XDP MISC.				
I174	XDP_BPM_L	XDP_PHY	XDP	XDP_BPM_L<7..0>	11 25
I175		XDP_PHY	XDP	CPU_CFG<17..16>	10 25
I257		XDP_PHY	XDP	CPU_CFG<11..0>	10 15 25
I266		XDP_PHY	XDP	XDP_DBRESET_L	11 25
I265		XDP_PHY	XDP	XDP_CPU_PWRGD	25
I264		XDP_PHY	XDP	XDP_CPU_PWRBTN_L	25
I263		XDP_PHY	XDP	XDP_CPU_CFG<0>	25
I262		XDP_PHY	XDP	XDP_VR_READY	25
I270		XDP_PHY	XDP	XDP_CPU_PLTRST_L	25 26
I269		XDP_PHY	XDP	XDP_PCH_PWRGD	25
I268		XDP_PHY	XDP	XDP_PCH_PWRBTN_L	25
I271		XDP_PHY	XDP	XDP_PCH_PLTRST_L	25 26
I176		CLK_PCIE_PHY	CLK_PCIE	ITPCPU_CLK100M_P	11 15
I177		CLK_PCIE_PHY	CLK_PCIE	ITPCPU_CLK100M_N	11 15
I178	ITP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPXDP_CLK100M_P	15 18 25
I180	ITP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPXDP_CLK100M_N	15 18 25
I179		CLK_PCIE_PHY	CLK_PCIE	XDP_CPU_CLK100M_P	25
I181		CLK_PCIE_PHY	CLK_PCIE	XDP_CPU_CLK100M_N	25
	CPU JTAG				
I182		XDP_PHY	CLK_JTAG	XDP_CPU_TCK	11 25
I183		XDP_PHY	XDP	XDP_CPU_TMS	11 25
I184		XDP_PHY	XDP	XDP_CPU_TDI	11 25
I185		XDP_PHY	XDP	XDP_CPU_TDO	11 25
I190		XDP_PHY	XDP	XDP_CPU_TRST_L	11 25
I258		XDP_PHY	XDP	XDP_CPU_PRDY_L	11 25
I259		XDP_PHY	XDP	XDP_CPU_PREQ_L	11 25
	PCH JTAG				
I186		XDP_PHY	CLK_JTAG	XDP_PCH_TCK	18 25
I187		XDP_PHY	XDP	XDP_PCH_TMS	18 25
I188		XDP_PHY	XDP	XDP_PCH_TDI	18 25
I189		XDP_PHY	XDP	XDP_PCH_TDO	18 25

Chipset Test Interface

	Electrical Contraint Set	Physical	Spacing	Net	Pages
	OBS PCH Side				
I191	XDP_PCH_OBS	XDP_PHY	XDP	USB_EXTA_OC_R_L	20 25
I192	XDP_PCH_OBS	XDP_PHY	XDP	USB_EXTB_OC_R_L	20 25
I207	XDP_PCH_OBS	XDP_PHY	XDP	USB_EXTC_OC_R_L	20 25
I206	XDP_PCH_OBS	XDP_PHY	XDP	USB_EXTD_OC_R_L	20 25
I205	XDP_PCH_OBS	XDP_PHY	XDP	USB_EXTB_OC_EHCI_R_L	20 25
I204	XDP_PCH_OBS	XDP_PHY	XDP	USB_EXTD_OC_EHCI_R_L	20 25
I203	XDP_PCH_OBS	XDP_PHY	XDP	AP_PWR_EN_R	20 25
I202	XDP_PCH_OBS	XDP_PHY	XDP	SDCONN_STATE_CHANGE_R	20 25
I201	XDP_PCH_OBS	XDP_PHY	OBS_STROBE	TBT_CIO_PLUG_EVENT_R	21 25
I200	XDP_PCH_OBS	XDP_PHY	XDP	ISOLATE_CPU_MEM_R_L	21 25
I199	XDP_PCH_OBS	XDP_PHY	XDP	GPU_GOOD_R	21 25
I198	XDP_PCH_OBS	XDP_PHY	XDP	DP_AUXCH_ISOL_R	18 25
I197	XDP_PCH_OBS	XDP_PHY	XDP	SATARDRVR_EN_R	18 25
I196	XDP_PCH_OBS	XDP_PHY	XDP	DP_TBT_SEL_R	21 25
I195	XDP_PCH_OBS	XDP_PHY	XDP	JTAG_TBT_TCK_R	21 25
I194	XDP_PCH_OBS	XDP_PHY	XDP	AUD_IPHS_SWITCH_EN_PCH_R	21 25
I193	XDP_PCH_OBS	XDP_PHY	XDP	ENET_LOW_PWR_PCH_R	21 25
I208	XDP_PCH_OBS_UNUSED	XDP_PHY	OBS_STROBE	XDP_PIN03	21 25
	OBS XDP Side				
I209		XDP_PHY	XDP	XDP_DA0_USB_EXTA_OC_L	25
I210		XDP_PHY	XDP	XDP_DA1_USB_EXTB_OC_L	25
I211		XDP_PHY	XDP	XDP_DA2_USB_EXTC_OC_L	25
I212		XDP_PHY	XDP	XDP_DA3_USB_EXTD_OC_L	25
I213		XDP_PHY	XDP	XDP_DB0_USB_EXTB_OC_EHCI_L	25
I214		XDP_PHY	XDP	XDP_DB1_USB_EXTD_OC_EHCI_L	25
I215		XDP_PHY	XDP	XDP_DB2_AP_PWR_EN	25
I216		XDP_PHY	XDP	XDP_DB3_SDCONN_STATE_CHANGE	25
I217		XDP_PHY	OBS_STROBE	XDP_FC1_TBT_CIO_PLUG_EVENT	25
I218		XDP_PHY	XDP	XDP_DC0_ISOLATE_CPU_MEM_L	25
I219		XDP_PHY	XDP	XDP_DC1_GPU_GOOD	25
I220		XDP_PHY	XDP	XDP_DC2_DP_AUXCH_ISOL	25
I221		XDP_PHY	XDP	XDP_DC3_SATARDRVR_EN	25
I222		XDP_PHY	XDP	XDP_DD0_DP_TBT_SEL	25
I223		XDP_PHY	XDP	XDP_DD1_JTAG_TBT_TCK	25
I224		XDP_PHY	XDP	XDP_DD2_AUD_IPHS_SWITCH_EN_PCH	25
I225		XDP_PHY	XDP	XDP_DD3_ENET_LOW_PWR_PCH	25
I226		XDP_PHY	OBS_STROBE	XDP_FC0_PCH_GPIO15	25
	Standard usage branch off from OBS				
I227		XDP_PHY	XDP	USB_EXTA_OC_L	15 20 45
I228		XDP_PHY	XDP	USB_EXTB_OC_L	15 20 45
I230		XDP_PHY	XDP	USB_EXTC_OC_L	15 20 46
I229		XDP_PHY	XDP	USB_EXTD_OC_L	15 20 46
I232		XDP_PHY	XDP	USB_EXTB_OC_EHCI_L	15 20
I231		XDP_PHY	XDP	USB_EXTD_OC_EHCI_L	15 20
I233		XDP_PHY	XDP	AP_PWR_EN	15 20
I235		XDP_PHY	XDP	SDCONN_STATE_CHANGE	15 20 41
I234		XDP_PHY	OBS_STROBE	TBT_CIO_PLUG_EVENT	15 21
I236		XDP_PHY	XDP	ISOLATE_CPU_MEM_L	21 28 34
I237		XDP_PHY	XDP	GPU_GOOD	5 21
I238		XDP_PHY	XDP	DP_AUXCH_ISOL	15 18
I240		XDP_PHY	XDP	SATARDRVR_EN	15 18
I239		XDP_PHY	XDP	DP_TBT_SEL	21 62 75
I241		XDP_PHY	XDP	JTAG_TBT_TCK	15 21
I273		XDP_PHY	XDP	JTAG_TBT_TCK_ISOL	15 36
I243		XDP_PHY	XDP	AUD_IPHS_SWITCH_EN_PCH	21 26
I242		XDP_PHY	XDP	ENET_LOW_PWR_PCH	15 21 26
I274		XDP_PHY	XDP	DP_AUXCH_ISOL_EN	15

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE
CPU MISC/DMI/FDI/XDP Constraints

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

SATA

SATA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SATA_PHY	*	SATA_90D
COMP_SATA_PHY	*	SATA_50S

SATA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_ISO	*	=7.2X_DIELECTRIC	?
COMP_SATA_ISO	*	=5.4X_DIELECTRIC	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA	*	*	SATA_ISO
COMP_SATA	*	*	COMP_SATA_ISO

SATA Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
15.2.1	90	95	20	23.62	SATA Gen2, SATA Gen3

SATA Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
15-3	50	50	15	15.75	SATA Gen2, SATA Gen3

SATA

Electrical Contraint Set	Physical	Spacing	Net	
PCH SATA Port 0 (HDD)				
I90 SATA_R2D	SATA_PHY	SATA	SATA_HDD_R2D_P	44
I92 SATA_R2D	SATA_PHY	SATA	SATA_HDD_R2D_N	44
I88	SATA_PHY	SATA	SATA_HDD_R2D_C_P	18 44
I89	SATA_PHY	SATA	SATA_HDD_R2D_C_N	18 44
I94 SATA_D2R	SATA_PHY	SATA	SATA_HDD_D2R_P	18 44
I95 SATA_D2R	SATA_PHY	SATA	SATA_HDD_D2R_N	18 44
I91	SATA_PHY	SATA	SATA_HDD_D2R_C_P	44
I93	SATA_PHY	SATA	SATA_HDD_D2R_C_N	44
PCH SATA Port 1 (SSD)				
I97 SATA_R2D_MUX_SSD	SATA_PHY	SATA	SATA_SSD_R2D_P	18 44
I99 SATA_R2D_MUX_SSD	SATA_PHY	SATA	SATA_SSD_R2D_N	18 44
I103 SATA_D2R_MUX_SSD	SATA_PHY	SATA	SATA_SSD_D2R_P	18 44
I102 SATA_D2R_MUX_SSD	SATA_PHY	SATA	SATA_SSD_D2R_N	18 44
PCH SATA Compensation				
I104	COMP_SATA_PHY	COMP_SATA	PCH_SATAICOMP	18
I106	COMP_SATA_PHY	COMP_SATA	PCH_SATA3COMP	18
I105	COMP_SATA_PHY	COMP_SATA	PCH_SATA3RBIAS	18

Unused

Electrical Contraint Set	Physical	Spacing

PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCH_ISO	*	=6.5X_DIELECTRIC	?
COMP_PCH_ISO	*	=2:1_SPACING	?
PCH_ISO	*	=3:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCH	*	*	CLK_PCH_ISO
COMP_PCH	*	*	COMP_PCH_ISO
PCH	*	*	PCH_ISO

PCI

PCI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_PCI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

PCI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCI_ISO	*	=3.6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCI	*	*	CLK_PCI_ISO

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC_ISO	*	=1.5:1_SPACING	?
CLK_LPC_ISO	*	=3.6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LPC	*	*	LPC_ISO
CLK_LPC	*	*	CLK_LPC_ISO

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA_ISO	*	=2X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
HDA	*	*	HDA_ISO

Crystal

Crystal-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_XTAL	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL_ISO	*	=4X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XTAL	*	*	XTAL_ISO

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI_ISO	*	=2X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SPI	*	*	SPI_ISO

PCI

Electrical Contraint Set	Physical	Spacing	Net	
PCI Clock				
I333	CLK_PCI_50S	CLK_PCI	PCH_CLK33M_PCIIN	18 26
I334	CLK_PCI_50S	CLK_PCI	PCH_CLK33M_PCIOUT	20 26

LPC

Electrical Contraint Set	Physical	Spacing	Net	
LPC				
I336	LPC_50S	LPC	LPC_AD<3..0>	18 47 49
I335	LPC_50S	LPC	LPC_R_AD<3..0>	18
I338	LPC_50S	LPC	LPC_FRAME_L	18 47 49
I337	LPC_50S	LPC	LFRAME_L	18
LPC Clocks				
I339	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_LPCPLUS	26 49
I340	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_LPCPLUS_R	20 26
I341	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC	26 47
I342	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC_R	20 26

PCH Clocks

Electrical Contraint Set	Physical	Spacing	Net	
PCH Reference Clock				
I385	CLK_PCH_50S	CLK_PCH	SYSCLK_CLK25M_SB	26
I406	CLK_PCH_50S	CLK_PCH	PCH_CLK25M_XTALIN	18 26
PCH Ref Clock Comp				
I346	PCH_50S	COMP_PCH	PCH_XCLK_RCOMP	18
PCH RTC 32K				
I347	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1	18 26
I348	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2	18 26
I352	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R	26
SMC 32K				
I349	CLK_PCH_50S	CLK_PCH	PM_CLK32K_SUSCLK_R	19 48
I350	CLK_PCH_50S	CLK_PCH	SMC_CLK32K	47 48

25 MHz Reference Clocks

Electrical Contraint Set	Physical	Spacing	Net	
25M Reference Crystal				
I379	CLK_XTAL	XTAL	SYSCLK_CLK25M_X1	26
I380	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2	26
I381	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2_R	26
25M Reference Clocks				
I384	CLK_PCH_50S	CLK_PCH	SYSCLK_CLK25M_ENET	26 39
I386	CLK_PCH_50S	CLK_PCH	SYSCLK_CLK25M_ENET_R	26
I388	CLK_PCH_50S	CLK_PCH	SYSCLK_CLK25M_TBT	26 36
I387	CLK_PCH_50S	CLK_PCH	SYSCLK_CLK25M_TBT_R	36

HDA

Electrical Contraint Set	Physical	Spacing	Net	
HDA				
I353	HDA_50S	HDA	HDA_BIT_CLK	18 56
I361	HDA_50S	HDA	HDA_BIT_CLK_R	18
I363	HDA_50S	HDA	HDA_RST_L	18 56
I362	HDA_50S	HDA	HDA_RST_R_L	18
I364	HDA_50S	HDA	HDA_SDOUT	18 56
I367	HDA_50S	HDA	HDA_SDOUT_R	15 18
I366	HDA_50S	HDA	HDA_SYNC	15 18 56
I368	HDA_50S	HDA	HDA_SYNC_R	18
I365	HDA_50S	HDA	HDA_SDIN0	18 56
I369	HDA_50S	HDA	AUD_SDI_R	56
I414	HDA_50S	HDA	SPI_DESCRIPTOR_OVERRIDE_R	15
SPDIF				
I370		HDA	AUD_SPDIF_CHIP	56
I371		HDA	AUD_SPDIF_OUT	56 60

SPI Bootrom

Electrical Contraint Set	Physical	Spacing	Net	
SPI ROM				
I391	SPI_50S	SPI	SPI_CLK_R	18 49
I392	SPI_50S	SPI	SPI_CLK	49
I393	SPI_50S	SPI	SPI_ALT_CLK	49
I408	SPI_50S	SPI	SPI_SMC_CLK	47 48
I407	SPI_50S	SPI	SPI_MLB_CLK	48 49
I394	SPI_50S	SPI	SPI_CS0_R_L	18 49
I395	SPI_50S	SPI	SPI_CS0_L	49
I396	SPI_50S	SPI	SPI_ALT_CS_L	49
I409	SPI_50S	SPI	SPI_SMC_CS_L	47 48
I397	SPI_50S	SPI	SPI_MLB_CS_L	48 49
I399	SPI_50S	SPI	SPI_MOSI_R	18 49
I400	SPI_50S	SPI	SPI_MOSI	49
I401	SPI_50S	SPI	SPI_ALT_MOSI	49
I410	SPI_50S	SPI	SPI_SMC_MOSI	47 48
I411	SPI_50S	SPI	SPI_MLB_MOSI	48 49
I404	SPI_50S	SPI	SPI_MISO	18 49
I403	SPI_50S	SPI	SPI_ALT_MISO	49
I412	SPI_50S	SPI	SPI_SMC_MISO	47 48
I413	SPI_50S	SPI	SPI_MLB_MISO	48 49
I405	SPI_50S	SPI	SPIROM_USE_MLB	21 49

USB

USB-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
USB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
USB2_PHY	*	USB_90D
USB3_PHY	*	USB_85D
USB_HUB_PHY	*	USB_50S

SMSC Hub Application Note 15.17
Single-ended impedance range from 45-80 ohm is acceptable

USB Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Comments
12.2.1	90	90	12/2.8 mils = 4.29:1	USB 2.0
13.3.1	85	85	20/2.8 mils = 7.14:1	USB 3.0
			50/2.8 mils = 17.9:1	USB 2.0/3.0

USB 2.0 Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB2_ISO	*	=4.4X_DIELECTRIC	?
USB2_ISO	TOP,BOTTOM	=4.4X_DIELECTRIC	?
USB2_CLK_ISO	*	=18X_DIELECTRIC	?
USB2_CLK_ISO	TOP,BOTTOM	=18X_DIELECTRIC	?
USB_HUB_ISO	*	=2:1_SPACING	?

USB 3.0 Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_ISO	*	=7.3X_DIELECTRIC	?
USB3_ISO	TOP,BOTTOM	=7.3X_DIELECTRIC	?
USB3_CLK_ISO	*	=18X_DIELECTRIC	?
USB3_CLK_ISO	TOP,BOTTOM	=18X_DIELECTRIC	?
USB3_USB3_ISO	*	=7.3X_DIELECTRIC	?
USB3_USB3_ISO	TOP,BOTTOM	=7.3X_DIELECTRIC	?
USB3_USB3_ISO	ISL10	=5X_DIELECTRIC	?

USB 2.0 Spacing Rules

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB2	*	*	USB2_ISO
USB2	*CLK*	*	USB2_CLK_ISO
USB2	DISPLAYPORT	*	USB2_CLK_ISO
USB2	*TBT*	*	USB2_CLK_ISO
USB2	*ENET*	*	USB2_CLK_ISO
USB2	*SD*	*	USB2_CLK_ISO
USB3	PCIE	*	USB3_CLK_ISO
USB3	SATA	*	USB3_CLK_ISO
USB_HUB	*	*	USB_HUB_ISO

USB 3.0 Spacing Rules

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3	*	*	USB3_ISO
USB3	*CLK*	*	USB3_CLK_ISO
USB3	DISPLAYPORT	*	USB3_CLK_ISO
USB3	*TBT*	*	USB3_CLK_ISO
USB3	*ENET*	*	USB3_CLK_ISO
USB3	*SD*	*	USB3_CLK_ISO
USB3	PCIE	*	USB3_CLK_ISO
USB3	SATA	*	USB3_CLK_ISO
USB3	USB3	*	USB3_USB3_ISO

CAMERA CONTROLLER

Camera Controller's SMIA Interface & MISC. Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMIA_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
CAM_SE	*	Y	0.2 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMIA_DIFF_PHY	*	SMIA_100D
CAM_PHY	*	CAM_SE

Camera Controller's SMIA Interface & MISC. Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMIA_DIFF_ISO	*	=6:1_SPACING	?
SMIA_DIFF2DIFF	*	=3:1_SPACING	?
CAM_ISO	*	=2:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMIA_DIFF	*	*	SMIA_DIFF_ISO
SMIA_DIFF	SMIA_DIFF	*	SMIA_DIFF2DIFF
CAM	*	*	CAM_ISO

USB 3.0 and USB 2.0 Trixies Muxing

	Electrical Contraint Set	Physical	Spacing	Net	
External Port A (J4600)					
I320	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTA_RX_P	45
I321	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTA_RX_N	45
I391		USB3_PHY	USB3	USB3_EXTA_RX_F_P	20 45
I392		USB3_PHY	USB3	USB3_EXTA_RX_F_N	20 45
I413	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTA_TX_P	20 45
I414	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTA_TX_N	20 45
I415		USB3_PHY	USB3	USB3_EXTA_TX_F_P	45
I416		USB3_PHY	USB3	USB3_EXTA_TX_F_N	45
I493		USB3_PHY	USB3	USB3_EXTA_TX_C_P	45
I494		USB3_PHY	USB3	USB3_EXTA_TX_C_N	45
I417	USB2_MUXED_MOJO_CONN	USB2_PHY	USB2	USB_PCH_0_P	20 45
I418	USB2_MUXED_MOJO_CONN	USB2_PHY	USB2	USB_PCH_0_N	20 45
I419		USB2_PHY	USB2	USB2_EXTA_MUXED_P	45
I420		USB2_PHY	USB2	USB2_EXTA_MUXED_N	45
I393		USB2_PHY	USB2	USB2_EXTA_MUXED_F_P	45
I394		USB2_PHY	USB2	USB2_EXTA_MUXED_F_N	45
External Port B (J4610)					
I421	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_P	45
I422	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_N	45
I423		USB3_PHY	USB3	USB3_EXTB_RX_F_P	20 45
I424		USB3_PHY	USB3	USB3_EXTB_RX_F_N	20 45
I425	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_P	20 45
I427	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_N	20 45
I426		USB3_PHY	USB3	USB3_EXTB_TX_F_P	45
I428		USB3_PHY	USB3	USB3_EXTB_TX_F_N	45
I495		USB3_PHY	USB3	USB3_EXTB_TX_C_P	45
I496		USB3_PHY	USB3	USB3_EXTB_TX_C_N	45
I429	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_1_P	20 45
I430	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_1_N	20 45
I436		USB2_PHY	USB2	USB_PCH_9_P	20 45
I435		USB2_PHY	USB2	USB_PCH_9_N	20 45
I432		USB2_PHY	USB2	USB2_EXTB_MUXED_P	45
I431		USB2_PHY	USB2	USB2_EXTB_MUXED_N	45
I434		USB2_PHY	USB2	USB2_EXTB_MUXED_F_P	45
I433		USB2_PHY	USB2	USB2_EXTB_MUXED_F_N	45
External Port C (J4700)					
I437	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTC_RX_P	46
I439	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTC_RX_N	46
I438		USB3_PHY	USB3	USB3_EXTC_RX_F_P	20 46
I440		USB3_PHY	USB3	USB3_EXTC_RX_F_N	20 46
I441	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTC_TX_P	20 46
I442	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTC_TX_N	20 46
I444		USB3_PHY	USB3	USB3_EXTC_TX_F_P	46
I443		USB3_PHY	USB3	USB3_EXTC_TX_F_N	46
I498		USB3_PHY	USB3	USB3_EXTC_TX_C_P	46
I497		USB3_PHY	USB3	USB3_EXTC_TX_C_N	46
I446	USB2_CONN	USB2_PHY	USB2	USB_PCH_2_P	20 46
I445	USB2_CONN	USB2_PHY	USB2	USB_PCH_2_N	20 46
I450		USB2_PHY	USB2	USB2_EXTC_F_P	46
I452		USB2_PHY	USB2	USB2_EXTC_F_N	46
External Port D (J4710)					
I453	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_P	46
I455	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_N	46
I454		USB3_PHY	USB3	USB3_EXTD_RX_F_P	20 46
I456		USB3_PHY	USB3	USB3_EXTD_RX_F_N	20 46
I457	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_P	20 46
I458	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_N	20 46
I460		USB3_PHY	USB3	USB3_EXTD_TX_F_P	46
I459		USB3_PHY	USB3	USB3_EXTD_TX_F_N	46
I499		USB3_PHY	USB3	USB3_EXTD_TX_C_P	46
I500		USB3_PHY	USB3	USB3_EXTD_TX_C_N	46
I462	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_3_P	20 46
I461	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_3_N	20 46
I463		USB2_PHY	USB2	USB_PCH_10_P	20 46
I464		USB2_PHY	USB2	USB_PCH_10_N	20 46
I465		USB2_PHY	USB2	USB2_EXTD_MUXED_P	46
I467		USB2_PHY	USB2	USB2_EXTD_MUXED_N	46
I466		USB2_PHY	USB2	USB2_EXTD_MUXED_F_P	46
I468		USB2_PHY	USB2	USB2_EXTD_MUXED_F_N	46
Camera (U4200)					
I324	USB2_CONN_INT	USB2_PHY	USB2	USB_CAMERA_P	42
I326	USB2_CONN_INT	USB2_PHY	USB2	USB_CAMERA_N	42
PCH USB Compensation					
I384		PCH_50S	COMP_PCH	PCH_USB_RBIAS	20

	Electrical Contraint Set	Physical	Spacing	Voltage	Net	
Camera Controller Local Ground						
I521		GND_PHY	GND	0V	CAM_AGND	42
I522		GND_PHY	GND	0V	CAM_PLLGND	42

USB Hub

	Electrical Contraint Set	Physical	Spacing	Net	
USB 2.0 Hub					
I401	USB2_HUB_PCH	USB2_PHY	USB2	USB_PCH_7_P	20 27
I402	USB2_HUB_PCH	USB2_PHY	USB2	USB_PCH_7_N	20 27
I409	USB2_HUB_BT	USB2_PHY	USB2	USB_BT_P	27 35
I410	USB2_HUB_BT	USB2_PHY	USB2	USB_BT_N	27 35
I505	USB2_HUB_BT	USB2_PHY	USB2	USB_BT_MUX_P	35
I506	USB2_HUB_BT	USB2_PHY	USB2	USB_BT_MUX_N	35
I541		USB_HUB_PHY	USB_HUB	USB_HUB_2P	27
I542		USB_HUB_PHY	USB_HUB	USB_HUB_2N	27
USB 2.0 Hub Misc.					
I516		USB_HUB_PHY	USB_HUB	USB_HUB_RESET_L	27
I520		USB_HUB_PHY	USB_HUB	USB_HUB_VBUS_DET	27
I519		USB_HUB_PHY	USB_HUB	USB_HUB_NON_REM0	27
I518		USB_HUB_PHY	USB_HUB	USB_HUB_NON_REM1	27
I517		USB_HUB_PHY	USB_HUB	USB_HUB_HS_IND	27
USB 2.0 Hub Compensation					
I471		PCH_50S	COMP_PCH	USB_HUB_RBIAS	27
USB 2.0 Hub Crystal					
I472		CLK_XTAL	XTAL	USB_HUB_XTAL1	27
I473		CLK_XTAL	XTAL	USB_HUB_XTAL2	27
I515		CLK_XTAL	XTAL	USB_HUB_XTAL2_R	27

Camera Controller

	Electrical Contraint Set	Physical	Spacing	Net	
SMIA					
I501	SMIA_DATA	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_DATA_P	42
I502	SMIA_DATA	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_DATA_N	42
I528	SMIA_CLK	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_CLK_P	42
I529	SMIA_CLK	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_CLK_N	42
MISC					
I507		SPI_50S	SPI	CAM_SF_CLK	42
I508		SPI_50S	SPI	CAM_SF_CLK_R	42
I509		SPI_50S	SPI	CAM_SF_DIN	42
I510		SPI_50S	SPI	CAM_SF_DIN_R	42
I511		SPI_50S	SPI	CAM_SF_CS_L	42
I513		SPI_50S	SPI	CAM_SF_WP_L	42
I512		SPI_50S	SPI	CAM_SF_DOUT	42
I514		SPI_50S	SPI	CAM_SF_DOUT_R	42
I540		SPI_50S	SPI	CAM_SF_HOLD_L	42
I524		CAM_PHY	CAM	CAM_USB_VRES	42
I523		CAM_PHY	CAM	MIPI_RESISTOR	42
I539			PM	CAM_EXT_BOOT_L	43
I538			PM	PCH_CAM_EXT_BOOT_R_L	21 43
I537			PM	CAM_P1V2_RST_HOLDOFF	43
I536			PM	CAM_P1V2_RST_HOLDOFF_L	43
I2C					
I534		SMB_PHY	SMB	I2C_CAMSENSOR_SDA	42
I535		SMB_PHY	SMB	I2C_CAMSENSOR_SCL	42
I532		SMB_PHY	SMB	SMB_ALS_F_SDA	42
I533		SMB_PHY	SMB	SMB_ALS_F_SCL	42
Camera Controller Crystal					
I525		CLK_XTAL	XTAL	CAM_XTAL_IN	42
I526		CLK_XTAL	XTAL	CAM_XTAL_OUT	42
I527		CLK_XTAL	XTAL	CAM_XTAL_OUT_R	42

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE
USB/Camera Constraints

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

SMBus

SMBus-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMB_PHY	*	SMB_55S

SMBus-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB_ISO	*	=2x_DIELECTRIC	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMB	*	*	SMB_ISO

Sensor

Sensor-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM
SNS_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SNS_DIFF_PHY	*	1:1_DIFFPAIR
SNS_PHY	*	SNS_50S

Sensor-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SENSE	*	*	SENSE_ISO
SENSE	POWER	*	PWR_P2MM
SENSE	GND	*	GND_P2MM

DC-DC

Power-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GND_P3MM	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
GND_P5MM	*	Y	0.500 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
POWER_P3MM	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P6MM	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P5MM	*	Y	0.500 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VR_CTL_PHY	*	POWER_P5MM
VR_CTL_PHY	BGA	STANDARD
VR_VID_PHY	*	POWER_50S
POWER_PHY	*	POWER_P6MM
GND_PHY	*	GND_P5MM

Power-specific Spacing Definitions
Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_ISO	*	=STANDARD	?

DC-DC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SWNODE_ISO	*	=8:1_SPACING	1000
SWNODE_SW2SW	*	=1:1_SPACING	?
SWNODE_SW2PWR	*	=2:1_SPACING	?
SWNODE_SW2GND	*	=2:1_SPACING	?
SWNODE_LG2SW	*	=3:1_SPACING	?

DC-DC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL_ISO	*	=3:1_SPACING	?
VR_VID_ISO	*	=4X_DIELECTRIC	?

Constraints
Power and Common

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	POWER_ISO
GND	*	*	GND_ISO

DC-DC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_SWITCH	*	*	SWNODE_ISO
VR_SWITCH	*	BGA	BGA_P1MM
VR_SWITCH	VR_SWITCH	*	SWNODE_SW2SW
VR_SWITCH	POWER	*	SWNODE_SW2PWR
VR_SWITCH	GND	*	SWNODE_SW2GND
VR_LGATE	*	*	SWNODE_ISO
VR_LGATE	VR_SWITCH	*	SWNODE_LG2SW

DC-DC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_CTL	*	*	VR_CTL_ISO
VR_VID	*	*	VR_VID_ISO

CPU VccSA

Port	Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
	Input Bus							
I24		POWER_PHY	POWER	5V			REG_VCC_U7500	70
I25		POWER_PHY	POWER	5V			REG_PVCC_U7500	70
	Local Ground							
I26		GND_PHY	GND	0V			AGND_VCCSAS0	70
	VCCIO							
I27		POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VCCSAS0	70
I28		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_VCCSAS0	70
I29		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_VCCSAS0_RC	70
I31		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VCCSAS0	70
I30		VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_VCCSAS0	70
I33		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_VCCSAS0	70
I32		VR_CTL_PHY	VR_CTL				REG_VCCSAS0_OCSET	70
I35		VR_CTL_PHY	VR_CTL				REG_VCCSAS0_VO	70
I34			SENSE				SNS_CPU_VCCSA	13 70
I37	VSNS_CPU_VCCSA	SNS_DIFF_PHY	SENSE				SNS_VCCSAS0_XW_P	70
I36	VSNS_CPU_VCCSA	SNS_DIFF_PHY	SENSE				SNS_VCCSAS0_XW_N	70
I38			SENSE				REG_VCCSAS0_FB	70
I39			SENSE				REG_VCCSAS0_RTN	70
I40		VR_CTL_PHY	VR_CTL				REG_VCCSAS0_SREF	70
I41		VR_CTL_PHY	VR_CTL				REG_VCCSAS0_FSEL	70
	Output Bus							
I42		POWER_PHY	POWER	0.925V			PPVCCSA_S0	6

CPU VccIO PCH 1.05V S0

Port	Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
	Input Bus							
I1		POWER_PHY	POWER	5V			REG_VCC_U7400	69
I2		POWER_PHY	POWER	5V			REG_PVCC_U7400	69
	Local Ground							
I3		GND_PHY	GND	0V			AGND_CPU_P1V05S0	69
	1.05V S0							
I5		POWER_PHY	VR_SWITCH	12V	TRUE		REG_CPU_PHASE_P1V05S0	69
I7		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_CPU_BOOT_P1V05S0	69
I6		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_CPU_BOOT_P1V05S0_RC	69
I8		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_CPU_UGATE_P1V05S0	69
I9		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_CPU_UGATE_P1V05S0_R	
I10		VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_CPU_LGATE_P1V05S0	69
I12		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_CPU_SNUBBER_P1V05S0	69
I11		VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_OCSET	69
I14		VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_VO	69
I13	VSNS_CPU_VCCIO	SNS_DIFF_PHY	SENSE				SNS_CPU_VCCIO_P	13 69
I15	VSNS_CPU_VCCIO	SNS_DIFF_PHY	SENSE				SNS_CPU_VCCIO_N	13 69
I16		SNS_DIFF_PHY	SENSE				SNS_CPU_P1V05S0_XW_P	69
I17		SNS_DIFF_PHY	SENSE				SNS_CPU_P1V05S0_XW_N	69
I19			SENSE				REG_CPU_P1V05S0_FB	69
I18			SENSE				REG_CPU_P1V05S0_RTN	69
I21		VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_SREF	69
I20		VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_FSEL	69
	Output Bus							
I22		POWER_PHY	POWER	1.05V			PP1V05_S0_CPU	6

PCH/GPU/TBT 1.05V S0

Port	Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
	Input Bus							
I44		POWER_PHY	POWER	5V			REG_VCC_UB700	99
	Local Ground							
I46		GND_PHY	GND	0V			P1V05_AGND	99
	1.05V S0							
I47		POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V05S0	99
I48		POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V05S0_L	99
I49		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P1V05S0	99
I50		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P1V05S0_RC	99
I51		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P1V05S0	99
I52		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P1V05S0_R	99
I53		VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_P1V05S0	99
I54		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_P1V05S0	99
I55		VR_CTL_PHY	VR_CTL				REG_P1V05S0_OCSET	99
I56		VR_CTL_PHY	VR_CTL				REG_P1V05S0_VO	99
I72		POWER_PHY	VR_SWITCH	12V			REG_P1V05S0_VO_R	99
I73		POWER_PHY	VR_SWITCH	12V			P1V05_OCSET_R	99
I61			SENSE				REG_P1V05S0_FB	99
I62			SENSE				REG_P1V05S0_RTN	99
I63		VR_CTL_PHY	VR_CTL				REG_P1V05S0_SREF	99
I64		VR_CTL_PHY	VR_CTL				REG_P1V05S0_FSEL	99
	Output Bus							
I68		POWER_PHY	POWER	1.05V			PP1V05_S0_PCH	6
I66		POWER_PHY	POWER	1.05V			PP1V05_S0	6
	FET Switched							
I67		POWER_PHY	POWER	1.05V			PP1V05_TBTLC	6 38
I69		POWER_PHY	POWER	1.05V			PP1V05_TBTCIO	6

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE
VReg Constraints

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

CPU Core Phases

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus							
	POWER_PHY	POWER	12V			PP12V_S0_CPUCORE_FLT	66 67 68
	POWER_PHY	POWER	5V			REG_VCC_U7100	66
Local Ground							
	GND_PHY	GND	0V			AGND_CPU	66 67 68
Phase 1							
	POWER_PHY	POWER	12V			REG_LVCC_U7210	67
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1	66 67
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1_R	66
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_1	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_1	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_1_RC	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_1	67
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_1	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_1	67
	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_1	67
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_1_P	66 67
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_1_N	67
		SENSE				REG_ISENCORE_1_NR	66 67
Phase 2							
	POWER_PHY	POWER	12V			REG_LVCC_U7230	67
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2	66 67
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2_R	66
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_2	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_2	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_2_RC	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_2	67
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_2	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_2	67
	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_2	67
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_2_P	66 67
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_2_N	67
		SENSE				REG_ISENCORE_2_NR	66 67
Phase 3							
	POWER_PHY	POWER	12V			REG_LVCC_U7250	67
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3	66 67
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3_R	66
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_3	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_3	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_3_RC	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_3	67
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_3	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_3	67
	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_3	67
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_3_P	66 67
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_3_N	67
		SENSE				REG_ISENCORE_3_NR	66 67
Phase 4							
	POWER_PHY	POWER	12V			REG_LVCC_U7310	68
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_4	66 68
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_4_R	66
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_4	68
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_4	68
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_4_RC	68
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_4	68
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_4	68
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_4	68
	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_4	68
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_4_P	66 68
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_4_N	68
		SENSE				REG_ISENCORE_4_NR	66 68
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CORE_XW_P	66
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CORE_XW_N	66
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CORE_R_P	66
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CORE_R_N	66
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CPU_VAXG_P	13 66
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CPU_VAXG_N	13 66
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CPU_VCORE_P	13 66
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CPU_VCORE_N	13 66
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_P1V05_IOVDD_XW_P	99
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_P1V05_IOVDD_XW_N	99

CPU AXG Phase and Core Controller

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
AXG							
	POWER_PHY	POWER	12V			REG_LVCC_U7330	68
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG	66 68
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG_R	66
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUAXG	68
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUAXG	68
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUAXG_RC	68
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUAXG	68
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUAXG	68
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUAXG	68
	POWER_PHY	POWER	1.1V			PPCPUAXG_S0_SENSE	68
ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE				REG_ISENAXG_P	68
ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE				REG_ISENAXG_N	68
		SENSE				REG_ISENAXG_PR	66 68
		SENSE				REG_ISENAXG_NR	66 68
ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE				SNS_AXG_R_P	66
ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE				SNS_AXG_R_N	66
ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE				SNS_AXG_XW_P	66
ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE				SNS_AXG_XW_N	66
ISL6364							
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_COMP	66
	VR_CTL_PHY	VR_CTL				CPUCORE_COMP_RC	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FB	66
	VR_CTL_PHY	VR_CTL				CPUCORE_FB_RC	66
	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_1	66
	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_2	66
	VR_CTL_PHY	VR_CTL				CPUCORE_PSICOMP_RC	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_PSICOMP	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_HFCOMP	66
		SENSE				REG_CPUCORE_VSEN	66
		SENSE				REG_CPUCORE_RGND	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IMON	51 66
	VR_CTL_PHY	VR_CTL				CPUCORE_IMON_R	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_TM	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SUTH	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_NPSI	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FDVID	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IAUTO	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SW_FREQ	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RAMPADJ	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_EN_PWR	66
	VR_CTL_PHY	VR_CTL				CPUCORE_EN_PWR_R	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RSET	66
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_COMP	66
	VR_CTL_PHY	VR_CTL				CPUAXG_COMP_RC	66
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_FB	66
	VR_CTL_PHY	VR_CTL				CPUAXG_FB_RC	66
	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_1	66
	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_2	66
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_HFCOMP	66
		SENSE				REG_CPUAXG_VSEN	66
		SENSE				REG_CPUAXG_RGND	66
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_IMON	51 66
	VR_CTL_PHY	VR_CTL				CPUAXG_IMON_R	66
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TM	66
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TCOMP	66
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_SW_FREQ	66
	VR_VID_PHY	VR_VID				CPU_VIDSCLK	13 66
	VR_VID_PHY	VR_VID				CPU_VIDSCLK_R	13
	VR_VID_PHY	VR_VID				CPU_VIDALERT_L	13 66
	VR_VID_PHY	VR_VID				CPU_VIDALERT_R_L	13
	VR_VID_PHY	VR_VID				CPU_VIDSOUT	13 66
	VR_VID_PHY	VR_VID				CPU_VIDSOUT_R	13
Output Bus							
	POWER_PHY	POWER	1.1V			PPVCORE_S0_CPU	6
	POWER_PHY	POWER	1.1V			PPVAXG_S0	6

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE: CPU VReg Constraints

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH

3.3V S5/5V S4

Physical	Spacing	Voltage	DIDT	NO_TEST	Net	
Input Bus						
POWER_PHY	POWER	12V			REG_VIN_U7600	71
POWER_PHY	POWER	5V			REG_VCC1_U7600	71
POWER_PHY	POWER	5V			REG_VCC2_U7600	71
3.3V S5						
POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P3V3S5	71
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P3V3S5	71
VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P3V3S5_RC	71
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P3V3S5	71
VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_P3V3S5	71
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P3V3S5	71
VR_CTL_PHY	VR_CTL				REG_P3V3S5_ISEN	71
VR_CTL_PHY	VR_CTL				REG_P3V3S5_OCSET	71
VR_CTL_PHY	VR_CTL				REG_P3V3S5_FSET	71
VR_CTL_PHY	VR_CTL				REG_P3V3S5_VOUT	71
VR_CTL_PHY	VR_CTL				REG_P3V3S5_VOUT_R	71
VR_CTL_PHY	VR_CTL				REG_P3V3S5_FB	71
5V S3						
POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P5VS4	71
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P5VS4	71
VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P5VS4_RC	71
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P5VS4	71
VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_P5VS4	71
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P5VS4	71
VR_CTL_PHY	VR_CTL				REG_P5VS4_ISEN	71
VR_CTL_PHY	VR_CTL				REG_P5VS4_OCSET	71
VR_CTL_PHY	VR_CTL				REG_P5VS4_FSET	71
VR_CTL_PHY	VR_CTL				REG_P5VS4_VOUT	71
VR_CTL_PHY	VR_CTL				REG_P5VS4_VOUT_R	71
VR_CTL_PHY	VR_CTL				REG_P5VS4_FB	71
Output Bus						
POWER_PHY	POWER	5V			PP5V_S5	6
POWER_PHY	POWER	5V			PP5V_S4	6
POWER_PHY	POWER	3.3V			PP3V3_S5	6
FET Switched						
POWER_PHY	POWER	5V			PP5V_S0	6
POWER_PHY	POWER	3.3V			PP3V3_S4	6 60
POWER_PHY	POWER	3.3V			PP3V3_S0	6
POWER_PHY	POWER	3.3V			PP3V3_S4_ENET	6
POWER_PHY	POWER	3.3V			PP3V3_TBTLC	6
POWER_PHY	POWER	3.3V			PP3V3_S4_AP	6
POWER_PHY	POWER	3.3V			PP3V3_S0_SSD	6

VDDQ S3 (1.5V)/VTT S0

Physical	Spacing	Voltage	DIDT	NO_TEST	Net	
Input Bus						
POWER_PHY	POWER	5V			REG_V5IN_U7700	72
Local Ground						
GND_PHY	GND	0V			AGND_VDDQS3	72
VDDQ S3						
POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3	72
POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3_L	72
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_VDDQS3	72
VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_VDDQS3_RC	72
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3	72
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3_R	72
VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_VDDQS3	72
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_VDDQS3	72
POWER_PHY	POWER	1.5V			PPVDDQ_S3_SENSE	
VR_CTL_PHY	VR_CTL				REG_VDDQS3_VDDQSNS	72
VR_CTL_PHY	VR_CTL				REG_VDDQS3_VREF	72
VR_CTL_PHY	VR_CTL				REG_VDDQS3_REFIN	72
VR_CTL_PHY	VR_CTL				REG_VDDQS3_MODE	72
VR_CTL_PHY	VR_CTL				REG_VDDQS3_TRIP	72
VR_CTL_PHY	VR_CTL				LDO_DDRVTTS0_SNS	72
Output Bus						
POWER_PHY	POWER	1.5V			PPVDDQ_S3	6
POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVTT_S0	6
FET Switched						
POWER_PHY	POWER	1.5V			PP1V5_S0	6
Sensed						
POWER_PHY	POWER	1.5V			PPVDDQ_S3_DDR	6
POWER_PHY	POWER	1.5V			PP1V5_S0_CPU_MEM	6
POWER_PHY	POWER	1.5V			PPFBVDDQ_S0_GPU	6

DDR3 Vref

Physical	Spacing	Voltage	DIDT	NO_TEST	Net	
Memory Vref						
POWER_PHY	POWER	3.3V			PP3V3_S4_VREFMRGN_DAC	34
POWER_PHY	POWER	3.3V			PP3V3_S4_VREFMRGN_CTRL	34
POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVREF_DQ_MEM_A_S3	6
POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVREF_DQ_MEM_B_S3	6
POWER_DDR_PHY	POWER_DDR	0.75V			CPU_DIMM_VREF_DAC_A	11 34
POWER_DDR_PHY	POWER_DDR	0.75V			CPU_DIMM_VREF_DAC_B	11 34
POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVREF_CA_MEM_A_S3	6
POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVREF_CA_MEM_B_S3	6
POWER_DDR_PHY	POWER_DDR	0.75V			CPU_DDR_VREF	11
POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVTT_S3	6

1.8V S0

Physical	Spacing	Voltage	DIDT	NO_TEST	Net	
1.8V S0						
POWER_PHY	VR_SWITCH	5V	TRUE		REG_PHASE_P1V8S0	72
VR_CTL_PHY	VR_CTL				REG_P1V8S0_VFB	72
VR_CTL_PHY	VR_CTL				REG_P1V8S0_SYNCH	72
Output Bus						
POWER_PHY	POWER	1.8V			PP1V8_S0	6

HDD S0

Physical	Spacing	Voltage	DIDT	NO_TEST	Net	
HDD S0						
POWER_PHY	POWER	5V			PP5V_S0_HDD	6

12V S5

Physical	Spacing	Voltage	DIDT	NO_TEST	Net	
Input Bus						
POWER_PHY	POWER	12V			PP12V_ACDC	6
FET Switched						
POWER_PHY	POWER	12V			PP12V_S0	6
Sensed						
POWER_PHY	POWER	12V			PP12V_S5	6
POWER_PHY	POWER	12V			PP12V_G3H	6
POWER_PHY	POWER	12V			PP12V_G3H_P3V42	73
POWER_PHY	POWER	12V			PP12V_S0_FBVDDQ	6
POWER_PHY	POWER	12V			PP12V_S0_CPU_P1V05	6
POWER_PHY	POWER	12V			PP12V_S0_VCCSA	6
POWER_PHY	POWER	12V			PP12V_S0_P1V05	6
POWER_PHY	POWER	12V			PP12V_S0_HDD	6
POWER_PHY	POWER	12V			PP12V_S0_BLC	6

Ground/Common

Physical	Spacing	Voltage	DIDT	NO_TEST	Net
Common					
GND_PHY	GND	0V			GND

3V42 S0

Physical	Spacing	Voltage	DIDT	NO_TEST	Net	
3V42 S0						
POWER_PHY	VR_SWITCH	5V	TRUE		P3V42G3H_SW	73
VR_CTL_PHY	VR_CTL				P3V42G3H_FB	73
VR_CTL_PHY	VR_CTL				P3V42G3H_SHDN_L	73
POWER_PHY	POWER	3.3V			PP3V3_G3	6
POWER_PHY	POWER	3.3V			PP3V42_G3H	6

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE
Platform VReg Constraints

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

Thunderbolt

Thunderbolt-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBT_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBTDP_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	0.075MM
TBT_GEN_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Thunderbolt-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_I2C_ISO	*	=2x_DIELECTRIC	?
TBT_SPI_ISO	*	=2x_DIELECTRIC	?
TBTDP_ISO	*	=5x_DIELECTRIC	?
TBTDP_ISO	TOP,BOTTOM	=7x_DIELECTRIC	?
TBT_GEN_ISO	*	=2X_DIELECTRIC	?
BGA_TBT_AREA	*	0.075MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_TBT	BGA_TBT_AREA
TBT_I2C	*	*	TBT_I2C_ISO
TBT_SPI	*	*	TBT_SPI_ISO
TBTDP	*	*	TBTDP_ISO
TBT_GEN	*	*	TBT_GEN_ISO

SOURCE: Bill Cornelius's T29 Routing Notes

DisplayPort

DP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

DP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_ISO	*	=6.7X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DISPLAYPORT	*	*	DP_ISO

MAX LENGTH OF DISPLAYPORT TRACES: 6 INCHES

DISPLAYPORT INTRA-PAIR MATCHING SHOULD BE 1 PS. INTER-PAIR MATCHING SHOULD BE WITHIN 150 PS.
DISPLAYPORT AUX CHANNEL INTRA-PAIR MATCHING SHOULD BE 5 PS.

TBT IC Net Properties

	Electrical Contraint Set	Physical	Spacing	Net		
I511		DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_P<3..0>	NO_TEST=TRUE	36 90
I513		DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_N<3..0>	NO_TEST=TRUE	36 90
I512	DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_P<3..0>	NO_TEST=TRUE	36
I514	DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_N<3..0>	NO_TEST=TRUE	36
I516		DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_P		36 90
I515		DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_N		36 90
I517	DP_TBTSNK0_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_P		36
I518	DP_TBTSNK0_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_N		36
I519		DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_P<3..0>	NO_TEST=TRUE	36 90
I521		DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_N<3..0>	NO_TEST=TRUE	36 90
I520	DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_P<3..0>	NO_TEST=TRUE	36
I523	DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_N<3..0>	NO_TEST=TRUE	36
I522		DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_P		36 90
I524		DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_N		36 90
I526	DP_TBTSNK1_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_P		36
I525	DP_TBTSNK1_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_N		36
I603	DP_INTPNL_TBT_ML_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_P<3..0>	NO_TEST=TRUE	75
I604	DP_INTPNL_TBT_ML_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_N<3..0>	NO_TEST=TRUE	75
I686	DP_INTPNL_TBT_ML_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_P<3..0>	NO_TEST=TRUE	75
I687	DP_INTPNL_TBT_ML_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_N<3..0>	NO_TEST=TRUE	75
I529	DP_INTPNL_TBT_AUX_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_P		75
I530	DP_INTPNL_TBT_AUX_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_N		75
I601		DP_85D	DISPLAYPORT	DP_TBTSRC_AUX_C_P		75
I602		DP_85D	DISPLAYPORT	DP_TBTSRC_AUX_C_N		75
I531		TBT_I2C_55S	TBT_I2C	I2C_TBTRTR_SCL		36 50
I532		TBT_I2C_55S	TBT_I2C	I2C_TBTRTR_SDA		36 50
I533	TBT_SPI_CLK	TBT_I2C_55S	TBT_SPI	TBT_SPI_CLK		36
I534	TBT_SPI_MOSI	TBT_I2C_55S	TBT_SPI	TBT_SPI_MOSI		36
I538	TBT_SPI_MISO	TBT_I2C_55S	TBT_SPI	TBT_SPI_MISO		36
I535	TBT_SPI_CS_L	TBT_I2C_55S	TBT_SPI	TBT_SPI_CS_L		36
I688		TBT_GEN_55S	TBT_GEN	TBT_A_CONFIG1_BUF		36 77
I689		TBT_GEN_55S	TBT_GEN	TBT_A_CONFIG2_RC		36 77
I691		TBT_GEN_55S	TBT_GEN	TBT_B_CONFIG1_BUF		36 79
I690		TBT_GEN_55S	TBT_GEN	TBT_B_CONFIG2_RC		36 79
I693		TBT_GEN_55S	TBT_GEN	TBT_A_LSTX		36 77
I692		TBT_GEN_55S	TBT_GEN	TBT_A_LSRX		36 77
I695		TBT_GEN_55S	TBT_GEN	TBT_B_LSTX		36 79
I694		TBT_GEN_55S	TBT_GEN	TBT_B_LSRX		36 79
I697		TBT_GEN_55S	TBT_GEN	DP_TBTSNK0_HPD		36 90
I696		TBT_GEN_55S	TBT_GEN	DP_TBTSNK1_HPD		36 90
I698		TBT_GEN_55S	TBT_GEN	DP_TBTSRC_HPD		36 75
I699		TBT_GEN_55S	TBT_GEN	DP_TBTSNK0_DDC_CLK		76 92
I701		TBT_GEN_55S	TBT_GEN	DP_TBTSNK0_DDC_DATA		76 92
I700		TBT_GEN_55S	TBT_GEN	DP_TBTSNK1_DDC_CLK		76 91
I702		TBT_GEN_55S	TBT_GEN	DP_TBTSNK1_DDC_DATA		76 91
I703		TBT_GEN_55S	TBT_GEN	VIDEO_ON		74 78
I704		TBT_GEN_55S	TBT_GEN	VIDEO_ON_L		5 78
I705		TBT_GEN_55S	TBT_GEN	BDV_BKL_PWM		48 75
I706		TBT_GEN_55S	TBT_GEN	GPU_LCD_BKLT_PWM		75 91
I707		TBT_GEN_55S	TBT_GEN	LCD_BL_PWM		75
I708		TBT_GEN_55S	TBT_GEN	LCD_BL_FILT		75
I709		TBT_GEN_55S	TBT_GEN	LCD_BKLT_PWM		75 80

*: Only used on hosts supporting T29 video-in

DisplayPort

	Electrical Contraint Set	Physical	Spacing	Net		
	Graphics Source					
I642	DP_INTPNL_EG_ML_MUX	DP_85D	DISPLAYPORT	DP_INT_EG_ML_P<3..0>	NO_TEST=TRUE	75 90
I643	DP_INTPNL_EG_ML_MUX	DP_85D	DISPLAYPORT	DP_INT_EG_ML_N<3..0>	NO_TEST=TRUE	75 90
I644	DP_INTPNL_EG_AUX_MUX	DP_85D	DISPLAYPORT	DP_INT_EG_AUX_P	NO_TEST=TRUE	75 90
I645	DP_INTPNL_EG_AUX_MUX	DP_85D	DISPLAYPORT	DP_INT_EG_AUX_N	NO_TEST=TRUE	75 90
I646		DP_85D	DISPLAYPORT	DP_INT_EG_AUX_C_P		75
I647		DP_85D	DISPLAYPORT	DP_INT_EG_AUX_C_N		75
	Internal Panel					
I605		DP_85D	DISPLAYPORT	DP_INTPNL_ML_C_P<3..0>	NO_TEST=TRUE	75
I606		DP_85D	DISPLAYPORT	DP_INTPNL_ML_C_N<3..0>	NO_TEST=TRUE	75
I607	DP_INTPNL_ML_CONN	DP_85D	DISPLAYPORT	DP_INTPNL_ML_P<3..0>	NO_TEST=TRUE	75 78
I608	DP_INTPNL_ML_CONN	DP_85D	DISPLAYPORT	DP_INTPNL_ML_N<3..0>	NO_TEST=TRUE	75 78
I609	DP_INTPNL_AUX_CONN	DP_85D	DISPLAYPORT	DP_INTPNL_AUX_P		75 78
I610	DP_INTPNL_AUX_CONN	DP_85D	DISPLAYPORT	DP_INTPNL_AUX_N		75 78
	Internal DP SPDIF					
I611			HDA	DP_INT_SPDIF_AUDIO		56 78

TBT/DP Net Properties

	Electrical Contraint Set	Physical	Spacing	Net		
	Port A					
I541	TBT_A_R2D	TBTDP_90D	TBTDP	TBT_A_R2D_C_P<0>	NO_TEST=TRUE	36 77
I542	TBT_A_R2D	TBTDP_90D	TBTDP	TBT_A_R2D_C_N<0>	NO_TEST=TRUE	36 77
I710	TBT_A_R2D_PINV	TBTDP_90D	TBTDP	TBT_A_R2D_C_N<1>	NO_TEST=TRUE	36 77
I711	TBT_A_R2D_PINV	TBTDP_90D	TBTDP	TBT_A_R2D_C_P<1>	NO_TEST=TRUE	36 77
I544		TBTDP_90D	TBTDP	TBT_A_R2D_P<1..0>	NO_TEST=TRUE	77
I543		TBTDP_90D	TBTDP	TBT_A_R2D_N<1..0>	NO_TEST=TRUE	77
I595	DP_TBTPA_ML1	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<1>	NO_TEST=TRUE	36 77
I596	DP_TBTPA_ML1	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<1>	NO_TEST=TRUE	36 77
I680	DP_TBTPA_ML3	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<3>	NO_TEST=TRUE	36 77
I681	DP_TBTPA_ML3	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<3>	NO_TEST=TRUE	36 77
I545		DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<1>	NO_TEST=TRUE	77
I547		DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<1>	NO_TEST=TRUE	77
I678		DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<3>	NO_TEST=TRUE	77
I679		DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<3>	NO_TEST=TRUE	77
I546		DP_85D	DISPLAYPORT	DP_A_LSX_ML_P<1>		77
I548		DP_85D	DISPLAYPORT	DP_A_LSX_ML_N<1>		77
I549		TBTDP_90D	TBTDP	TBT_A_D2R_C_P<1..0>	NO_TEST=TRUE	77
I550		TBTDP_90D	TBTDP	TBT_A_D2R_C_N<1..0>	NO_TEST=TRUE	77
I564	TBT_A_D2R1	TBTDP_90D	TBTDP	TBT_A_D2R_P<1>	NO_TEST=TRUE	36 77
I563	TBT_A_D2R1	TBTDP_90D	TBTDP	TBT_A_D2R_N<1>	NO_TEST=TRUE	36 77
I675	TBT_A_D2R0	TBTDP_90D	TBTDP	TBT_A_D2R_P<0>	NO_TEST=TRUE	36 77
I674	TBT_A_D2R0	TBTDP_90D	TBTDP	TBT_A_D2R_N<0>	NO_TEST=TRUE	36 77
I599	DP_TBT_A_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_C_P		36 77
I600	DP_TBT_A_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_C_N		36 77
I565		DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_P		77
I566		DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_N		77
I648		DP_85D	DISPLAYPORT	DP_A_AUXCH_DDC_P		77
I649		DP_85D	DISPLAYPORT	DP_A_AUXCH_DDC_N		77
I650		TBTDP_90D	TBTDP	TBT_A_D2R1_AUXDDC_P	NO_TEST=TRUE	77
I651		TBTDP_90D	TBTDP	TBT_A_D2R1_AUXDDC_N	NO_TEST=TRUE	77
	Port B					
I652	TBT_B_R2D	TBTDP_90D	TBTDP	TBT_B_R2D_C_P<1..0>	NO_TEST=TRUE	36 79
I653	TBT_B_R2D	TBTDP_90D	TBTDP	TBT_B_R2D_C_N<1..0>	NO_TEST=TRUE	36 79
I655		TBTDP_90D	TBTDP	TBT_B_R2D_P<1..0>	NO_TEST=TRUE	79
I654		TBTDP_90D	TBTDP	TBT_B_R2D_N<1..0>	NO_TEST=TRUE	79
I657	DP_TBTPB_ML1	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<1>	NO_TEST=TRUE	36 79
I658	DP_TBTPB_ML1	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<1>	NO_TEST=TRUE	36 79
I685	DP_TBTPB_ML3	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<3>	NO_TEST=TRUE	36 79
I684	DP_TBTPB_ML3	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<3>	NO_TEST=TRUE	36 79
I683		DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<1>	NO_TEST=TRUE	79
I682		DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<1>	NO_TEST=TRUE	79
I656		DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<3>	NO_TEST=TRUE	79
I660		DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<3>	NO_TEST=TRUE	79
I659		DP_85D	DISPLAYPORT	DP_B_LSX_ML_P<1>	NO_TEST=TRUE	79
I662		DP_85D	DISPLAYPORT	DP_B_LSX_ML_N<1>	NO_TEST=TRUE	79
I661		TBTDP_90D	TBTDP	TBT_B_D2R_C_P<1..0>	NO_TEST=TRUE	79
I663		TBTDP_90D	TBTDP	TBT_B_D2R_C_N<1..0>	NO_TEST=TRUE	79
I665	TBT_B_D2R1	TBTDP_90D	TBTDP	TBT_B_D2R_P<1>	NO_TEST=TRUE	36 79
I664	TBT_B_D2R1	TBTDP_90D	TBTDP	TBT_B_D2R_N<1>	NO_TEST=TRUE	36 79
I676	TBT_B_D2R0	TBTDP_90D	TBTDP	TBT_B_D2R_P<0>	NO_TEST=TRUE	36 79
I677	TBT_B_D2R0	TBTDP_90D	TBTDP	TBT_B_D2R_N<0>	NO_TEST=TRUE	36 79
I666	DP_TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_C_P		36 79
I667	DP_TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_C_N		36 79
I669		DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_P		79
I670		DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_N		79
I668		DP_85D	DISPLAYPORT	DP_B_AUXCH_DDC_P		79
I672		DP_85D	DISPLAYPORT	DP_B_AUXCH_DDC_N		79
I671		TBTDP_90D	TBTDP	TBT_B_D2R1_AUXDDC_P	NO_TEST=TRUE	79
I673		TBTDP_90D	TBTDP	TBT_B_D2R1_AUXDDC_N	NO_TEST=TRUE	79

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE: TBT/DP Constraints

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH

GDDR5

GDDR5-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12 MM	=STANDARD	=STANDARD
GDDR_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GDDR_MA_PHY	*	GDDR_45S
GDDR_ADBI_PHY	*	GDDR_45S
GDDR_CTRL_PHY	*	GDDR_45S
GDDR_CLK_PHY	*	GDDR_80D
GDDR_DQ_PHY	*	GDDR_45S
GDDR_EDC_PHY	*	GDDR_45S
GDDR_DBI_PHY	*	GDDR_45S
GDDR_WCK_PHY	*	GDDR_80D

Main Segment Min Spacing Rules for 4.5 Gbps or Less (AMD Doc# 49919)

Table	Trace-to-Trace Micro	Trace-to-Trace Design	Strip	Design	Isolation Micro	Isolation Design	Strip	Design	Comments
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Memory address (MA). Implented 4.5 Gbps or less rules for K70.
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Address dynamic bus inversion (ADBI)
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Control (CTRL)
5-6/5-7	5:1	5:1	5:1	5:1	5:1	5:1	5:1	5:1	Clock (CLK)
5-6/5-7	3:1	3:1	3:1	3:1	5:1	5:1	5:1	5:1	Data (DQ)
5-6/5-7	7:1	7:1	7:1	7:1	7:1	7:1	7:1	7:1	Error detection pins (EDC). Using larger isolation rules,
5-6/5-7	3:1	3:1	3:1	3:1	5:1	5:1	5:1	5:1	Data dynamic bus inversion (DBI)
5-6/5-7	5:1	5:1	5:1	5:1	5:1	5:1	5:1	5:1	Forwarded clock (WCK)

GDDR5-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_ISO	*	=3X_DIELECTRIC	?
GDDR_ISO	TOP,BOTTOM	=5x_DIELECTRIC	?
GDDR_MA2MA	*	=2x_DIELECTRIC	?
GDDR_MA2MA	TOP,BOTTOM	=3X_DIELECTRIC	?
GDDR_ADBI2ADBI	*	=2x_DIELECTRIC	?
GDDR_ADBI2ADBI	TOP,BOTTOM	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	*	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	TOP,BOTTOM	=2x_DIELECTRIC	?
GDDR_CLK2CLK	*	=3X_DIELECTRIC	?
GDDR_CLK2CLK	TOP,BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_DQ2DQ	*	=3x_DIELECTRIC	?
GDDR_DQ2DQ	TOP,BOTTOM	=3x_DIELECTRIC	?
GDDR_EDC_ISO	*	=3X_DIELECTRIC	?
GDDR_EDC_ISO	TOP,BOTTOM	=5X_DIELECTRIC	?
GDDR_EDC2EDC	*	=3X_DIELECTRIC	?
GDDR_EDC2EDC	TOP,BOTTOM	=5X_DIELECTRIC	?
GDDR_DBI2DBI	*	=3x_DIELECTRIC	?
GDDR_DBI2DBI	TOP,BOTTOM	=3x_DIELECTRIC	?
GDDR_WCK2WCK	*	=3X_DIELECTRIC	?
GDDR_WCK2WCK	TOP,BOTTOM	=5x_DIELECTRIC	?

Constraints (x in {A, B}, y in {0, 1})

Memory Address: MAxy[8:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_MA	*	*	GDDR_ISO
GDDR_*_*_MA	=SAME	*	GDDR_MA2MA

Data: DQxy[31:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_DQ	*	*	GDDR_ISO
GDDR_*_*_DQ	=SAME	*	GDDR_DQ2DQ

Address Dynamic Bus Inversion: ADBIxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_ADBI	*	*	GDDR_ISO
GDDR_*_*_ADBI	=SAME	*	GDDR_ADBI2ADBI

Error Detection: EDCxy[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_EDC	*	*	GDDR_EDC_ISO
GDDR_*_*_EDC	=SAME	*	GDDR_EDC2EDC

Control: Reset, CKExy, CSxy, WExy, RASxy, CASxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	=SAME	*	GDDR_CTRL2CTRL

Data Dynamic Bus Inversion: DDBIxy[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_DBI	*	*	GDDR_ISO
GDDR_*_*_DBI	=SAME	*	GDDR_DBI2DBI

Clock: CKxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_CLK	*	*	GDDR_ISO
GDDR_*_*_CLK	=SAME	*	GDDR_CLK2CLK

Forwarded Clock: WCKxy[1:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_WCK	*	*	GDDR_ISO
GDDR_*_*_WCK	=SAME	*	GDDR_WCK2WCK

GPU

GPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_GPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

GPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_GPU_ISO	*	=4:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_GPU	*	*	CLK_GPU_ISO

GDDR5 Frame Buffer A

	Electrical Contraint Set	Physical	Spacing	Net		Pages
	Memory Address					
I642	GDDR_A0_MA	GDDR_MA_PHY	GDDR_A_0_MA	FB_A0_A<8..0>	NO_TEST=TRUE	84 86
I643	GDDR_A1_MA	GDDR_MA_PHY	GDDR_A_1_MA	FB_A1_A<8..0>	NO_TEST=TRUE	84 86
	Address Dynamic Bus Inv					
I644	GDDR_A0_ADBI	GDDR_ADBI_PHY	GDDR_A_0_ADBI	FB_A0_ABI_L	NO_TEST=TRUE	84 86
I645	GDDR_A1_ADBI	GDDR_ADBI_PHY	GDDR_A_1_ADBI	FB_A1_ABI_L	NO_TEST=TRUE	84 86
	Control					
I647	GDDR_A0_CKE	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_CKE_L	NO_TEST=TRUE	84 86
I648	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_CS_L	NO_TEST=TRUE	84 86
I649	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_WE_L	NO_TEST=TRUE	84 86
I650	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_CAS_L	NO_TEST=TRUE	84 86
I651	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_RAS_L	NO_TEST=TRUE	84 86
I652	GDDR_A1_CKE	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_CKE_L	NO_TEST=TRUE	84 86
I654	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_CS_L	NO_TEST=TRUE	84 86
I653	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_WE_L	NO_TEST=TRUE	84 86
I655	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_CAS_L	NO_TEST=TRUE	84 86
I656	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_RAS_L	NO_TEST=TRUE	84 86
	Clock					
I657	GDDR_A0_CLK	GDDR_CLK_PHY	GDDR_A_0_CLK	FB_A0_CLK_P	NO_TEST=TRUE	84 86
I658	GDDR_A0_CLK	GDDR_CLK_PHY	GDDR_A_0_CLK	FB_A0_CLK_N	NO_TEST=TRUE	84 86
I660	GDDR_A1_CLK	GDDR_CLK_PHY	GDDR_A_1_CLK	FB_A1_CLK_P	NO_TEST=TRUE	84 86
I659	GDDR_A1_CLK	GDDR_CLK_PHY	GDDR_A_1_CLK	FB_A1_CLK_N	NO_TEST=TRUE	84 86
	Data					
I661	GDDR_A0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<7..0>	NO_TEST=TRUE	84 86
I663	GDDR_A0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<15..8>	NO_TEST=TRUE	84 86
I662	GDDR_A0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<23..16>	NO_TEST=TRUE	84 86
I664	GDDR_A0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<31..24>	NO_TEST=TRUE	84 86
I666	GDDR_A1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<7..0>	NO_TEST=TRUE	84 86
I665	GDDR_A1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<15..8>	NO_TEST=TRUE	84 86
I667	GDDR_A1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<23..16>	NO_TEST=TRUE	84 86
I668	GDDR_A1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<31..24>	NO_TEST=TRUE	84 86
	Error Detection					
I670	GDDR_A0_EDC0	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<0>	NO_TEST=TRUE	84 86
I669	GDDR_A0_EDC1	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<1>	NO_TEST=TRUE	84 86
I671	GDDR_A0_EDC2	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<2>	NO_TEST=TRUE	84 86
I672	GDDR_A0_EDC3	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<3>	NO_TEST=TRUE	84 86
I673	GDDR_A1_EDC0	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<0>	NO_TEST=TRUE	84 86
I675	GDDR_A1_EDC1	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<1>	NO_TEST=TRUE	84 86
I674	GDDR_A1_EDC2	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<2>	NO_TEST=TRUE	84 86
I676	GDDR_A1_EDC3	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<3>	NO_TEST=TRUE	84 86
	Data Dynamic Bus Inv					
I677	GDDR_A0_DBI0	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<0>	NO_TEST=TRUE	84 86
I678	GDDR_A0_DBI1	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<1>	NO_TEST=TRUE	84 86
I680	GDDR_A0_DBI2	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<2>	NO_TEST=TRUE	84 86
I679	GDDR_A0_DBI3	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<3>	NO_TEST=TRUE	84 86
I682	GDDR_A1_DBI0	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<0>	NO_TEST=TRUE	84 86
I681	GDDR_A1_DBI1	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<1>	NO_TEST=TRUE	84 86
I683	GDDR_A1_DBI2	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<2>	NO_TEST=TRUE	84 86
I684	GDDR_A1_DBI3	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<3>	NO_TEST=TRUE	84 86
	Forwarded Clock					
I685	GDDR_A0_WCK0	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_P<0>	NO_TEST=TRUE	84 86
I687	GDDR_A0_WCK0	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_N<0>	NO_TEST=TRUE	84 86
I686	GDDR_A0_WCK1	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_P<1>	NO_TEST=TRUE	84 86
I688	GDDR_A0_WCK1	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_N<1>	NO_TEST=TRUE	84 86
I689	GDDR_A1_WCK0	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_P<0>	NO_TEST=TRUE	84 86
I690	GDDR_A1_WCK0	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_N<0>	NO_TEST=TRUE	84 86
I692	GDDR_A1_WCK1	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_P<1>	NO_TEST=TRUE	84 86
I691	GDDR_A1_WCK1	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_N<1>	NO_TEST=TRUE	84 86

GPU

Electrical Contraint Set	Physical	Spacing	Net		Pages
Clocks					
	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_O_PL		83
	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_O_NG		83
	CLK_PCIE_PHY	CLK_PCIE	GPU_TESTMODE		83
	CLK_PCIE_PHY	CLK_PCIE	GPU_PEX_TERMP		83
	CLK_GPU_55S	CLK_GPU	FB_A0_CK_MID	NO_TEST=TRUE	86
	CLK_GPU_55S	CLK_GPU	FB_A1_CK_MID	NO_TEST=TRUE	86
	CLK_GPU_55S	CLK_GPU	FB_B0_CK_MID	NO_TEST=TRUE	87
	CLK_GPU_55S	CLK_GPU	FB_B1_CK_MID	NO_TEST=TRUE	87
	CLK_GPU_55S	CLK_GPU	FB_C0_CK_MID	NO_TEST=TRUE	88
	CLK_GPU_55S	CLK_GPU	FB_C1_CK_MID	NO_TEST=TRUE	88
	CLK_GPU_55S	CLK_GPU	FB_D0_CK_MID	NO_TEST=TRUE	89
	CLK_GPU_55S	CLK_GPU	FB_D1_CK_MID	NO_TEST=TRUE	89
	CLK_GPU_55S	CLK_GPU	GPU_JTAG_TCK		91
	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK		91
	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK_R		91
	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTAL_P		91
	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTAL_N		91
SMB					
	SMB_PHY	SMB	GPU_SMB_CLK		91
	SMB_PHY	SMB	GPU_SMB_DAT		91
	SMB_PHY	SMB	GPU_SMB_CLK_R		50 91
	SMB_PHY	SMB	GPU_SMB_DAT_R		50 91
PCIe Compensation					
	PCIE_50S	COMP_PCIE	FB_CAL_PD_VDDQ		94
	PCIE_50S	COMP_PCIE	FB_CAL_PU_GND		94
	PCIE_50S	COMP_PCIE	FB_CAL_TERM_GND		94

GDDR5 Frame Buffer B

	Electrical Contraint Set	Physical	Spacing	Net		Pages
	Memory Address					
I695	GDDR_B0_MA	GDDR_MA_PHY	GDDR_B_0_MA	FB_B0_A<8..0>	NO_TEST=TRUE	84 87
I694	GDDR_B1_MA	GDDR_MA_PHY	GDDR_B_1_MA	FB_B1_A<8..0>	NO_TEST=TRUE	84 87
	Address Dynamic Bus Inv					
I696	GDDR_B0_ADBI	GDDR_ADBI_PHY	GDDR_B_0_ADBI	FB_B0_ABI_L	NO_TEST=TRUE	84 87
I697	GDDR_B1_ADBI	GDDR_ADBI_PHY	GDDR_B_1_ADBI	FB_B1_ABI_L	NO_TEST=TRUE	84 87
	Control					
I698	GDDR_B0_CKE	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CKE_L	NO_TEST=TRUE	84 87
I701	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CS_L	NO_TEST=TRUE	84 87
I700	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_WE_L	NO_TEST=TRUE	84 87
I702	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CAS_L	NO_TEST=TRUE	84 87
I703	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_RAS_L	NO_TEST=TRUE	84 87
I704	GDDR_B1_CKE	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CKE_L	NO_TEST=TRUE	84 87
I705	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CS_L	NO_TEST=TRUE	84 87
I706	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_WE_L	NO_TEST=TRUE	84 87
I708	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CAS_L	NO_TEST=TRUE	84 87
I707	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_RAS_L	NO_TEST=TRUE	84 87
	Clock					
I693	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_P	NO_TEST=TRUE	84 87
I710	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_N	NO_TEST=TRUE	84 87
I709	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_P	NO_TEST=TRUE	84 87
I711	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_N	NO_TEST=TRUE	84 87
	Data					
I713	GDDR_B0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<7..0>	NO_TEST=TRUE	84 87
I712	GDDR_B0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<15..8>	NO_TEST=TRUE	84 87
I714	GDDR_B0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<23..16>	NO_TEST=TRUE	84 87
I715	GDDR_B0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<31..24>	NO_TEST=TRUE	84 87
I716	GDDR_B1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<7..0>	NO_TEST=TRUE	84 87
I718	GDDR_B1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<15..8>	NO_TEST=TRUE	84 87
I717	GDDR_B1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<23..16>	NO_TEST=TRUE	84 87
I719	GDDR_B1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<31..24>	NO_TEST=TRUE	84 87
	Error Detection					
I720	GDDR_B0_EDC0	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<0>	NO_TEST=TRUE	84 87
I721	GDDR_B0_EDC1	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<1>	NO_TEST=TRUE	84 87
I723	GDDR_B0_EDC2	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<2>	NO_TEST=TRUE	84 87
I722	GDDR_B0_EDC3	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<3>	NO_TEST=TRUE	84 87
I725	GDDR_B1_EDC0	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<0>	NO_TEST=TRUE	84 87
I724	GDDR_B1_EDC1	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<1>	NO_TEST=TRUE	84 87
I726	GDDR_B1_EDC2	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<2>	NO_TEST=TRUE	84 87
I727	GDDR_B1_EDC3	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<3>	NO_TEST=TRUE	84 87
	Data Dynamic Bus Inv					
I729	GDDR_B0_DBI0	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<0>	NO_TEST=TRUE	84 87
I728	GDDR_B0_DBI1	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<1>	NO_TEST=TRUE	84 87
I730	GDDR_B0_DBI2	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<2>	NO_TEST=TRUE	84 87
I731	GDDR_B0_DBI3	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<3>	NO_TEST=TRUE	84 87
I732	GDDR_B1_DBI0	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<0>	NO_TEST=TRUE	84 87
I733	GDDR_B1_DBI1	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<1>	NO_TEST=TRUE	84 87
I734	GDDR_B1_DBI2	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<2>	NO_TEST=TRUE	84 87
I735	GDDR_B1_DBI3	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<3>	NO_TEST=TRUE	84 87
	Forwarded Clock					
I736	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<0>	NO_TEST=TRUE	84 87
I737	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<0>	NO_TEST=TRUE	84 87
I739	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<1>	NO_TEST=TRUE	84 87
I738	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<1>	NO_TEST=TRUE	84 87
I740	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<0>	NO_TEST=TRUE	84 87
I741	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<0>	NO_TEST=TRUE	84 87
I743	GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<1>	NO_TEST=TRUE	84 87
I742	GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<1>	NO_TEST=TRUE	84 87

Frame Buffer Reset

Electrical Contraint Set	Physical	Spacing	Net	Pages
Reset				
GDDR_A0_RESET	GDDR_50S	GDDR_CTRL	FB_A0_RESET_L	84 86
GDDR_A1_RESET	GDDR_50S	GDDR_CTRL	FB_A1_RESET_L	84 86
GDDR_B0_RESET	GDDR_50S	GDDR_CTRL	FB_B0_RESET_L	84 87
GDDR_B1_RESET	GDDR_50S	GDDR_CTRL	FB_B1_RESET_L	84 87
GDDR_C0_RESET	GDDR_50S	GDDR_CTRL	FB_C0_RESET_L	85 88
GDDR_C1_RESET	GDDR_50S	GDDR_CTRL	FB_C1_RESET_L	85 88
GDDR_D0_RESET	GDDR_50S	GDDR_CTRL	FB_D0_RESET_L	85 89
GDDR_D1_RESET	GDDR_50S	GDDR_CTRL	FB_D1_RESET_L	85 89

SYNC_MASTER=D8_MLB SYNC_DATE=01/11/2012

PAGE TITLE: GDDR5/GPU Constraints

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH

GDDR5 FRAME BUFFER C

Ref	Electrical Contraint Set	Physical	Spacing	Net	Property	Pages
	Memory Address					
I642	GDDR_C0_MA	GDDR_MA_PHY	GDDR_C_0_MA	FB_C0_A<8..0>	NO_TEST=TRUE	85 88
I643	GDDR_C1_MA	GDDR_MA_PHY	GDDR_C_1_MA	FB_C1_A<8..0>	NO_TEST=TRUE	85 88
	Address Dynamic Bus Inv					
I644	GDDR_C0_ADBI	GDDR_ADBI_PHY	GDDR_C_0_ADBI	FB_C0_ABI_L	NO_TEST=TRUE	85 88
I645	GDDR_C1_ADBI	GDDR_ADBI_PHY	GDDR_C_1_ADBI	FB_C1_ABI_L	NO_TEST=TRUE	85 88
	Control					
I647	GDDR_C0_CKE	GDDR_CTRL_PHY	GDDR_C_0_CTRL	FB_C0_CKE_L	NO_TEST=TRUE	85 88
I648	GDDR_C0_CTRL	GDDR_CTRL_PHY	GDDR_C_0_CTRL	FB_C0_CS_L	NO_TEST=TRUE	85 88
I649	GDDR_C0_CTRL	GDDR_CTRL_PHY	GDDR_C_0_CTRL	FB_C0_WE_L	NO_TEST=TRUE	85 88
I650	GDDR_C0_CTRL	GDDR_CTRL_PHY	GDDR_C_0_CTRL	FB_C0_CAS_L	NO_TEST=TRUE	85 88
I651	GDDR_C0_CTRL	GDDR_CTRL_PHY	GDDR_C_0_CTRL	FB_C0_RAS_L	NO_TEST=TRUE	85 88
I652	GDDR_C1_CKE	GDDR_CTRL_PHY	GDDR_C_1_CTRL	FB_C1_CKE_L	NO_TEST=TRUE	85 88
I654	GDDR_C1_CTRL	GDDR_CTRL_PHY	GDDR_C_1_CTRL	FB_C1_CS_L	NO_TEST=TRUE	85 88
I653	GDDR_C1_CTRL	GDDR_CTRL_PHY	GDDR_C_1_CTRL	FB_C1_WE_L	NO_TEST=TRUE	85 88
I655	GDDR_C1_CTRL	GDDR_CTRL_PHY	GDDR_C_1_CTRL	FB_C1_CAS_L	NO_TEST=TRUE	85 88
I656	GDDR_C1_CTRL	GDDR_CTRL_PHY	GDDR_C_1_CTRL	FB_C1_RAS_L	NO_TEST=TRUE	85 88
	Clock					
I657	GDDR_C0_CLK	GDDR_CLK_PHY	GDDR_C_0_CLK	FB_C0_CLK_P	NO_TEST=TRUE	85 88
I658	GDDR_C0_CLK	GDDR_CLK_PHY	GDDR_C_0_CLK	FB_C0_CLK_N	NO_TEST=TRUE	85 88
I660	GDDR_C1_CLK	GDDR_CLK_PHY	GDDR_C_1_CLK	FB_C1_CLK_P	NO_TEST=TRUE	85 88
I659	GDDR_C1_CLK	GDDR_CLK_PHY	GDDR_C_1_CLK	FB_C1_CLK_N	NO_TEST=TRUE	85 88
	Data					
I661	GDDR_C0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_C_0_DQ	FB_C0_DQ<7..0>	NO_TEST=TRUE	85 88
I663	GDDR_C0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_C_0_DQ	FB_C0_DQ<15..8>	NO_TEST=TRUE	85 88
I662	GDDR_C0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_C_0_DQ	FB_C0_DQ<23..16>	NO_TEST=TRUE	85 88
I664	GDDR_C0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_C_0_DQ	FB_C0_DQ<31..24>	NO_TEST=TRUE	85 88
I666	GDDR_C1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_C_1_DQ	FB_C1_DQ<7..0>	NO_TEST=TRUE	85 88
I665	GDDR_C1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_C_1_DQ	FB_C1_DQ<15..8>	NO_TEST=TRUE	85 88
I667	GDDR_C1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_C_1_DQ	FB_C1_DQ<23..16>	NO_TEST=TRUE	85 88
I668	GDDR_C1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_C_1_DQ	FB_C1_DQ<31..24>	NO_TEST=TRUE	85 88
	Error Detection					
I670	GDDR_C0_EDC0	GDDR_EDC_PHY	GDDR_C_0_EDC	FB_C0_EDC<0>	NO_TEST=TRUE	85 88
I669	GDDR_C0_EDC1	GDDR_EDC_PHY	GDDR_C_0_EDC	FB_C0_EDC<1>	NO_TEST=TRUE	85 88
I671	GDDR_C0_EDC2	GDDR_EDC_PHY	GDDR_C_0_EDC	FB_C0_EDC<2>	NO_TEST=TRUE	85 88
I672	GDDR_C0_EDC3	GDDR_EDC_PHY	GDDR_C_0_EDC	FB_C0_EDC<3>	NO_TEST=TRUE	85 88
I673	GDDR_C1_EDC0	GDDR_EDC_PHY	GDDR_C_1_EDC	FB_C1_EDC<0>	NO_TEST=TRUE	85 88
I675	GDDR_C1_EDC1	GDDR_EDC_PHY	GDDR_C_1_EDC	FB_C1_EDC<1>	NO_TEST=TRUE	85 88
I674	GDDR_C1_EDC2	GDDR_EDC_PHY	GDDR_C_1_EDC	FB_C1_EDC<2>	NO_TEST=TRUE	85 88
I676	GDDR_C1_EDC3	GDDR_EDC_PHY	GDDR_C_1_EDC	FB_C1_EDC<3>	NO_TEST=TRUE	85 88
	Data Dynamic Bus Inv					
I677	GDDR_C0_DBI0	GDDR_DBI_PHY	GDDR_C_0_DBI	FB_C0_DBI_L<0>	NO_TEST=TRUE	85 88
I678	GDDR_C0_DBI1	GDDR_DBI_PHY	GDDR_C_0_DBI	FB_C0_DBI_L<1>	NO_TEST=TRUE	85 88
I680	GDDR_C0_DBI2	GDDR_DBI_PHY	GDDR_C_0_DBI	FB_C0_DBI_L<2>	NO_TEST=TRUE	85 88
I679	GDDR_C0_DBI3	GDDR_DBI_PHY	GDDR_C_0_DBI	FB_C0_DBI_L<3>	NO_TEST=TRUE	85 88
I682	GDDR_C1_DBI0	GDDR_DBI_PHY	GDDR_C_1_DBI	FB_C1_DBI_L<0>	NO_TEST=TRUE	85 88
I681	GDDR_C1_DBI1	GDDR_DBI_PHY	GDDR_C_1_DBI	FB_C1_DBI_L<1>	NO_TEST=TRUE	85 88
I683	GDDR_C1_DBI2	GDDR_DBI_PHY	GDDR_C_1_DBI	FB_C1_DBI_L<2>	NO_TEST=TRUE	85 88
I684	GDDR_C1_DBI3	GDDR_DBI_PHY	GDDR_C_1_DBI	FB_C1_DBI_L<3>	NO_TEST=TRUE	85 88
	Forwarded Clock					
I685	GDDR_C0_WCK0	GDDR_WCK_PHY	GDDR_C_0_WCK	FB_C0_WCLK_P<0>	NO_TEST=TRUE	85 88
I687	GDDR_C0_WCK0	GDDR_WCK_PHY	GDDR_C_0_WCK	FB_C0_WCLK_N<0>	NO_TEST=TRUE	85 88
I686	GDDR_C0_WCK1	GDDR_WCK_PHY	GDDR_C_0_WCK	FB_C0_WCLK_P<1>	NO_TEST=TRUE	85 88
I688	GDDR_C0_WCK1	GDDR_WCK_PHY	GDDR_C_0_WCK	FB_C0_WCLK_N<1>	NO_TEST=TRUE	85 88
I689	GDDR_C1_WCK0	GDDR_WCK_PHY	GDDR_C_1_WCK	FB_C1_WCLK_P<0>	NO_TEST=TRUE	85 88
I690	GDDR_C1_WCK0	GDDR_WCK_PHY	GDDR_C_1_WCK	FB_C1_WCLK_N<0>	NO_TEST=TRUE	85 88
I692	GDDR_C1_WCK1	GDDR_WCK_PHY	GDDR_C_1_WCK	FB_C1_WCLK_P<1>	NO_TEST=TRUE	85 88
I691	GDDR_C1_WCK1	GDDR_WCK_PHY	GDDR_C_1_WCK	FB_C1_WCLK_N<1>	NO_TEST=TRUE	85 88
	Memory Address					
I744	GDDR_A0_MA	GDDR_MA_PHY	GDDR_A_0_MA	FB_A0_A<12>	NO_TEST=TRUE	84 86
I745	GDDR_A1_MA	GDDR_MA_PHY	GDDR_A_1_MA	FB_A1_A<12>	NO_TEST=TRUE	84 86
I746	GDDR_B0_MA	GDDR_MA_PHY	GDDR_B_0_MA	FB_B0_A<12>	NO_TEST=TRUE	84 87
I747	GDDR_B1_MA	GDDR_MA_PHY	GDDR_B_1_MA	FB_B1_A<12>	NO_TEST=TRUE	84 87
I748	GDDR_C0_MA	GDDR_MA_PHY	GDDR_C_0_MA	FB_C0_A<12>	NO_TEST=TRUE	85 88
I749	GDDR_C1_MA	GDDR_MA_PHY	GDDR_C_1_MA	FB_C1_A<12>	NO_TEST=TRUE	85 88
I750	GDDR_D0_MA	GDDR_MA_PHY	GDDR_D_0_MA	FB_D0_A<12>	NO_TEST=TRUE	85 89
I751	GDDR_D1_MA	GDDR_MA_PHY	GDDR_D_1_MA	FB_D1_A<12>	NO_TEST=TRUE	85 89

GDDR5 FRAME BUFFER D

Ref	Electrical Contraint Set	Physical	Spacing	Net	Property	Pages
	Memory Address					
I695	GDDR_D0_MA	GDDR_MA_PHY	GDDR_D_0_MA	FB_D0_A<8..0>	NO_TEST=TRUE	85 89
I694	GDDR_D1_MA	GDDR_MA_PHY	GDDR_D_1_MA	FB_D1_A<8..0>	NO_TEST=TRUE	85 89
	Address Dynamic Bus Inv					
I696	GDDR_D0_ADBI	GDDR_ADBI_PHY	GDDR_D_0_ADBI	FB_D0_ABI_L	NO_TEST=TRUE	85 89
I697	GDDR_D1_ADBI	GDDR_ADBI_PHY	GDDR_D_1_ADBI	FB_D1_ABI_L	NO_TEST=TRUE	85 89
	Control					
I698	GDDR_D0_CKE	GDDR_CTRL_PHY	GDDR_D_0_CTRL	FB_D0_CKE_L	NO_TEST=TRUE	85 89
I701	GDDR_D0_CTRL	GDDR_CTRL_PHY	GDDR_D_0_CTRL	FB_D0_CS_L	NO_TEST=TRUE	85 89
I700	GDDR_D0_CTRL	GDDR_CTRL_PHY	GDDR_D_0_CTRL	FB_D0_WE_L	NO_TEST=TRUE	85 89
I702	GDDR_D0_CTRL	GDDR_CTRL_PHY	GDDR_D_0_CTRL	FB_D0_CAS_L	NO_TEST=TRUE	85 89
I703	GDDR_D0_CTRL	GDDR_CTRL_PHY	GDDR_D_0_CTRL	FB_D0_RAS_L	NO_TEST=TRUE	85 89
I704	GDDR_D1_CKE	GDDR_CTRL_PHY	GDDR_D_1_CTRL	FB_D1_CKE_L	NO_TEST=TRUE	85 89
I705	GDDR_D1_CTRL	GDDR_CTRL_PHY	GDDR_D_1_CTRL	FB_D1_CS_L	NO_TEST=TRUE	85 89
I706	GDDR_D1_CTRL	GDDR_CTRL_PHY	GDDR_D_1_CTRL	FB_D1_WE_L	NO_TEST=TRUE	85 89
I708	GDDR_D1_CTRL	GDDR_CTRL_PHY	GDDR_D_1_CTRL	FB_D1_CAS_L	NO_TEST=TRUE	85 89
I707	GDDR_D1_CTRL	GDDR_CTRL_PHY	GDDR_D_1_CTRL	FB_D1_RAS_L	NO_TEST=TRUE	85 89
	Clock					
I693	GDDR_D0_CLK	GDDR_CLK_PHY	GDDR_D_0_CLK	FB_D0_CLK_P	NO_TEST=TRUE	85 89
I710	GDDR_D0_CLK	GDDR_CLK_PHY	GDDR_D_0_CLK	FB_D0_CLK_N	NO_TEST=TRUE	85 89
I709	GDDR_D1_CLK	GDDR_CLK_PHY	GDDR_D_1_CLK	FB_D1_CLK_P	NO_TEST=TRUE	85 89
I711	GDDR_D1_CLK	GDDR_CLK_PHY	GDDR_D_1_CLK	FB_D1_CLK_N	NO_TEST=TRUE	85 89
	Data					
I713	GDDR_D0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_D_0_DQ	FB_D0_DQ<7..0>	NO_TEST=TRUE	85 89
I712	GDDR_D0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_D_0_DQ	FB_D0_DQ<15..8>	NO_TEST=TRUE	85 89
I714	GDDR_D0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_D_0_DQ	FB_D0_DQ<23..16>	NO_TEST=TRUE	85 89
I715	GDDR_D0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_D_0_DQ	FB_D0_DQ<31..24>	NO_TEST=TRUE	85 89
I716	GDDR_D1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_D_1_DQ	FB_D1_DQ<7..0>	NO_TEST=TRUE	85 89
I718	GDDR_D1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_D_1_DQ	FB_D1_DQ<15..8>	NO_TEST=TRUE	85 89
I717	GDDR_D1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_D_1_DQ	FB_D1_DQ<23..16>	NO_TEST=TRUE	85 89
I719	GDDR_D1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_D_1_DQ	FB_D1_DQ<31..24>	NO_TEST=TRUE	85 89
	Error Detection					
I720	GDDR_D0_EDC0	GDDR_EDC_PHY	GDDR_D_0_EDC	FB_D0_EDC<0>	NO_TEST=TRUE	85 89
I721	GDDR_D0_EDC1	GDDR_EDC_PHY	GDDR_D_0_EDC	FB_D0_EDC<1>	NO_TEST=TRUE	85 89
I723	GDDR_D0_EDC2	GDDR_EDC_PHY	GDDR_D_0_EDC	FB_D0_EDC<2>	NO_TEST=TRUE	85 89
I722	GDDR_D0_EDC3	GDDR_EDC_PHY	GDDR_D_0_EDC	FB_D0_EDC<3>	NO_TEST=TRUE	85 89
I725	GDDR_D1_EDC0	GDDR_EDC_PHY	GDDR_D_1_EDC	FB_D1_EDC<0>	NO_TEST=TRUE	85 89
I724	GDDR_D1_EDC1	GDDR_EDC_PHY	GDDR_D_1_EDC	FB_D1_EDC<1>	NO_TEST=TRUE	85 89
I726	GDDR_D1_EDC2	GDDR_EDC_PHY	GDDR_D_1_EDC	FB_D1_EDC<2>	NO_TEST=TRUE	85 89
I727	GDDR_D1_EDC3	GDDR_EDC_PHY	GDDR_D_1_EDC	FB_D1_EDC<3>	NO_TEST=TRUE	85 89
	Data Dynamic Bus Inv					
I729	GDDR_D0_DBI0	GDDR_DBI_PHY	GDDR_D_0_DBI	FB_D0_DBI_L<0>	NO_TEST=TRUE	85 89
I728	GDDR_D0_DBI1	GDDR_DBI_PHY	GDDR_D_0_DBI	FB_D0_DBI_L<1>	NO_TEST=TRUE	85 89
I730	GDDR_D0_DBI2	GDDR_DBI_PHY	GDDR_D_0_DBI	FB_D0_DBI_L<2>	NO_TEST=TRUE	85 89
I731	GDDR_D0_DBI3	GDDR_DBI_PHY	GDDR_D_0_DBI	FB_D0_DBI_L<3>	NO_TEST=TRUE	85 89
I732	GDDR_D1_DBI0	GDDR_DBI_PHY	GDDR_D_1_DBI	FB_D1_DBI_L<0>	NO_TEST=TRUE	85 89
I733	GDDR_D1_DBI1	GDDR_DBI_PHY	GDDR_D_1_DBI	FB_D1_DBI_L<1>	NO_TEST=TRUE	85 89
I734	GDDR_D1_DBI2	GDDR_DBI_PHY	GDDR_D_1_DBI	FB_D1_DBI_L<2>	NO_TEST=TRUE	85 89
I735	GDDR_D1_DBI3	GDDR_DBI_PHY	GDDR_D_1_DBI	FB_D1_DBI_L<3>	NO_TEST=TRUE	85 89
	Forwarded Clock					
I736	GDDR_D0_WCK0	GDDR_WCK_PHY	GDDR_D_0_WCK	FB_D0_WCLK_P<0>	NO_TEST=TRUE	85 89
I737	GDDR_D0_WCK0	GDDR_WCK_PHY	GDDR_D_0_WCK	FB_D0_WCLK_N<0>	NO_TEST=TRUE	85 89
I739	GDDR_D0_WCK1	GDDR_WCK_PHY	GDDR_D_0_WCK	FB_D0_WCLK_P<1>	NO_TEST=TRUE	85 89
I738	GDDR_D0_WCK1	GDDR_WCK_PHY	GDDR_D_0_WCK	FB_D0_WCLK_N<1>	NO_TEST=TRUE	85 89
I740	GDDR_D1_WCK0	GDDR_WCK_PHY	GDDR_D_1_WCK	FB_D1_WCLK_P<0>	NO_TEST=TRUE	85 89
I741	GDDR_D1_WCK0	GDDR_WCK_PHY	GDDR_D_1_WCK	FB_D1_WCLK_N<0>	NO_TEST=TRUE	85 89
I743	GDDR_D1_WCK1	GDDR_WCK_PHY	GDDR_D_1_WCK	FB_D1_WCLK_P<1>	NO_TEST=TRUE	85 89
I742	GDDR_D1_WCK1	GDDR_WCK_PHY	GDDR_D_1_WCK	FB_D1_WCLK_N<1>	NO_TEST=TRUE	85 89

SYNC_MASTER=D8_MLB SYNC_DATE=12/20/2011

PAGE TITLE
GDDR5 FB C/D CONSTRAINTS

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

Backlight Controller

BLC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
BLC_P6MM	*	Y	0.600 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD
BLC_P3MM	*	Y	0.300 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_BLC	*	BLC_P6MM
POWER_BLC_RET	*	BLC_P3MM
BLC_CTL_PHY	*	BLC_P3MM

BLC-specific Spacing Definitions
BLC High Voltage Output

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BLC_HV_ISO	*	1.00MM	1000

Constraints
BLC High Voltage Output

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_HV	BLC_HV	*	BLC_CTL_ISO
BLC_HV	*	*	BLC_HV_ISO

BLC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PHASE_ISO	*	=8:1_SPACING	2000
PHASE_SW2SW	*	=1:1_SPACING	?
PHASE_SW2PWR	*	=2:1_SPACING	?
PHASE_SW2GND	*	=2:1_SPACING	?

BLC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_PHASE	*	*	PHASE_ISO
BLC_PHASE	BLC_PHASE	*	PHASE_SW2SW
BLC_PHASE	POWER	*	PHASE_SW2PWR
BLC_PHASE	GND	*	PHASE_SW2GND

BLC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BLC_CTL_ISO	*	=3:1_SPACING	?

BLC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_CTL	*	*	BLC_CTL_ISO

BKLT MISCELLANEOUS

Electrical Contraint Set	Physical	Spacing	Net	Pages
SPI				
I564	SMB_PHY	SMB	SMB_PCH_BLC_SCL	50 80
I563	SMB_PHY	SMB	SMB_PCH_BLC_SDA	50 80
I812	SMB_PHY	SMB	SMB_TCON_BLC_SCL	50 80
I811	SMB_PHY	SMB	SMB_TCON_BLC_SDA	50 80
12M REFERENCE CRYSTAL				
I842	CLK_XTAL	XTAL	BLC_MCU_XTAL_IN	80
I843	CLK_XTAL	XTAL	BLC_MCU_XTAL_OUT	80
I844	CLK_XTAL	XTAL	BLC_MCU_XTAL_OUT_R	80
250K REFERENCE CLOCKS				
I845	CLK_PCH_50S	CLK_PCH	STRCLK_R1	80 81
I846	CLK_PCH_50S	CLK_PCH	LED_DRVR_CLK	81

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus							
I750	POWER_PHY	POWER	12V			PP12V_S0_BLC_VIN2	80 81 82
I772	POWER_PHY	POWER	12V			PP12V_S0_BLC_VINP	80 82
I806	POWER_PHY	POWER	14V			PRE_REG_OUT	80 81
I807	POWER_PHY	POWER	3.3V			BLC_P3V3S	80 82
I867	POWER_PHY	POWER	3.3V			BLC_P3V3_REF	80 82
I809	POWER_PHY	POWER	3.3V			BLC_P3V3	
I864	POWER_PHY	POWER	3.3V			PP3V3_S0_BLC_R	80
I841	POWER_PHY	POWER	8V			SPTX_VIN	81
I810	POWER_PHY	POWER	8V			PP8V_BLC	80 81
I847	POWER_PHY	POWER	8V			BLC_VIN2	81 82
I862	POWER_PHY	POWER	14V			BOOST_FET_DRAIN	80
I863	POWER_PHY	POWER	12V			BOOST_VDD	80
I865	POWER_PHY	POWER	5V			PP5V_S0_BLC_R	80 81 82
I866	POWER_PHY	POWER	12V			PP12V_S0_BLC_F	82
Local Ground							
I751	BLC_CTL_PHY	BLC_PHASE	0V			BLC_GND_1	81 82
I804	BLC_CTL_PHY	BLC_PHASE	0V			BLC_GND_2	81 82
I805	BLC_CTL_PHY	BLC_PHASE	0V			BLC_GND_3	81 82
I816	GND_PHY	GND	0V			AGND_BLC	81 82
Backlight							
I755	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE1	81
I756	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE1_R	81
I798	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE2	81
I797	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE2_R	81
I800	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE3	81
I799	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE3_R	81
I795	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_RC_1	81
I794	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_RC_2	81
I796	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_RC_3	81
I801	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS1	81
I803	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS2	81
I802	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS3	81
I828	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_C1	81
I827	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_C2	81
I826	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_C3	81
I833	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_1	81
I832	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_2	81
I831	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_3	81
I830	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK1	81
I834	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK2	81
I829	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK3	81
I835	BLC_CTL_PHY	BLC_CTL				BLC_PWM_1_R	80 81
I836	BLC_CTL_PHY	BLC_CTL				BLC_PWM_2_R	80 81
I837	BLC_CTL_PHY	BLC_CTL				BLC_PWM_3_R	80 81
I840	BLC_CTL_PHY	BLC_CTL				BLC_PWM_1	81 82
I839	BLC_CTL_PHY	BLC_CTL				BLC_PWM_2	81 82
I838	BLC_CTL_PHY	BLC_CTL				BLC_PWM_3	81 82
I819	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF1	81 82
I818	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF2	81 82
I817	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF3	81 82
I758	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP1	81
I759	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP2	81
I781	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP3	81
I822	BLC_CTL_PHY	BLC_CTL				BCOMP1	81
I821	BLC_CTL_PHY	BLC_CTL				BCOMP2	81
I820	BLC_CTL_PHY	BLC_CTL				BCOMP3	81
I849	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT1	81
I848	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT2	81
I850	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT3	81
I851	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_1	81
I852	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_2	81
I853	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_3	81
I856	BLC_CTL_PHY	BLC_CTL				PRE_REG_OUT_R	80
I855	BLC_CTL_PHY	BLC_CTL				BOOST_FB	80
I854	BLC_CTL_PHY	BLC_CTL				BOOST_COMP	80
I857	BLC_CTL_PHY	BLC_CTL				BOOST_COMP_C	80
I860	BLC_CTL_PHY	BLC_CTL			TRUE	BOOST_GDRV	80
I861	BLC_CTL_PHY	BLC_CTL			TRUE	BOOST_GDRV_R	80
I859	BLC_CTL_PHY	BLC_CTL				BOOST_ISNS	80
I858	BLC_CTL_PHY	BLC_CTL				BOOST_ISNS_R	80
I752	BLC_CTL_PHY	BLC_HV	80V			LED_DRVR_DRAIN_1	81
I753	BLC_CTL_PHY	BLC_HV	80V			LED_DRVR_DRAIN_2	81
I754	BLC_CTL_PHY	BLC_HV	80V			LED_DRVR_DRAIN_3	81
OUTPUT BUS							
I789	POWER_BLC_RET	BLC_HV	80V			IS1_BLC_F	82
I788	POWER_BLC_RET	BLC_HV	80V			IS2_BLC_F	82
I790	POWER_BLC_RET	BLC_HV	80V			IS3_BLC_F	82
I791	POWER_BLC_RET	BLC_HV	80V			IS1_BLC	81 82
I792	POWER_BLC_RET	BLC_HV	80V			IS2_BLC	81 82
I793	POWER_BLC_RET	BLC_HV	80V			IS3_BLC	81 82
I783	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_1	82
I782	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_1	82
I784	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_2	82
I785	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_2	82
I786	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_3	82
I787	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_3	82
I825	POWER_BLC	BLC_HV	80V			BLC_VOUT1	81 82
I824	POWER_BLC	BLC_HV	80V			BLC_VOUT2	81 82
I823	POWER_BLC	BLC_HV	80V			BLC_VOUT3	81 82

GPU CORE PHASES

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus							
	POWER_PHY	POWER	12V			PP12V_S0_GPUCORE_FLT	96 97 98
	POWER_PHY	POWER	5V			PP5V_S0_GPU_VCORE_VCC	96
Local Ground							
	GND_PHY	GND	0V			AGND_GPU	96
Phase 1							
	POWER_PHY	POWER	12V			REG_LVCC_UB510	97
	POWER_PHY	POWER	12V			REG_UVCC_UB510	97
	VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_1	96 97
	VR_CTL_PHY	VR_CTL				VR_GPU_PWM1_R	96
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_1	97
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_1	97
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_1_RC	97
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_1	97
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_GPUCORE_1	97
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_1	97
	POWER_PHY	POWER	0.9V			PPGPUCORE_S0_SENSE_1	97
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISEN_GCORE_1_P	96 97
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISEN_GCORE_1_N	96 97
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS1_R_P	96
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS1_R_N	96
ISNS_GPU_CORE		SENSE				VR_GPU_ISNS1_RR_2	96
Phase 2							
	POWER_PHY	POWER	12V			REG_LVCC_UB530	97
	VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_2	96 97
	VR_CTL_PHY	VR_CTL				VR_GPU_PWM2_R	96
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_2	97
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_2	97
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_2_RC	97
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_2	97
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_GPUCORE_2	97
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_2	97
	POWER_PHY	POWER	0.9V			PPGPUCORE_S0_SENSE_2	97
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISEN_GCORE_2_P	96 97
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISEN_GCORE_2_N	96 97
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS2_R_P	96
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS2_R_N	96
ISNS_GPU_CORE		SENSE				VR_GPU_ISNS2_RR_2	96
Phase 3							
	POWER_PHY	POWER	12V			REG_VCC_UB550	97
	POWER_PHY	POWER	12V			REG_UVCC_UB550	97
	POWER_PHY	POWER	12V			REG_LVCC_UB550	97
	VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_3	96 97
	VR_CTL_PHY	VR_CTL				VR_GPU_PWM3_R	96
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_3	97
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_3	97
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_3_RC	97
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_3	97
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_GPUCORE_3	97
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_3	97
	POWER_PHY	POWER	0.9V			PPGPUCORE_S0_SENSE_3	97
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISEN_GCORE_3_P	96 97
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISEN_GCORE_3_N	96 97
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS3_R_P	96
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS3_R_N	96
ISNS_GPU_CORE		SENSE				VR_GPU_ISNS3_RR_2	96
Phase 4							
	POWER_PHY	POWER	12V			REG_LVCC_UB650	98
	VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_4	96 98
	VR_CTL_PHY	VR_CTL				VR_GPU_PWM4_R	96
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_4	98
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_4	98
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_4_RC	98
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_4	98
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_GPUCORE_4	98
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_4	98
	POWER_PHY	POWER	0.9V			PPGPUCORE_S0_SENSE_4	98
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISEN_GCORE_4_P	96 98
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISEN_GCORE_4_N	96 98
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS4_R_P	96
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS4_R_N	96
ISNS_GPU_CORE		SENSE				VR_GPU_ISNS4_RR_2	96

GPU CORE CONTROLLER

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
ISL6334							
	VR_CTL_PHY	VR_CTL				VR_GPU_COMP	96
	VR_CTL_PHY	VR_CTL				VR_GPU_COMP_R	96
	VR_CTL_PHY	VR_CTL				VR_GPU_COMP_RC	96
	VR_CTL_PHY	VR_CTL				VR_GPU_FB	96
	VR_CTL_PHY	VR_CTL				VR_GPU_FB_R	96
	VR_CTL_PHY	VR_CTL				VR_GPU_VDIFF	96
	VR_CTL_PHY	VR_CTL				VR_VDF_R1	96
	VR_CTL_PHY	VR_CTL				VR_VDF_R2	96
	VR_CTL_PHY	VR_CTL				VR_GPU_TCOMP	96
	VR_CTL_PHY	VR_CTL				VR_GPU_OFS	96
	VR_CTL_PHY	VR_CTL				VR_GPU_FS	96
		VR_CTL				VR_GPU_EN_VTT	96
		VR_CTL				PM_EN_REG_GPUCORE_S0	64 96
		VR_CTL				PM_PGOOD_REG_GPUCORE_S0	5 64 96
		VR_CTL				GPU_PSI_L	91 96
	VR_CTL_PHY	VR_CTL				VR_GPU_IOUT	
	VR_CTL_PHY	VR_CTL				VR_GPU_IMON	51 96 115
	VR_CTL_PHY	VR_CTL				VR_GPU_FAN	96 115
		VR_CTL				VR_GPU_VRDHOT	96
		VR_CTL				VR_GPU_EN_PWR	
	VR_CTL_PHY	VR_CTL				VR_GPU_SS	96
	VR_CTL_PHY	VR_CTL				VR_GPU_DAC	96
	VR_CTL_PHY	VR_CTL				VR_GPU_REF	96
	VR_CTL_PHY	VR_CTL				VR_GPU_TM	96
	VR_CTL_PHY	VR_CTL				VR_GPU_IMON	51 96 115
	VR_CTL_PHY	VR_CTL				VR_GPU_FAN	96 115
	VR_CTL_PHY	VR_CTL				VR_GPU_VRHOT	
		VR_CTL				GPU_PSI_R	
		VR_CTL				GPU_PSI_L_R	91
	VR_CTL_PHY	VR_CTL				VR_GPU_IMON_R	96
VSNS_GPU_VDD	SNS_DIFF_PHY	SENSE				VSNS_GPU_VDD	93 96
VSNS_GPU_VSS	SNS_DIFF_PHY	SENSE				VSNS_GPU_VSS	93 96
	SNS_DIFF_PHY	SENSE				VR_GPU_VSEN	96
	SNS_DIFF_PHY	SENSE				VR_GPU_RGND	96
GPU VIDS							
		VR_VID				REG_GPUCORE_VID7	91 96
		VR_VID				REG_GPUCORE_VID6	91 96
		VR_VID				REG_GPUCORE_VID5	91 96
		VR_VID				REG_GPUCORE_VID4	91 96
		VR_VID				REG_GPUCORE_VID3	91 96
		VR_VID				REG_GPUCORE_VID2	91 96
		VR_VID				REG_GPUCORE_VID1	91 96
		VR_VID				REG_GPUCORE_VID0	96
		VR_VID				GPU_VCORE_VID6	91
		VR_VID				GPU_VCORE_VID5	91
		VR_VID				GPU_VCORE_VID4	91
		VR_VID				GPU_VCORE_VID3	91
		VR_VID				GPU_VCORE_VID2	91
		VR_VID				GPU_VCORE_VID1	91
		VR_VID				GPU_VCORE_VID0	91
Output Bus							
	POWER_PHY	POWER	0.9V			PPVCORE_S0_GPU	6

GPU FBVDDQ

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus							
	POWER_PHY	POWER	5V			REG_VCC_UB750	99
	POWER_PHY	POWER	5V			REG_PVCC_UB750	99
Local Ground							
	GND_PHY	GND	0V			AGND_FBVDDQ	99
FBVDDQ							
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_FBVDDQ	99
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_FBVDDQ	99
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_FBVDDQ_RC	99
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_FBVDDQ	99
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_FBVDDQ_R	
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_FBVDDQ	99
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_FBVDDQ	99
		SENSE				VSNS_FBVDDQ	94
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VSNS_FBVDDQ_P	94 99
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VSNS_FBVDDQ_N	94 99
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				SNS_FBVDDQ_XW_P	99
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				SNS_FBVDDQ_XW_N	99
		SENSE				FBVDDQ_SENSE_R	94
	VR_CTL_PHY	VR_CTL				REG_FBVDDQ_OCSET	
	VR_CTL_PHY	VR_CTL				REG_FBVDDQ_VO	99
		SENSE				REG_FBVDDQ_FB	99
		SENSE				REG_FBVDDQ_RTN	99
Output Bus							
	POWER_PHY	POWER	1.5V			PP1V5R1V35_S0_GPU_REG	6 99

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE
GPU VREG CONSTRAINTS

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

Caesar IV (Ethernet/SD)

CIV-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
ENET_COMP_PHY	*	ENET_50S
ENET_DIFF_PHY	*	ENET_100D
SD_PHY	*	SD_50S

CIV-specific Spacing Definitions
Ethernet

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF_ISO	*	=6:1_SPACING	?
ENET_DIFF2DIFF	*	=3:1_SPACING	?
ENET_TRANS_ISO	*	1.27 MM	?
COMP_ENET_ISO	*	=4:1_SPACING	?

SD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_ISO	*	=3:1_SPACING	?

Constraints
Ethernet

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	
ENET_DIFF	*	*	ENET_DIFF_ISO	
ENET_DIFF	ENET_DIFF	*	ENET_DIFF2DIFF	
ENET_TRANS	*	*	ENET_TRANS_ISO	2 kV isolation
COMP_ENET	*	*	COMP_ENET_ISO	
ENET_TRANS	ENET_TRANS	*	ENET_DIFF2DIFF	

SD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SD	*	*	SD_ISO

VENI, VIDI, VICI!

	Electrical Contraint Set	Physical	Spacing	Net		
	Ethernet					
I474	ENET_MDI_NORM	ENET_DIFF_PHY	ENET_DIFF	ENETCONN_MDI_P<3..0>		39 40
I475	ENET_MDI_NORM	ENET_DIFF_PHY	ENET_DIFF	ENETCONN_MDI_N<3..0>		39 40
I476		ENET_DIFF_PHY	ENET_TRANS	ENETCONN_MDI_T_P<3..0>		40
I477		ENET_DIFF_PHY	ENET_TRANS	ENETCONN_MDI_T_N<3..0>		40
I504			ENET_TRANS	ENETCONN_MCT3		40
I503			ENET_TRANS	ENETCONN_MCT2		40
I502			ENET_TRANS	ENETCONN_MCT1		40
I501			ENET_TRANS	ENETCONN_MCT0		40
I478		ENET_COMP_PHY	COMP_ENET	ENET_RDAC		39
	SD					
I479	SD_DATA	SD_PHY	SD	ENET_CR_DATA<7..0>	NO_TEST=TRUE	39
I480		SD_PHY	SD	SDCONN_DATA<7..0>		39 41
I499		SD_PHY	SD	SDCONN_DATA_R<7..0>	NO_TEST=TRUE	41
I481	SD_CMD	SD_PHY	SD	ENET_SD_CMD	NO_TEST=TRUE	39
I482		SD_PHY	SD	SDCONN_CMD	NO_TEST=TRUE	39 41
I500		SD_PHY	SD	SDCONN_CMD_R	NO_TEST=TRUE	41
I483	SD_CLK	SD_PHY	SD	ENET_SD_CLK	NO_TEST=TRUE	39
I484		SD_PHY	SD	SDCONN_CLK	NO_TEST=TRUE	39 41
I486		SD_PHY	SD	SDCONN_CLK_R	NO_TEST=TRUE	41
I485		SD_PHY	SD	ENET_MEDIA_SENSE		15 18 39
I488		SD_PHY	SD	ENET_SD_DETECT_L		39 41
I487		SD_PHY	SD	SDCONN_WP		39 41
	CIV SPI					
I490		SPI_50S	SPI	ENET_SCLK		39
I489		SPI_50S	SPI	ENET_MISO		39
I492		SPI_50S	SPI	ENET_MOSI		39
I491		SPI_50S	SPI	ENET_CS_L		39

SYNC_MASTER=D8_MLB SYNC_DATE=07/03/2012

PAGE TITLE: ETHERNET/SD CONSTRAINTS

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH

AUTO-CONSTRAINTS PG 1

4V5_*

	Physical	Spacing	Netname	
I1		PM	4V5_REG_EN	56
I2	POWER_PHY	POWER	4V5_REG_IN	56

ACDC_*

	Spacing	Netname	
I3	PM	ACDC_BURST_EN	48
I4	PM	ACDC_BURST_EN_L	48

ALL_*

	Spacing	Netname	
I5	PM	ALL_SYS_PWRGD	5 47 65

AP_*

	Spacing	Netname	
I6	GENERIC_ISO	AP_CLKREQ_L	15 21
I7	GENERIC_ISO	AP_CLKREQ_L_ISO	15 35
I8	GENERIC_ISO	AP_CLKREQ_Q_L	35
I9	GENERIC_ISO	AP_EVENT_L	35 47 48
I10	GENERIC_ISO	AP_PWR_EN_ISO	15 35
I11	PM	AP_RESET_CONN_L	35
I12	PM	AP_RESET_L	26 35
I13	PM	AP_WAKE_L	35

BLC12V_*

	Spacing	Netname
I14	PM	BLC12V_FAULT
I15	PM	BLC12V_FAULT_L
I16	PM	BLC12V_UVLO
I17	PM	BLC12V_UVLO_OUT
I18	PM	BLC12V_UVLO_REF

BLC_*

	Physical	Spacing	Voltage	Netname	
I19		GENERIC_ISO		BLC_BL	80 82
I20		GENERIC_ISO		BLC_BL_GATE	82
I21		GENERIC_ISO		BLC_BST	80 82
I22		GENERIC_ISO		BLC_BST_R	80
I23		GENERIC_ISO		BLC_BYPASS_GATE	82
I24		GENERIC_ISO		BLC_DIM_MCU	80
I25		PM		BLC_EN	78 80 81
I26		GENERIC_ISO		BLC_ENA	80
I27		GENERIC_ISO		BLC_ENA1	80 82
I28		PM		BLC_EN_DELAY	78
I29		PM		BLC_EN_R	80

BLC_*

	Physical	Spacing	Voltage	Netname	
I30		PM		BLC_EXT_BOOT	48 80
I31		PM		BLC_EXT_BOOT_L	80
I32	BLC_CTL_PHY	BLC_CTL	5V	BLC_MCU_AOUT_R	82
I33		GENERIC_ISO		BLC_MCU_BV	80 82
I34		GENERIC_ISO		BLC_MCU_BV_D	82
I35		GENERIC_ISO		BLC_MCU_BV_R	82
I36	SMB_PHY	SMB		BLC_MCU_B_SDA_CONN	80
I37		GENERIC_ISO		BLC_MCU_FLAG_V	80
I38		GENERIC_ISO		BLC_MCU_PWM5	80
I39		GENERIC_ISO		BLC_MCU_PWM5_R	80
I40		PM		BLC_MCU_RESET	80
I41		PM		BLC_MCU_RESET_L	80
I42		PM		BLC_MCU_RESET_R_L	80
I43	XDP_PHY	CLK_JTAG		BLC_MCU_RTCK	80
I44		GENERIC_ISO		BLC_MCU_RXD0	80
I45	XDP_PHY	XDP		BLC_MCU_TCK	80
I46	XDP_PHY	XDP		BLC_MCU_TDI	80
I47	XDP_PHY	XDP		BLC_MCU_TDO	80
I48	XDP_PHY	XDP		BLC_MCU_TMS	80
I49	XDP_PHY	XDP		BLC_MCU_TRST	80
I50	XDP_PHY	XDP		BLC_MCU_TXD0	80
I51		GENERIC_ISO		BLC_MCU_UVLO	80
I52		GENERIC_ISO		BLC_ON	82
I53		GENERIC_ISO		BLC_ON_DRAIN	82
I54		GENERIC_ISO		BLC_ON_R	82
I55		GENERIC_ISO		BLC_P_ON	80 81 82
I56		GENERIC_ISO		BLC_P_ON_BYPASS	82
I57		GENERIC_ISO		BLC_P_ON_D	82
I58		GENERIC_ISO		BLC_P_ON_DRAIN	82
I59		GENERIC_ISO		BLC_P_ON_D_R	82
I60		GENERIC_ISO		BLC_P_ON_GATE	82
I61		GENERIC_ISO		BLC_P_ON_R	82
I62		GENERIC_ISO		BLC_SKIP	81
I63		GENERIC_ISO		BLC_SNUB_1	81
I64		GENERIC_ISO		BLC_SNUB_2	81
I65		GENERIC_ISO		BLC_SNUB_3	81
I66		GENERIC_ISO		BLC_UVLO	80 82
I67		GENERIC_ISO		BLC_VIN2_GATE	82
I68		GENERIC_ISO		BLC_VIN2_SRC	82
I69		GENERIC_ISO		BLC_VINP_GATE	82
I70		GENERIC_ISO		BLC_VIN_SNS	81
I71		GENERIC_ISO		BLC_VSYNC	78 80
I72		GENERIC_ISO		BLC_VSYNC_R	80

BOOST_*

	Spacing	Netname	
I73	GENERIC_ISO	BOOST_BP	80
I74	GENERIC_ISO	BOOST_BYPASS	80 82
I75	GENERIC_ISO	BOOST_BYPASS_GATE	80
I76	PM	BOOST_EN_GATE	80
I77	PM	BOOST_EN_L	80
I78	GENERIC_ISO	BOOST_RC	80
I79	GENERIC_ISO	BOOST_SS	80

BT_*

	Spacing	Netname	
I80	PM	BT_PWR_EN	35
I81	PM	BT_PWR_RST_L	15 20 35
I82	PM	BT_PWR_RST_L_Q	35

BURSTMODE_*

	Spacing	Netname	
I83	PM	BURSTMODE_EN	71
I84	PM	BURSTMODE_EN_L	6 48 71

CAM_*

	Spacing	Netname	
I85	PM	CAM_EXT_BOOT	42 43
I86	PM	CAM_PROC_RESET	43
I87	PM	CAM_PROC_RESET_L	42 43

CPU_*

	Physical	Spacing	Netname	
I88		PM	CPU_PWRGD_1V05_R	28
I89		PM	CPU_PWRGD_3V3	28
I90		PM	CPU_PWRGD_3V3_R	28
I91		GENERIC_ISO	CPU_SKTOCC	64
I92	CPU_PHY	CPU	CPU_THRMTRIP_3V3	47 48
I93	CPU_PHY	CPU	CPU_THRMTRIP_R_L	48

DEBUG_*

	Spacing	Netname	
I94	PM	DEBUG_RESET_L	26 49

DP_*

	Physical	Spacing	Netname	
I95		PM	DP_AUXIO_EN	15 77 79
I96		PM	DP_GPU_MUX_EN	75
I97		GENERIC_ISO	DP_INTPNL_HPD	75 78
I98		GENERIC_ISO	DP_INT_EG_HPD	75 90
I99	TBT_GEN_55S	TBT_GEN	DP_TBTPA_DDC_CLK	76 77
I100	TBT_GEN_55S	TBT_GEN	DP_TBTPA_DDC_DATA	76 77
I101		GENERIC_ISO	DP_TBTPA_HPD	36 77
I102	TBT_GEN_55S	TBT_GEN	DP_TBTPB_DDC_CLK	76 79
I103	TBT_GEN_55S	TBT_GEN	DP_TBTPB_DDC_DATA	76 79
I104		GENERIC_ISO	DP_TBTPB_HPD	36 79

AUTO-CONSTRAINTS GENERATED Fri 04/06/12 07:46:02 PM

AUTO-CONSTRAINTS PG 2

ENETCONN_*

	Spacing	Netname	
I1	GENERIC_ISO	ENETCONN_TCT	40

ENET_*

	Physical	Spacing	Netname	
I2		GENERIC_ISO	ENET_ACT	40
I3		GENERIC_ISO	ENET_ASF_GPIO	47 48
I4		GENERIC_ISO	ENET_CLKREQ_L	15 18 40
I5		PM	ENET_CLKREQ_L_Q	39 40
I6		PM	ENET_CR_1V8_EN	
I7		PM	ENET_CR_1V8_EN_R	
I8		PM	ENET_CR_3V3_EN_L	
I9		PM	ENET_CR_3V3_EN_L_R	
I10		PM	ENET_CR_PWREN	39 41
I11	XDP_PHY	XDP	ENET_LOW_PWR	26 39 41
I12		PM	ENET_PWR_EN_L	40
I13		PM	ENET_PWR_EN_L_R	40
I14		PM	ENET_RESET_L	39 41
I15		PM	ENET_SD_RESET_L	26 41
I16		GENERIC_ISO	ENET_SR_DISABLE	39
I17		GENERIC_ISO	ENET_SR_LX	39 40
I18		PM	ENET_WAKE_L	40

FAN_*

	Spacing	Netname	
I19	GENERIC_ISO	FAN_0_PWM_FET	54
I20	GENERIC_ISO	FAN_0_PWM_FILT	54
I21	GENERIC_ISO	FAN_0_TACH_FET	54
I22	GENERIC_ISO	FAN_0_TACH_FILT	54

FBVDD_*

	Spacing	Netname	
I23	GENERIC_ISO	FBVDD_ALTVO	91 99

FB_*

	Spacing	Netname	
I24	GENERIC_ISO	FB_A0_VREFC	86
I25	GENERIC_ISO	FB_A0_VREFD	86
I26	GENERIC_ISO	FB_A1_VREFC	86
I27	GENERIC_ISO	FB_A1_VREFD	86
I28	GENERIC_ISO	FB_B0_VREFC	87
I29	GENERIC_ISO	FB_B0_VREFD	87
I30	GENERIC_ISO	FB_B1_VREFC	87
I31	GENERIC_ISO	FB_B1_VREFD	87
I32	GENERIC_ISO	FB_C0_VREFC	88
I33	GENERIC_ISO	FB_C0_VREFD	88
I34	GENERIC_ISO	FB_C1_VREFC	88
I35	GENERIC_ISO	FB_C1_VREFD	88
I36	GENERIC_ISO	FB_D0_VREFC	89
I37	GENERIC_ISO	FB_D0_VREFD	89
I38	GENERIC_ISO	FB_D1_VREFC	89
I39	GENERIC_ISO	FB_D1_VREFD	89
I40	GENERIC_ISO	FB_SW_LEG_A	84 86

FB_*

	Spacing	Netname	
I41	GENERIC_ISO	FB_SW_LEG_B	84 87
I42	GENERIC_ISO	FB_SW_LEG_C	85 88
I43	GENERIC_ISO	FB_SW_LEG_D	85 89
I44	GENERIC_ISO	FB_VREF_GPU	94

FET_*

	Physical	Spacing	Voltage	Netname	
I45		GENERIC_ISO		FET_EN_P12V_S0	74
I46		GENERIC_ISO		FET_EN_P12V_S0_BLC	74
I47		GENERIC_ISO		FET_EN_P12V_S0_BLC_R	74
I48		GENERIC_ISO		FET_EN_P12V_S0_R	74
I49		GENERIC_ISO		FET_EN_P12V_S5	74
I50		GENERIC_ISO		FET_EN_P12V_S5_R	74
I51		GENERIC_ISO		FET_EN_VDDQ_S0	74
I52		GENERIC_ISO		FET_HDD_SLGSW	52
I53	POWER_PHY	POWER	12V	FET_VCC_U7950	74
I54	POWER_PHY	POWER	12V	FET_VCC_U7970	74
I55	POWER_PHY	POWER	12V	FET_VCC_U7980	74

FLAG_*

	Spacing	Netname	
I56	GENERIC_ISO	FLAG_V	80 81 82

G3_*

	Spacing	Netname	
I57	GENERIC_ISO	G3_POWERON_L	47 48

GND_*

	Spacing	Netname	
I58	SENSE	GND_SMC_AVSS	47 48 51 55

GPU_*

	Physical	Spacing	Netname	
I59		GENERIC_ISO	GPU_ALT_VREF	84 85 91
I60		GENERIC_ISO	GPU_BUFRSTN	91
I61		GENERIC_ISO	GPU_IFPAB_PLLVDD	90
I62		GENERIC_ISO	GPU_IFPA_IOVDD	90
I63		GENERIC_ISO	GPU_IFPB_IOVDD	90
I64		GENERIC_ISO	GPU_IFPC_IOVDD	90
I65		GENERIC_ISO	GPU_IFPC_PLLVDD	90
I66	XDP_PHY	XDP	GPU_JTAG_TDI	91
I67	XDP_PHY	XDP	GPU_JTAG_TDO	91
I68	XDP_PHY	XDP	GPU_JTAG_TMS	91
I69	XDP_PHY	XDP	GPU_JTAG_TRST_L	91
I70		GENERIC_ISO	GPU_MLS_STRAP0	91
I71		GENERIC_ISO	GPU_MLS_STRAP1	91

GPU_*

	Physical	Spacing	Netname	
I72		GENERIC_ISO	GPU_MLS_STRAP2	91
I73		GENERIC_ISO	GPU_MLS_STRAP3	91
I74		GENERIC_ISO	GPU_MLS_STRAP4	91
I75		GENERIC_ISO	GPU_RESET_L	26 83 91
I76	SPI_50S	SPI	GPU_ROM_CS_L	91
I77	SPI_50S	SPI	GPU_ROM_CS_L_R	91
I78		GENERIC_ISO	GPU_ROM_HOLD_L	91
I79	SPI_50S	SPI	GPU_ROM_SI	91
I80	SPI_50S	SPI	GPU_ROM_SI_R	91
I81	SPI_50S	SPI	GPU_ROM_SO	91
I82	SPI_50S	SPI	GPU_ROM_SO_R	91
I83	SPI_50S	SPI	GPU_ROM_WP_L	91

HDD_*

	Spacing	Netname	
I84	GENERIC_ISO	HDD_12V_S0_GATE	52
I85	GENERIC_ISO	HDD_OOB_1V00_REF	52
I86	PM	HDD_PWR_EN	21 52
I87	GENERIC_ISO	HDD_PWR_EN_L	52
I88	GENERIC_ISO	HDD_PWR_EN_R	52

I2C_*

	Physical	Spacing	Netname	
I89	SMB_PHY	SMB	I2C_TCON_MAS_SCL	50 78
I90	SMB_PHY	SMB	I2C_TCON_MAS_SDA	50 78

IFPD_*

	Spacing	Netname	
I91	GENERIC_ISO	IFPD_RSET	90

IFPEF_*

	Spacing	Netname	
I92	GENERIC_ISO	IFPEF_RSET	90

ISNSA_*

	Electrical	Physical	Spacing	Netname	
I93	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VG3H_N	51
I94	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VG3H_P	51
I95	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_P1V05_N	55
I96	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_P1V05_P	55
I97	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_VCCSA_N	55
I98	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_VCCSA_P	55
I99	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_FBVDDQ_N	51
I100	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_FBVDDQ_P	51
I101	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_HDD_N	51
I102	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_HDD_P	51

AUTO-CONSTRAINTS GENERATED Fri 04/06/12 07:46:02 PM

PAGE TITLE
AUTO-CONSTRAINTS 2

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D
REVISION 6.1.0
BRANCH

AUTO-CONSTRAINTS PG 3

ISNSA_*

	Electrical	Physical	Spacing	Netname	
I1	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_P1V05_N	55
I2	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_P1V05_P	55
I3	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V05S0_PCH_N	51
I4	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V05S0_PCH_P	51
I5	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V5S0_CPU_MEM_N	51
I6	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V5S0_CPU_MEM_P	51
I7	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S0_SSD_N	51
I8	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S0_SSD_P	51
I9	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S4_AP_N	55
I10	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S4_AP_P	55
I11	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P5VS0_HDD_N	51
I12	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P5VS0_HDD_P	51
I13	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_PVDDQS3_DDR_N	51
I14	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_PVDDQS3_DDR_P	51

ISNS_*

	Physical	Spacing	Netname	
I15	SNS_PHY	SENSE	ISNS_CPUAXG_FB	51
I16	SNS_PHY	SENSE	ISNS_CPUAXG_N	51
I17	SNS_PHY	SENSE	ISNS_CPUAXG_P	51
I18	SNS_PHY	SENSE	ISNS_CPUCORE_FB	51
I19	SNS_PHY	SENSE	ISNS_CPUCORE_N	51
I20	SNS_PHY	SENSE	ISNS_CPUCORE_P	51
I21	SNS_PHY	SENSE	ISNS_GPUCORE_FB	51
I22	SNS_PHY	SENSE	ISNS_GPUCORE_N	51
I23	SNS_PHY	SENSE	ISNS_GPUCORE_P	51
I24	SNS_PHY	SENSE	ISNS_P12VG3H_R	51
I25	SNS_PHY	SENSE	ISNS_P12VS0_CPU_P1V05_R	55
I26	SNS_PHY	SENSE	ISNS_P12VS0_CPU_VCCSA_R	55
I27	SNS_PHY	SENSE	ISNS_P12VS0_FBVDDQ_R	51
I28	SNS_PHY	SENSE	ISNS_P12VS0_HDD_R	51
I29	SNS_PHY	SENSE	ISNS_P12VS0_P1V05_R	55
I30	SNS_PHY	SENSE	ISNS_P1V05S0_PCH_R	51
I31	SNS_PHY	SENSE	ISNS_P1V5S0_CPU_MEM_R	51
I32	SNS_PHY	SENSE	ISNS_P3V3S0_SSD_R	51
I33	SNS_PHY	SENSE	ISNS_P3V3S4_AP_R	55
I34	SNS_PHY	SENSE	ISNS_P5VS0_HDD_R	51
I35	SNS_PHY	SENSE	ISNS_PVDDQS3_DDR_R	51

ISOLATE_*

	Spacing	Netname	
I36	PM	ISOLATE_CPU_MEM_5V	28
I37	PM	ISOLATE_CPU_MEM_5V_L	28

JTAG_*

	Physical	Spacing	Netname	
I38	XDP_PHY	XDP	JTAG_GMUX_TMS	20
I39	XDP_PHY	XDP	JTAG_TBT_TDI	15 21
I40	XDP_PHY	XDP	JTAG_TBT_TDI_ISOL	15 36
I41	XDP_PHY	XDP	JTAG_TBT_TDO	15 21
I42	XDP_PHY	XDP	JTAG_TBT_TDO_ISOL	15 36
I43	XDP_PHY	XDP	JTAG_TBT_TMS	15 18
I44	XDP_PHY	XDP	JTAG_TBT_TMS_ISOL	15 36

LED_*

	Spacing	Netname	
I45	PM	LED_DRIVER_EN	81 82
I46	PM	LED_DRIVER_EN_L	82
I47	PM	LED_DRIVER_EN_L_R	82
I48	GENERIC_ISO	LED_DRIVER_OVP1	81 82
I49	GENERIC_ISO	LED_DRIVER_OVP1P	80 81
I50	GENERIC_ISO	LED_DRIVER_OVP1_OUT	81
I51	GENERIC_ISO	LED_DRIVER_OVP2	81 82
I52	GENERIC_ISO	LED_DRIVER_OVP2P	80 81
I53	GENERIC_ISO	LED_DRIVER_OVP2_OUT	81
I54	GENERIC_ISO	LED_DRIVER_OVP3	81 82
I55	GENERIC_ISO	LED_DRIVER_OVP3P	80 81
I56	GENERIC_ISO	LED_DRIVER_OVP3_OUT	81

LPCPLUS_*

	Spacing	Netname	
I57	GENERIC_ISO	LPCPLUS_GPIO	21 49

LPC_*

	Spacing	Netname	
I58	GENERIC_ISO	LPC_PWRDWN_L	19 26 47 49
I59	GENERIC_ISO	LPC_SERIRQ	18 47 49

MEMVTT_*

	Spacing	Netname	
I60	PM	MEMVTT_EN	28 64
I61	PM	MEMVTT_EN_L	28

MEM_*

	Spacing	Netname	
I62	GENERIC_ISO	MEM_EVENT_L	29 30 31 32 47 48

MOJO_*

	Physical	Spacing	Netname	
I63	XDP_PHY	XDP	MOJO_RX_L	45 47 48
I64	XDP_PHY	XDP	MOJO_TX_L	45 47 48

OCA_*

	Spacing	Netname	
I65	GENERIC_ISO	OCA_FET_DRAIN	82

OVP_*

	Spacing	Netname	
I66	GENERIC_ISO	OVP_OREF	82
I67	GENERIC_ISO	OVP_OUT1	82
I68	GENERIC_ISO	OVP_OUT1_R	82
I69	GENERIC_ISO	OVP_OUT2	82
I70	GENERIC_ISO	OVP_OUT2_R	82
I71	GENERIC_ISO	OVP_OUT3	82
I72	GENERIC_ISO	OVP_OUT3_R	82

P1V2_*

	Spacing	Netname	
I73	PM	P1V2_S4_EN	43

P1V8_*

	Spacing	Netname	
I74	PM	P1V8_S4_EN	43

P3V3AP_*

	Spacing	Netname	
I75	GENERIC_ISO	P3V3AP_VMON	35

P3V3_*

	Spacing	Netname	
I76	GENERIC_ISO	P3V3_S0_EN_G	74
I77	GENERIC_ISO	P3V3_S3_EN_G	74

P3V42G3H_*

	Spacing	Netname	
I78	GENERIC_ISO	P3V42G3H_BOOST	73

P5V_*

	Spacing	Netname	
I79	GENERIC_ISO	P5V_S0_EN_G	74

PCA9557D_*

	Spacing	Netname	
I80	GENERIC_ISO	PCA9557D_RESET_L	26 28 34

AUTO-CONSTRAINTS GENERATED Fri 04/06/12 07:46:02 PM

8 7 6 5 4 3 2 1

D C B A

PAGE TITLE
AUTO-CONSTRAINTS 3

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D

REVISION 6.1.0

BRANCH

AUTO-CONSTRAINTS PG 4

PCH_*

Port	Physical	Spacing	Netname	Pages
I1		PM	PCH_BLC_EXT_BOOT	21 80
I2		PM	PCH_BLC_EXT_BOOT_R	15 21
I3		PM	PCH_BLC_MCU_RESET	21 80
I4		PM	PCH_BLC_MCU_RESET_R	15 21
I5		PM	PCH_CAM_RESET	15 21
I6		PM	PCH_CAM_RESET_R	21 43
I7		GENERIC_ISO	PCH_DSWVRMEN	19
I8	CPU_PHY	CPU	PCH_PECI	21
I9	CPU_PHY	CPU	PCH_PROCPWRGD	21
I10		GENERIC_ISO	PCH_RCIN_L	21
I11		GENERIC_ISO	PCH_RI_L	19
I12		GENERIC_ISO	PCH_SMBALERT_L	15 18
I13		GENERIC_ISO	PCH_SRTCRST_L	18
I14		GENERIC_ISO	PCH_STRP_TOPBLK_SWP_L	20
I15		GENERIC_ISO	PCH_SUSWARN_L	15 19

PCIE_*

Port	Spacing	Netname	Pages
I16	GENERIC_ISO	PCIE_CLKREQ5_GPIO44_L	18
I17	PM	PCIE_WAKE_L	19 35 40

PEG_*

Port	Spacing	Netname	Pages
I18	GENERIC_ISO	PEG_CLKREQ_L	15 18 83

PGOOD_*

Port	Spacing	Netname	Pages
I19	PM	PGOOD_P1V5_S0_DLY	28

PLT_*

Port	Spacing	Netname	Pages
I20	PCH	PLT_RESET_L	20 26
I21	PCH	PLT_RST_BUF_L	26

PM_*

Port	Spacing	Netname	Pages
I22	PM	PM_CLKRUN_L	47 48 49
I23	PM	PM_DSW_PWRGD	47 48 65
I24	PM	PM_EN_FET_P12V_S0	64 74
I25	PM	PM_EN_FET_P12V_S0_R	74
I26	PM	PM_EN_FET_P3V3_S0	64 74
I27	PM	PM_EN_FET_P3V3_S4	64 74
I28	PM	PM_EN_FET_P5V_S0	64 74
I29	PM	PM_EN_FET_VDDQ_S0	64 74
I30	PM	PM_EN_LDO_DDRVTT_S0	64 72
I31	PM	PM_EN_REG_CPUCORE_S0	64 66

PM_*

Port	Spacing	Netname	Pages
I32	PM	PM_EN_REG_CPU_P1V05_S0	64 69
I33	PM	PM_EN_REG_FBVDDQ_S0	64 99
I34	PM	PM_EN_REG_GPUCORE_S0_R	96
I35	PM	PM_EN_REG_P1V05_S0	64 99
I36	PM	PM_EN_REG_P1V8_S0	64 72
I37	PM	PM_EN_REG_P3V3_S5	64 71 74
I38	PM	PM_EN_REG_P5V_S4	64 71
I39	PM	PM_EN_REG_VCCSA_S0	64 70
I40	PM	PM_EN_REG_VDDQ_S3	64 72
I41	PM	PM_EN_S0	64
I42	PM	PM_EN_S4	64
I43	PM	PM_EN_USB_PWR	45 46 64
I44	PM	PM_LED_A_ALL_SYS_PWRGD	5
I45	PM	PM_LED_A_BLC_GOOD	5
I46	PM	PM_LED_A_CPUAXG_PGOOD	5
I47	PM	PM_LED_A_GPU_GOOD	5
I48	PM	PM_LED_A_PGOOD_CPUCORE_S0	5
I49	PM	PM_LED_A_PGOOD_CPU_P1V05_S0	5
I50	PM	PM_LED_A_PGOOD_REG_FBVDDQ_S0	5
I51	PM	PM_LED_A_PGOOD_REG_GPUCORE_S0	5
I52	PM	PM_LED_A_PGOOD_REG_P1V05	5
I53	PM	PM_LED_A_PGOOD_REG_VDDQ_S3	5
I54	PM	PM_LED_A_S4	5
I55	PM	PM_LED_A_S5	5
I56	PM	PM_LED_A_SLP_S3	5
I57	PM	PM_LED_A_VIDEO_ON	5
I58	PM	PM_LED_K_ALL_SYS_PWRGD	5
I59	PM	PM_LED_K_BLC_GOOD	5
I60	PM	PM_LED_K_CPUAXG_PGOOD	5
I61	PM	PM_LED_K_GPU_GOOD	5
I62	PM	PM_LED_K_PGOOD_CPUCORE_S0	5
I63	PM	PM_LED_K_PGOOD_CPU_P1V05_S0	5
I64	PM	PM_LED_K_PGOOD_REG_FBVDDQ_S0	5
I65	PM	PM_LED_K_PGOOD_REG_GPUCORE_S0	5
I66	PM	PM_LED_K_PGOOD_REG_P1V05	5
I67	PM	PM_LED_K_PGOOD_REG_VDDQ_S3	5
I68	PM	PM_LED_K_SLP_S3	5
I69	PM	PM_MEM_PWRGD_L	28
I70	PM	PM_PCH_APWROK	19 65
I71	PM	PM_PCH_PWROK	15 19 26 35 43 65 80
I72	PM	PM_PCH_PWROK_APWROK	65
I73	PM	PM_PCH_SYS_PWROK	19 48 65
I74	PM	PM_PGOOD_FBVDDQ_VDDQ_S0	64
I75	PM	PM_PGOOD_FET_P12V_S0	74
I76	PM	PM_PGOOD_FET_P12V_S0_BLC	74 82
I77	PM	PM_PGOOD_FET_P12V_S5	74
I78	PM	PM_PGOOD_FET_P3V3_S0	64 74
I79	PM	PM_PGOOD_FET_P5V_S0	64 74
I80	PM	PM_PGOOD_FET_VDDQ_S0	28 64 74
I81	PM	PM_PGOOD_P3V3_S4_FET	27 35 64 74
I82	PM	PM_PGOOD_REG_ALL_P1V05_S0	64 65
I83	PM	PM_PGOOD_REG_ALL_P1V05_S0_R	28 64
I84	PM	PM_PGOOD_REG_CPUCORE_S0	5 25 65 66
I85	PM	PM_PGOOD_REG_CPU_P1V05_S0	5 64 69
I86	PM	PM_PGOOD_REG_FBVDDQ_S0	5 64 99
I87	PM	PM_PGOOD_REG_P1V05_S0	5 64 99
I88	PM	PM_PGOOD_REG_P1V8_S0	64 72
I89	PM	PM_PGOOD_REG_P3V3_S5	65 71
I90	PM	PM_PGOOD_REG_P5V_S4	64 71
I91	PM	PM_PGOOD_REG_VCCSA_S0	64 65 70
I92	PM	PM_PGOOD_REG_VDDQ_S3	5 64 72
I93	PM	PM_PWRBTN_L	15 19 25 47
I94	PM	PM_RSMRST_PCH_L	19 65
I95	PM	PM_RSMRST_PCH_L_R	65
I96	PM	PM_SLP_S3_L	5 15 19 28 40 47 48 64
I97	PM	PM_SLP_S4_L	15 19 47 64
I98	PM	PM_SLP_S5_L	15 19 47 64
I99	PCH	PM_SYSRST_L	19 25 26 47
I100	PCH	PM_THRMTRIP_L	21 48
I118	PM	PGOOD_P12V_S0_R	65
I119	PM	PGOOD_P12V_S0	64 65 74

PP12V_*

Port	Physical	Spacing	Voltage	Netname	Pages
I101	POWER_PHY	POWER	12V	PP12V_LCD	78
I102	POWER_PHY	POWER	12V	PP12V_LCD_EXT	78
I103	POWER_PHY	POWER	12V	PP12V_S0_FAN_0_FILT	54
I104	POWER_PHY	POWER	12V	PP12V_S0_HDD_FET	44 52

PP1V05_*

Port	Physical	Spacing	Voltage	Netname	Pages
I105	POWER_PHY	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLA_F	17 22
I106	POWER_PHY	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLB_F	17 22

PP1V2_*

Port	Physical	Spacing	Voltage	Netname	Pages
I107	POWER_PHY	POWER	1.2V	PP1V2_ENET_INTREG	40
I108	POWER_PHY	POWER	1.2V	PP1V2_G3H_SMC_VDDC	47
I109	POWER_PHY	POWER	1.2V	PP1V2_S4_ENET_PHY_AVDDL	39
I110	POWER_PHY	POWER	1.2V	PP1V2_S4_ENET_PHY_GPHYPLL	39
I111	POWER_PHY	POWER	1.2V	PP1V2_S4_ENET_PHY_PCIEPLL	39
I112	POWER_PHY	POWER	1.2V	PP1V2_USB_HUB_CRFILT	27
I113	POWER_PHY	POWER	1.2V	PP1V2_USB_HUB_PLLFILT	27

PP1V5_*

Port	Physical	Spacing	Voltage	Netname	Pages
I114	POWER_PHY	POWER	1.5V	PP1V5_S0_DP_BIAS	75

PP1V8_*

Port	Physical	Spacing	Voltage	Netname	Pages
I115	POWER_PHY	POWER	1.8V	PP1V8_S0_PCH_VCCVRM_F	22 24

PP3V3R1V8_*

Port	Physical	Spacing	Netname	Pages
I117	POWER_PHY	POWER	PP3V3R1V8_ENET_LR_OUT_REG	39

AUTO-CONSTRAINTS GENERATED Fri 04/06/12 07:46:02 PM

PAGE TITLE: AUTO-CONSTRAINTS 4

Apple Inc.

DRAWING NUMBER: 051-9505 SIZE: D

REVISION: 6.1.0

BRANCH:

AUTO-CONSTRAINTS PG 5

PP3V3RHV_*

Physical	Spacing	Voltage	Netname	
POWER_PHY	POWER	12V	PP3V3RHV_SW_TBTAPWR	77
POWER_PHY	POWER	12V	PP3V3RHV_SW_TBTBPWR	79

PP3V3_*

Physical	Spacing	Voltage	Netname	
POWER_PHY	POWER	3.3V	PP3V3_DMIC_CONN	42
POWER_PHY	POWER	3.3V	PP3V3_G3H_AVREF_SMC	47 48
POWER_PHY	POWER	3.3V	PP3V3_G3H_BT_FET	35
POWER_PHY	POWER	3.3V	PP3V3_G3H_BT_FLT	35
POWER_PHY	POWER	3.3V	PP3V3_G3H_SMC_USBMUX_R	45
POWER_PHY	POWER	3.3V	PP3V3_G3H_SMC_VDDA	47
POWER_PHY	POWER	3.3V	PP3V3_G3_RTC	6 26
POWER_PHY	POWER	3.3V	PP3V3_PVDDQS3_ISNS	51
POWER_PHY	POWER	3.3V	PP3V3_S0_PCH_VCCA_DAC_F	17 22
POWER_PHY	POWER	3.3V	PP3V3_S0_SSD_FLT	44
POWER_PHY	POWER	3.3V	PP3V3_S0_SW_SD_PWR	41
POWER_PHY	POWER	3.3V	PP3V3_S4_ALS_F	42
POWER_PHY	POWER	3.3V	PP3V3_S4_AP_FET	35
POWER_PHY	POWER	3.3V	PP3V3_S4_AP_FLT	35
POWER_PHY	POWER	3.3V	PP3V3_S4_ENET_FET_AVDDH	39
POWER_PHY	POWER	3.3V	PP3V3_S4_ENET_FET_BIASVDDH	39
POWER_PHY	POWER	3.3V	PP3V3_S4_ENET_FET_SRVDD	39
POWER_PHY	POWER	3.3V	PP3V3_S4_ENET_FET_XTALVDDH	39
POWER_PHY	POWER	3.3V	PP3V3_S4_USB_HUB_VDD	27
POWER_PHY	POWER	3.3V	PP3V3_S5_XDP_R	25

PP4V5_*

Physical	Spacing	Voltage	Netname	
POWER_PHY	POWER	4.5V	PP4V5_AUDIO_ANALOG	56 60 62

PP5V_*

Physical	Spacing	Voltage	Netname	
POWER_PHY	POWER	5V	PP5V_AUDIO_HPAMP	56 57
POWER_PHY	POWER	5V	PP5V_S0_HDD_FET	44 52
POWER_PHY	POWER	5V	PP5V_S0_PCH_VSREF	24
POWER_PHY	POWER	5V	PP5V_S4_EXTA	45
POWER_PHY	POWER	5V	PP5V_S4_EXTA_F	45
POWER_PHY	POWER	5V	PP5V_S4_EXTB	45
POWER_PHY	POWER	5V	PP5V_S4_EXTB_F	45
POWER_PHY	POWER	5V	PP5V_S4_EXTC	46
POWER_PHY	POWER	5V	PP5V_S4_EXTC_F	46
POWER_PHY	POWER	5V	PP5V_S4_EXTD	46
POWER_PHY	POWER	5V	PP5V_S4_EXTD_F	46
POWER_PHY	POWER	5V	PP5V_S5_PCH_V5REFSUS	24

PPHV_*

Physical	Spacing	Voltage	Netname	
POWER_PHY	POWER	12V	PPHV_SW_TBTAPWR	77
POWER_PHY	POWER	12V	PPHV_SW_TBTBPWR	79

PPVBATT_*

Physical	Spacing	Voltage	Netname	
POWER_PHY	POWER	3.3V	PPVBATT_G3_RTC	26
POWER_PHY	POWER	3.3V	PPVBATT_G3_RTC_R	26

PU_*

Spacing	Netname	
GENERIC_ISO	PU_U6900	64

PWR_*

Spacing	Netname	
PM	PWR_BTN	6 48
PM	PWR_BTN_R	6

REG_*

Physical	Spacing	Netname	
	PM	REG_CPUAXG_PGOOD	5 66
	PM	REG_CPUCORE_PGOOD	66
CPU_PHY	CPU	REG_CPUCORE_VRHOT_L	66
	PM	REG_CPU_P1V05S0_PGOOD	69
VR_CTL_PHY	VR_CTL	REG_FBVDDQ_SET0	99
VR_CTL_PHY	VR_CTL	REG_FBVDDQ_SET1	99
VR_CTL_PHY	VR_CTL	REG_FBVDDQ_SET1_R	99
VR_CTL_PHY	VR_CTL	REG_FBVDDQ_SREF	99
	PM	REG_P1V8S0_PGOOD	72
	PM	REG_P3V3S5_PGOOD	71
	PM	REG_P5VS4_PGOOD	71
	PM	REG_VCCSAS0_PGOOD	70
	PM	REG_VDDQS3_PGOOD	72

RTC_*

Spacing	Netname	
PM	RTC_RESET_L	18 26 48
PM	RTC_RESET_L_R	48

S5_*

Spacing	Netname	
PM	S5_PWRGD	47 65

SATALED_*

Spacing	Netname	
GENERIC_ISO	SATALED_L	44

SDCONN_*

Spacing	Netname	
GENERIC_ISO	SDCONN_DETECT	41
GENERIC_ISO	SDCONN_ILIM	41
GENERIC_ISO	SDCONN_OC_L	41

SD_*

Spacing	Netname	
GENERIC_ISO	SD_DETECT_LVL	41

SLG_*

Spacing	Netname	
PM	SLG_ENET_RESET_L	41
PM	SLG_ENET_RESET_R_L	41

SMBUS_*

Physical	Spacing	Netname	
SMB_PHY	SMB	SMBUS_PCH_CLK	18 50
SMB_PHY	SMB	SMBUS_PCH_CLK_R	50
SMB_PHY	SMB	SMBUS_PCH_DATA	18 50
SMB_PHY	SMB	SMBUS_PCH_DATA_R	50
SMB_PHY	SMB	SMBUS_SMC_0_S0_SCL	47 50
SMB_PHY	SMB	SMBUS_SMC_0_S0_SDA	47 50
SMB_PHY	SMB	SMBUS_SMC_1_S0_SCL	47 50
SMB_PHY	SMB	SMBUS_SMC_1_S0_SDA	47 50
SMB_PHY	SMB	SMBUS_SMC_2_S4_SCL	47 50
SMB_PHY	SMB	SMBUS_SMC_2_S4_SDA	47 50
SMB_PHY	SMB	SMBUS_SMC_3_SCL	47 50
SMB_PHY	SMB	SMBUS_SMC_3_SDA	47 50
SMB_PHY	SMB	SMBUS_SMC_4_ASF_SCL	47 48 50
SMB_PHY	SMB	SMBUS_SMC_4_ASF_SDA	47 48 50
SMB_PHY	SMB	SMBUS_SMC_5_G3H_SCL	47 48 50
SMB_PHY	SMB	SMBUS_SMC_5_G3H_SDA	47 48 50

AUTO-CONSTRAINTS GENERATED Fri 04/06/12 07:46:02 PM

AUTO-CONSTRAINTS PG 6

SMCISNS_*

	Physical	Spacing	Netname	
I1	SNS_PHY	SENSE	SMCISNS_CPUAXG	48 51
I2	SNS_PHY	SENSE	SMCISNS_CPUCORE	48 51
I3	SNS_PHY	SENSE	SMCISNS_GPUCORE	48 51
I4	SNS_PHY	SENSE	SMCISNS_P12VG3H	48 51
I5	SNS_PHY	SENSE	SMCISNS_P12VS0_CPU_P1V05	48 55
I6	SNS_PHY	SENSE	SMCISNS_P12VS0_CPU_VCCSA	48 55
I7	SNS_PHY	SENSE	SMCISNS_P12VS0_FBVDDQ	48 51
I8	SNS_PHY	SENSE	SMCISNS_P12VS0_HDD	48 51
I9	SNS_PHY	SENSE	SMCISNS_P12VS0_P1V05	48 55
I10	SNS_PHY	SENSE	SMCISNS_P1V05S0_PCH	48 51
I11	SNS_PHY	SENSE	SMCISNS_P1V5S0_CPU_MEM	48 51
I12	SNS_PHY	SENSE	SMCISNS_P3V3S0_SSD	48 51
I13	SNS_PHY	SENSE	SMCISNS_P3V3S4_AP	48 55
I14	SNS_PHY	SENSE	SMCISNS_P5VS0_HDD	48 51
I15	SNS_PHY	SENSE	SMCISNS_PVDDQS3_DDR	48 51

SMC_*

	Physical	Spacing	Netname	
I59		GENERIC_ISO	SMC_ROMBOOT	48 49
I60		GENERIC_ISO	SMC_RUNTIME_SCI_L	15 21 47
I61		GENERIC_ISO	SMC_RX_L	47 48 49
I62		PM	SMC_S4_WAKESRC_EN	35 47 48
I63		GENERIC_ISO	SMC_TCK	47 48 49
I64		GENERIC_ISO	SMC_TDI	47 48 49
I65		GENERIC_ISO	SMC_TDO	47 48 49
I66		PM	SMC_THRMTRIP	47 48
I67		GENERIC_ISO	SMC_TMS	47 48 49
I68		GENERIC_ISO	SMC_TO_BLC_RX_L	48
I69		GENERIC_ISO	SMC_TO_BLC_TX_L	48
I70		GENERIC_ISO	SMC_TX_L	47 48 49
I71		PM	SMC_WAKE_L	47
I72		GENERIC_ISO	SMC_WAKE_SCI_L	15 21 47
I73	CLK_XTAL	XTAL	SMC_XTAL	47 48
I74	CLK_XTAL	XTAL	SMC_XTAL_R	48

TBT_*

	Physical	Spacing	Voltage	Netname	
I82		GENERIC_ISO	3.3V	TBT_A_BIAS	77
I83	TBT_GEN_55S	TBT_GEN		TBT_A_CONFIG1_RC	77
I84		PM		TBT_A_HV_EN	36 77
I85		GENERIC_ISO	3.3V	TBT_B_BIAS	79
I86	TBT_GEN_55S	TBT_GEN		TBT_B_CONFIG1_RC	79
I87		PM		TBT_B_HV_EN	36 79
I88		GENERIC_ISO		TBT_CLKREQ_ISOL_L	38
I89		GENERIC_ISO		TBT_CLKREQ_L	15 38
I90		GENERIC_ISO		TBT_DDC_XBAR_EN_L	36 76
I91		GENERIC_ISO		TBT_EN_CIO_PWR	38
I92		PM		TBT_EN_CIO_PWR_L	36 38
I93		GENERIC_ISO		TBT_EN_LC_ISOL	38
I94		GENERIC_ISO		TBT_EN_LC_PWR	36 38
I95		GENERIC_ISO		TBT_PCH_CLKREQ_L	15 21
I96		PM		TBT_PWR_EN	15 26 36
I97		PM		TBT_PWR_EN_PCH	18 26
I98		PM		TBT_PWR_ON_POC_RST_L	36 38
I99		PM		TBT_PWR_REQ_L	15 20 36
I100		PM		TBT_S0_EN	64
I101		PM		TBT_SW_RESET_L	21 38

SMCVSNS_*

	Physical	Spacing	Netname	
I16	SNS_PHY	SENSE	SMCVSNS_CPUAXG	48 51
I17	SNS_PHY	SENSE	SMCVSNS_CPUCORE	48 51
I18	SNS_PHY	SENSE	SMCVSNS_GPUCORE	48 51
I19	SNS_PHY	SENSE	SMCVSNS_P12VG3H	48 51
I20	SNS_PHY	SENSE	SMCVSNS_P1V05S0_PCH	48 51
I21	SNS_PHY	SENSE	SMCVSNS_P1V5S0_CPU_MEM	48 51
I22	SNS_PHY	SENSE	SMCVSNS_P3V3S0	48 51
I23	SNS_PHY	SENSE	SMCVSNS_P5VS0_HDD	48 51
I24	SNS_PHY	SENSE	SMCVSNS_PVDDQS3_DDR	48 51

SML_*

	Physical	Spacing	Netname	
I75	SMB_PHY	SMB	SML_PCH_0_CLK	18 50
I76	SMB_PHY	SMB	SML_PCH_0_DATA	18 50
I77	SMB_PHY	SMB	SML_PCH_1_CLK	18 50
I78	SMB_PHY	SMB	SML_PCH_1_DATA	18 50

TSNS_*

	Electrical	Physical	Spacing	Netname	
I102	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_1_N	53
I103	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_1_P	53
I104	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_2_N	53
I105	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_2_P	53
I106	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_3_N	53
I107	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_3_P	53
I108	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_1_N	53
I109	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_1_P	53
I110	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_2_N	53
I111	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_2_P	53
I112	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_3_N	53
I113	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_3_P	53
I114	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_4_N	53
I115	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_4_P	53
I116	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_5_N	53
I117	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_5_P	53
I118	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_6_N	53
I119	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_6_P	53
I120	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_7_N	53
I121	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_7_P	53
I122	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_ACDC_N	6 53
I123	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_ACDC_P	6 53
I124	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_SKIN_N	53
I125	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_SKIN_P	53

SPI_*

	Spacing	Netname	
I79	GENERIC_ISO	SPI_DESCRIPTOR_OVERRIDE_L	15 47

SMC_*

	Physical	Spacing	Netname	
I25		GENERIC_ISO	SMC_ACDC_ID	6 48
I26		GENERIC_ISO	SMC_ACDC_ID_R	6
I27		PM	SMC_ASSERT_RTCRST	48
I28		GENERIC_ISO	SMC_BLC_MUX_RX_L	48 80
I29		GENERIC_ISO	SMC_BLC_MUX_TX_L	48 80
I30		GENERIC_ISO	SMC_CPU_CATERR_L	47 48
I31	CPU_PHY	CPU	SMC_CPU_PECI	48
I32		PM	SMC_DELAYED_PWRGD	38 47 48 65
I33		GENERIC_ISO	SMC_DP_HPD_L	47 48
I34	CLK_XTAL	XTAL	SMC_EXTAL	47 48
I35		GENERIC_ISO	SMC_FAN_0_CTL	47 54
I36		GENERIC_ISO	SMC_FAN_0_TACH	47 54
I37		GENERIC_ISO	SMC_GFX_OVERTEMP	47 48 91
I38		GENERIC_ISO	SMC_GFX_OVERTEMP_Q	91
I39		GENERIC_ISO	SMC_GFX_OVERTEMP_R_L	91
I40		PM	SMC_GFX_THROTTLE_L	47 91
I41		PM	SMC_GFX_THROTTLE_R_L	91
I42		PM	SMC_LRESET_L	26 47
I43		PM	SMC_MANUAL_RST_L	48
I44		PM	SMC_ONOFF_L	47 48
I45		SENSE	SMC_OOB1_RX_CN	44
I46		SENSE	SMC_OOB1_RX_FILT	44 52
I47		SENSE	SMC_OOB1_RX_L	47 52
I48		SENSE	SMC_OOB1_RX_R	52
I49		SENSE	SMC_OOB1_TX_L	47 48
I50		SENSE	SMC_OOB2_RX_L	44 48
I51		SENSE	SMC_OOB2_TX_L	44 48
I52	CPU_PHY	CPU	SMC_PECI_L	47 48
I53	CPU_PHY	CPU	SMC_PECI_L_R	48
I54		PM	SMC_PME_S4_WAKE_L	35 47 48
I55		GENERIC_ISO	SMC_PM_G2_EN	47 48 74
I56		PM	SMC_PM_PCH_SYS_PWROK	47 48
I57		CPU	SMC_PROCHOT	47 48
I58		PM	SMC_RESET_L	47 48 49

SSD_*

	Spacing	Netname	
I80	GENERIC_ISO	SSD_CLKREQ_L	15 18

SYS_*

	Spacing	Netname	
I81	PM	SYS_PWROK_R	65

UVP_*

	Spacing	Netname	
I126	PM	UVP_IN_1	82
I127	PM	UVP_IN_1_REF	82
I128	PM	UVP_IN_2	82
I129	PM	UVP_IN_3	82
I130	PM	UVP_IN_4	82
I131	PM	UVP_REF	82

AUTO-CONSTRAINTS GENERATED Fri 04/06/12 07:46:02 PM

PAGE TITLE
AUTO-CONSTRAINTS 6

Apple Inc.

DRAWING NUMBER 051-9505 SIZE D
REVISION 6.1.0
BRANCH
PAGE 143 OF 144
SHEET 122 OF 123

AUTO-CONSTRAINTS PG 7

VREFMRGN_*

Port	Spacing	Netname	Pages
I1	GENERIC_ISO	VREFMRGN_CA_SODIMMA_EN	34
I2	GENERIC_ISO	VREFMRGN_CA_SODIMMB_EN	34

VTTCLAMP_*

Port	Spacing	Netname	Pages
I3	GENERIC_ISO	VTTCLAMP_EN	28
I4	GENERIC_ISO	VTTCLAMP_L	28

WOL_*

Port	Spacing	Netname	Pages
I5	GENERIC_ISO	WOL_EN	15 21 40

AUTO-CONSTRAINTS GENERATED Fri 04/06/12 07:46:02 PM

PAGE TITLE
AUTO-CONSTRAINTS 7

Apple Inc.

DRAWING NUMBER 051-9505
SIZE D
REVISION 6.1.0
BRANCH